

Integration and measure theory

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Abstract

We develop measure and integration from the abstract framework of measurable spaces and measures and then move toward the classical Lebesgue theory on $\mathbb{R}^N$. After introducing σ -algebras, Borel sets, and measures, we study measurable functions, simple functions, positive integration, and the main convergence theorems. We then turn to several fundamental applications: integrals with parameters, product measures, the Tonelli and Fubini theorems, modes of convergence, absolute continuity of measures and the Radon–Nikodym theorem, and finally the fundamental theorem of calculus together with one-dimensional and multidimensional change-of-variables formulas. In the Euclidean setting we discuss Lebesgue measure from two complementary viewpoints: the classical Carathéodory construction from rectangles and, later on, a construction through Radon measures and the Riesz representation theorem.

The notes also develop the structural side of measure theory. We introduce signed measures, Hahn and Jordan decompositions, total variation, absolute continuity and singularity, Lebesgue decomposition, and the Radon–Nikodym theorem. We then study positive and bounded Radon measures, the Riesz representation theorem, Lusin’s theorem, weak convergence of measures, and the Lebesgue spaces $L^p(\Omega, \mu)$. This includes density results, convolution and mollifiers in the Euclidean setting, duality, weak topologies, and weak compactness statements. The aim is to provide a first coherent course in measure theory at bachelor level, while already preparing later developments in probability, partial differential equations, and functional analysis. We refer to the classical textbooks [5, 2, 7, 6, 3, 1] for complementary expositions.

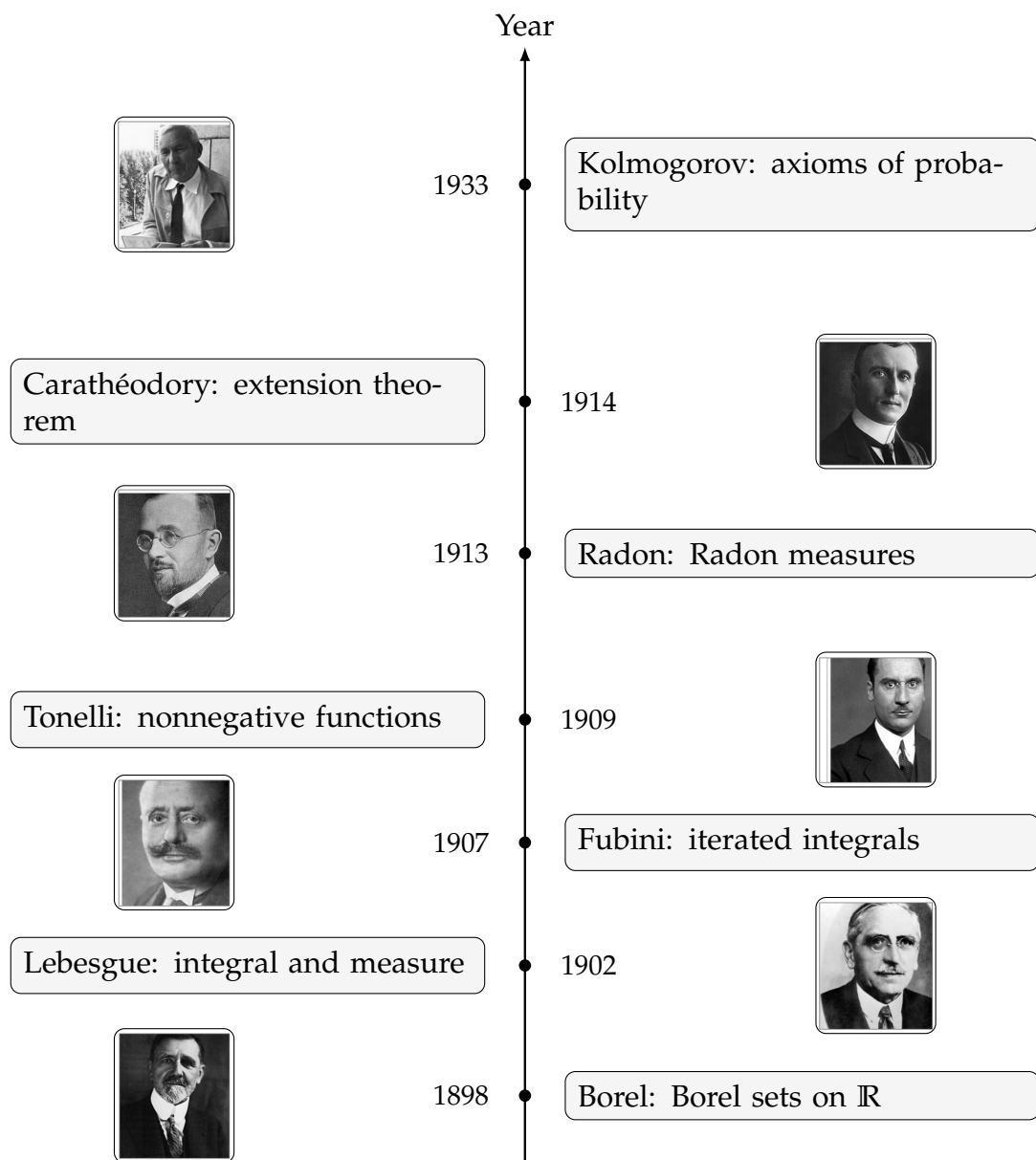
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Some milestones in the development of measure theory



A few landmarks and protagonists that will reappear throughout the course.

Chapter 1

σ -algebras

We introduce the collections of subsets that will serve as measurable sets throughout the notes. The chapter starts from the abstract definition, then explains how σ -algebras are generated, how Borel σ -algebras arise from topology, and why the completed real line will be useful for extended-valued functions.

1.1 First definitions

This is the first fundamental notion in measure theory. The elements of a σ -algebra are precisely the subsets of a set that will be declared measurable. A measure will later assign sizes only to the sets in such a collection.

In the following, if X is a set, we denote by $\mathcal{P}(X)$ its power set, that is, the set of all subsets of X , the empty set $\emptyset$ and the set X being included. If $A \subset X$, we denote by $A^c := X \setminus A$ its complement in X .

Definition 1.1 (σ -algebras). *Let X be a set. A collection $\mathcal{A} \subset \mathcal{P}(X)$ is a σ -algebra on X if the following properties are satisfied:*

- 1) $X \in \mathcal{A}$,
- 2) if $A \in \mathcal{A}$ then $A^c \in \mathcal{A}$,
- 3) for every family $(A_i)_{i \in I}$ of elements of $\mathcal{A}$ indexed by a countable set I , we have $\bigcup_{i \in I} A_i \in \mathcal{A}$.

Definition 1.2 (Measurable spaces). *Let X be a set and let $\mathcal{A}$ be a σ -algebra on X . The pair $(X, \mathcal{A})$ is called a measurable space.*

The following picture illustrates the distinction between the underlying set X and a σ -algebra on X , which is a family of subsets of X .

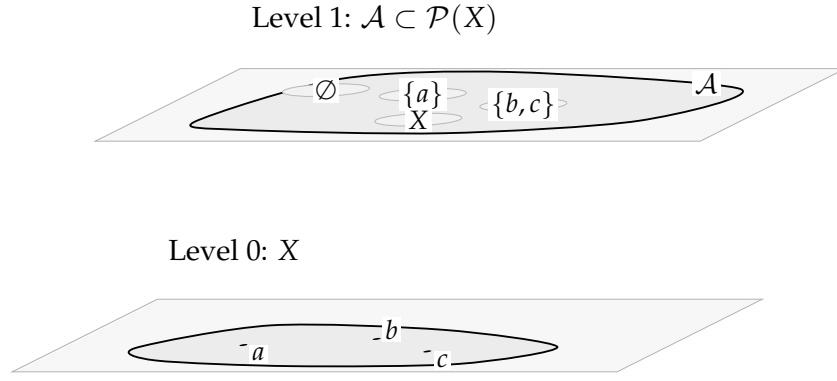


Figure 1.1: A set X with three elements a , b , and c on Level 0, and a σ -algebra $\mathcal{A} := \{\emptyset, \{a\}, \{b, c\}, X\} \subset \mathcal{P}(X)$ on Level 1.

Remark 1.3 (Countable and uncountable sets). Recall that a set E is *countable* if it is finite or if there exists a bijection from $\mathbb{N}$ onto E . Equivalently, E injects into $\mathbb{N}$. Otherwise, E is said to be *uncountable*. Every subset of $\mathbb{N}$ is countable. Finite products of countable sets, countable unions of countable sets, and the set of rational numbers $\mathbb{Q}$ are countable. On the other hand, Cantor's theorem shows that the set of real numbers $\mathbb{R}$ is uncountable. Consequently, the set $\mathbb{R} \setminus \mathbb{Q}$ of irrational numbers is uncountable. Finally, note that if E is a countable infinite set, then there exists a bijection $\varphi: \mathbb{N} \rightarrow E$, so that the elements of E can be enumerated as $E = \{e_1, e_2, \dots, e_n, \dots\}$.

Let us give two examples of σ -algebras available for any set X :

- the set $\mathcal{P}(X)$ is a σ -algebra on X ,
- the set $\{\emptyset, X\}$ is also a σ -algebra on X .

1.2 Main properties

Here are a few properties easily derived from the definition.

Lemma 1.4 (Elementary properties of σ -algebras). *Let X be a set and $\mathcal{A}$ a σ -algebra on X . We have the following:*

- (a) *The empty set $\emptyset$ is in $\mathcal{A}$,*
- (b) *any countable intersection of elements of $\mathcal{A}$ is still an element of $\mathcal{A}$,*
- (c) *if $A, B \in \mathcal{A}$ then $A \setminus B \in \mathcal{A}$.*

Proof. To prove (a), combine axioms 1) and 2). For (b), if $(A_i)_{i \in I}$ is a countable family in $\mathcal{A}$, then De Morgan's formula gives

$$\bigcap_{i \in I} A_i = \left(\bigcup_{i \in I} A_i^c \right)^c,$$

and axioms 2) and 3) apply. For (c), note that $A \setminus B = A \cap B^c$ and use (b) and axiom 2). $\square$

Operation / axiom	σ -algebra $\mathcal{A}$	Topology $\mathcal{T}$
$\emptyset$ and X belong to the family	Yes	Yes
Closed under complements	Yes	No (not required, not true in general)
Closed under finite unions	Yes (follows)	Yes (follows from arbitrary unions)
Closed under countable unions	Yes (axiom)	Yes (since arbitrary unions are allowed)
Closed under arbitrary unions	No in general	Yes (axiom)
Closed under finite intersections	Yes (follows)	Yes (axiom)
Closed under countable intersections	Yes (follows)	No in general

Table 1.1: A comparison between the axioms of a σ -algebra and those of a topology.

The following result states that any intersection of σ -algebras is a σ -algebra.

Lemma 1.5 (Intersections of σ -algebras). *Let X be a set. Let $I \neq \emptyset$ be any set of indices and let $(\mathcal{A}_i)_{i \in I}$ be a family of σ -algebras on X , then the intersection*

$$\bigcap_{i \in I} \mathcal{A}_i := \{A \subset X : \forall i \in I, A \in \mathcal{A}_i\},$$

is a σ -algebra on X .

Proof. The set X belongs to every $\mathcal{A}_i$, hence to the intersection. If A belongs to the intersection, then A belongs to every $\mathcal{A}_i$, and therefore A^c belongs to every $\mathcal{A}_i$. Finally, if $(A_n)_{n \geq 1}$ is a sequence of elements of the intersection, then each A_n belongs to every $\mathcal{A}_i$, so $\bigcup_{n \geq 1} A_n$ belongs to every $\mathcal{A}_i$. Thus the intersection is a σ -algebra. $\square$

In particular, this result allows us to prove the following consequence.

Lemma 1.6 (Generated σ -algebra). *Let X be a set and let $\mathcal{E} \subset \mathcal{P}(X)$. Let $\mathcal{S}$ denote the set of all σ -algebras on X containing $\mathcal{E}$. We set*

$$\sigma(\mathcal{E}) := \bigcap_{\mathcal{B} \in \mathcal{S}} \mathcal{B}.$$

Then $\sigma(\mathcal{E})$ is the smallest σ -algebra on X containing $\mathcal{E}$.

The σ -algebra $\sigma(\mathcal{E})$ is called the σ -algebra generated by $\mathcal{E}$ on X . To say that $\sigma(\mathcal{E})$ is the smallest σ -algebra on X containing $\mathcal{E}$ means that $\sigma(\mathcal{E})$ is a σ -algebra on X , that $\mathcal{E} \subset \sigma(\mathcal{E})$, and that every σ -algebra $\mathcal{A}$ on X containing $\mathcal{E}$ satisfies $\sigma(\mathcal{E}) \subset \mathcal{A}$.

Proof. We observe that $\mathcal{S} \neq \emptyset$ because $\mathcal{P}(X) \in \mathcal{S}$. By the previous lemma, $\sigma(\mathcal{E})$ is a σ -algebra on X . Moreover, for every $\mathcal{B}$ in $\mathcal{S}$, we have $\mathcal{E} \subset \mathcal{B}$. Therefore $\mathcal{E} \subset \sigma(\mathcal{E})$. Thus $\sigma(\mathcal{E})$ is a σ -algebra on X and contains $\mathcal{E}$.

Furthermore, if $\mathcal{A}$ is another σ -algebra on X containing $\mathcal{E}$, then $\mathcal{A} \in \mathcal{S}$. Since $\sigma(\mathcal{E})$ is the intersection of all elements of $\mathcal{S}$, we have $\sigma(\mathcal{E}) \subset \mathcal{A}$. $\square$

1.3 σ -algebra associated with a topological space

It is important not to mix the axioms defining the notion of σ -algebra with those defining the notion of topology, especially since the following definition canonically associates a σ -algebra with any topology.

Definition 1.7 (Borel σ -algebra). *Let (X, τ) be a topological space (i.e. τ forms the collection of open sets of X). The Borel σ -algebra of X relative to the topology τ is*

$$\mathcal{B}(X) := \sigma(\tau).$$

An element of $\mathcal{B}(X)$ is a Borel set of X (relative to τ). When the topology is clear from the context, we suppress it from the notation.

The following properties are easy to obtain.

Proposition 1.8 (Elementary properties of Borel σ -algebras). *Let (X, τ) be a topological space, and let $\mathcal{B}$ be the associated Borel σ -algebra. We have the following properties.*

- (i) *Any closed subset of X is an element of $\mathcal{B}$,*
- (ii) *Any countable union of closed subsets of X is an element of $\mathcal{B}$,*
- (iii) *Any countable intersection of open sets of X is an element of $\mathcal{B}$.*

Proof. If F is closed, then F^c is open and therefore belongs to $\tau \subset \mathcal{B}$. Since $\mathcal{B}$ is stable by complements, F belongs to $\mathcal{B}$. Countable unions of closed sets then belong to $\mathcal{B}$ by countable stability. Finally, if $(U_n)_{n \geq 1}$ is a sequence of open sets, then

$$\bigcap_{n=1}^{\infty} U_n = \left(\bigcup_{n=1}^{\infty} U_n^c \right)^c,$$

and the right-hand side is Borel. □

In the special case of the Euclidean space $\mathbb{R}^n$, for $n \geq 1$, let $\mathcal{D}_n$ be the family of open blocks

$$D((x_i)_i, (y_i)_i) := \{(z_1, z_2, \dots, z_n) \in \mathbb{R}^n : \forall i, x_i < z_i < y_i\},$$

where the pairs $((x_i)_i, (y_i)_i)$ vary among the n -tuples of real numbers satisfying $x_i < y_i$ for $1 \leq i \leq n$. Then

$$\mathcal{B}(\mathbb{R}^n) = \sigma(\mathcal{D}_n).$$

Indeed, the subfamily of blocks with rational endpoints is countable and forms a basis of the Euclidean topology: for every open set U and every x in U , there is a rational block D such that x is in $D \subset U$. Hence every open set is the union of the rational blocks contained in it, and this is a countable union. Since every block is open, this proves that the open sets generate exactly $\sigma(\mathcal{D}_n)$. In particular, in the case $n = 1$, the Borel σ -algebra of $\mathbb{R}$ is $\sigma(\mathcal{I})$, where $\mathcal{I}$ is the family of open intervals of $\mathbb{R}$. Recall that the usual absolute value $|\cdot|$ equips $\mathbb{R}$ with a metric topology that we call the *standard topology* of $\mathbb{R}$. If we fix any norm $\|\cdot\|$ on $\mathbb{R}^n$, then $\mathbb{R}^n$ is endowed with the associated metric topology. This topology is independent of the chosen norm, since all norms are equivalent on the same finite-dimensional space. We call it the *standard topology* of $\mathbb{R}^n$.

1.4 A few words on the completed real line

In what follows, we will have to consider functions that can take on infinite values (think, for example, of the function $x \mapsto 1/\sqrt{x}$, which is integrable on $]0, 1]$ and tends to $+\infty$ as $x \rightarrow 0^+$). We define

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\},$$

and generalized “intervals”:

$$]a, +\infty] =]a, +\infty[\cup \{+\infty\}, \quad [-\infty, b[=]-\infty, b[\cup \{-\infty\},$$

$$[a, +\infty] = [a, +\infty[\cup \{+\infty\}, \quad [-\infty, b] =]-\infty, b[\cup \{-\infty\},$$

for all $a, b \in \mathbb{R}$. We endow $\overline{\mathbb{R}}$ with the topology generated by the basis consisting of the intervals

$$]a, b[, \quad [-\infty, b[, \quad]a, +\infty],$$

with a, b in $\mathbb{R}$ and $a < b$. Equivalently, it is enough to use rational endpoints; this gives a countable basis. This topology is called the *standard topology* of $\overline{\mathbb{R}}$. If $\mathcal{G}$ denotes the family consisting of the open intervals of $\mathbb{R}$ and the intervals $]a, +\infty]$ and $[-\infty, b[$, for all a, b in $\mathbb{R}$, then

$$\mathcal{B}(\overline{\mathbb{R}}) = \sigma(\mathcal{G}).$$

Indeed, the topology is generated by $\mathcal{G}$, and the rational-endpoint subfamily shows that every open subset of $\overline{\mathbb{R}}$ is a countable union of elements of $\mathcal{G}$.

Next chapter introduces the concept of a measure on a measurable space. Such measures can take infinite values, *i.e.* values in $[0, +\infty] \subset \overline{\mathbb{R}}$. To do this, we set out some conventions for calculations in $[0, \infty]$: for $a \in \mathbb{R}_{\geq 0}$,

$$a + \infty := \infty, \quad \infty + a := \infty.$$

Also,

$$a \times \infty := \infty, \quad \infty \times a := \infty \quad \text{for } a > 0,$$

whereas

$$0 \times \infty := 0, \quad \infty \times 0 := 0.$$

With these conventions, the algebraic identities used below for nonnegative extended quantities remain valid on $[0, \infty]$. Another convenient consequence of “adding ∞ ” to $[0, \infty[$ is that any *increasing* sequence in $[0, \infty]$ converges. We can even extend the usual properties, as follows.

Proposition 1.9 (Increasing limits in $\{0 \leq x \leq \infty\}$). *If $(a_n)_{n \geq 1}$ and $(b_n)_{n \geq 1}$ are two increasing sequences in $[0, \infty]$ with limits ℓ_a, ℓ_b in $[0, \infty]$ respectively, then the sequences $(a_n + b_n)_{n \geq 1}$ and $(a_n b_n)_{n \geq 1}$ converge in $[0, +\infty]$ and*

$$\lim_{n \rightarrow \infty} (a_n + b_n) = \ell_a + \ell_b, \quad \lim_{n \rightarrow \infty} (a_n b_n) = \ell_a \times \ell_b.$$

Proof. The sequence $(a_n + b_n)_{n \geq 1}$ is increasing and has limit $\ell_a + \ell_b$. If both limits are finite, this is the usual rule for limits in $\mathbb{R}$. If one limit is $+\infty$, the conclusion follows from monotonicity and the definition of convergence to $+\infty$.

For the product, the sequence $(a_n b_n)_{n \geq 1}$ is increasing. If ℓ_a and ℓ_b are finite, the usual real-valued rule applies. If, for instance, $\ell_a = +\infty$ and $\ell_b > 0$, choose $c > 0$ and n_0 such that $b_n \geq c$ for $n \geq n_0$; then $a_n b_n \geq c a_n$ tends to $+\infty$. The case $\ell_b = +\infty$ and $\ell_a > 0$ is symmetric. Finally, if $\ell_b = 0$, then $b_n = 0$ for every n because (b_n) is increasing and nonnegative; hence $a_n b_n = 0$ for every n . The case $\ell_a = 0$ is symmetric. This proves the formula in all cases, with the convention $0 \times \infty = 0$. $\square$

Chapter 2

Measures

We now equip measurable spaces with countably additive set functions. After establishing the basic calculus of measures, we introduce negligible sets, completion, and regularity notions, and we conclude with a first look at the Lebesgue measure, which will remain the model example throughout the course.

2.1 First definitions

Definition 2.1 (Measures). *Let $(X, \mathcal{A})$ be a measurable space. A measure on $(X, \mathcal{A})$ is a map $\mu: \mathcal{A} \rightarrow [0, \infty]$ verifying:*

- 1) $\mu(\emptyset) = 0$,
- 2) *for every sequence $(A_i)_{i \geq 1}$ of pairwise disjoint elements of $\mathcal{A}$, we have*

$$\mu \left(\bigcup_{i=1}^{\infty} A_i \right) = \sum_{i=1}^{\infty} \mu(A_i).$$

A measure space is a triple $(X, \mathcal{A}, \mu)$ where $(X, \mathcal{A})$ is a measurable space and μ is a measure on $(X, \mathcal{A})$.

The following picture illustrates that a measure assigns a value in $[0, +\infty]$ to each measurable set.

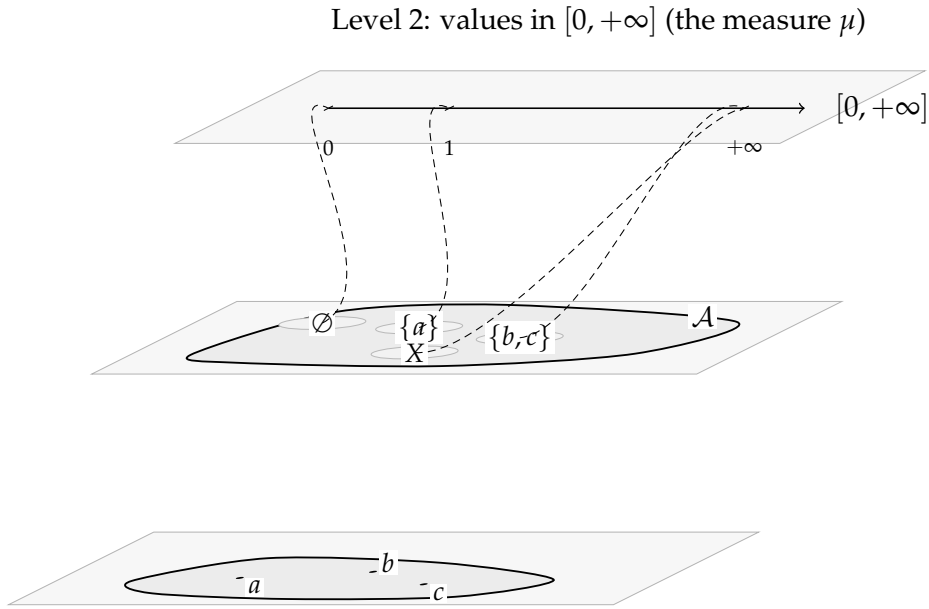


Figure 2.1: A three-level picture: the initial set X , a σ -algebra $\mathcal{A}$ of measurable sets, and a measure μ which assigns the values $\mu(\emptyset) = 0$, $\mu(\{a\}) = 1$, $\mu(\{b, c\}) = +\infty$, and $\mu(X) = +\infty$.

Let us look at a few examples.

- Firstly, $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$ is a measurable space, corresponding to the case of the so-called discrete σ -algebra. The *counting measure* is defined by

$$\#(A) = \text{card}(A)$$

when A is finite, and by $\#(A) = +\infty$ when A is infinite. It is a measure on $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$.

- Let $(X, \mathcal{A})$ be a measurable space, and let x_0 be in X . We define $\delta_{x_0}: \mathcal{A} \rightarrow [0, \infty]$ by $\delta_{x_0}(A) = 1$ if x_0 is in A and $\delta_{x_0}(A) = 0$, otherwise. The map δ_{x_0} is a measure, called the *Dirac measure at x_0* .

2.2 Main properties

We now give the main properties of measures.

Theorem 2.2 (Basic properties of measures). *Let $(X, \mathcal{A}, \mu)$ be a measure space.*

1) *Let n in $\mathbb{N}_{\geq 1}$ and let $A_1, \dots, A_n$ be pairwise disjoint elements of $\mathcal{A}$. Then*

$$\mu\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \mu(A_i).$$

2) *If $B_1, \dots, B_n$ are elements of $\mathcal{A}$, then*

$$\mu\left(\bigcup_{i=1}^n B_i\right) \leq \sum_{i=1}^n \mu(B_i).$$

3) *Let A, B in $\mathcal{A}$ such that $A \subset B$. Then $\mu(A) \leq \mu(B)$.*

4) *If $(B_i)_{i \geq 1}$ is a sequence of elements of $\mathcal{A}$, then*

$$\mu\left(\bigcup_{i=1}^{\infty} B_i\right) \leq \sum_{i=1}^{\infty} \mu(B_i).$$

Proof. 1) We complete the family by setting $A_i = \emptyset$ for $i \geq n + 1$. The sequence $(A_i)_{i \geq 1}$ is pairwise disjoint, so countable additivity gives

$$\mu\left(\bigcup_{i=1}^n A_i\right) = \mu\left(\bigcup_{i \geq 1} A_i\right) = \sum_{i \geq 1} \mu(A_i) = \sum_{i=1}^n \mu(A_i).$$

2) We introduce an auxiliary family $(C_i)_{1 \leq i \leq n}$ by setting

$$C_1 := B_1, \quad C_i := B_i \setminus \bigcup_{j=1}^{i-1} B_j, \quad (2 \leq i \leq n).$$

Then each C_i is in $\mathcal{A}$, the sets C_i are pairwise disjoint, and

$$\bigcup_{i=1}^n B_i = \bigcup_{i=1}^n C_i.$$

Therefore, by 1),

$$\mu\left(\bigcup_{i=1}^n B_i\right) = \mu\left(\bigcup_{i=1}^n C_i\right) = \sum_{i=1}^n \mu(C_i).$$

Moreover, for $i = 1$ we have $\mu(C_1) = \mu(B_1)$. For $i \geq 2$, we have $C_i \subset B_i$ and

$$B_i = C_i \cup \left(B_i \cap \bigcup_{j=1}^{i-1} B_j\right),$$

where the union is disjoint. Another application of 1) therefore gives

$$\mu(B_i) = \mu(C_i) + \mu\left(B_i \cap \bigcup_{j=1}^{i-1} B_j\right) \geq \mu(C_i).$$

By summing over i we conclude that

$$\mu\left(\bigcup_{i=1}^n B_i\right) = \sum_{i=1}^n \mu(C_i) \leq \sum_{i=1}^n \mu(B_i).$$

3) Since $B \setminus A$ is in $\mathcal{A}$ and

$$B = A \cup (B \setminus A),$$

with a disjoint union, 1) gives

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A).$$

4) Define

$$C_1 := B_1, \quad C_i := B_i \setminus \bigcup_{j=1}^{i-1} B_j \quad (i \geq 2).$$

Then the sets C_i are pairwise disjoint, $C_i \subset B_i$, and $\bigcup_{i \geq 1} B_i = \bigcup_{i \geq 1} C_i$. By countable additivity and monotonicity,

$$\mu\left(\bigcup_{i=1}^{\infty} B_i\right) = \sum_{i=1}^{\infty} \mu(C_i) \leq \sum_{i=1}^{\infty} \mu(B_i).$$

□

One should pay attention to the lack of symmetry in the assumptions of the following couple of results.

Theorem 2.3 (Continuity properties of measures). *Let $(X, \mathcal{A}, \mu)$ be a measure space.*

1) *Let $(A_i)_{i \geq 1}$ be an increasing sequence of elements of $\mathcal{A}$ (i.e. $A_i \subset A_{i+1}$ for all i), then the sequence $(\mu(A_i))_i$ converges in $[0, \infty]$ and*

$$\lim_{N \rightarrow \infty} \mu(A_N) = \mu\left(\bigcup_{i \geq 1} A_i\right).$$

2) *Let $(A_i)_{i \geq 1}$ be a decreasing sequence of elements of $\mathcal{A}$ (i.e. $A_{i+1} \subset A_i$ for all i), then the sequence $(\mu(A_i))_i$ converges in $[0, \infty]$. Moreover, if there exists $i_0 \geq 1$ such that $\mu(A_{i_0}) < \infty$, then*

$$\lim_{i \rightarrow \infty} \mu(A_i) = \mu\left(\bigcap_{i \geq 1} A_i\right).$$

Proof. **1) Step 1. Increasing sequences.** The convergence of $(\mu(A_i))_i$ is justified by its growth in $[0, \infty]$. We will associate an auxiliary sequence $(B_i)_{i \geq 1}$ with the sequence $(A_i)_{i \geq 1}$ as follows: set $B_1 = A_1$, and $B_i = A_i \setminus A_{i-1}$, if $i \geq 2$. We see that B_i are elements of $\mathcal{A}$ and that by construction A_N is the disjoint union of B_k for $1 \leq k \leq N$, so that

$$\sum_{k=1}^N \mu(B_k) = \mu(A_N).$$

Finally,

$$\bigcup_{i \geq 1} A_i = \bigcup_{k \geq 1} B_k.$$

Therefore

$$\mu\left(\bigcup_{i \geq 1} A_i\right) = \mu\left(\bigcup_{k \geq 1} B_k\right) = \sum_{k \geq 1} \mu(B_k) = \lim_{N \rightarrow \infty} \sum_{k=1}^N \mu(B_k) = \lim_{N \rightarrow \infty} \mu(A_N).$$

2) Step 2. Decreasing sequences. Replacing the sequence by the tail $(A_{i_0+k})_{k \geq 0}$, one may suppose that $i_0 = 1$. We set

$$A := \bigcap_{i \geq 1} A_i, \quad C_i := A_i \setminus A_{i+1}, \quad (i \geq 1).$$

Then $A \in \mathcal{A}$ and $C_i \in \mathcal{A}$ for all i . We claim that, for every $n \geq 1$,

$$A_n = A \cup \left(\bigcup_{i \geq n} C_i \right).$$

Let us prove this equality carefully. If x is in A , then x is in A_n . If x is in C_i for some $i \geq n$, then x is in A_i , and since the sequence $(A_i)_i$ is decreasing we have $A_i \subset A_n$. Therefore

$$A \cup \left(\bigcup_{i \geq n} C_i \right) \subset A_n.$$

Conversely, let x be in A_n . If x is in A , there is nothing to prove. Assume that x is not in A . Since $A = \bigcap_{i \geq 1} A_i$, there exists $j \geq 1$ such that $x \notin A_j$. Because the sequence is decreasing and $x \in A_n$, necessarily $j > n$. Hence the set

$$I_x := \{i \geq n : x \in A_i\}$$

is nonempty and bounded above by $j - 1$, so it admits a largest element p . We then have $x \in A_p$ and $x \notin A_{p+1}$, therefore $x \in C_p$ with $p \geq n$. This proves the reverse inclusion, and therefore the desired equality.

We now check that the union is disjoint. For every $i \geq 1$, we have $A \subset A_{i+1}$ and $C_i \subset A_i \setminus A_{i+1}$, hence $A \cap C_i = \emptyset$. Moreover, if $n \leq i < j$, then $C_j \subset A_j \subset A_{i+1}$, whereas $C_i \subset A_i \setminus A_{i+1}$, so $C_i \cap C_j = \emptyset$. Therefore countable additivity yields

$$\mu(A_n) = \mu(A) + \sum_{i \geq n} \mu(C_i), \quad (n \geq 1). \quad (2.1)$$

We have $A \subset A_1$ and $\mu(A_1) < \infty$. Since $A \subset A_1$ and $\mu(A_1) < \infty$, equality (2.1) for $n = 1$ proves that

$$\sum_{i \geq 1} \mu(C_i) = \mu(A_1) - \mu(A) < \infty.$$

Therefore the sequence of remainders

$$\left(\sum_{i \geq n} \mu(C_i) \right)_n$$

converges to 0. Letting $n \rightarrow +\infty$ in (2.1), we obtain

$$\mu(A_n) \rightarrow \mu(A) \quad \text{as } n \rightarrow +\infty.$$

□

Remark. The necessity of the assumption $\mu(A_1) < \infty$ in (2) above can be seen on the measurable space $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$ equipped with the counting measure $\#$. If

$$A_n := \{n, n+1, n+2, \dots\},$$

then $A_n \downarrow \emptyset$ and $\#(A_n) = +\infty$ for every $n \geq 1$, whereas $\#(\emptyset) = 0$.

2.3 Negligible sets, completion and regularity notions

Before turning to the Lebesgue measure itself, we introduce a few notions that will be used repeatedly in the sequel.

Definition 2.4 (Negligible sets). *Let $(X, \mathcal{A}, \mu)$ be a measure space.*

- 1) *A subset $N \subset X$ is said to be μ -negligible if there exists M in $\mathcal{A}$ such that $N \subset M$ and $\mu(M) = 0$.*
- 2) *If $\mathcal{P}$ is a property depending on x in X , we say that $\mathcal{P}$ holds μ -almost everywhere if the set of points where $\mathcal{P}$ fails is μ -negligible.*
- 3) *The measure space $(X, \mathcal{A}, \mu)$ is said to be complete if every μ -negligible set belongs to $\mathcal{A}$.*

Remark. If we only ask N itself to be measurable and of measure 0, then we recover the usual notion of null set. The formulation above is slightly more flexible, and on a complete measure space the two points of view coincide.

Proposition 2.5 (Countable unions of negligible sets). *Let $(X, \mathcal{A}, \mu)$ be a measure space. Then every countable union of μ -negligible sets is again μ -negligible.*

Proof. Let $(N_i)_{i \geq 1}$ be a sequence of μ -negligible sets. For each $i \geq 1$, there exists M_i in $\mathcal{A}$ such that $N_i \subset M_i$ and $\mu(M_i) = 0$. Then $\bigcup_{i \geq 1} M_i$ is in $\mathcal{A}$ and

$$\mu\left(\bigcup_{i \geq 1} M_i\right) \leq \sum_{i \geq 1} \mu(M_i) = 0.$$

Hence $\mu(\bigcup_{i \geq 1} M_i) = 0$, and since

$$\bigcup_{i \geq 1} N_i \subset \bigcup_{i \geq 1} M_i,$$

the set $\bigcup_{i \geq 1} N_i$ is μ -negligible. □

Remark. If $\mathcal{P}_n$ holds μ -almost everywhere on X for every $n \geq 1$, then the conjunction

$$\forall n \geq 1, \quad \mathcal{P}_n$$

is again true μ -almost everywhere. Indeed, we only have to remove the countable union of the exceptional negligible sets.

Remark. Later on, the Lebesgue measure initially considered on the Borel σ -algebra will provide a standard example of a measure which is not complete. Passing to the completion is therefore a natural operation.

Proposition 2.6 (Completion of a measure). *Let $(X, \mathcal{A}, \mu)$ be a measure space. We set*

$$\overline{\mathcal{A}} := \left\{ A \cup N : A \in \mathcal{A}, \exists M \in \mathcal{A}, N \subset M, \mu(M) = 0 \right\}.$$

For $E = A \cup N$ in $\overline{\mathcal{A}}$, we define

$$\overline{\mu}(E) := \mu(A).$$

Then $\overline{\mathcal{A}}$ is a σ -algebra on X , the mapping $\overline{\mu}$ is a measure on $\overline{\mathcal{A}}$, the measure space $(X, \overline{\mathcal{A}}, \overline{\mu})$ is complete, and $\overline{\mu}$ extends μ .

Proof. Step 1. Well-definedness of the measure. We first check that $\bar{\mu}$ is well defined. Assume that

$$A \cup N = A' \cup N',$$

with A, A' in $\mathcal{A}$, $N \subset M$, $N' \subset M'$, M, M' in $\mathcal{A}$, and

$$\mu(M) = \mu(M') = 0.$$

Then

$$A \setminus A' \subset M'$$

because every point of $A \setminus A'$ belongs to $A \cup N = A' \cup N'$ and cannot belong to A' , hence belongs to $N' \subset M'$. Therefore $\mu(A \setminus A') = 0$. Since

$$A = (A \setminus A') \cup (A \cap A')$$

is a disjoint union, we obtain $\mu(A) = \mu(A \cap A') \leq \mu(A')$. By symmetry, $\mu(A') \leq \mu(A)$, hence $\mu(A) = \mu(A')$.

Step 2. The completed σ -algebra. We now prove that $\bar{\mathcal{A}}$ is a σ -algebra. We have $X = X \cup \emptyset$, so X is in $\bar{\mathcal{A}}$. Let $E = A \cup N$ be in $\bar{\mathcal{A}}$, with $N \subset M$, M in $\mathcal{A}$, and $\mu(M) = 0$. Since $N \subset M$, we have

$$E^c = (A \cup N)^c = A^c \cap N^c = (A^c \setminus M) \cup (A^c \cap (M \setminus N)).$$

The set $A^c \setminus M$ is in $\mathcal{A}$, whereas $A^c \cap (M \setminus N) \subset M$. Hence E^c is in $\bar{\mathcal{A}}$. Finally, if $(E_i)_{i \geq 1}$ is a sequence in $\bar{\mathcal{A}}$ and

$$E_i = A_i \cup N_i, \quad N_i \subset M_i, \quad \mu(M_i) = 0,$$

then

$$\bigcup_{i \geq 1} E_i = \left(\bigcup_{i \geq 1} A_i \right) \cup \left(\bigcup_{i \geq 1} N_i \right),$$

the set $\bigcup_{i \geq 1} A_i$ is in $\mathcal{A}$, and $\bigcup_{i \geq 1} N_i \subset \bigcup_{i \geq 1} M_i$ with

$$\mu\left(\bigcup_{i \geq 1} M_i\right) \leq \sum_{i \geq 1} \mu(M_i) = 0.$$

Therefore $\bigcup_{i \geq 1} E_i$ is in $\bar{\mathcal{A}}$.

Step 3. Countable additivity. We now verify the measure property. Clearly,

$$\bar{\mu}(\emptyset) = \mu(\emptyset) = 0.$$

Let $(E_i)_{i \geq 1}$ be a sequence of pairwise disjoint elements of $\bar{\mathcal{A}}$, and write

$$E_i = A_i \cup N_i, \quad N_i \subset M_i, \quad \mu(M_i) = 0.$$

Since $A_i \subset E_i$, the sets A_i are pairwise disjoint. Moreover,

$$\bigcup_{i \geq 1} E_i = \left(\bigcup_{i \geq 1} A_i \right) \cup \left(\bigcup_{i \geq 1} N_i \right),$$

and $\bigcup_{i \geq 1} N_i$ is contained in the measurable set $\bigcup_{i \geq 1} M_i$ of measure 0. Thus

$$\begin{aligned} \bar{\mu}\left(\bigcup_{i \geq 1} E_i\right) &= \mu\left(\bigcup_{i \geq 1} A_i\right) \\ &= \sum_{i \geq 1} \mu(A_i) \\ &= \sum_{i \geq 1} \bar{\mu}(E_i). \end{aligned}$$

Step 4. Completeness. It remains to prove completeness. Let $N \subset X$ be $\bar{\mu}$ -negligible. Then there exists E in $\bar{\mathcal{A}}$ such that

$$N \subset E, \quad \bar{\mu}(E) = 0.$$

Write $E = A \cup Z$ with A in $\mathcal{A}$, $Z \subset M$, M in $\mathcal{A}$, and $\mu(M) = 0$. Since $\bar{\mu}(E) = \mu(A) = 0$, the measurable set $A \cup M$ has measure 0 and contains E , hence also N . Therefore $N = \emptyset \cup N$ belongs to $\bar{\mathcal{A}}$.

Step 5. Extension of the original measure. Finally, if A is in $\mathcal{A}$, then $A = A \cup \emptyset$ is in $\bar{\mathcal{A}}$ and

$$\bar{\mu}(A) = \mu(A).$$

Thus $\bar{\mu}$ extends μ . □

We now record a few notions of regularity on topological spaces. We denote by $\mathcal{B}(X)$ the Borel σ -algebra of a topological space X .

Definition 2.7 (Local finiteness and finiteness). *Let X be a topological space and let μ be a measure on $\mathcal{B}(X)$.*

1) *The measure μ is said to be locally finite if every point of X admits an open neighbourhood U such that $\mu(U) < \infty$.*

2) *The measure μ is said to be inner regular on open sets if, for every open set U of X ,*

$$\mu(U) = \sup\{\mu(K) : K \subset U, K \text{ compact}\}.$$

3) *The measure μ is said to be outer regular on Borel sets if, for every Borel set E of X ,*

$$\mu(E) = \inf\{\mu(U) : E \subset U, U \text{ open}\}.$$

Remark. When X is an open subset of $\mathbb{R}^N$, local finiteness is equivalent to finiteness on compact sets. Indeed, local finiteness gives a finite subcover of every compact set by finite-measure neighbourhoods. Conversely, if compact sets have finite measure, every point has a closed ball contained in X , hence a finite-measure neighbourhood.

Definition 2.8 (Radon measures on Euclidean open sets). *Let Ω be an open subset of $\mathbb{R}^N$. A positive Radon measure on Ω is a Borel measure on Ω which is locally finite.*

Remark. In these notes, on an open subset of $\mathbb{R}^N$, a positive Radon measure means a locally finite Borel measure. The inner and outer regularity properties introduced above are the ones that many authors include in the definition of a Radon measure. In the present course we first isolate local finiteness, and later Theorem 14.2 will show that positive Radon measures on open subsets of $\mathbb{R}^N$ also satisfy these regularity properties.

Remark. Once the Lebesgue measure has been constructed, it will provide a fundamental example of a positive Radon measure. Dirac masses are also examples. Later on, we will also meet the Riesz representation theorem, which identifies positive Radon measures with positive linear functionals on $\mathcal{C}_c(\Omega)$.

2.4 First encounter with the Lebesgue measure

Another fundamental example of measure is given by the Lebesgue measure on $\mathbb{R}^n$, which we shall now consider. Let n in $\mathbb{N}_{\geq 1}$ and let $\mathcal{B}(\mathbb{R}^n)$ be the Borel σ -algebra of $\mathbb{R}^n$.

Given B in $\mathcal{B}(\mathbb{R}^n)$ and x in $\mathbb{R}^n$, we define

$$x + B := \{x + y : y \in B\},$$

and $x + B$ is again in $\mathcal{B}(\mathbb{R}^n)$ because the translation $y \mapsto x + y$ is a homeomorphism of $\mathbb{R}^n$.

Definition 2.9 (Translation-invariant measures). *A measure μ on $\mathcal{B}(\mathbb{R}^n)$ is said to be translation invariant if*

$$\forall B \in \mathcal{B}(\mathbb{R}^n), \forall x \in \mathbb{R}^n, \quad \mu(x + B) = \mu(B).$$

The Borel Lebesgue measure λ_n on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ is characterized by the following result.

Theorem 2.10 (Existence and uniqueness of Lebesgue measure). *There exists a unique measure λ_n on $\mathcal{B}(\mathbb{R}^n)$ such that*

1) λ_n is translation invariant,

2) $\lambda_n([0, 1]^n) = 1$, where

$$[0, 1]^n = \{(a_1, \dots, a_n) : 0 \leq a_i \leq 1 \text{ for every } i\}.$$

We will later justify Theorem 2.10 in two different ways. The next chapter presents the classical construction from outer measures and Carathéodory's extension theorem, which yields the Borel Lebesgue measure. Later on, in Section 14.3, we will recover the same Borel measure again from the Riesz representation theorem and then pass to its completion.

In particular, the following supplement indicates that the Lebesgue measure λ_1 should be considered as an extension of the notion of length, λ_2 as an extension of the notion of area, and λ_3 as an extension of the notion of volume.

Theorem 2.11 (Lebesgue measure of rectangles). *The Lebesgue measure λ_n of a cartesian product of intervals, open, closed, or half-open, bounded or unbounded, is the product of the lengths of its sides, with value $+\infty$ allowed.*

This statement will follow from the construction in the next chapter. As an example, let us note that $\lambda_1(\mathbb{Q}) = 0$. Indeed, if $\mathbb{Q} = \{q_k : k \geq 1\}$, then countable subadditivity gives

$$\lambda_1(\mathbb{Q}) \leq \sum_{k=1}^{\infty} \lambda_1(\{q_k\}) = 0.$$

Example 2.12 (Dirichlet function). *Since $\lambda_1(\mathbb{Q}) = 0$, the function $\mathbf{1}_{\mathbb{Q}}$ is zero λ_1 -almost everywhere on $\mathbb{R}$. Indeed, outside the negligible set $\mathbb{Q}$, it is identically equal to 0.*

2.5 Comparison of measures

We record two basic ways of comparing positive measures on the same measurable space. These notions already appear in elementary examples and will become structural in the Lebesgue decomposition and Radon–Nikodym theorems.

Definition 2.13 (Absolute continuity of positive measures). *Let μ and ν be positive measures on $(X, \mathcal{A})$. We say that ν is absolutely continuous with respect to μ , and we write $\nu \ll \mu$, if, for every E in $\mathcal{A}$,*

$$\mu(E) = 0 \quad \implies \quad \nu(E) = 0.$$

Definition 2.14 (Mutual singularity of positive measures). Let μ and ν be positive measures on $(X, \mathcal{A})$. We say that ν and μ are mutually singular, and we write $\nu \perp \mu$, if there exists S in $\mathcal{A}$ such that

$$\nu(S^c) = 0, \quad \mu(S) = 0.$$

Such a set S is called a mutually singular set associated with the ordered pair (ν, μ) . Thus ν is concentrated on S , while μ is concentrated on S^c .

Example 2.15 (Restricted measures on a partition). Let μ be a positive measure on $(X, \mathcal{A})$ and let A be in $\mathcal{A}$. Define two positive measures by

$$\mu_A(E) := \mu(E \cap A), \quad \mu_{A^c}(E) := \mu(E \cap A^c),$$

for every E in $\mathcal{A}$. Then $\mu_A \ll \mu$ and $\mu_{A^c} \ll \mu$. Moreover, $\mu_A \perp \mu_{A^c}$, and A is a mutually singular set associated with the ordered pair (μ_A, μ_{A^c}) .

Example 2.16 (Singular Dirac measures). Assume that x and y are two distinct points of X and that the singletons $\{x\}$ and $\{y\}$ belong to $\mathcal{A}$. Then the Dirac measures δ_x and δ_y are mutually singular: the set $\{x\}$ is a mutually singular set associated with the ordered pair (δ_x, δ_y) . Moreover, δ_x is not absolutely continuous with respect to δ_y , because

$$\delta_y(\{x\}) = 0, \quad \delta_x(\{x\}) = 1.$$

The following elementary form of absolute continuity will be used later, in particular in the Baire-category argument for set functions.

Lemma 2.17 (Finite absolute continuity lemma). Let $(X, \mathcal{A}, \mu)$ be a finite measure space and let α be a finite positive measure such that $\alpha \ll \mu$. Then for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\mu(E) < \delta \implies \alpha(E) < \varepsilon$$

for every E in $\mathcal{A}$.

Proof. Assume by contradiction that the conclusion fails for some $\varepsilon > 0$. Then, for every $n \geq 1$, there exists E_n in $\mathcal{A}$ such that

$$\mu(E_n) < 2^{-n}, \quad \alpha(E_n) \geq \varepsilon.$$

Set

$$F_m := \bigcup_{n \geq m} E_n, \quad F := \bigcap_{m \geq 1} F_m.$$

Then F_m decreases to F and

$$\mu(F_m) \leq \sum_{n \geq m} 2^{-n} \rightarrow 0 \quad \text{as } m \rightarrow +\infty.$$

Hence $\mu(F) = 0$, and absolute continuity gives $\alpha(F) = 0$. On the other hand, $E_n \subset F_m$ for all $n \geq m$, so $\alpha(F_m) \geq \varepsilon$ for every m . By continuity from above for the finite measure α ,

$$\alpha(F_m) \rightarrow \alpha(F) = 0,$$

which is a contradiction. □

2.6 Signed measures

Although the integral will first be developed for positive measures, it is convenient to record here the basic structural facts on finite signed measures. They belong naturally to the measure-theoretic foundations and will later be used in the proof of Lebesgue decomposition, in the signed Radon–Nikodym theorem, in the Baire argument for set functions, and in the study of bounded Radon measures.

Definition 2.18 (Finite signed measures). *Let $(X, \mathcal{A})$ be a measurable space. A finite signed measure is a map $\nu : \mathcal{A} \rightarrow \mathbb{R}$ such that $\nu(\emptyset) = 0$ and, for every sequence $(E_n)_{n \geq 1}$ of pairwise disjoint sets in $\mathcal{A}$, the series $\sum_{n=1}^{\infty} \nu(E_n)$ converges in $\mathbb{R}$ and*

$$\nu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \nu(E_n).$$

2.7 Hahn decomposition

Definition 2.19 (Positive and negative sets for signed measures). *Let ν be a finite signed measure on $(X, \mathcal{A})$. A measurable set P is called positive for ν if*

$$\nu(E) \geq 0$$

for every measurable set $E \subset P$. A measurable set N is called negative for ν if

$$\nu(E) \leq 0$$

for every measurable set $E \subset N$.

Lemma 2.20 (Stability of positive and negative sets). *Let ν be a finite signed measure on $(X, \mathcal{A})$. If $A \in \mathcal{A}$ satisfies $\nu(A) > 0$, then A contains a positive measurable subset P such that $\nu(P) > 0$.*

Proof. Set $A_0 := A$. If A_0 is positive, then we may take $P = A_0$. Otherwise there is a measurable subset of A_0 with strictly negative measure. Let m_0 be the smallest positive integer such that some measurable $C \subset A_0$ satisfies

$$\nu(C) < -\frac{1}{m_0},$$

and choose $B_0 \subset A_0$ with $\nu(B_0) < -1/m_0$.

Assume that A_k has been constructed. If A_k is positive, then we stop. Otherwise, let m_k be the smallest positive integer such that some measurable $C \subset A_k$ satisfies

$$\nu(C) < -\frac{1}{m_k},$$

choose $B_k \subset A_k$ with $\nu(B_k) < -1/m_k$, and set

$$A_{k+1} := A_k \setminus B_k.$$

If the construction stops at rank k , then the last set A_k is positive. Moreover, only sets of negative measure have been removed, hence

$$\nu(A_k) \geq \nu(A) > 0.$$

Thus $P := A_k$ is suitable.

Assume now that the construction never stops. The sets B_k are pairwise disjoint. Since ν is finite-valued and countably additive, the series

$$\sum_{k=0}^{\infty} \nu(B_k)$$

converges in $\mathbb{R}$. All its terms are negative, and therefore $m_k \rightarrow +\infty$. Define

$$P := A \setminus \bigcup_{k=0}^{\infty} B_k = \bigcap_{k=0}^{\infty} A_k.$$

By countable additivity,

$$\nu(P) = \nu(A) - \sum_{k=0}^{\infty} \nu(B_k) \geq \nu(A) > 0.$$

We claim that P is positive. Indeed, suppose that a measurable set $C \subset P$ satisfies $\nu(C) < 0$. Choose an integer $m \geq 1$ such that

$$\nu(C) < -\frac{1}{m}.$$

For k large enough we have $m_k > m$. Since $C \subset P \subset A_k$ and $\nu(C) < -1/m$, the integer m satisfies the defining property used in the construction of m_k . Hence $m_k \leq m$, a contradiction. Thus P is positive, and the lemma is proved. $\square$

Lemma 2.21 (Countable unions of positive and negative sets). *Let ν be a finite signed measure on $(X, \mathcal{A})$. A countable union of positive measurable sets is positive.*

Proof. Let $(P_n)_{n \geq 1}$ be positive measurable sets and set $P := \bigcup_{n \geq 1} P_n$. If $E \subset P$ is measurable, decompose E into the disjoint union

$$E = \bigcup_{n=1}^{\infty} E_n, \quad E_n := E \cap \left(P_n \setminus \bigcup_{j < n} P_j \right).$$

Since $E_n \subset P_n$, positivity of P_n gives $\nu(E_n) \geq 0$ for every n . Countable additivity gives

$$\nu(E) = \sum_{n=1}^{\infty} \nu(E_n) \geq 0.$$

Thus P is positive. $\square$

Theorem 2.22 (Hahn decomposition theorem). *Let ν be a finite signed measure on $(X, \mathcal{A})$. There exist two measurable sets P and N such that*

$$P \cap N = \emptyset, \quad P \cup N = X,$$

and:

- P is positive for ν ,
- N is negative for ν .

Any such pair (P, N) is called a Hahn decomposition of ν .

Proof. **Step 1. The supremum of the masses of positive sets is finite.** The empty set is positive, and therefore the family of positive measurable sets is nonempty. Let

$$s := \sup\{\nu(P) : P \text{ is a positive measurable set}\}.$$

We have $s \geq 0$. This supremum is finite in $\mathbb{R}$. Indeed, if the measures of positive sets were unbounded from above, we could choose positive sets $(P_n)_{n \geq 1}$ with $\nu(P_n) > n$. By Lemma 2.21, the set $P_* := \bigcup_{n \geq 1} P_n$ would be positive. Since $P_n \subset P_*$ and P_* is positive, the set $P_* \setminus P_n$ is positive, and therefore

$$\nu(P_*) = \nu(P_n) + \nu(P_* \setminus P_n) \geq \nu(P_n) > n$$

for every n , which is impossible because $\nu(P_*)$ is a real number.

Step 2. Construction of a maximal positive set. Choose positive measurable sets $(P_n)_{n \geq 1}$ such that

$$\nu(P_n) > s - \frac{1}{n}.$$

Set

$$P := \bigcup_{n=1}^{\infty} P_n.$$

By Lemma 2.21, the set P is positive. Since P is positive, $\nu(P) \leq s$. On the other hand, for every n we have $P_n \subset P$ and $P \setminus P_n \subset P$, hence

$$\nu(P) = \nu(P_n) + \nu(P \setminus P_n) \geq \nu(P_n) > s - \frac{1}{n}.$$

Letting $n \rightarrow +\infty$, we obtain $\nu(P) = s$.

Step 3. The complement is negative. Set $N := X \setminus P$. If N were not negative, there would exist a measurable set $A \subset N$ with $\nu(A) > 0$. By Lemma 2.20, the set A would contain a positive measurable subset Q with $\nu(Q) > 0$. Then $P \cup Q$ would be positive by Lemma 2.21, and, since $P \cap Q = \emptyset$,

$$\nu(P \cup Q) = \nu(P) + \nu(Q) > s,$$

contradicting the definition of s . Therefore N is negative, and (P, N) is a Hahn decomposition of ν . $\square$

Proposition 2.23 (Comparison of Hahn decompositions). *Let ν be a finite signed measure on $(X, \mathcal{A})$. Let (P, N) and (P', N') be two Hahn decompositions of ν . Then, for every E in $\mathcal{A}$,*

$$\nu(E \cap P) = \nu(E \cap P'), \quad \nu(E \cap N) = \nu(E \cap N').$$

Moreover, if F is a measurable subset of $P \triangle P'$ or of $N \triangle N'$, then $\nu(F) = 0$.

Proof. The set $E \cap P \cap N'$ is contained in the positive set P and in the negative set N' . Hence

$$\nu(E \cap P \cap N') \geq 0 \quad \text{and} \quad \nu(E \cap P \cap N') \leq 0,$$

and therefore

$$\nu(E \cap P \cap N') = 0.$$

Similarly,

$$\nu(E \cap N \cap P') = 0.$$

It follows that

$$\begin{aligned} \nu(E \cap P) &= \nu(E \cap P \cap P') + \nu(E \cap P \cap N') \\ &= \nu(E \cap P \cap P'), \end{aligned}$$

and

$$\begin{aligned}\nu(E \cap P') &= \nu(E \cap P' \cap P) + \nu(E \cap P' \cap N) \\ &= \nu(E \cap P \cap P').\end{aligned}$$

This proves

$$\nu(E \cap P) = \nu(E \cap P').$$

The equality involving N and N' is proved in the same way.

Let now F be a measurable subset of $P \Delta P'$. Then F is the disjoint union of $F \cap P \cap N'$ and $F \cap N \cap P'$. Both sets have signed measure 0 by the preceding argument, hence $\nu(F) = 0$. The proof for measurable subsets of $N \Delta N'$ is identical. $\square$

Remark 2.24 (Non-uniqueness of Hahn decompositions). *A Hahn decomposition is not unique in general. For instance, if $\nu = 0$, every measurable set P gives a Hahn decomposition (P, P^c) . Proposition 2.23 is the uniqueness statement that is actually needed: two Hahn decompositions give the same positive and negative parts of the signed measure.*

2.8 Jordan decomposition

Theorem 2.25 (Jordan decomposition theorem). *Let ν be a finite signed measure on $(X, \mathcal{A})$ and let (P, N) be a Hahn decomposition of ν . We define*

$$\nu^+(E) := \nu(E \cap P), \quad \nu^-(E) := -\nu(E \cap N),$$

for every E in $\mathcal{A}$. Then ν^+ and ν^- are finite positive measures, they are mutually singular, and

$$\nu = \nu^+ - \nu^-.$$

Moreover, the pair (ν^+, ν^-) is uniquely determined by these properties, and in particular it does not depend on the chosen Hahn decomposition.

Proof. Step 1. Construction of the two positive measures. We first check that ν^+ and ν^- are positive measures. If $E \in \mathcal{A}$, then $E \cap P \subset P$ and $E \cap N \subset N$. Therefore

$$\nu^+(E) = \nu(E \cap P) \geq 0, \quad \nu^-(E) = -\nu(E \cap N) \geq 0.$$

If $(E_j)_{j \geq 1}$ is a sequence of pairwise disjoint measurable sets, then the sets $E_j \cap P$ are pairwise disjoint, and hence

$$\begin{aligned}\nu^+\left(\bigcup_{j=1}^{\infty} E_j\right) &= \nu\left(\bigcup_{j=1}^{\infty} (E_j \cap P)\right) \\ &= \sum_{j=1}^{\infty} \nu(E_j \cap P) = \sum_{j=1}^{\infty} \nu^+(E_j).\end{aligned}$$

The same argument applied on N , with the sign changed, proves countable additivity of ν^- . Moreover, $\nu^+(X) = \nu(P)$ and $\nu^-(X) = -\nu(N)$ are finite real numbers, so both positive measures are finite.

Step 2. Reconstruction and singularity. For every $E \in \mathcal{A}$, the decomposition

$$E = (E \cap P) \cup (E \cap N)$$

is disjoint, and therefore

$$\nu(E) = \nu(E \cap P) + \nu(E \cap N) = \nu^+(E) - \nu^-(E).$$

The measures ν^+ and ν^- are mutually singular because they are separated by P :

$$\nu^+(P^c) = \nu^+(N) = 0, \quad \nu^-(P) = 0.$$

Step 3. Uniqueness. It remains to prove uniqueness. Assume that another pair of finite positive measures satisfies the same properties:

$$\nu = \eta^+ - \eta^-, \quad \eta^+ \perp \eta^-.$$

By the definition of mutual singularity, there exists a measurable set A such that

$$\eta^+(A^c) = 0, \quad \eta^-(A) = 0.$$

Thus η^+ is concentrated on A , while η^- is concentrated on A^c . Consequently, if $F \subset A$ is measurable, then

$$\nu(F) = \eta^+(F) - \eta^-(F) = \eta^+(F) \geq 0,$$

and if $F \subset A^c$ is measurable, then

$$\nu(F) = \eta^+(F) - \eta^-(F) = -\eta^-(F) \leq 0.$$

Hence (A, A^c) is also a Hahn decomposition of ν . By Proposition 2.23, applied to the two Hahn decompositions (P, N) and (A, A^c) , for every E in $\mathcal{A}$ we have

$$\nu(E \cap P) = \nu(E \cap A), \quad \nu(E \cap N) = \nu(E \cap A^c).$$

We can now identify the two decompositions. Since $\eta^+(A^c) = 0$ and $\eta^-(A) = 0$, for every E in $\mathcal{A}$ we have

$$\eta^+(E) = \eta^+(E \cap A) = \nu(E \cap A), \quad \eta^-(E) = \eta^-(E \cap A^c) = -\nu(E \cap A^c).$$

Using the comparison identities above, we obtain

$$\eta^+(E) = \nu(E \cap P) = \nu^+(E), \quad \eta^-(E) = -\nu(E \cap N) = \nu^-(E).$$

Thus $\eta^+ = \nu^+$ and $\eta^- = \nu^-$. The Jordan decomposition is unique. $\square$

2.9 Total variation

Definition 2.26 (Total variation measure and total variation mass). *Let ν be a finite signed measure on $(X, \mathcal{A})$ and let $\nu = \nu^+ - \nu^-$ be its Jordan decomposition. The total variation measure of ν is the positive measure*

$$|\nu| := \nu^+ + \nu^-.$$

The total variation mass of ν is the number

$$\|\nu\| := |\nu|(X).$$

Proposition 2.27 (Partition formula for the total variation). *Let ν be a finite signed measure on $(X, \mathcal{A})$. For every E in $\mathcal{A}$ we have*

$$|\nu|(E) = \sup \left\{ \sum_{k=1}^m |\nu(E_k)| : m \geq 1, (E_k)_{1 \leq k \leq m} \text{ is a measurable partition of } E \right\}.$$

In particular, for every E in $\mathcal{A}$ we have $|\nu(E)| \leq |\nu|(E)$.

Proof. Let $\nu = \nu^+ - \nu^-$ be the Jordan decomposition. Let E be in $\mathcal{A}$ and let $(E_k)_{1 \leq k \leq m}$ be a measurable partition of E . By additivity and the triangle inequality,

$$\sum_{k=1}^m |\nu(E_k)| \leq \sum_{k=1}^m (\nu^+(E_k) + \nu^-(E_k)) = \nu^+(E) + \nu^-(E) = |\nu|(E).$$

Taking the supremum over all partitions yields

$$\sup \left\{ \sum_{k=1}^m |\nu(E_k)| \right\} \leq |\nu|(E).$$

Conversely, we use the partition $E = (E \cap P) \cup (E \cap N)$ associated with a Hahn decomposition (P, N) . Then

$$|\nu(E \cap P)| + |\nu(E \cap N)| = \nu^+(E) + \nu^-(E) = |\nu|(E),$$

which gives the reverse inequality. $\square$

Proposition 2.28 (Total variation norm). *Let $(X, \mathcal{A})$ be a measurable space. The set of finite signed measures on $(X, \mathcal{A})$ is a real vector space, and the map*

$$\nu \mapsto \|\nu\| := |\nu|(X)$$

is a norm on this vector space.

Proof. The fact that finite signed measures form a real vector space follows directly from the definition.

Let ν be a finite signed measure. Since $|\nu|$ is a positive measure, we have

$$\|\nu\| = |\nu|(X) \geq 0.$$

If $\|\nu\| = 0$, then for every E in $\mathcal{A}$,

$$|\nu(E)| \leq |\nu|(E) \leq |\nu|(X) = 0.$$

Thus $\nu(E) = 0$ for every E in $\mathcal{A}$, and therefore $\nu = 0$.

Let a be in $\mathbb{R}$. If $a = 0$, then $\|a\nu\| = 0 = |a|\|\nu\|$. If $a \neq 0$, Proposition 2.27 gives

$$\begin{aligned} \|a\nu\| &= \sup \left\{ \sum_{k=1}^m |a\nu(E_k)| : m \geq 1, (E_k)_{1 \leq k \leq m} \text{ is a measurable partition of } X \right\} \\ &= |a| \sup \left\{ \sum_{k=1}^m |\nu(E_k)| : m \geq 1, (E_k)_{1 \leq k \leq m} \text{ is a measurable partition of } X \right\} \\ &= |a| \|\nu\|. \end{aligned}$$

Finally, let ν and η be finite signed measures. For every finite measurable partition $(E_k)_{1 \leq k \leq m}$ of X , we have

$$\begin{aligned} \sum_{k=1}^m |(\nu + \eta)(E_k)| &\leq \sum_{k=1}^m |\nu(E_k)| + \sum_{k=1}^m |\eta(E_k)| \\ &\leq |\nu|(X) + |\eta|(X). \end{aligned}$$

Taking the supremum over all finite measurable partitions of X and using Proposition 2.27 gives

$$\|\nu + \eta\| \leq \|\nu\| + \|\eta\|.$$

Thus $\|\cdot\|$ is a norm. $\square$

We can therefore call $\|v\| = |v|(X)$ the *total variation norm* of v .

Definition 2.29 (Absolute continuity of finite signed measures). Let μ be a positive measure on $(X, \mathcal{A})$ and let v be a finite signed measure on $(X, \mathcal{A})$. We say that v is *absolutely continuous* with respect to μ , and we write $v \ll \mu$, if the total variation measure satisfies

$$|v| \ll \mu.$$

Equivalently, if $v = v^+ - v^-$ is the Jordan decomposition, then $v \ll \mu$ means that both $v^+ \ll \mu$ and $v^- \ll \mu$. This is also equivalent to saying that $v(E) = 0$ for every μ -negligible set E , but the formulation through $|v|$ keeps track of both signs at once.

Chapter 3

Outer measures and Carathéodory's extension theorem

This chapter presents the classical extension procedure which starts from a set function defined on a small family of sets and produces an outer measure. The Carathéodory measurable sets then form a complete measure space, and the restriction to the generated σ -algebra gives the desired extension. The chapter complements the abstract material of the previous chapter, explains how the Lebesgue measure may also be constructed from rectangles by Carathéodory's method, and introduces the Dynkin π - λ theorem and the associated uniqueness principle that will be used repeatedly later on.

3.1 Outer measures

Definition 3.1 (Outer measures). *Let X be a set. An outer measure on X is a map*

$$\mu^*: \mathcal{P}(X) \rightarrow [0, \infty]$$

such that

- 1) $\mu^*(\emptyset) = 0$,
- 2) if $E \subset F \subset X$, then $\mu^*(E) \leq \mu^*(F)$,
- 3) for every sequence $(E_n)_{n \geq 1}$ of subsets of X we have

$$\mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} \mu^*(E_n).$$

Proposition 3.2 (Outer measures from coverings). *Let X be a set, let $\mathcal{E}$ be a family of subsets of X , and let $m: \mathcal{E} \rightarrow [0, \infty]$. For every $A \subset X$, we define*

$$\mu^*(A) := \inf \left\{ \sum_{j \in J} m(E_j) : J \text{ finite or countable, } A \subset \bigcup_{j \in J} E_j, E_j \in \mathcal{E} \right\},$$

where the empty family is allowed for the empty set, the corresponding sum is 0, and the infimum of the empty set is understood as $+\infty$. Then μ^ is an outer measure on X .*

Proof. We obviously have $\mu^*(\emptyset) = 0$, since the empty family covers $\emptyset$ with total cost 0. If $A \subset B$, then every covering of B by sets of $\mathcal{E}$ is also a covering of A , hence $\mu^*(A) \leq \mu^*(B)$. Let now

$(A_k)_{k \geq 1}$ be a sequence of subsets of X . If $\sum_{k=1}^{\infty} \mu^*(A_k) = +\infty$, then countable subadditivity is immediate. We may therefore assume that this sum is finite. Let $\varepsilon > 0$. For every $k \geq 1$, we choose a finite or countable covering $(E_{k,j})_{j \in J_k}$ of A_k by sets of $\mathcal{E}$ such that

$$\sum_{j \in J_k} m(E_{k,j}) \leq \mu^*(A_k) + \frac{\varepsilon}{2^k}.$$

Then the family $(E_{k,j})_{k \geq 1, j \in J_k}$ is a countable covering of $\bigcup_{k \geq 1} A_k$, so

$$\begin{aligned} \mu^*\left(\bigcup_{k=1}^{\infty} A_k\right) &\leq \sum_{k=1}^{\infty} \sum_{j \in J_k} m(E_{k,j}) \\ &\leq \sum_{k=1}^{\infty} \mu^*(A_k) + \varepsilon. \end{aligned}$$

Letting $\varepsilon \rightarrow 0$ yields the countable subadditivity. □

3.2 Carathéodory measurable sets

Definition 3.3 (Carathéodory measurable sets). Let μ^* be an outer measure on a set X . A subset A of X is said to be μ^* -measurable in the sense of Carathéodory if, for every subset E of X ,

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \setminus A).$$

Since subadditivity always gives

$$\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \setminus A),$$

Carathéodory measurability is equivalent to the reverse inequality for every subset E of X .

Theorem 3.4 (Caratheodory measurability theorem). Let μ^* be an outer measure on a set X , and let

$$\mathcal{M} := \{A \subset X : A \text{ is } \mu^*\text{-measurable}\}.$$

Then $\mathcal{M}$ is a σ -algebra on X , and the restriction of μ^* to $\mathcal{M}$ is a complete measure.

Proof. **Step 1. Elementary stability properties.** We first note that X belongs to $\mathcal{M}$, since for every $E \subset X$,

$$\mu^*(E) = \mu^*(E \cap X) + \mu^*(E \setminus X) = \mu^*(E) + 0.$$

Moreover, if A belongs to $\mathcal{M}$, then A^c also belongs to $\mathcal{M}$, because

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \setminus A) = \mu^*(E \setminus A^c) + \mu^*(E \cap A^c)$$

for every $E \subset X$.

Step 2. Finite intersections. We now prove that $\mathcal{M}$ is stable by finite intersections. Let A and B be in $\mathcal{M}$, and let $E \subset X$. Since A is measurable, we have

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \setminus A).$$

We apply the measurability of B to the sets $E \cap A$ and $E \setminus A$:

$$\begin{aligned} \mu^*(E \cap A) &= \mu^*(E \cap A \cap B) + \mu^*(E \cap A \setminus B), \\ \mu^*(E \setminus A) &= \mu^*(E \cap B \setminus A) + \mu^*(E \setminus (A \cup B)). \end{aligned}$$

Hence

$$\begin{aligned}\mu^*(E) &= \mu^*(E \cap A \cap B) + \mu^*(E \cap A \setminus B) \\ &\quad + \mu^*(E \cap B \setminus A) + \mu^*(E \setminus (A \cup B)).\end{aligned}$$

By countable subadditivity,

$$\mu^*(E \setminus (A \cap B)) \leq \mu^*(E \cap A \setminus B) + \mu^*(E \cap B \setminus A) + \mu^*(E \setminus (A \cup B)).$$

Therefore

$$\mu^*(E) \geq \mu^*(E \cap A \cap B) + \mu^*(E \setminus (A \cap B)).$$

The reverse inequality always follows from subadditivity, so $A \cap B$ is measurable. Since $\mathcal{M}$ is stable by complements and finite intersections, it is also stable by finite unions.

Step 3. Countable disjoint unions and additivity. We next prove that $\mathcal{M}$ is stable by countable disjoint unions, and that μ^* is countably additive on such unions. Let $(A_n)_{n \geq 1}$ be a sequence of pairwise disjoint sets in $\mathcal{M}$, and let

$$A := \bigcup_{n=1}^{\infty} A_n.$$

For every $E \subset X$ and every $N \geq 1$, we use the measurability of $A_1, \dots, A_N$ successively to obtain

$$\mu^*(E) = \sum_{n=1}^N \mu^*(E \cap A_n) + \mu^*\left(E \setminus \bigcup_{n=1}^N A_n\right).$$

Since

$$E \setminus A \subset E \setminus \bigcup_{n=1}^N A_n,$$

monotonicity gives

$$\mu^*(E) \geq \sum_{n=1}^N \mu^*(E \cap A_n) + \mu^*(E \setminus A).$$

Letting $N \rightarrow \infty$, we obtain

$$\mu^*(E) \geq \sum_{n=1}^{\infty} \mu^*(E \cap A_n) + \mu^*(E \setminus A).$$

On the other hand,

$$E = (E \cap A) \cup (E \setminus A),$$

and

$$E \cap A = \bigcup_{n=1}^{\infty} (E \cap A_n)$$

is a disjoint union, so countable subadditivity yields

$$\mu^*(E) \leq \sum_{n=1}^{\infty} \mu^*(E \cap A_n) + \mu^*(E \setminus A).$$

Since

$$\mu^*(E \cap A) \leq \sum_{n=1}^{\infty} \mu^*(E \cap A_n),$$

we deduce that

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \setminus A).$$

The reverse inequality follows from subadditivity, so A belongs to $\mathcal{M}$. The two opposite inequalities above give equality. Taking $E = A$ in this identity, and noting that $A \cap A_n = A_n$, we obtain

$$\mu^*(A) = \sum_{n=1}^{\infty} \mu^*(A_n).$$

Step 4. Arbitrary countable unions. Finally, let $(A_n)_{n \geq 1}$ be an arbitrary sequence in $\mathcal{M}$. We define

$$B_1 := A_1, \quad B_n := A_n \setminus \bigcup_{k=1}^{n-1} A_k \quad (n \geq 2).$$

Since $\mathcal{M}$ is stable by complements and finite intersections, every B_n belongs to $\mathcal{M}$. The sets B_n are pairwise disjoint and

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n,$$

so the previous step shows that $\bigcup_{n \geq 1} A_n$ belongs to $\mathcal{M}$. Therefore $\mathcal{M}$ is a σ -algebra.

Step 5. Completeness. It remains to prove completeness. Let $N \subset X$ be such that $\mu^*(N) = 0$. For every $E \subset X$, we have

$$\mu^*(E \cap N) \leq \mu^*(N) = 0,$$

so

$$\mu^*(E \cap N) = 0.$$

Since

$$E = (E \cap N) \cup (E \setminus N),$$

subadditivity gives

$$\mu^*(E) \leq \mu^*(E \cap N) + \mu^*(E \setminus N) = \mu^*(E \setminus N).$$

The reverse inequality follows from monotonicity, hence N belongs to $\mathcal{M}$. In particular, if M belongs to $\mathcal{M}$, $\mu^*(M) = 0$, and $N \subset M$, then $\mu^*(N) = 0$, hence N belongs to $\mathcal{M}$. Thus the restricted measure is complete in the usual sense. $\square$

3.3 Extension from a premeasure

Definition 3.5 (Algebras and premeasures). *Let X be a set.*

- 1) *A family $\mathcal{E} \subset \mathcal{P}(X)$ is called an algebra of sets if X belongs to $\mathcal{E}$, and if $\mathcal{E}$ is stable by finite unions and complements. An algebra is then automatically stable by finite intersections and by set differences.*
- 2) *A map $\mu_0: \mathcal{E} \rightarrow [0, \infty]$ is called a premeasure if $\mu_0(\emptyset) = 0$ and if, for every sequence $(E_n)_{n \geq 1}$ of pairwise disjoint sets in $\mathcal{E}$ such that $\bigcup_{n \geq 1} E_n$ still belongs to $\mathcal{E}$, we have*

$$\mu_0\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu_0(E_n).$$

Theorem 3.6 (Caratheodory extension theorem). Let X be a set, let $\mathcal{E}$ be an algebra of subsets of X , and let μ_0 be a premeasure on $\mathcal{E}$. We define an outer measure μ^* on X by

$$\mu^*(A) := \inf \left\{ \sum_{n=1}^{\infty} \mu_0(E_n) : A \subset \bigcup_{n=1}^{\infty} E_n, E_n \in \mathcal{E} \right\},$$

where finite coverings may be padded by empty sets, and the empty set is covered by the constant sequence $E_n = \emptyset$. Then every set of $\mathcal{E}$ is μ^* -measurable, and

$$\mu^*(E) = \mu_0(E)$$

for every E in $\mathcal{E}$. Consequently, the restriction of μ^* to $\sigma(\mathcal{E})$ is a measure extending μ_0 .

Proof. **Step 1. Measurability of the algebra.** The previous proposition shows that μ^* is an outer measure. We shall use the elementary consequences that a premeasure on an algebra is finitely additive, monotone, and finitely subadditive. Finite additivity follows by applying the definition of premeasure to a finite disjoint family completed by empty sets; monotonicity and finite subadditivity then follow as in Theorem 2.2. Let A be in $\mathcal{E}$, and let $B \subset X$. For every covering

$$B \subset \bigcup_{n=1}^{\infty} E_n, \quad E_n \in \mathcal{E},$$

we have

$$B \cap A \subset \bigcup_{n=1}^{\infty} (E_n \cap A), \quad B \setminus A \subset \bigcup_{n=1}^{\infty} (E_n \setminus A),$$

and all the sets $E_n \cap A$ and $E_n \setminus A$ belong to $\mathcal{E}$. Moreover,

$$E_n = (E_n \cap A) \cup (E_n \setminus A)$$

is a disjoint union, so finite additivity on the algebra gives

$$\mu_0(E_n) = \mu_0(E_n \cap A) + \mu_0(E_n \setminus A).$$

Therefore

$$\mu^*(B \cap A) + \mu^*(B \setminus A) \leq \sum_{n=1}^{\infty} \mu_0(E_n).$$

Taking the infimum over all coverings of B , we obtain

$$\mu^*(B \cap A) + \mu^*(B \setminus A) \leq \mu^*(B).$$

The converse inequality follows from subadditivity, so A is μ^* -measurable.

Step 2. Equality on the algebra. We now prove that $\mu^*(A) = \mu_0(A)$. The inequality $\mu^*(A) \leq \mu_0(A)$ is immediate, since A covers itself. Conversely, let

$$A \subset \bigcup_{n=1}^{\infty} E_n, \quad E_n \in \mathcal{E}.$$

For every $N \geq 1$, we define

$$C_N := A \cap \bigcup_{n=1}^N E_n.$$

Then C_N belongs to $\mathcal{E}$, the sequence $(C_N)_{N \geq 1}$ is increasing, and

$$\bigcup_{N=1}^{\infty} C_N = A.$$

We write

$$D_1 := C_1, \quad D_N := C_N \setminus C_{N-1} \quad (N \geq 2).$$

The sets D_N belong to $\mathcal{E}$, they are pairwise disjoint, and

$$A = \bigcup_{N=1}^{\infty} D_N.$$

Since A belongs to $\mathcal{E}$ and μ_0 is a premeasure,

$$\mu_0(A) = \sum_{N=1}^{\infty} \mu_0(D_N) = \lim_{N \rightarrow \infty} \mu_0(C_N).$$

For each N , we have

$$C_N \subset \bigcup_{n=1}^N (A \cap E_n),$$

so finite subadditivity on the algebra yields

$$\mu_0(C_N) \leq \sum_{n=1}^N \mu_0(A \cap E_n) \leq \sum_{n=1}^N \mu_0(E_n).$$

Letting $N \rightarrow \infty$, we obtain

$$\mu_0(A) \leq \sum_{n=1}^{\infty} \mu_0(E_n).$$

Taking the infimum over all coverings of A gives $\mu_0(A) \leq \mu^*(A)$. Hence $\mu^*(A) = \mu_0(A)$.

Step 3. Restriction to the generated σ -algebra. We have proved that $\mathcal{E} \subset \mathcal{M}$, where $\mathcal{M}$ denotes the Carathéodory σ -algebra associated with μ^* . Since $\mathcal{M}$ is a σ -algebra, it contains $\sigma(\mathcal{E})$. Therefore the restriction of μ^* to $\sigma(\mathcal{E})$ is a measure extending μ_0 . $\square$

3.4 Dynkin systems and uniqueness on generators

Definition 3.7 (Dynkin systems). *Let X be a set.*

1) A family $\mathcal{D} \subset \mathcal{P}(X)$ is called a Dynkin system if

- (a) X belongs to $\mathcal{D}$,
- (b) if A and B belong to $\mathcal{D}$ and $A \subset B$, then $B \setminus A$ belongs to $\mathcal{D}$,
- (c) if $(A_n)_{n \geq 1}$ is an increasing sequence in $\mathcal{D}$, then $\bigcup_{n \geq 1} A_n$ belongs to $\mathcal{D}$.

2) A family $\mathcal{P} \subset \mathcal{P}(X)$ is called a π -system if it is stable by finite intersections.

Remark. This definition is equivalent to the common definition in which a Dynkin system is required to be stable under countable disjoint unions. Indeed, increasing unions follow by taking partial disjoint unions, and conversely the difference axiom turns an increasing union into a countable disjoint union of successive differences.

Theorem 3.8 (Dynkin pi-lambda theorem). *Let $\mathcal{P}$ be a π -system on a set X . Then $\sigma(\mathcal{P})$ is the smallest Dynkin system containing $\mathcal{P}$.*

Proof. **Step 1. The minimal Dynkin system.** Let $\mathcal{D}$ be the smallest Dynkin system containing $\mathcal{P}$; it is obtained as the intersection of all Dynkin systems containing $\mathcal{P}$.

We first prove that, for every A in $\mathcal{P}$, the class

$$\mathcal{D}_A := \{B \subset X : A \cap B \in \mathcal{D}\}$$

is a Dynkin system. Since $A \cap X = A$ belongs to $\mathcal{P} \subset \mathcal{D}$, we have $X \in \mathcal{D}_A$. If B and C belong to $\mathcal{D}_A$ and $B \subset C$, then

$$A \cap (C \setminus B) = (A \cap C) \setminus (A \cap B),$$

so $C \setminus B$ belongs to $\mathcal{D}_A$ because $\mathcal{D}$ is a Dynkin system. If $(B_n)_{n \geq 1}$ is an increasing sequence in $\mathcal{D}_A$, then

$$A \cap \bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} (A \cap B_n),$$

and the sequence $(A \cap B_n)_{n \geq 1}$ is increasing in $\mathcal{D}$. Hence $\bigcup_{n \geq 1} B_n$ belongs to $\mathcal{D}_A$. Thus $\mathcal{D}_A$ is a Dynkin system.

Since $\mathcal{P}$ is a π -system, for every B in $\mathcal{P}$ we have

$$A \cap B \in \mathcal{P} \subset \mathcal{D},$$

so $\mathcal{P} \subset \mathcal{D}_A$. By minimality of $\mathcal{D}$, it follows that $\mathcal{D} \subset \mathcal{D}_A$. Therefore, for every A in $\mathcal{P}$ and every B in $\mathcal{D}$, we have

$$A \cap B \in \mathcal{D}.$$

Step 2. Stability under intersections with elements of $\mathcal{D}$. We now consider the class

$$\mathcal{E} := \{A \subset X : A \cap B \in \mathcal{D} \text{ for every } B \in \mathcal{D}\}.$$

We claim that $\mathcal{E}$ is a Dynkin system. Since $X \cap B = B$ belongs to $\mathcal{D}$ for every B in $\mathcal{D}$, we have $X \in \mathcal{E}$. If A and C belong to $\mathcal{E}$ and $A \subset C$, then for every B in $\mathcal{D}$,

$$(C \setminus A) \cap B = (C \cap B) \setminus (A \cap B),$$

and the right-hand side belongs to $\mathcal{D}$ because $\mathcal{D}$ is a Dynkin system. Hence $C \setminus A$ belongs to $\mathcal{E}$. If $(A_n)_{n \geq 1}$ is an increasing sequence in $\mathcal{E}$, then for every B in $\mathcal{D}$,

$$\left(\bigcup_{n=1}^{\infty} A_n \right) \cap B = \bigcup_{n=1}^{\infty} (A_n \cap B),$$

and the sequence $(A_n \cap B)_{n \geq 1}$ is increasing in $\mathcal{D}$. Therefore $\bigcup_{n \geq 1} A_n$ belongs to $\mathcal{E}$. Thus $\mathcal{E}$ is a Dynkin system.

The first step shows that $\mathcal{P} \subset \mathcal{E}$. By minimality of $\mathcal{D}$, we obtain $\mathcal{D} \subset \mathcal{E}$. Hence, if A and B belong to $\mathcal{D}$, then

$$A \cap B \in \mathcal{D}.$$

So $\mathcal{D}$ is stable by finite intersections.

Step 3. The Dynkin system is a σ -algebra. A Dynkin system is stable by complements, because

$$X \setminus A \in \mathcal{D}$$

whenever A belongs to $\mathcal{D}$. Since $\mathcal{D}$ is also stable by finite intersections, it is stable by finite unions. If $(A_n)_{n \geq 1}$ is any sequence in $\mathcal{D}$, the partial unions

$$U_N := \bigcup_{n=1}^N A_n$$

belong to $\mathcal{D}$ and form an increasing sequence. Hence

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{N=1}^{\infty} U_N$$

belongs to $\mathcal{D}$. Therefore $\mathcal{D}$ is a σ -algebra.

Step 4. Identification with $\sigma(\mathcal{P})$. Since $\mathcal{D}$ contains $\mathcal{P}$, we have

$$\sigma(\mathcal{P}) \subset \mathcal{D}.$$

Conversely, every σ -algebra is a Dynkin system, so the minimality of $\mathcal{D}$ gives

$$\mathcal{D} \subset \sigma(\mathcal{P}).$$

Thus $\mathcal{D} = \sigma(\mathcal{P})$. □

Definition 3.9 (Monotone classes). *Let X be a set. A family $\mathcal{M} \subset \mathcal{P}(X)$ is called a monotone class if*

- 1) *whenever $(A_n)_{n \geq 1}$ is an increasing sequence in $\mathcal{M}$, then $\bigcup_{n \geq 1} A_n$ belongs to $\mathcal{M}$,*
- 2) *whenever $(A_n)_{n \geq 1}$ is a decreasing sequence in $\mathcal{M}$, then $\bigcap_{n \geq 1} A_n$ belongs to $\mathcal{M}$.*

Theorem 3.10 (Monotone class theorem). *Let $\mathcal{E}$ be an algebra of subsets of a set X . Then $\sigma(\mathcal{E})$ is the smallest monotone class containing $\mathcal{E}$.*

Proof. **Step 1. Stability under complements.** Let $\mathcal{M}$ be the smallest monotone class containing $\mathcal{E}$; it is obtained as the intersection of all monotone classes containing $\mathcal{E}$. We first prove that $\mathcal{M}$ is stable by complements. Let

$$\mathcal{C} := \{A \in \mathcal{M} : A^c \in \mathcal{M}\}.$$

Since $\mathcal{E}$ is an algebra, we have $\mathcal{E} \subset \mathcal{C}$. If $(A_n)_{n \geq 1}$ is an increasing sequence in $\mathcal{C}$, then $A := \bigcup_{n \geq 1} A_n$ belongs to $\mathcal{M}$ and

$$A^c = \bigcap_{n=1}^{\infty} A_n^c$$

belongs to $\mathcal{M}$ because $(A_n^c)_{n \geq 1}$ is decreasing in $\mathcal{M}$. Thus A belongs to $\mathcal{C}$. Similarly, if $(A_n)_{n \geq 1}$ is a decreasing sequence in $\mathcal{C}$, then $A := \bigcap_{n \geq 1} A_n$ belongs to $\mathcal{M}$ and

$$A^c = \bigcup_{n=1}^{\infty} A_n^c$$

belongs to $\mathcal{M}$ because $(A_n^c)_{n \geq 1}$ is increasing in $\mathcal{M}$. Hence A belongs to $\mathcal{C}$. Therefore $\mathcal{C}$ is a monotone class containing $\mathcal{E}$, so by minimality of $\mathcal{M}$ we obtain

$$\mathcal{C} = \mathcal{M}.$$

Thus $\mathcal{M}$ is stable by complements.

Step 2. Intersections with elements of the algebra. We next prove that $\mathcal{M}$ is stable by finite intersections. For every A in $\mathcal{E}$, we define

$$\mathcal{M}_A := \{B \in \mathcal{M} : A \cap B \in \mathcal{M}\}.$$

Since $\mathcal{E}$ is an algebra, if B belongs to $\mathcal{E}$, then $A \cap B$ also belongs to $\mathcal{E} \subset \mathcal{M}$. Hence $\mathcal{E} \subset \mathcal{M}_A$. If $(B_n)_{n \geq 1}$ is an increasing sequence in $\mathcal{M}_A$, then

$$A \cap \bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} (A \cap B_n)$$

belongs to $\mathcal{M}$, because $(A \cap B_n)_{n \geq 1}$ is increasing in $\mathcal{M}$. Thus $\bigcup_{n \geq 1} B_n$ belongs to $\mathcal{M}_A$. Similarly, if $(B_n)_{n \geq 1}$ is a decreasing sequence in $\mathcal{M}_A$, then

$$A \cap \bigcap_{n=1}^{\infty} B_n = \bigcap_{n=1}^{\infty} (A \cap B_n)$$

belongs to $\mathcal{M}$. Hence $\mathcal{M}_A$ is a monotone class containing $\mathcal{E}$, so $\mathcal{M}_A = \mathcal{M}$. Therefore, for every A in $\mathcal{E}$ and every B in $\mathcal{M}$, we have $A \cap B$ in $\mathcal{M}$.

Step 3. Intersections of arbitrary elements of $\mathcal{M}$. We now define

$$\mathcal{N} := \{A \in \mathcal{M} : A \cap B \in \mathcal{M} \text{ for every } B \in \mathcal{M}\}.$$

The previous step gives $\mathcal{E} \subset \mathcal{N}$. If $(A_n)_{n \geq 1}$ is an increasing sequence in $\mathcal{N}$ and if B belongs to $\mathcal{M}$, then

$$\left(\bigcup_{n=1}^{\infty} A_n \right) \cap B = \bigcup_{n=1}^{\infty} (A_n \cap B)$$

belongs to $\mathcal{M}$. Similarly, if $(A_n)_{n \geq 1}$ is a decreasing sequence in $\mathcal{N}$, then

$$\left(\bigcap_{n=1}^{\infty} A_n \right) \cap B = \bigcap_{n=1}^{\infty} (A_n \cap B)$$

belongs to $\mathcal{M}$. Hence $\mathcal{N}$ is a monotone class containing $\mathcal{E}$, so $\mathcal{N} = \mathcal{M}$. Thus $\mathcal{M}$ is stable by finite intersections.

Step 4. Identification with the generated σ -algebra. Since $\mathcal{M}$ is stable by complements and finite intersections, it is stable by finite unions. If $(A_n)_{n \geq 1}$ is any sequence in $\mathcal{M}$, then the partial unions

$$U_N := \bigcup_{n=1}^N A_n$$

belong to $\mathcal{M}$ and form an increasing sequence. Therefore

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{N=1}^{\infty} U_N$$

belongs to $\mathcal{M}$. So $\mathcal{M}$ is a σ -algebra containing $\mathcal{E}$, hence

$$\sigma(\mathcal{E}) \subset \mathcal{M}.$$

Conversely, every σ -algebra is a monotone class, so the minimality of $\mathcal{M}$ yields

$$\mathcal{M} \subset \sigma(\mathcal{E}).$$

Thus $\mathcal{M} = \sigma(\mathcal{E})$. □

Corollary 3.11 (Approximation by a generating algebra). *Let $(X, \mathcal{A}, \mu)$ be a finite measure space, and let $\mathcal{E}$ be an algebra of subsets of X such that*

$$\sigma(\mathcal{E}) = \mathcal{A}.$$

Then for every E in $\mathcal{A}$ and every $\eta > 0$, there exists A in $\mathcal{E}$ such that

$$\mu(E \Delta A) < \eta.$$

Proof. We define

$$\mathcal{D} := \{E \in \mathcal{A} : \forall \eta > 0, \exists A \in \mathcal{E} \text{ such that } \mu(E \Delta A) < \eta\}.$$

Clearly $\mathcal{E} \subset \mathcal{D}$.

We claim that $\mathcal{D}$ is a monotone class. Let $(E_n)_{n \geq 1}$ be an increasing sequence in $\mathcal{D}$, and set

$$E := \bigcup_{n=1}^{\infty} E_n.$$

Since $\mu(X) < \infty$, continuity from below gives

$$\mu(E \setminus E_n) \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Given $\eta > 0$, we choose n such that

$$\mu(E \setminus E_n) < \eta/2.$$

Since E_n belongs to $\mathcal{D}$, there exists A in $\mathcal{E}$ such that

$$\mu(E_n \Delta A) < \eta/2.$$

Then

$$\mu(E \Delta A) \leq \mu(E \setminus E_n) + \mu(E_n \Delta A) < \eta.$$

Hence E belongs to $\mathcal{D}$.

If $(E_n)_{n \geq 1}$ is decreasing in $\mathcal{D}$ and

$$E := \bigcap_{n=1}^{\infty} E_n,$$

then continuity from above gives

$$\mu(E_n \setminus E) \rightarrow 0 \quad \text{as } n \rightarrow +\infty$$

because $\mu(E_1) \leq \mu(X) < \infty$. Given $\eta > 0$, choose n such that $\mu(E_n \setminus E) < \eta/2$. Since E_n belongs to $\mathcal{D}$, choose A in $\mathcal{E}$ such that $\mu(E_n \Delta A) < \eta/2$. Then

$$\mu(E \Delta A) \leq \mu(E_n \setminus E) + \mu(E_n \Delta A) < \eta.$$

Thus E belongs to $\mathcal{D}$. Thus $\mathcal{D}$ is a monotone class.

Since $\mathcal{D}$ is a monotone class containing the algebra $\mathcal{E}$, the monotone class theorem (Theorem 3.10) yields

$$\mathcal{A} = \sigma(\mathcal{E}) \subset \mathcal{D}.$$

This proves the corollary. □

Corollary 3.12 (Uniqueness on a pi-system). *Let $\mathcal{P}$ be a π -system on a set X such that X belongs to $\mathcal{P}$. Let μ and ν be finite measures on $\sigma(\mathcal{P})$. If*

$$\mu(E) = \nu(E)$$

for every E in $\mathcal{P}$, then $\mu = \nu$ on $\sigma(\mathcal{P})$.

Proof. We consider the class

$$\mathcal{D} := \{A \in \sigma(\mathcal{P}) : \mu(A) = \nu(A)\}.$$

Since X belongs to $\mathcal{P}$, we have

$$\mu(X) = \nu(X),$$

so X belongs to $\mathcal{D}$. If A and B belong to $\mathcal{D}$ and $A \subset B$, then

$$\mu(B \setminus A) = \mu(B) - \mu(A) = \nu(B) - \nu(A) = \nu(B \setminus A),$$

so $B \setminus A$ belongs to $\mathcal{D}$. If $(A_n)_{n \geq 1}$ is an increasing sequence in $\mathcal{D}$, continuity from below gives

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n) = \lim_{n \rightarrow \infty} \nu(A_n) = \nu\left(\bigcup_{n=1}^{\infty} A_n\right),$$

so $\bigcup_{n \geq 1} A_n$ belongs to $\mathcal{D}$. Thus $\mathcal{D}$ is a Dynkin system. Since it contains $\mathcal{P}$, Theorem 3.8 implies that

$$\mathcal{D} = \sigma(\mathcal{P}).$$

Hence $\mu = \nu$ on $\sigma(\mathcal{P})$. □

Lemma 3.13 (Half-open intervals as a pi-system). *Let J be a bounded interval of $\mathbb{R}$. Define*

$$\mathcal{P}_J := \{\emptyset, J\} \cup \{(s, t] \subset J : s < t\}.$$

Then $\mathcal{P}_J$ is a π -system on J and

$$\sigma(\mathcal{P}_J) = \mathcal{B}(J).$$

Proof. The intersection of two intervals of the form $(s, t]$ is either empty or again of the same form, and the empty set belongs to $\mathcal{P}_J$. Intersections with J do not leave the family. Thus $\mathcal{P}_J$ is a π -system. Every element of $\mathcal{P}_J$ is Borel in J , hence $\sigma(\mathcal{P}_J) \subset \mathcal{B}(J)$. Conversely, for every $a < b$ with $(a, b) \subset J$, we have

$$(a, b) = \bigcup_{n \geq 1} \left(a, b - \frac{1}{n}\right]$$

where terms with $b - 1/n \leq a$ are simply omitted. Hence every open interval contained in J belongs to $\sigma(\mathcal{P}_J)$. Since the open intervals contained in J generate the Borel σ -algebra of J , we obtain $\mathcal{B}(J) \subset \sigma(\mathcal{P}_J)$. □

Corollary 3.14 (Uniqueness of σ -finite extensions). *Let $\mathcal{E}$ be an algebra on X , and let μ_0 be a premeasure on $\mathcal{E}$. Assume that μ_0 is σ -finite on $\mathcal{E}$, that is, that there exists a sequence $(E_n)_{n \geq 1}$ in $\mathcal{E}$ such that*

$$X = \bigcup_{n=1}^{\infty} E_n, \quad \mu_0(E_n) < \infty$$

for every $n \geq 1$. Then the extension of μ_0 to $\sigma(\mathcal{E})$ is unique.

Proof. Let μ and ν be two measures on $\sigma(\mathcal{E})$ extending μ_0 . Replacing E_n by

$$F_n := \bigcup_{k=1}^n E_k,$$

we may assume that the sequence $(E_n)_{n \geq 1}$ is increasing.

For every $n \geq 1$, we set

$$\mathcal{P}_n := \{A \cap E_n : A \in \mathcal{E}\}.$$

Since $\mathcal{E}$ is an algebra, $\mathcal{P}_n$ is a π -system on E_n and E_n belongs to $\mathcal{P}_n$. Let

$$\mathcal{A}_n := \{B \cap E_n : B \in \sigma(\mathcal{E})\}.$$

Then $\mathcal{A}_n$ is a σ -algebra on E_n containing $\mathcal{P}_n$, so

$$\sigma(\mathcal{P}_n) \subset \mathcal{A}_n.$$

Conversely, the class

$$\mathcal{G}_n := \{B \in \sigma(\mathcal{E}) : B \cap E_n \in \sigma(\mathcal{P}_n)\}$$

is a σ -algebra on X which contains $\mathcal{E}$. Hence

$$\mathcal{G}_n = \sigma(\mathcal{E}),$$

which means that

$$\mathcal{A}_n \subset \sigma(\mathcal{P}_n).$$

Therefore

$$\mathcal{A}_n = \sigma(\mathcal{P}_n).$$

We now define two finite measures on $\mathcal{A}_n$ by

$$\mu_n(C) := \mu(C), \quad \nu_n(C) := \nu(C)$$

for every C in $\mathcal{A}_n$. Since μ and ν extend μ_0 , for every A in $\mathcal{E}$ we have

$$\mu_n(A \cap E_n) = \mu(A \cap E_n) = \mu_0(A \cap E_n) = \nu(A \cap E_n) = \nu_n(A \cap E_n).$$

Thus μ_n and ν_n coincide on $\mathcal{P}_n$. Corollary 3.12 applied on E_n yields

$$\mu(B \cap E_n) = \nu(B \cap E_n)$$

for every B in $\sigma(\mathcal{E})$.

Since $E_n \uparrow X$, continuity from below gives, for every B in $\sigma(\mathcal{E})$,

$$\mu(B) = \lim_{n \rightarrow \infty} \mu(B \cap E_n) = \lim_{n \rightarrow \infty} \nu(B \cap E_n) = \nu(B).$$

Hence $\mu = \nu$ on $\sigma(\mathcal{E})$. □

3.5 Application to the Lebesgue measure

For $N \geq 1$, let $\mathcal{I}$ be the family consisting of $\emptyset$, $\mathbb{R}$, and all intervals of the forms

$$(a, b], \quad (-\infty, b], \quad (a, +\infty),$$

where a, b are real numbers and $a < b$. Let $\mathcal{E}_N$ be the family of all finite disjoint unions of rectangles

$$R = I_1 \times \cdots \times I_N, \quad I_j \in \mathcal{I}.$$

This family is an algebra on $\mathbb{R}^N$: finite intersections of such rectangles are rectangles of the same type, and complements are finite disjoint unions of such rectangles.

For I in $\mathcal{I}$, define

$$\ell(\emptyset) = 0, \quad \ell((a, b]) = b - a, \quad \ell((-\infty, b]) = \ell((a, +\infty)) = \ell(\mathbb{R}) = +\infty.$$

For a rectangle $R = I_1 \times \cdots \times I_N$, set

$$|R| := \prod_{j=1}^N \ell(I_j),$$

with the conventions of Section 1.4. If E in $\mathcal{E}_N$ is written as a finite disjoint union

$$E = \bigcup_{j=1}^m R_j,$$

we define

$$\mu_0(E) := \sum_{j=1}^m |R_j|.$$

The finite-refinement argument shows that this definition is independent of the chosen disjoint representation. Indeed, two finite decompositions admit the common refinement formed by all nonempty intersections of rectangles from the two decompositions, and the volume of a rectangle is the sum of the volumes of the pieces obtained by cutting its sides at finitely many endpoints. The same refinement argument gives finite additivity on $\mathcal{E}_N$.

The map μ_0 is a premeasure on $\mathcal{E}_N$. For finite-volume sets this follows from the standard compactness argument for rectangles: if $E = \bigcup_{k \geq 1} E_k$ is a disjoint union in $\mathcal{E}_N$, with E also in $\mathcal{E}_N$, then finite additivity gives $\sum_{k=1}^m \mu_0(E_k) \leq \mu_0(E)$ for every m , and the reverse inequality is obtained by shrinking the finitely many rectangles in a decomposition of E and enlarging the finitely many rectangles in decompositions of the E_k , then extracting a finite subcover. The infinite-volume case follows by intersecting with the boxes $(-r, r]^N$ and letting $r \rightarrow +\infty$.

The Carathéodory extension theorem therefore yields a measure on $\sigma(\mathcal{E}_N)$ extending this notion of volume. Since every element of $\mathcal{E}_N$ is Borel, we have $\sigma(\mathcal{E}_N) \subset \mathcal{B}(\mathbb{R}^N)$. Conversely, bounded half-open rectangles with rational endpoints belong to $\mathcal{E}_N$ and generate the Euclidean Borel σ -algebra. Hence

$$\sigma(\mathcal{E}_N) = \mathcal{B}(\mathbb{R}^N).$$

This produces the classical Borel Lebesgue measure and proves the existence part of Theorem 2.10. Later, in Section 14.3, we will recover the same Borel measure from the Riesz representation theorem and then describe its completion.

3.6 Stieltjes measures and absolute continuity of functions

The phrase “absolute continuity” has already been used for measures in Section 2.5. We now record the corresponding notion for functions on a compact interval and explain the link between the two notions through Stieltjes measures. This material will be used later in the proof of the converse part of the fundamental theorem of calculus.

Definition 3.15 (Absolutely continuous functions on a compact interval). Let $a < b$ be real numbers and let $F: [a, b] \rightarrow \mathbb{R}$ be a function. We say that F is absolutely continuous on $[a, b]$ if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that, for every finite family of pairwise disjoint intervals (α_k, β_k) with

$$a \leq \alpha_k < \beta_k \leq b,$$

if

$$\sum_{k=1}^m (\beta_k - \alpha_k) < \delta,$$

then

$$\sum_{k=1}^m |F(\beta_k) - F(\alpha_k)| < \varepsilon.$$

We denote by $AC([a, b])$ the set of absolutely continuous functions on $[a, b]$.

Proposition 3.16 (Basic properties of absolutely continuous functions). Let $a < b$ be real numbers. Every function in $AC([a, b])$ is continuous on $[a, b]$. Moreover, $AC([a, b])$ is a real vector space for the pointwise operations.

Proof. Let F be in $AC([a, b])$. Let $\varepsilon > 0$ and choose $\delta > 0$ as in Definition 3.15. If x, y are in $[a, b]$ and $|x - y| < \delta$, then, assuming for instance that $x < y$, the family consisting of the single interval (x, y) gives

$$|F(y) - F(x)| < \varepsilon.$$

Thus F is uniformly continuous, and in particular continuous.

Let F and G be in $AC([a, b])$. For $F + G$, given $\varepsilon > 0$, choose numbers $\delta_F > 0$ and $\delta_G > 0$ associated respectively with F and G and with the tolerance $\varepsilon/2$. If

$$\sum_{k=1}^m (\beta_k - \alpha_k) < \min(\delta_F, \delta_G),$$

then

$$\begin{aligned} \sum_{k=1}^m |(F + G)(\beta_k) - (F + G)(\alpha_k)| &\leq \sum_{k=1}^m |F(\beta_k) - F(\alpha_k)| + \sum_{k=1}^m |G(\beta_k) - G(\alpha_k)| \\ &< \varepsilon. \end{aligned}$$

Hence $F + G$ is absolutely continuous.

Let c be in $\mathbb{R}$. If $c = 0$, then cF is absolutely continuous. If $c \neq 0$, we choose $\delta > 0$ associated with F and with the tolerance $\varepsilon/|c|$. Then

$$\sum_{k=1}^m |cF(\beta_k) - cF(\alpha_k)| < \varepsilon$$

whenever the total length of the intervals is smaller than δ . Thus cF is absolutely continuous. Therefore $AC([a, b])$ is a real vector space. $\square$

Theorem 3.17 (Lebesgue–Stieltjes measure on a compact interval). *Let $a < b$ be real numbers and let $F: [a, b] \rightarrow \mathbb{R}$ be increasing and right-continuous at every point of $[a, b[$. Then there exists a unique finite Borel measure μ_F on $[a, b]$ such that*

$$\mu_F(\{a\}) = 0$$

and, for every $a \leq s < t \leq b$,

$$\mu_F((s, t]) = F(t) - F(s).$$

The measure μ_F is called the Stieltjes measure associated with F .

Proof. Let $\mathcal{E}$ be the algebra of subsets of $[a, b]$ consisting of finite disjoint unions of the singleton $\{a\}$ and intervals of the form $(s, t]$, where $a \leq s < t \leq b$. On this algebra, define μ_0 by

$$\mu_0(\{a\}) = 0, \quad \mu_0((s, t]) = F(t) - F(s),$$

and extend this formula additively to finite disjoint unions. The usual finite-refinement argument shows that μ_0 is well defined and finitely additive.

We prove that μ_0 is a premeasure. Since $\mu_0([a, b]) = F(b) - F(a) < +\infty$, it is enough to prove continuity from above at the empty set. Let $(A_n)_{n \geq 1}$ be a decreasing sequence in $\mathcal{E}$ such that

$$\bigcap_{n=1}^{\infty} A_n = \emptyset.$$

Fix $\varepsilon > 0$. For every $n \geq 1$, we choose a compact set $K_n \subset A_n$ and a set B_n in $\mathcal{E}$ such that

$$A_n \setminus K_n \subset B_n, \quad \mu_0(B_n) < \varepsilon 2^{-n}.$$

This is obtained as follows. Write A_n as a finite disjoint union of a possible singleton $\{a\}$ and half-open intervals $(s, t]$. The singleton is kept in the compact part. On each interval $(s, t]$, we approximate from inside by $[s + r, t]$ and cover the discarded part by $(s, s + r]$. Since F is right-continuous at s , the quantity

$$\mu_0((s, s + r]) = F(s + r) - F(s)$$

can be made arbitrarily small. Since the decomposition is finite, the discarded pieces can be chosen so that their total μ_0 -mass is smaller than $\varepsilon 2^{-n}$.

Since $\bigcap_{n \geq 1} K_n = \emptyset$ and the sets K_n are compact subsets of $[a, b]$, compactness gives a finite set of indices $n_1, \dots, n_q$ such that

$$K_{n_1} \cap \dots \cap K_{n_q} = \emptyset.$$

Let $N := \max(n_1, \dots, n_q)$. Since the sequence (A_n) is decreasing, if x belongs to A_N , then x belongs to every A_{n_j} . If x belonged to none of the sets B_{n_j} , the inclusions $A_{n_j} \setminus K_{n_j} \subset B_{n_j}$ would imply $x \in K_{n_j}$ for every j , contradicting the emptiness of $K_{n_1} \cap \dots \cap K_{n_q}$. Hence

$$A_N \subset \bigcup_{j=1}^q B_{n_j}.$$

Since A_N and the B_{n_j} belong to $\mathcal{E}$, finite subadditivity gives

$$\mu_0(A_N) \leq \sum_{j=1}^q \mu_0(B_{n_j}) < \varepsilon.$$

Thus $\mu_0(A_n) \rightarrow 0$, so μ_0 is a finite premeasure on $\mathcal{E}$.

The Carathéodory extension theorem gives a finite measure on $\sigma(\mathcal{E})$. Since the intervals $(s, t]$ and the singleton $\{a\}$ generate the Borel σ -algebra of $[a, b]$, this gives a finite Borel measure μ_F with the desired properties.

The uniqueness follows from Corollary 3.12, applied to the π -system

$$\mathcal{P} := \{[a, t] : a \leq t \leq b\}.$$

This π -system contains $[a, b]$ and generates $\mathcal{B}([a, b])$. The prescribed identities determine the value of the measure on $\mathcal{P}$, since

$$\mu_F([a, t]) = \mu_F(\{a\}) + \mu_F((a, t]) = F(t) - F(a).$$

□

Proposition 3.18 (Absolute continuity and the associated Stieltjes measure). *Let $a < b$ be real numbers and let $F: [a, b] \rightarrow \mathbb{R}$ be increasing and right-continuous at every point of $[a, b]$. Let μ_F be the associated Stieltjes measure. Then*

$$F \in AC([a, b]) \iff \mu_F \ll \lambda_1|_{[a, b]}.$$

Proof. Assume first that F is absolutely continuous. By Proposition 3.16, F is continuous on $[a, b]$. Let E be a Borel subset of $[a, b]$ such that $\lambda_1(E) = 0$, and let $\varepsilon > 0$. Choose $\delta > 0$ in the definition of absolute continuity of F associated with ε . By the outer-measure construction of the Lebesgue measure, there exists an open set U in $\mathbb{R}$ such that

$$E \subset U, \quad \lambda_1(U) < \delta.$$

Write U as a countable disjoint union of open intervals. Intersecting these intervals with $]a, b]$ and replacing each nonempty intersection by the corresponding half-open interval of the form $(\alpha_j, \beta_j]$ gives a countable disjoint family such that

$$E \cap]a, b] \subset \bigcup_j (\alpha_j, \beta_j], \quad \sum_j (\beta_j - \alpha_j) \leq \lambda_1(U) < \delta.$$

The point a has μ_F -measure zero by the definition of the Stieltjes measure. Therefore, for every finite subset J of indices, the absolute continuity of F gives

$$\sum_{j \in J} \mu_F((\alpha_j, \beta_j]) = \sum_{j \in J} (F(\beta_j) - F(\alpha_j)) < \varepsilon.$$

By countable additivity, equivalently by continuity from below for measures, the whole countable sum is at most ε , and hence $\mu_F(E) \leq \varepsilon$. Since ε is arbitrary, $\mu_F(E) = 0$. Thus $\mu_F \ll \lambda_1|_{[a, b]}$.

Conversely, assume that $\mu_F \ll \lambda_1|_{[a, b]}$. Since both measures are finite, Lemma 2.17 gives, for every $\varepsilon > 0$, a number $\delta > 0$ such that

$$\lambda_1(A) < \delta \implies \mu_F(A) < \varepsilon$$

for every Borel set $A \subset [a, b]$. Let $(\alpha_k, \beta_k)_{1 \leq k \leq m}$ be a finite family of pairwise disjoint intervals with $a \leq \alpha_k < \beta_k \leq b$ and

$$\sum_{k=1}^m (\beta_k - \alpha_k) < \delta.$$

Set

$$A := \bigcup_{k=1}^m (\alpha_k, \beta_k].$$

Then $\lambda_1(A) < \delta$, and therefore

$$\begin{aligned}\sum_{k=1}^m |F(\beta_k) - F(\alpha_k)| &= \sum_{k=1}^m (F(\beta_k) - F(\alpha_k)) \\ &= \mu_F(A) < \varepsilon.\end{aligned}$$

Thus F is absolutely continuous on $[a, b]$. □

Chapter 4

Measurable functions

Having defined measurable sets and measures, we now pass to measurable functions. The chapter recalls the role of preimages, gives practical criteria for measurability, and develops the stability properties that let us build large classes of measurable functions from simpler ones and from continuous functions.

4.1 Gentle reminder on the notion of preimage

Let us start with a few reminders and notations in the general case of set maps. Let $f : X \rightarrow Y$ be a map from a set X to a set Y . For any $B \subset Y$, we define the preimage (or reciprocal image) of B by f :

$$f^{-1}(B) := \{x \in X : f(x) \in B\}.$$

Remember that this is simple notation and does not imply that f is invertible. Of course, we have $f^{-1}(Y) = X$, $f^{-1}(\emptyset) = \emptyset$, and recall the formulas, valid for any $A, B \subset Y$, where complements are taken in Y and X respectively:

$$\begin{aligned} f^{-1}(A \cap B) &= f^{-1}(A) \cap f^{-1}(B), \\ f^{-1}(A \cup B) &= f^{-1}(A) \cup f^{-1}(B), \\ f^{-1}(A^c) &= (f^{-1}(A))^c. \end{aligned}$$

If I is any set of indices, the first two formulas generalize to $\cap_{i \in I} A_i$ and $\cup_{i \in I} A_i$, where $A_i \subset Y$ for all i .

4.2 Definition of a measurable function

Definition 4.1 (Measurable maps). *Let $(X, \mathcal{A}_X)$ and $(Y, \mathcal{A}_Y)$ be measurable spaces and let $f : X \rightarrow Y$. We say that f is $(\mathcal{A}_X, \mathcal{A}_Y)$ -measurable (or simply measurable if there is no ambiguity on the σ -algebras involved) if for all B in $\mathcal{A}_Y$ we have $f^{-1}(B)$ in $\mathcal{A}_X$.*

Let us look at some basic examples of measurable functions.

- Let $(X, \mathcal{A}_X)$ be a measurable space. Then $\text{Id} : X \rightarrow X$, defined by $\text{Id}(x) = x$, is measurable.
- If $(X, \mathcal{A}_X)$ and $(Y, \mathcal{A}_Y)$ are measurable spaces and y_0 is in Y , then the constant map $x \mapsto y_0$ is measurable, because the preimage of any measurable set in Y is either X or $\emptyset$.

Proposition 4.2 (Measurability of indicator functions). *Let $(X, \mathcal{A})$ be a measurable space and let $A \subset X$. Then A is in $\mathcal{A}$ if and only if $\mathbf{1}_A : X \rightarrow \mathbb{R}$ is $(\mathcal{A}, \mathcal{B}(\mathbb{R}))$ -measurable.*

Proof. Suppose first that A is in $\mathcal{A}$. Let B in $\mathcal{B}(\mathbb{R})$. Since $\mathbf{1}_A$ only takes the values 0 and 1, the set $\mathbf{1}_A^{-1}(B)$ is equal to $\emptyset$, A , A^c , or X . In each case it belongs to $\mathcal{A}$. Thus $\mathbf{1}_A$ is measurable. Conversely, if $\mathbf{1}_A$ is measurable, then

$$A = \mathbf{1}_A^{-1}(\{1\}).$$

Since $\{1\}$ is closed in $\mathbb{R}$, it is Borel, hence A is in $\mathcal{A}$. □

4.3 A maximal σ -algebra on the target and a related criterion

The following proposition associates with an arbitrary map $f : X \rightarrow Y$ the largest target σ -algebra for which f is measurable. This is the final σ -algebra on Y induced by f and $\mathcal{A}_X$. It is useful for proving the following generator criterion.

Proposition 4.3 (Measurable sets defined by level conditions). *Let $(X, \mathcal{A}_X)$ and $(Y, \mathcal{A}_Y)$ be two measurable spaces. Let $f : X \rightarrow Y$ be a map, which is not necessarily assumed to be measurable. We define*

$$f_*\mathcal{A}_X := \{B \subset Y : f^{-1}(B) \in \mathcal{A}_X\},$$

which is a subset of $\mathcal{P}(Y)$. Then

- 1) $f_*\mathcal{A}_X$ is a σ -algebra on Y ,
- 2) moreover, a map $f : X \rightarrow Y$ is $(\mathcal{A}_X, \mathcal{A}_Y)$ -measurable if and only if $\mathcal{A}_Y \subset f_*\mathcal{A}_X$,
- 3) and finally, $f_*\mathcal{A}_X$ is the largest σ -algebra $\mathcal{S}$ on Y such that f is measurable from $(X, \mathcal{A}_X)$ to $(Y, \mathcal{S})$.

Proof. We first prove 1). Since $f^{-1}(Y) = X$ and X is in $\mathcal{A}_X$, we have Y in $f_*\mathcal{A}_X$. If B is in $f_*\mathcal{A}_X$, then

$$f^{-1}(B^c) = (f^{-1}(B))^c,$$

and the right-hand side is in $\mathcal{A}_X$. Hence B^c is in $f_*\mathcal{A}_X$. Finally, if $(B_n)_{n \geq 1}$ is a sequence of elements of $f_*\mathcal{A}_X$, then

$$f^{-1}\left(\bigcup_{n \geq 1} B_n\right) = \bigcup_{n \geq 1} f^{-1}(B_n),$$

and the right-hand side belongs to $\mathcal{A}_X$. Therefore $\bigcup_{n \geq 1} B_n$ is in $f_*\mathcal{A}_X$. We have proved that $f_*\mathcal{A}_X$ is a σ -algebra on Y .

Property 2) is just a reformulation of the definition of measurability: saying that f is $(\mathcal{A}_X, \mathcal{A}_Y)$ -measurable means exactly that for every B in $\mathcal{A}_Y$, the preimage $f^{-1}(B)$ belongs to $\mathcal{A}_X$, i.e. that every B in $\mathcal{A}_Y$ belongs to $f_*\mathcal{A}_X$.

We finally prove 3). By 1), $f_*\mathcal{A}_X$ is a σ -algebra on Y , and by 2) the inclusion $f_*\mathcal{A}_X \subset f_*\mathcal{A}_X$ shows that f is measurable from $(X, \mathcal{A}_X)$ to $(Y, f_*\mathcal{A}_X)$. Conversely, if $\mathcal{S}$ is a σ -algebra on Y such that f is measurable from $(X, \mathcal{A}_X)$ to $(Y, \mathcal{S})$, then 2) gives $\mathcal{S} \subset f_*\mathcal{A}_X$. Thus $f_*\mathcal{A}_X$ is the largest such σ -algebra. □

Proposition 4.4 (Criteria for measurable functions). *Let $(X, \mathcal{A}_X)$ be a measurable space, let Y be a set, let S be a collection of subsets of Y , and let $\mathcal{A}_Y := \sigma(S)$. Then a function $f : X \rightarrow Y$ is $(\mathcal{A}_X, \mathcal{A}_Y)$ -measurable if and only if*

$$\forall B \in S, f^{-1}(B) \in \mathcal{A}_X.$$

Proof. Only the reciprocal implication requires a proof. We have $S \subset f_*\mathcal{A}_X$ by hypothesis, so, since $f_*\mathcal{A}_X$ is a σ -algebra on Y , we deduce that $\mathcal{A}_Y \subset f_*\mathcal{A}_X$. Thus f is $(\mathcal{A}_X, \mathcal{A}_Y)$ -measurable. □

4.4 Measurability versus continuity

When topological spaces are involved, the Borel σ -algebras provide a natural bridge between continuity and measurability.

Definition 4.5 (Borel measurable maps). *Let (X, τ_X) and (Y, τ_Y) be topological spaces and let $\mathcal{B}_X, \mathcal{B}_Y$ be the Borel σ -algebras associated with τ_X and τ_Y respectively. We say that $f: X \rightarrow Y$ is a Borel function if f is $(\mathcal{B}_X, \mathcal{B}_Y)$ -measurable.*

Theorem 4.6 (Borel measurability of continuous maps). *Let (X, τ_X) and (Y, τ_Y) be topological spaces and let $\mathcal{B}_X, \mathcal{B}_Y$ be the Borel σ -algebras associated with τ_X and τ_Y respectively. If $f: X \rightarrow Y$ is continuous then f is $(\mathcal{B}_X, \mathcal{B}_Y)$ -measurable.*

! The converse implication is false in general. For instance, $\mathbf{1}_{]0,1[}: \mathbb{R} \rightarrow \mathbb{R}$ is Borel measurable but discontinuous at 0 and 1.

Proof. By Proposition 4.4, it is enough to check the measurability of the preimages of the open subsets of Y . Let U be an open subset of Y . Since f is continuous, we have $f^{-1}(U) \in \tau_X \subset \mathcal{B}_X$. Thus f is $(\mathcal{B}_X, \mathcal{B}_Y)$ -measurable. $\square$

4.5 Stability of measurable functions

The following result proves that the composition of two measurable functions gives a measurable function.

Proposition 4.7 (Composition of measurable maps). *Let $(X, \mathcal{A}_X)$, $(Y, \mathcal{A}_Y)$, and $(Z, \mathcal{A}_Z)$ be measurable spaces. If $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are measurable, then the composition*

$$g \circ f: X \rightarrow Z$$

is measurable.

Proof. Let C in $\mathcal{A}_Z$. Since g is measurable, $g^{-1}(C)$ is in $\mathcal{A}_Y$. Since f is measurable, we deduce that

$$(g \circ f)^{-1}(C) = f^{-1}(g^{-1}(C))$$

belongs to $\mathcal{A}_X$. Hence $g \circ f$ is measurable. $\square$

Proposition 4.8 (Vector-valued measurability criterion). *Let $(X, \mathcal{A})$ be a measurable space. We endow each $\mathbb{R}^k$, $k \geq 1$, with its Borel σ -algebra $\mathcal{B}(\mathbb{R}^k)$. Let $n \in \mathbb{N}_{\geq 1}$ and let*

$$f = (f_1, \dots, f_n): X \rightarrow \mathbb{R}^n, \quad x \mapsto (f_1(x), \dots, f_n(x)),$$

be a map. Then f is measurable if and only if every $f_i: X \rightarrow \mathbb{R}$ is measurable.

Proof. For any $1 \leq i \leq n$, let $\pi_i: \mathbb{R}^n \rightarrow \mathbb{R}$ be the projection onto the i -th coordinate. The map π_i is continuous and therefore Borel. So, if f is measurable, then $f_i = \pi_i \circ f$ is $(\mathcal{A}, \mathcal{B}(\mathbb{R}))$ -measurable. Conversely, assume that each f_i is measurable. If $\mathcal{R}_n$ denotes the family of open rectangles of $\mathbb{R}^n$, then $\mathcal{B}(\mathbb{R}^n) = \sigma(\mathcal{R}_n)$, where the rectangles are

$$D((x_i)_i, (y_i)_i) := \{(z_1, \dots, z_n) \in \mathbb{R}^n: \forall i, x_i < z_i < y_i\}.$$

For such a rectangle we have

$$f^{-1}(D((x_i)_i, (y_i)_i)) = \bigcap_{k=1}^n f_k^{-1}(]x_k, y_k[),$$

which belongs to $\mathcal{A}$. Proposition 4.4 therefore gives the measurability of f . $\square$

Proposition 4.9 (Stability by continuous functional calculus). *Let $(X, \mathcal{A})$ be a measurable space and let $u, v: X \rightarrow \mathbb{R}$ be two measurable maps, where $\mathbb{R}$ is provided with its Borel σ -algebra. Let $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}$ be Borel measurable, in particular continuous. Then the map*

$$h: X \rightarrow \mathbb{R}, \quad x \mapsto \phi(u(x), v(x))$$

is measurable.

Proof. Let $F: X \rightarrow \mathbb{R}^2$ be defined by

$$F(x) := (u(x), v(x)).$$

Since u and v are measurable, Proposition 4.8 shows that F is measurable. Since ϕ is Borel, Proposition 4.7 implies that

$$h = \phi \circ F$$

is measurable. $\square$

We can now state the following theorem, which is constantly used in practice. We recall that, by definition, for a sequence of real numbers $(a_n)_n$,

$$\limsup_n a_n := \lim_{n \rightarrow +\infty} \sup_{k \geq n} a_k \quad \text{and} \quad \liminf_n a_n := \lim_{n \rightarrow +\infty} \inf_{k \geq n} a_k.$$

The same definitions apply to functions; then the previous formulas are applied pointwise.

Theorem 4.10 (Stability of measurable functions under limits). *Let $(X, \mathcal{A})$ be a measurable space. Let $\mathbb{R}$ and $\overline{\mathbb{R}}$ be given their respective Borel σ -algebras.*

1) *If $f: X \rightarrow \mathbb{R}$ is measurable, then $|f|$ is also measurable.*

2) *If $f, g: X \rightarrow \mathbb{R}$ are measurable, then the maps*

$$f + g, \quad f \times g, \quad \max(f, g), \quad \min(f, g)$$

are measurable.

3) *If $f_k: X \rightarrow \overline{\mathbb{R}}$ is the general term of a sequence $(f_k)_{k \geq 1}$ of measurable functions, then the following maps from X to $\overline{\mathbb{R}}$:*

$$\inf_k f_k, \quad \sup_k f_k, \quad \limsup_k f_k, \quad \liminf_k f_k$$

are measurable.

4) *If $(f_k)_{k \geq 1}$ is a sequence of measurable functions from X to $\overline{\mathbb{R}}$ and if f_k converges pointwise to a function $f: X \rightarrow \overline{\mathbb{R}}$, then f is measurable.*

Proof. For 1), we apply Proposition 4.9 with $u = v = f$ and

$$\phi(x, y) = |x|.$$

The function ϕ is continuous on $\mathbb{R}^2$, hence $|f|$ is measurable.

For 2), we apply Proposition 4.9 with the continuous functions

$$\phi_1(x, y) = x + y, \quad \phi_2(x, y) = xy, \quad \phi_3(x, y) = \max(x, y), \quad \phi_4(x, y) = \min(x, y).$$

This gives the measurability of $f + g$, $f \times g$, $\max(f, g)$, and $\min(f, g)$.

For 3), we use Proposition 4.4. The family $\{[-\infty, a[: a \in \mathbb{R}\}$ generates $\mathcal{B}(\overline{\mathbb{R}})$: indeed, $[-\infty, a] = \bigcap_{m \geq 1} [-\infty, a + 1/m[$, while $]a, +\infty] = [-\infty, a]^c$ and $]a, b[= [-\infty, b[\cap [-\infty, a]^c$. Thus it is enough to consider these generating intervals. Let

$$g(x) := \sup_{k \geq 1} f_k(x) \quad \text{and} \quad h(x) := \inf_{k \geq 1} f_k(x).$$

For every a in $\mathbb{R}$, we have

$$\begin{aligned} g^{-1}([-\infty, a[) &= \bigcap_{k \geq 1} f_k^{-1}([-\infty, a[), \\ h^{-1}([-\infty, a[) &= \bigcup_{k \geq 1} f_k^{-1}([-\infty, a[). \end{aligned}$$

The right-hand sides are measurable, so g and h are measurable.

For each $n \geq 1$, let

$$g_n := \sup_{k \geq n} f_k \quad \text{and} \quad h_n := \inf_{k \geq n} f_k.$$

By the first part of 3), applied to the tails $(f_k)_{k \geq n}$, each g_n and each h_n is measurable. Since

$$\limsup_k f_k = \inf_{n \geq 1} g_n \quad \text{and} \quad \liminf_k f_k = \sup_{n \geq 1} h_n,$$

another application of the same argument shows that $\limsup_k f_k$ and $\liminf_k f_k$ are measurable. This proves 3). If $\lim_k f_k(x)$ exists for all x in X , then it coincides pointwise with both $\limsup_k f_k$ and $\liminf_k f_k$. Hence the limit function is measurable, which proves 4). $\square$

Chapter 5

Simple functions and their Lebesgue integral

Simple functions are the first class for which the integral can be defined directly and computed explicitly. We describe their decompositions, characterize when they are measurable, and establish the algebraic properties of their integral, which will be the starting point for the general construction.

5.1 Simple functions and their decompositions

We will see that defining the Lebesgue integral of an indicator function of a measurable part is straightforward. In order to define the integrals of positive measurable functions more generally, we therefore seek to approximate them by linear combinations of indicators of measurable parts. This point of view motivates the introduction of the following so-called simple functions.

Definition 5.1 (Simple functions). *Let X be a set. A function $s: X \rightarrow \mathbb{R}$ is a simple function if s takes only a finite number of values, i.e. if $s(X)$ is a finite part of $\mathbb{R}$. The empty image is allowed when $X = \emptyset$.*

Proposition 5.2 (Canonical decomposition of a simple function). *Let X be a set and let $s: X \rightarrow \mathbb{R}$ be a simple function. If $X = \emptyset$, then s admits the empty decomposition, and all sums below are understood as empty sums. If $X \neq \emptyset$, write*

$$s(X) = \{\alpha_1, \dots, \alpha_r\}$$

and, for each $1 \leq i \leq r$, let

$$A_i = s^{-1}(\{\alpha_i\}).$$

Then

$$s = \sum_{i=1}^r \alpha_i \mathbf{1}_{A_i}. \quad (5.1)$$

Moreover, the family $(A_i)_{1 \leq i \leq r}$ satisfies the three following properties:

- (i) $\bigcup_{i=1}^r A_i = X$,*
- (ii) if $i \neq j$, then $A_i \cap A_j = \emptyset$,*
- (iii) for every $1 \leq i \leq r$, we have $A_i \neq \emptyset$.*

This decomposition is called the canonical decomposition of s .

Proof. If $X = \emptyset$, the statement follows from the convention on the empty sum. Assume that

$X \neq \emptyset$. Fix x in X . Since $s(x)$ belongs to $s(X)$, there exists i such that $s(x) = \alpha_i$, hence x is in A_i . This proves (i). If x is in $A_i \cap A_j$, then $s(x) = \alpha_i$ and $s(x) = \alpha_j$, so $\alpha_i = \alpha_j$, and therefore $i = j$. This proves (ii). Finally, since α_i is a value taken by s , there exists x in X such that $s(x) = \alpha_i$, hence x is in A_i . This proves (iii). For every x in X , property (i) gives an index i such that x is in A_i , and property (ii) shows that this index is unique. Therefore

$$\sum_{i=1}^r \alpha_i \mathbf{1}_{A_i}(x) = \alpha_i = s(x),$$

which proves (5.1). □

Lemma 5.3 (Refinement of finite partitions). *Let X be a set and let $(A_i)_{1 \leq i \leq r}$ be subsets of X such that $\bigcup_{i=1}^r A_i = X$ and $A_i \cap A_j = \emptyset$ whenever $i \neq j$. Then*

$$\mathbf{1}_X = \sum_{i=1}^r \mathbf{1}_{A_i}.$$

Proof. Let x in X . Since the sets A_i cover X , there exists i such that x is in A_i . Since they are pairwise disjoint, this index is unique. Hence exactly one term in the sum is equal to 1, while all the others are equal to 0. Therefore

$$\sum_{i=1}^r \mathbf{1}_{A_i}(x) = 1 = \mathbf{1}_X(x).$$

□

Proposition 5.4 (Finite range of simple functions). *Let X be a set. Let $r \in \mathbb{N}_{\geq 1}$, let $\alpha_1, \dots, \alpha_r$ be real numbers, and let $(A_i)_{1 \leq i \leq r}$ be subsets of X such that $\bigcup_{i=1}^r A_i = X$ and $A_i \cap A_j = \emptyset$ whenever $i \neq j$. If*

$$s = \sum_{i=1}^r \alpha_i \mathbf{1}_{A_i},$$

then

$$s(X) \subset \{\alpha_1, \dots, \alpha_r\}.$$

In particular, s is a simple function.

Proof. Let x in X . Since the sets A_i cover X and are pairwise disjoint, there exists a unique index i such that x is in A_i . Therefore

$$s(x) = \sum_{j=1}^r \alpha_j \mathbf{1}_{A_j}(x) = \alpha_i,$$

which proves that $s(x)$ belongs to $\{\alpha_1, \dots, \alpha_r\}$. □

Any simple function admits an expression of the type (5.1), and the canonical decomposition is uniquely determined by the values taken by s . Yet there can be other decompositions of s of the form (5.1). For example, if we consider the function s from $\mathbb{R}$ to $\mathbb{R}$ given by

$$s := \mathbf{1}_{[0,1]},$$

then

$$s = 0 \cdot \mathbf{1}_{[0,1]^c} + 1 \cdot \mathbf{1}_{[0,1]},$$

is its canonical decomposition. But we also have

$$s = 0 \cdot \mathbf{1}_{[0,1]^c} + \mathbf{1}_{[0,1/2[} + \mathbf{1}_{[1/2,1]},$$

which is another decomposition and also satisfies the three properties above.

Let us also recall that a finite collection of subsets $(A_i)_{1 \leq i \leq r}$ of X is called a partition if it satisfies (i), (ii), and (iii). A finite family satisfying only the covering and pairwise-disjointness properties will be called a finite disjoint cover.

5.2 Stability properties

Proposition 5.5 (Stability of simple functions). *Let X be a set, and let $s, t: X \rightarrow \mathbb{R}$ be two simple functions. Then $s + t$, st , $\sup(s, t)$, and $\inf(s, t)$ are also simple functions.*

The following proof even gives explicit formulas.

Lemma 5.6 (Product of indicators). *Let X be a set and let $A, B \subset X$. Then*

$$\mathbf{1}_A \mathbf{1}_B = \mathbf{1}_{A \cap B}.$$

Proof. Let x in X . If x belongs to $A \cap B$, then both sides are equal to 1. Otherwise at least one of the two indicators on the left-hand side vanishes, and both sides are equal to 0. $\square$

Proof. If $X = \emptyset$, then all the functions under consideration have empty image, hence are simple. We therefore assume that $X \neq \emptyset$. For integers $n, m \geq 1$, let

$$s = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i}, \quad t = \sum_{j=1}^m \beta_j \mathbf{1}_{B_j},$$

where $(A_i)_{1 \leq i \leq n}$ and $(B_j)_{1 \leq j \leq m}$ are partitions of X , and α_i, β_j are real numbers. By Lemma 5.3, we have

$$\mathbf{1}_X = \sum_{j=1}^m \mathbf{1}_{B_j} = \sum_{i=1}^n \mathbf{1}_{A_i}.$$

Hence

$$\begin{aligned} s &= s \mathbf{1}_X = s \sum_{j=1}^m \mathbf{1}_{B_j} \\ &= \sum_{j=1}^m s \mathbf{1}_{B_j} \\ &= \sum_{i=1}^n \sum_{j=1}^m \alpha_i \mathbf{1}_{A_i} \mathbf{1}_{B_j}, \\ t &= t \mathbf{1}_X = t \sum_{i=1}^n \mathbf{1}_{A_i} \\ &= \sum_{i=1}^n t \mathbf{1}_{A_i} \\ &= \sum_{i=1}^n \sum_{j=1}^m \beta_j \mathbf{1}_{A_i} \mathbf{1}_{B_j}. \end{aligned}$$

Using Lemma 5.6, we have $\mathbf{1}_{A_i}\mathbf{1}_{B_j} = \mathbf{1}_{A_i \cap B_j}$, and therefore

$$\begin{aligned} s + t &= \sum_{i=1}^n \sum_{j=1}^m (\alpha_i + \beta_j) \mathbf{1}_{A_i \cap B_j}, \\ s t &= \sum_{i=1}^n \sum_{j=1}^m \alpha_i \beta_j \mathbf{1}_{A_i \cap B_j}, \\ \sup(s, t) &= \sum_{i=1}^n \sum_{j=1}^m \sup(\alpha_i, \beta_j) \mathbf{1}_{A_i \cap B_j}, \\ \inf(s, t) &= \sum_{i=1}^n \sum_{j=1}^m \inf(\alpha_i, \beta_j) \mathbf{1}_{A_i \cap B_j}. \end{aligned}$$

Each of these functions therefore takes only finitely many values, hence is simple. $\square$

Remark 5.7 (Refined decompositions of simple functions). *In the notation of the preceding proof, the sets*

$$(C_{i,j} := A_i \cap B_j)_{1 \leq i \leq n, 1 \leq j \leq m}$$

satisfy

$$C_{i,j} \cap C_{i',j'} = \emptyset$$

for $(i, j) \neq (i', j')$, and

$$\bigcup_{1 \leq i \leq n, 1 \leq j \leq m} C_{i,j} = X.$$

Yet the decompositions in the proof above are not canonical in general.

Proposition 5.8 (Simple functions as finite combinations of indicators). *Let X be a set. Let $N \in \mathbb{N}_{\geq 1}$ and let $s_1, \dots, s_N: X \rightarrow \mathbb{R}$ be simple functions. Then*

$$\sum_{\ell=1}^N s_{\ell}, \quad \prod_{\ell=1}^N s_{\ell}, \quad \max(s_1, \dots, s_N), \quad \min(s_1, \dots, s_N)$$

are simple functions.

Proof. We proceed by induction on N and repeatedly apply Proposition 5.5. $\square$

5.3 A characterization of the simple functions which are measurable

Proposition 5.9 (Measurability criterion for simple functions). *Let $(X, \mathcal{A})$ be a measurable space and let $s: X \rightarrow \mathbb{R}$. Then s is a simple measurable function if and only if there exist $r \in \mathbb{N}_{\geq 1}$, measurable sets A_i in $\mathcal{A}$ such that $\bigcup_{i=1}^r A_i = X$ and $A_i \cap A_j = \emptyset$ whenever $i \neq j$, and real numbers α_i ($1 \leq i \leq r$) such that*

$$s = \sum_{i=1}^r \alpha_i \mathbf{1}_{A_i}.$$

Proof. Suppose that s is a simple measurable function. If $X = \emptyset$, we may take $r = 1$, $A_1 = \emptyset$, and $\alpha_1 = 0$. This gives the required representation. Assume now that $X \neq \emptyset$. Consider its canonical

decomposition as in (5.1). Then, since the singletons of $\mathbb{R}$ are closed and therefore Borel, we see that their preimages by s are measurable. Thus, for $1 \leq i \leq r$, we have $A_i = s^{-1}(\{\alpha_i\})$ in $\mathcal{A}$. Conversely, assume that there exist $r \in \mathbb{N}_{\geq 1}$, measurable sets A_i in $\mathcal{A}$ such that $\bigcup_{i=1}^r A_i = X$ and $A_i \cap A_j = \emptyset$ whenever $i \neq j$, and real numbers α_i such that

$$s = \sum_{i=1}^r \alpha_i \mathbf{1}_{A_i}.$$

Proposition 5.4 shows that s is simple. Let B in $\mathcal{B}(\mathbb{R})$. Then

$$s^{-1}(B) = \bigcup_{\{i: \alpha_i \in B\}} A_i,$$

which is in $\mathcal{A}$. Hence s is measurable. □

Corollary 5.10 (Simple functions and algebraic stability). *Let $(X, \mathcal{A})$ be a measurable space and let $s, t: X \rightarrow \mathbb{R}$ be simple measurable functions. Then*

$$s + t, \quad st, \quad \sup(s, t), \quad \inf(s, t)$$

are simple measurable functions. More generally, finite sums, finite products, finite pointwise maxima, and finite pointwise minima of simple measurable functions are again simple measurable functions.

Proof. It is enough to combine Proposition 5.5, Proposition 5.8, and Theorem 4.10. □

5.4 Lebesgue integral of a simple measurable function

Definition 5.11 (Integral of a nonnegative simple function). *Let $(X, \mathcal{A}, \mu)$ be a measure space. Let E in $\mathcal{A}$. Let $s: X \rightarrow \mathbb{R}$ be a nonnegative simple measurable function. If $X = \emptyset$, we define*

$$\int_E s \, d\mu := 0.$$

If $X \neq \emptyset$, let

$$s = \sum_{i=1}^r \alpha_i \mathbf{1}_{A_i}$$

denote the canonical decomposition of s , with $\alpha_i \in \mathbb{R}_{\geq 0}$ and A_i in $\mathcal{A}$ for all i . Then the Lebesgue integral of s on E is defined to be

$$\int_E s \, d\mu := \sum_{i=1}^r \alpha_i \mu(A_i \cap E).$$

We use the convention $0 \cdot \infty = 0$.

The following property is quite important, as many decompositions similar to the canonical one may compete. Fortunately enough, they all give the same value.

Proposition 5.12 (Independence of the decomposition). Let $(X, \mathcal{A}, \mu)$ be a measure space. Let E in $\mathcal{A}$ and let s be a nonnegative simple measurable function. Assume that

$$s = \sum_{i=1}^r \alpha_i \mathbf{1}_{A_i}$$

with $\alpha_i \geq 0$, $A_i \in \mathcal{A}$ for every i , $\bigcup_{i=1}^r A_i = X$, and the sets $(A_i)_{1 \leq i \leq r}$ pairwise disjoint. Then

$$\sum_{i=1}^r \alpha_i \mu(A_i \cap E) = \int_E s \, d\mu,$$

where the integral is defined using the canonical decomposition of s .

Proof. **Step 1. Common refinement of the two decompositions.** Let s be a nonnegative simple measurable function. Let

$$s = \sum_{k=1}^m \beta_k \mathbf{1}_{B_k}$$

be the canonical decomposition, so that $(B_k)_{1 \leq k \leq m}$ are pairwise disjoint, $\bigcup_{k=1}^m B_k = X$, and β_k are the distinct values taken by s . By definition,

$$\int_E s \, d\mu = \sum_{k=1}^m \beta_k \mu(B_k \cap E).$$

Assume that we also have a decomposition

$$s = \sum_{i=1}^r \alpha_i \mathbf{1}_{A_i},$$

where $\alpha_i \geq 0$, $A_i \in \mathcal{A}$ for every i , $\bigcup_{i=1}^r A_i = X$, and the sets $(A_i)_{1 \leq i \leq r}$ are pairwise disjoint. For each pair (i, k) , we set

$$C_{i,k} := A_i \cap B_k.$$

Then each $C_{i,k}$ is in $\mathcal{A}$, the family $(C_{i,k})_{1 \leq i \leq r, 1 \leq k \leq m}$ is pairwise disjoint, and

$$X = \bigcup_{i=1}^r \bigcup_{k=1}^m C_{i,k}.$$

Moreover, if $C_{i,k} \neq \emptyset$ and x is in $C_{i,k}$, then $s(x) = \alpha_i$ (since x is in A_i) and also $s(x) = \beta_k$ (since x is in B_k), hence $\alpha_i = \beta_k$. Therefore, on each $C_{i,k}$ we have the identity

$$\alpha_i \mathbf{1}_{C_{i,k}} = \beta_k \mathbf{1}_{C_{i,k}}.$$

Step 2. Comparison of the two sums. We now compute using additivity of μ on disjoint unions. For fixed i ,

$$A_i \cap E = \bigcup_{k=1}^m (A_i \cap B_k \cap E) = \bigcup_{k=1}^m (C_{i,k} \cap E),$$

and the union is disjoint, hence

$$\mu(A_i \cap E) = \sum_{k=1}^m \mu(C_{i,k} \cap E).$$

Thus

$$\begin{aligned}\sum_{i=1}^r \alpha_i \mu(A_i \cap E) &= \sum_{i=1}^r \alpha_i \sum_{k=1}^m \mu(C_{i,k} \cap E) \\ &= \sum_{i=1}^r \sum_{k=1}^m \alpha_i \mu(C_{i,k} \cap E).\end{aligned}$$

Similarly, for fixed k ,

$$B_k \cap E = \bigcup_{i=1}^r (A_i \cap B_k \cap E) = \bigcup_{i=1}^r (C_{i,k} \cap E),$$

disjoint union, hence

$$\mu(B_k \cap E) = \sum_{i=1}^r \mu(C_{i,k} \cap E),$$

and therefore

$$\begin{aligned}\int_E s \, d\mu &= \sum_{k=1}^m \beta_k \mu(B_k \cap E) \\ &= \sum_{k=1}^m \beta_k \sum_{i=1}^r \mu(C_{i,k} \cap E) \\ &= \sum_{i=1}^r \sum_{k=1}^m \beta_k \mu(C_{i,k} \cap E).\end{aligned}$$

Finally, for each pair (i, k) we have $\alpha_i = \beta_k$ whenever $C_{i,k} \neq \emptyset$, and if $C_{i,k} = \emptyset$ then $\mu(C_{i,k} \cap E) = 0$, so in all cases

$$\alpha_i \mu(C_{i,k} \cap E) = \beta_k \mu(C_{i,k} \cap E).$$

Summing over (i, k) yields

$$\sum_{i=1}^r \alpha_i \mu(A_i \cap E) = \int_E s \, d\mu,$$

which proves the independence of the decomposition. $\square$

5.5 Properties of the Lebesgue integral of a simple measurable function

The following proposition establishes that the integral of simple functions is additive not only with respect to the integrand, but also with respect to the domain of integration. For the latter, the Chasles-type decomposition can even be performed with a countable family of parts.

Proposition 5.13 (Linearity of the simple integral). *Let $(X, \mathcal{A}, \mu)$ be a measure space. We have the following properties.*

1) Let $s, t: X \rightarrow \mathbb{R}_{\geq 0}$ be measurable simple functions. For all E in $\mathcal{A}$,

$$\int_E (s + t) d\mu = \int_E s d\mu + \int_E t d\mu.$$

2) Let $s: X \rightarrow \mathbb{R}_{\geq 0}$ be a measurable simple function. Let $(E_n)_{n \geq 1}$ be a sequence of pairwise disjoint elements of $\mathcal{A}$. Then

$$\int_E s d\mu = \sum_{n=1}^{\infty} \int_{E_n} s d\mu, \quad \text{where } E := \bigcup_{n=1}^{\infty} E_n.$$

Proof. If $X = \emptyset$, both assertions are immediate from the empty-domain convention. We therefore assume that $X \neq \emptyset$.

1) Let

$$s = \sum_{i=1}^r \alpha_i \mathbf{1}_{A_i}, \quad t = \sum_{j=1}^m \beta_j \mathbf{1}_{B_j},$$

be the canonical decompositions of s and t . By Lemma 5.3, we have

$$\mathbf{1}_X = \sum_{j=1}^m \mathbf{1}_{B_j} = \sum_{i=1}^r \mathbf{1}_{A_i}.$$

Therefore

$$\begin{aligned} s &= s \mathbf{1}_X = s \sum_{j=1}^m \mathbf{1}_{B_j} \\ &= \sum_{j=1}^m s \mathbf{1}_{B_j} \\ &= \sum_{i=1}^r \sum_{j=1}^m \alpha_i \mathbf{1}_{A_i} \mathbf{1}_{B_j}, \\ t &= t \mathbf{1}_X = t \sum_{i=1}^r \mathbf{1}_{A_i} \\ &= \sum_{i=1}^r t \mathbf{1}_{A_i} \\ &= \sum_{i=1}^r \sum_{j=1}^m \beta_j \mathbf{1}_{A_i} \mathbf{1}_{B_j}. \end{aligned}$$

By Lemma 5.6, we have $\mathbf{1}_{A_i} \mathbf{1}_{B_j} = \mathbf{1}_{A_i \cap B_j}$, hence

$$s + t = \sum_{i=1}^r \sum_{j=1}^m (\alpha_i + \beta_j) \mathbf{1}_{A_i \cap B_j}.$$

The family $(A_i \cap B_j)_{1 \leq i \leq r, 1 \leq j \leq m}$ is pairwise disjoint and its union is X . Proposition 5.12

therefore gives

$$\begin{aligned}
\int_E (s+t) d\mu &= \sum_{i=1}^r \sum_{j=1}^m (\alpha_i + \beta_j) \mu(A_i \cap B_j \cap E) \\
&= \sum_{i=1}^r \sum_{j=1}^m \alpha_i \mu(A_i \cap B_j \cap E) + \sum_{i=1}^r \sum_{j=1}^m \beta_j \mu(A_i \cap B_j \cap E) \\
&= \sum_{i=1}^r \alpha_i \sum_{j=1}^m \mu(A_i \cap B_j \cap E) + \sum_{j=1}^m \beta_j \sum_{i=1}^r \mu(A_i \cap B_j \cap E) \\
&= \sum_{i=1}^r \alpha_i \mu\left(A_i \cap \left(\bigcup_{j=1}^m B_j\right) \cap E\right) + \sum_{j=1}^m \beta_j \mu\left(\left(\bigcup_{i=1}^r A_i\right) \cap B_j \cap E\right) \\
&= \sum_{i=1}^r \alpha_i \mu(A_i \cap E) + \sum_{j=1}^m \beta_j \mu(B_j \cap E) \\
&= \int_E s d\mu + \int_E t d\mu.
\end{aligned}$$

2) Let

$$s = \sum_{i=1}^r \alpha_i \mathbf{1}_{A_i}$$

be the canonical decomposition of s . For each i , the family $(A_i \cap E_n)_{n \geq 1}$ is pairwise disjoint, because the $(E_n)_{n \geq 1}$ are pairwise disjoint. Therefore

$$\begin{aligned}
\int_E s d\mu &= \sum_{i=1}^r \alpha_i \mu(A_i \cap E) \\
&= \sum_{i=1}^r \alpha_i \mu\left(A_i \cap \bigcup_{n=1}^{\infty} E_n\right) \\
&= \sum_{i=1}^r \alpha_i \mu\left(\bigcup_{n=1}^{\infty} (A_i \cap E_n)\right) \\
&= \sum_{i=1}^r \alpha_i \sum_{n=1}^{\infty} \mu(A_i \cap E_n) \\
&= \sum_{n=1}^{\infty} \sum_{i=1}^r \alpha_i \mu(A_i \cap E_n) \\
&= \sum_{n=1}^{\infty} \int_{E_n} s d\mu.
\end{aligned}$$

The interchange between the finite sum over i and the nonnegative series over n is justified because only finitely many outer terms are involved and all summands are nonnegative.

□

Chapter 6

Lebesgue integral of positive measurable functions

We now extend the integral from simple functions to arbitrary positive measurable functions by approximation from below. The chapter establishes the fundamental properties of this definition, explains its behavior on negligible sets, and proves the approximation principle by simple functions that will drive the convergence theorems.

6.1 Definition

We will now define the integral of positive measurable functions from the definition of the integral of simple positive measurable functions.

Definition 6.1 (Positive integral). Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $E \in \mathcal{A}$. Let $f: X \rightarrow [0, \infty]$ be a measurable function (let $[0, \infty]$ be endowed with its Borel σ -algebra). The Lebesgue integral of f on E is defined as:

$$\int_E f d\mu := \sup_{s \in \mathcal{S}} \int_E s d\mu,$$

where the set $\mathcal{S}$ is defined as the set of all simple positive measurable functions $s: X \rightarrow \mathbb{R}_{\geq 0}$ such that $s(x) \leq f(x)$ for all $x \in X$. The set $\mathcal{S}$ is nonempty, since it contains the zero function, and the supremum is taken in $[0, +\infty]$.

Proposition 6.2 (Compatibility with simple functions). Let $(X, \mathcal{A}, \mu)$ be a measure space, let E be in $\mathcal{A}$, and let $s: X \rightarrow \mathbb{R}_{\geq 0}$ be a simple measurable function. Then the definition of $\int_E s d\mu$ as a supremum over simple minorants agrees with the simple-function integral of Chapter 5.

Proof. Let I_s denote the simple-function integral of s on E . Since s is one of its own simple minorants, the supremum definition gives

$$\int_E s d\mu \geq I_s.$$

Conversely, let t be a simple positive measurable function such that $t \leq s$ on X . Then $s - t$ is a simple positive measurable function. By the additivity of the simple-function integral,

$$I_s = I_t + I_{s-t} \geq I_t.$$

Taking the supremum over all such t gives $\int_E s d\mu \leq I_s$. Hence the two definitions agree. $\square$

6.2 Main properties

The definition above satisfies several basic properties, familiar for instance from the Riemann integral, as established below. Let us start with some comparison properties.

Theorem 6.3 (First properties of the positive integral). *Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $f, g: X \rightarrow [0, \infty]$ be measurable functions, and let $E \in \mathcal{A}$. We have:*

1) if $f(x) \leq g(x)$ for all x in E , then

$$\int_E f \, d\mu \leq \int_E g \, d\mu,$$

2) if $A \in \mathcal{A}$ and $A \subset E$ then

$$\int_A f \, d\mu \leq \int_E f \, d\mu.$$

Proof. 1) Let s be a simple positive measurable function such that $s \leq f$ on X . Then $s\mathbf{1}_E \leq g$ on X , and $s\mathbf{1}_E$ is again simple, positive, and measurable. Since s and $s\mathbf{1}_E$ coincide on E , we have

$$\int_E s \, d\mu = \int_E s\mathbf{1}_E \, d\mu \leq \int_E g \, d\mu.$$

We conclude by taking the supremum over all such s .

2) Let s be a simple positive measurable function such that $s \leq f$ on X , and let

$$s = \sum_{i=1}^r \alpha_i \mathbf{1}_{A_i}$$

be its canonical decomposition. Since $A \subset E$, we have $A_i \cap A \subset A_i \cap E$ for every i , hence

$$\mu(A_i \cap A) \leq \mu(A_i \cap E).$$

Therefore

$$\int_A s \, d\mu = \sum_{i=1}^r \alpha_i \mu(A_i \cap A) \leq \sum_{i=1}^r \alpha_i \mu(A_i \cap E) = \int_E s \, d\mu \leq \int_E f \, d\mu.$$

We conclude by taking the supremum over all such s . □

Theorem 6.4 (Additivity and monotonicity of the positive integral). *Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $f, g: X \rightarrow [0, \infty]$ be measurable functions, and let $E \in \mathcal{A}$. We have:*

1) $\int_E f \, d\mu = \int_X (f \times \mathbf{1}_E) \, d\mu$, where again $0 \cdot \infty = 0$,

2) if $\mu(E) = 0$ then $\int_E f \, d\mu = 0$,

3) if there exist parts $E_0 \subset X$ and $E_1 \in \mathcal{A}$ such that

$$E_0 \subset E_1, \quad \mu(E_1) = 0, \quad (\forall x \in E \setminus E_0, f(x) = 0)$$

then

$$\int_E f \, d\mu = 0.$$

4) for all $c \geq 0$, we have

$$\int_E (c \cdot f) \, d\mu = c \left(\int_E f \, d\mu \right).$$

Proof. 1) We first prove that

$$\int_E f d\mu \leq \int_X (f \mathbf{1}_E) d\mu.$$

Let s be a simple positive measurable function such that $s \leq f$ on X . Then $s \mathbf{1}_E$ is a simple positive measurable function by Corollary 5.10, and

$$s \mathbf{1}_E \leq f \mathbf{1}_E$$

on X . Moreover, by the definition of the integral of a simple function,

$$\int_E s d\mu = \int_X s \mathbf{1}_E d\mu.$$

Therefore

$$\int_E s d\mu \leq \int_X (f \mathbf{1}_E) d\mu.$$

By taking the supremum over all such s , we obtain the desired inequality.

Conversely, let t be a simple positive measurable function such that

$$t \leq f \mathbf{1}_E$$

on X . Then $t \leq f$ on X and, since $t \geq 0$ while $f \mathbf{1}_E = 0$ on $X \setminus E$, we also have $t = t \mathbf{1}_E$. In particular, t vanishes on $X \setminus E$, and the definition of the integral of a simple function gives

$$\int_X t d\mu = \int_E t d\mu \leq \int_E f d\mu.$$

By taking the supremum over all such t , we deduce that

$$\int_X (f \mathbf{1}_E) d\mu \leq \int_E f d\mu.$$

This proves 1).

Step 2. The case where $\mu(E) = 0$. Let s be a simple positive measurable function such that $s(x) \leq f(x)$ for all x in X . Then the definition of the integral of a simple function gives

$$\int_E s d\mu = 0$$

because $\mu(E) = 0$. We conclude by taking the supremum over all such s .

2) Let s be a simple positive measurable function such that $s \leq f$ on X , and let

$$s = \sum_{i=1}^r \alpha_i \mathbf{1}_{A_i}$$

be its canonical decomposition. If $\alpha_i > 0$ and x is in $A_i \cap E$, then $s(x) > 0$. Since $s \leq f$, we have $f(x) > 0$, and therefore x cannot belong to $E \setminus E_0$. Hence

$$A_i \cap E \subset E_0 \subset E_1$$

for every i such that $\alpha_i > 0$. Since $\mu(E_1) = 0$, this implies

$$\mu(A_i \cap E) = 0$$

whenever $\alpha_i > 0$. Therefore

$$\int_E s d\mu = \sum_{i=1}^r \alpha_i \mu(A_i \cap E) = 0.$$

We conclude by taking the supremum over all such s .

Step 4. Positive homogeneity. If $c = 0$, the result is immediate. Assume now that $c > 0$. We first note that if

$$u = \sum_{i=1}^r \gamma_i \mathbf{1}_{G_i}$$

is the canonical decomposition of a simple positive measurable function, then

$$cu = \sum_{i=1}^r (c\gamma_i) \mathbf{1}_{G_i},$$

and therefore

$$\int_E cud\mu = c \int_E ud\mu.$$

Let s be a simple positive measurable function such that $s \leq f$ on X . Then cs is a simple positive measurable function such that

$$cs \leq cf$$

on X . By the observation above, we have

$$\int_E csd\mu = c \int_E sd\mu.$$

Taking the supremum over all such s yields

$$c \int_E f d\mu \leq \int_E (cf) d\mu.$$

Conversely, let t be a simple positive measurable function such that $t \leq cf$ on X . Then

$$\frac{1}{c}t \leq f$$

on X , and $\frac{1}{c}t$ is again a simple positive measurable function. Applying the observation above with $u = \frac{1}{c}t$, we obtain

$$\int_E t d\mu = c \int_E \frac{1}{c}t d\mu \leq c \int_E f d\mu.$$

Taking the supremum over all such t gives

$$\int_E (cf) d\mu \leq c \int_E f d\mu.$$

This concludes the proof. □

6.3 Functions vanishing almost everywhere

The notion of negligible set was introduced in Section 2.3. When measurable functions are involved, it is convenient to use the following language.

Definition 6.5 (Almost-everywhere vanishing). *Let $(X, \mathcal{A}, \mu)$ be a measure space and let $f: X \rightarrow \overline{\mathbb{R}}$ be a measurable function. We say that f is zero μ -almost everywhere if there exists a μ -negligible set $E_0 \subset X$ such that $f(x) = 0$ for all $x \in X \setminus E_0$.*

Using this vocabulary, we can deduce from item 3) of Theorem 6.4 that the integral over X of a positive measurable function which is zero almost everywhere is equal to 0.

6.4 Approximation property

We also want the integral thus constructed to be *linear* (that is to say, “the integral of $f + g$ is the sum of the integral of f and of the integral of g ”). We will see in the following chapter that this point is not easy to establish and requires an important result: the *monotone convergence theorem*, as well as the following result which specifies the idea according to which we can approach positive measurable functions by simple measurable functions.

Theorem 6.6 (Approximation by simple functions). *Let $(X, \mathcal{A})$ be a measurable space and let $f: X \rightarrow [0, \infty]$. We equip $[0, \infty]$ with the Borel σ -algebra. Then f is measurable if and only if there exists a sequence $(s_n)_{n \geq 1}$ of simple measurable functions $s_n: X \rightarrow \mathbb{R}_{\geq 0}$ such that*

1) *for all n in $\mathbb{N}_{\geq 1}$ and for all x in X ,*

$$0 \leq s_n(x) \leq s_{n+1}(x) \leq f(x),$$

2) *for all x in X ,*

$$\lim_{n \rightarrow \infty} s_n(x) = f(x).$$

Proof. We first prove the reciprocal implication. Assume that there exists a sequence $(s_n)_{n \geq 1}$ with the stated properties. For each x in X , the sequence $(s_n(x))_{n \geq 1}$ is increasing in $[0, \infty]$ and converges to $f(x)$. Hence

$$f = \limsup_{n \rightarrow \infty} s_n = \liminf_{n \rightarrow \infty} s_n.$$

Theorem 4.10 therefore implies that f is measurable.

We now prove the direct implication. Assume that f is measurable. For each n in $\mathbb{N}_{\geq 1}$, we define

$$\phi_n := \left(\sum_{k=0}^{n2^n-1} \frac{k}{2^n} \cdot \mathbf{1}_{\left[\frac{k}{2^n}, \frac{k+1}{2^n}\right[} \right) + n \cdot \mathbf{1}_{[n, \infty]}.$$

Each $\phi_n: [0, \infty] \rightarrow \mathbb{R}$ takes only finitely many values, hence is simple. Moreover, all the intervals which appear in this decomposition are Borel subsets of $[0, \infty]$ for the topology introduced on the completed half-line, so Proposition 5.9 shows that ϕ_n is measurable. We then set

$$s_n := \phi_n \circ f: X \rightarrow \mathbb{R}.$$

By Proposition 4.7, each s_n is measurable, and since ϕ_n has finite image, s_n is simple.

Let y in $[0, \infty]$. If $0 \leq y < n$, then

$$\phi_n(y) = \frac{\lfloor 2^n y \rfloor}{2^n},$$

so that

$$0 \leq y - \phi_n(y) < \frac{1}{2^n}.$$

If $y \geq n$, then $\phi_n(y) = n \leq y$. Thus

$$0 \leq \phi_n(y) \leq y$$

for every y in $[0, \infty]$. We also claim that

$$\phi_n(y) \leq \phi_{n+1}(y)$$

for every y in $[0, \infty]$. If $0 \leq y < n$, then

$$\phi_n(y) = \frac{\lfloor 2^n y \rfloor}{2^n} \leq \frac{\lfloor 2^{n+1} y \rfloor}{2^{n+1}} = \phi_{n+1}(y).$$

If $y \geq n$, then $\phi_n(y) = n$, whereas $\phi_{n+1}(y) \geq n$ by construction. Hence the claim follows. By applying these inequalities to $y = f(x)$, we obtain

$$0 \leq s_n(x) \leq s_{n+1}(x) \leq f(x)$$

for all n in $\mathbb{N}_{\geq 1}$ and all x in X .

Finally, let x in X . If $f(x) < \infty$, then for all $n > f(x)$ we have

$$0 \leq f(x) - s_n(x) < \frac{1}{2^n},$$

so $s_n(x) \rightarrow f(x)$ as $n \rightarrow +\infty$. If $f(x) = +\infty$, then $s_n(x) = n$ for all n , hence $s_n(x) \rightarrow +\infty$ as $n \rightarrow +\infty$. Thus

$$\lim_{n \rightarrow \infty} s_n(x) = f(x)$$

for every x in X . This concludes the proof. □

Chapter 7

Lebesgue's monotone convergence theorem

This chapter contains the first major convergence theorem of the course. Once monotone convergence has been proved, the linearity of the integral for positive functions, the series version, the generalized Chasles property, and Fatou's lemma follow in a systematic way.

7.1 Monotone convergence theorem

An ingredient for establishing the linearity of the integral of positive measurable functions, which is also one of the most important results of this course, is the following theorem.

Theorem 7.1 (Monotone convergence theorem, or Beppo Levi theorem). *Let $(X, \mathcal{A}, \mu)$ be a measure space and let $(f_n)_{n \geq 1}$ be a sequence of measurable functions from X to $[0, \infty]$ such that*

$$\forall n \geq 1, \forall x \text{ in } X, \quad f_n(x) \leq f_{n+1}(x).$$

For every x in X , we define

$$f(x) := \lim_{n \rightarrow \infty} f_n(x).$$

Then f is measurable, the sequence $(\int_X f_n d\mu)_{n \geq 1}$ converges in $[0, \infty]$, and

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

Proof. Step 1. The easy inequality. The function f is measurable by Theorem 4.10, since it is the pointwise limit of measurable functions.

For every $n \geq 1$, we have $0 \leq f_n \leq f$ on X , so Theorem 6.3 gives

$$\int_X f_n d\mu \leq \int_X f d\mu.$$

Moreover, since $f_n \leq f_{n+1}$, the sequence $(\int_X f_n d\mu)_{n \geq 1}$ is increasing. Hence its limit

$$\ell := \lim_{n \rightarrow \infty} \int_X f_n d\mu$$

exists in $[0, \infty]$ and satisfies

$$\ell \leq \int_X f d\mu.$$

Step 2. Testing against a simple function below f . Let now s be a simple positive measurable function such that $s \leq f$ on X , and let c in $]0, 1[$. For every $n \geq 1$, we define

$$X_n := \{x \in X : f_n(x) \geq cs(x)\}.$$

Each set X_n belongs to $\mathcal{A}$, and the sequence $(X_n)_{n \geq 1}$ is increasing. Moreover,

$$\bigcup_{n \geq 1} X_n = X.$$

Indeed, let x in X . If $s(x) = 0$, then x belongs to X_n for every $n \geq 1$. If $s(x) > 0$, then

$$f_n(x) \rightarrow f(x) \quad \text{as } n \rightarrow +\infty, \quad f(x) \geq s(x) > cs(x),$$

so there exists $n \geq 1$ such that x belongs to X_n .

Let

$$s = \sum_{i=1}^r \alpha_i \mathbf{1}_{A_i}$$

be the canonical decomposition of s . For every $n \geq 1$, we have

$$\int_{X_n} s \, d\mu = \sum_{i=1}^r \alpha_i \mu(A_i \cap X_n).$$

For every $i \in \{1, \dots, r\}$, the sequence $(A_i \cap X_n)_{n \geq 1}$ is increasing and

$$\bigcup_{n \geq 1} (A_i \cap X_n) = A_i.$$

The first item of Theorem 2.3 therefore yields

$$\mu(A_i \cap X_n) \rightarrow \mu(A_i) \quad \text{as } n \rightarrow +\infty.$$

Since the sum is finite, we deduce that

$$\int_{X_n} s \, d\mu \rightarrow \int_X s \, d\mu \quad \text{as } n \rightarrow +\infty. \quad (7.1)$$

Step 3. Passage to the supremum over simple functions. For every $n \geq 1$, we have

$$\int_X f_n \, d\mu \geq \int_{X_n} f_n \, d\mu \geq c \int_{X_n} s \, d\mu.$$

Passing to the limit and using (7.1), we obtain

$$\ell \geq c \int_X s \, d\mu.$$

Letting c tend to 1, which is valid in $[0, +\infty]$ also when $\int_X s \, d\mu = +\infty$, we get

$$\ell \geq \int_X s \, d\mu.$$

Since this is true for every simple positive measurable function s such that $s \leq f$, Definition 6.1 yields

$$\ell \geq \int_X f \, d\mu.$$

Therefore

$$\ell = \int_X f \, d\mu,$$

which concludes the proof. □

Example 7.2 (Monotone approximation by simple truncations). Let $(X, \mathcal{A}, \mu)$ be a measure space and let $f: X \rightarrow [0, \infty]$ be measurable. For every $n \geq 1$, we define

$$f_n := \min(f, n).$$

Then each f_n is measurable, the sequence $(f_n)_{n \geq 1}$ is increasing, and

$$0 \leq f_n \leq f$$

on X . Moreover, for every x in X , the sequence $(f_n(x))_{n \geq 1}$ increases to $f(x)$. This is the most basic monotone approximation of a positive measurable function. By the monotone convergence theorem just proved, it follows that

$$\int_X f_n d\mu \uparrow \int_X f d\mu.$$

7.2 Linearity of the integral of positive measurable functions

The monotone convergence theorem combined with Theorem 6.6 and Lemma 5.13 finally allows us to obtain the desired general linearity properties.

Corollary 7.3 (Linearity of the integral). Let $(X, \mathcal{A}, \mu)$ be a measure space. If $f, g: X \rightarrow [0, \infty]$ are measurable, then for all $E \in \mathcal{A}$,

$$\int_E (f + g) d\mu = \int_E f d\mu + \int_E g d\mu.$$

Proof. By Theorem 6.6, there exist two sequences $(s_k)_{k \geq 1}$ and $(t_k)_{k \geq 1}$ of simple positive measurable functions such that

$$s_k \uparrow f, \quad t_k \uparrow g$$

pointwise on X . For every $k \geq 1$, the function $s_k + t_k$ is simple, positive, and measurable, and

$$s_k + t_k \uparrow f + g$$

pointwise on X .

By Lemma 5.13, for every $k \geq 1$,

$$\int_X (s_k + t_k) d\mu = \int_X s_k d\mu + \int_X t_k d\mu.$$

Applying the monotone convergence theorem to the three increasing sequences $(s_k)_{k \geq 1}$, $(t_k)_{k \geq 1}$, and $(s_k + t_k)_{k \geq 1}$, we obtain

$$\int_X (f + g) d\mu = \int_X f d\mu + \int_X g d\mu.$$

Finally, item 1 of Theorem 6.4 gives

$$\begin{aligned} \int_E (f + g) d\mu &= \int_X (f + g) \mathbf{1}_E d\mu \\ &= \int_X (f \mathbf{1}_E + g \mathbf{1}_E) d\mu \\ &= \int_X f \mathbf{1}_E d\mu + \int_X g \mathbf{1}_E d\mu \\ &= \int_E f d\mu + \int_E g d\mu. \end{aligned}$$

This is the desired identity. The same identity extends by induction to every finite sum of positive measurable functions. $\square$

7.3 Monotone convergence theorem for series of positive functions

Another corollary of the monotone convergence theorem concerns a series of positive functions.

Corollary 7.4 (Integration of normally convergent series). *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $(f_n)_{n \geq 1}$ be a sequence of measurable functions from X to $[0, \infty]$. Then the function*

$$f := \sum_{n=1}^{\infty} f_n$$

is measurable, and for all $E \in \mathcal{A}$, we have

$$\int_E f d\mu = \sum_{n=1}^{\infty} \left(\int_E f_n d\mu \right).$$

Proof. For every $N \geq 1$, we define the partial sum

$$S_N := \sum_{n=1}^N f_n.$$

Each function S_N is measurable, the sequence $(S_N)_{N \geq 1}$ is increasing, and

$$S_N \uparrow f$$

pointwise on X . By the previous corollary, for every $N \geq 1$ we have

$$\int_E S_N d\mu = \sum_{n=1}^N \int_E f_n d\mu.$$

Applying the monotone convergence theorem to the sequence $(S_N \mathbf{1}_E)_{N \geq 1}$ on X yields

$$\int_E f d\mu = \lim_{N \rightarrow \infty} \int_E S_N d\mu = \lim_{N \rightarrow \infty} \sum_{n=1}^N \int_E f_n d\mu = \sum_{n=1}^{\infty} \int_E f_n d\mu.$$

□

7.4 Chasles property for positive functions

We can also deduce a Chasles-type result, which generalizes the statement seen for simple functions.

Corollary 7.5 (Integrals over increasing domains). *Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $f: X \rightarrow [0, \infty]$ be a measurable function. If $(E_n)_{n \geq 1}$ is a sequence of elements of $\mathcal{A}$ which are pairwise disjoint, then*

$$\int_E f d\mu = \sum_{n=1}^{\infty} \int_{E_n} f d\mu, \quad \text{where } E := \bigcup_{n=1}^{\infty} E_n.$$

Proof. For every $n \geq 1$, we define

$$g_n := f \mathbf{1}_{E_n}.$$

The functions g_n are positive and measurable. Since the sets E_n are pairwise disjoint, we have

$$\sum_{n=1}^{\infty} g_n = f \mathbf{1}_E$$

pointwise on X . By Corollary 7.4 and by item 1 of Theorem 6.4, we obtain

$$\begin{aligned} \int_E f d\mu &= \int_X f \mathbf{1}_E d\mu \\ &= \int_X \sum_{n=1}^{\infty} g_n d\mu \\ &= \sum_{n=1}^{\infty} \int_X g_n d\mu \\ &= \sum_{n=1}^{\infty} \int_{E_n} f d\mu. \end{aligned}$$

□

7.5 Fatou's Lemma

Finally, let us end this chapter with the following corollary of the monotone convergence theorem which does not require considering a convergent sequence but only allows us to prove an inequality.

Corollary 7.6 (Fatou's lemma). *Let $(X, \mathcal{A}, \mu)$ be a measure space and let $(f_n)_{n \geq 1}$ be a sequence of measurable functions from X to $[0, \infty]$. Then*

$$\int_X \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int_X f_n d\mu.$$

Proof. For every $n \geq 1$, we define

$$g_n(x) := \inf_{i \geq n} f_i(x).$$

Each function g_n is positive and measurable by Theorem 4.10, and the sequence $(g_n)_{n \geq 1}$ is increasing. Moreover, for every x in X , we have

$$g_n(x) \uparrow \liminf_{n \rightarrow \infty} f_n(x).$$

The monotone convergence theorem therefore yields

$$\int_X \liminf_{n \rightarrow \infty} f_n d\mu = \lim_{n \rightarrow \infty} \int_X g_n d\mu.$$

Now let $n \geq 1$. For every $i \geq n$, we have $g_n \leq f_i$, hence

$$\int_X g_n d\mu \leq \int_X f_i d\mu.$$

Taking the infimum over all $i \geq n$, we obtain

$$\int_X g_n d\mu \leq \inf_{i \geq n} \int_X f_i d\mu.$$

Passing to the limit in n , we conclude that

$$\int_X \liminf_{n \rightarrow \infty} f_n \, d\mu \leq \lim_{n \rightarrow \infty} \inf_{i \geq n} \int_X f_i \, d\mu = \liminf_{n \rightarrow \infty} \int_X f_n \, d\mu.$$

□

Chapter 8

Dominated convergence, $\mathcal{L}^1(\mu)$ and $L^1(\mu)$

We now gather in a single chapter the material on integrable functions, dominated convergence, negligible sets, and the passage from $\mathcal{L}^1(\mu)$ to $L^1(\mu)$. Since the notions of negligible set and of property holding μ -almost everywhere were introduced earlier in Section 2.3, the almost everywhere form of the dominated convergence theorem can now be proved directly from the point-wise form.

8.1 Definition of Lebesgue integrability

We fix a measure space $(X, \mathcal{A}, \mu)$.

Definition 8.1 (Integrable functions). *We say that a function f from X to $\mathbb{R}$ is in $\mathcal{L}^1(\mu)$ if*

- 1) f is $(\mathcal{A}, \mathcal{B}(\mathbb{R}))$ -measurable,
- 2) $\int_X |f| d\mu < \infty$.

We denote

$$\mathcal{L}^1(\mu) := \mathcal{L}^1(X, \mathcal{A}, \mu) := \{f: X \rightarrow \mathbb{R}: f \text{ is } \mu\text{-integrable}\}.$$

For E in $\mathcal{A}$, we define for all f in $\mathcal{L}^1(\mu)$,

$$\int_E f d\mu := \int_E f^+ d\mu - \int_E f^- d\mu,$$

where

$$f^+(x) := \max\{f(x), 0\}, \quad f^-(x) := \max\{-f(x), 0\}$$

for all x in X .

We observe that for any function f we have the decompositions

$$f = f^+ - f^-, \quad |f| = f^+ + f^-, \tag{8.1}$$

and the properties

$$f^+ \geq 0, \quad f^- \geq 0, \quad f^+(x)f^-(x) = 0 \text{ for all } x \text{ in } X. \tag{8.2}$$

Remark. Using Theorem 6.3 and Theorem 6.4 from the previous chapter, we see that

$$\int_E f^+ d\mu \leq \int_X |f| d\mu < \infty, \quad \int_E f^- d\mu \leq \int_X |f| d\mu < \infty,$$

since f is in $\mathcal{L}^1(\mu)$. This shows that $\int_E f d\mu$ is indeed defined as the difference of two finite quantities.

8.2 Properties of the Lebesgue integral

For the following results, we keep the notation above.

Theorem 8.2 (Vector space and linearity of the integral). *The set $\mathcal{L}^1(\mu)$ is a $\mathbb{R}$ -vector space and, for all E in $\mathcal{A}$, the map*

$$f \in \mathcal{L}^1(\mu) \mapsto \int_E f \, d\mu \in \mathbb{R}$$

is linear. In other words, for all f, g in $\mathcal{L}^1(\mu)$ and for all c in $\mathbb{R}$, we have

$$\int_E (f + g) \, d\mu = \int_E f \, d\mu + \int_E g \, d\mu, \quad \int_E (cf) \, d\mu = c \int_E f \, d\mu.$$

Proof. Let f, g in $\mathcal{L}^1(\mu)$ and let c in $\mathbb{R}$. The measurability of $f + g$ and cf follows from Theorem 4.10. Since

$$|f + g| \leq |f| + |g|, \quad |cf| = |c| |f|,$$

the functions $f + g$ and cf belong to $\mathcal{L}^1(\mu)$.

For the additivity, we use the pointwise identity

$$(f + g)^+ + f^- + g^- = (f + g)^- + f^+ + g^+,$$

which follows from (8.1). All the functions involved are positive and measurable, so the finite-sum form of Corollary 7.3 gives

$$\begin{aligned} \int_E (f + g)^+ \, d\mu + \int_E f^- \, d\mu + \int_E g^- \, d\mu &= \int_E ((f + g)^+ + f^- + g^-) \, d\mu \\ &= \int_E ((f + g)^- + f^+ + g^+) \, d\mu \\ &= \int_E (f + g)^- \, d\mu + \int_E f^+ \, d\mu + \int_E g^+ \, d\mu. \end{aligned}$$

After rearranging, we obtain

$$\int_E (f + g) \, d\mu = \int_E f \, d\mu + \int_E g \, d\mu.$$

For the homogeneity, if $c \geq 0$, then

$$(cf)^+ = cf^+, \quad (cf)^- = cf^-,$$

hence item 4 of Theorem 6.4 yields

$$\int_E cf \, d\mu = c \int_E f^+ \, d\mu - c \int_E f^- \, d\mu = c \int_E f \, d\mu.$$

If $c < 0$, then

$$(cf)^+ = (-c)f^-, \quad (cf)^- = (-c)f^+,$$

and therefore

$$\int_E cf \, d\mu = (-c) \int_E f^- \, d\mu - (-c) \int_E f^+ \, d\mu = c \int_E f \, d\mu.$$

This completes the proof. □

Theorem 8.3 (Order properties of the integral). 1) For all f, g in $\mathcal{L}^1(\mu)$ and for all E in $\mathcal{A}$,

$$(\forall x \in E, f(x) \leq g(x)) \Rightarrow \int_E f \, d\mu \leq \int_E g \, d\mu.$$

2) For all f in $\mathcal{L}^1(\mu)$ and for all E in $\mathcal{A}$,

$$\left| \int_E f \, d\mu \right| \leq \int_E |f| \, d\mu.$$

Proof. 1) Let $h := g - f$. Then h is in $\mathcal{L}^1(\mu)$ and $h(x) \geq 0$ for all x in E . Therefore

$$\int_E h \, d\mu = \int_E h^+ \, d\mu \geq 0,$$

because $h^- = 0$ on E and item 3 of Theorem 6.4 gives $\int_E h^- \, d\mu = 0$. Using the linearity proved above, we obtain

$$0 \leq \int_E h \, d\mu = \int_E g \, d\mu - \int_E f \, d\mu,$$

which gives the result.

2) Since

$$-|f| \leq f \leq |f|$$

on E , item 1 gives

$$-\int_E |f| \, d\mu \leq \int_E f \, d\mu \leq \int_E |f| \, d\mu,$$

hence

$$\left| \int_E f \, d\mu \right| \leq \int_E |f| \, d\mu.$$

□

Theorem 8.4 (Countable additivity over domains). For any f in $\mathcal{L}^1(\mu)$ and for any sequence $(E_k)_{k \geq 1}$ of pairwise disjoint elements of $\mathcal{A}$, if

$$E = \bigcup_{k \geq 1} E_k,$$

then the series $\sum_{k \geq 1} \int_{E_k} f \, d\mu$ converges absolutely and

$$\int_E f \, d\mu = \sum_{k \geq 1} \int_{E_k} f \, d\mu.$$

Proof. The functions f^+ and f^- are positive, measurable, and integrable. By the generalized Chasles property for positive functions,

$$\int_E f^+ \, d\mu = \sum_{k \geq 1} \int_{E_k} f^+ \, d\mu, \quad \int_E f^- \, d\mu = \sum_{k \geq 1} \int_{E_k} f^- \, d\mu.$$

Both positive series are convergent because their sums are finite. Moreover,

$$\sum_{k \geq 1} \left| \int_{E_k} f \, d\mu \right| \leq \sum_{k \geq 1} \int_{E_k} |f| \, d\mu = \int_E |f| \, d\mu < \infty,$$

so the signed series is absolutely convergent. Hence

$$\begin{aligned}
\int_E f \, d\mu &= \int_E f^+ \, d\mu - \int_E f^- \, d\mu \\
&= \sum_{k \geq 1} \int_{E_k} f^+ \, d\mu - \sum_{k \geq 1} \int_{E_k} f^- \, d\mu \\
&= \sum_{k \geq 1} \left(\int_{E_k} f^+ \, d\mu - \int_{E_k} f^- \, d\mu \right) \\
&= \sum_{k \geq 1} \int_{E_k} f \, d\mu.
\end{aligned}$$

□

Theorem 8.5 (Integration over measurable subsets). *For all E in $\mathcal{A}$ and for all f in $\mathcal{L}^1(\mu)$,*

$$\int_E f \, d\mu = \int_X f \mathbf{1}_E \, d\mu.$$

Proof. Since $|f \mathbf{1}_E| \leq |f|$, the function $f \mathbf{1}_E$ belongs to $\mathcal{L}^1(\mu)$. Moreover,

$$(f \mathbf{1}_E)^+ = f^+ \mathbf{1}_E, \quad (f \mathbf{1}_E)^- = f^- \mathbf{1}_E.$$

Applying item 1 of Theorem 6.4 to the positive functions f^+ and f^- , we obtain

$$\begin{aligned}
\int_X f \mathbf{1}_E \, d\mu &= \int_X (f \mathbf{1}_E)^+ \, d\mu - \int_X (f \mathbf{1}_E)^- \, d\mu \\
&= \int_X f^+ \mathbf{1}_E \, d\mu - \int_X f^- \mathbf{1}_E \, d\mu \\
&= \int_E f^+ \, d\mu - \int_E f^- \, d\mu \\
&= \int_E f \, d\mu.
\end{aligned}$$

□

Lemma 8.6 (Markov's inequality). *Let f be in $\mathcal{L}^1(\mu)$, let E be in $\mathcal{A}$, and let $\alpha > 0$. Then*

$$\alpha \mu(E \cap \{|f| > \alpha\}) \leq \int_E |f| \, d\mu.$$

In particular,

$$\mu(\{|f| > \alpha\}) \leq \frac{1}{\alpha} \int_X |f| \, d\mu.$$

Proof. Since

$$\alpha \mathbf{1}_{E \cap \{|f| > \alpha\}} \leq |f| \mathbf{1}_E$$

on X , the comparison theorem and the indicator formulation give

$$\begin{aligned}
\alpha \mu(E \cap \{|f| > \alpha\}) &= \alpha \int_X \mathbf{1}_{E \cap \{|f| > \alpha\}} \, d\mu \\
&= \int_X \alpha \mathbf{1}_{E \cap \{|f| > \alpha\}} \, d\mu \\
&\leq \int_X |f| \mathbf{1}_E \, d\mu \\
&= \int_E |f| \, d\mu.
\end{aligned}$$

The particular case corresponds to $E = X$. □

Theorem 8.7 (Integrals on negligible sets). *Let E in $\mathcal{A}$ and let f in $\mathcal{L}^1(\mu)$. If one of the following two conditions holds:*

(a) $\mu(E) = 0$,

(b) *there exists $E_0 \subset E$ and E_1 in $\mathcal{A}$ such that $E_0 \subset E_1$, $\mu(E_1) = 0$, and $f(x) = 0$ for all x in $E \setminus E_0$,*

then

$$\int_E f \, d\mu = 0.$$

Proof. If $\mu(E) = 0$, item 2 of Theorem 6.4 applied to f^+ and f^- yields

$$\int_E f^+ \, d\mu = 0, \quad \int_E f^- \, d\mu = 0,$$

hence

$$\int_E f \, d\mu = 0.$$

Assume now that item (b) holds. Since f^+ and f^- vanish on $E \setminus E_0$, item 3 of Theorem 6.4 applied to f^+ and f^- gives

$$\int_E f^+ \, d\mu = 0, \quad \int_E f^- \, d\mu = 0.$$

Therefore

$$\int_E f \, d\mu = 0.$$

□

8.3 Equality almost everywhere

The notions of negligible set and of property holding almost everywhere now become essential.

Lemma 8.8 (Positive integrable functions are finite almost everywhere). *Let $h: X \rightarrow [0, +\infty]$ be measurable. If*

$$\int_X h \, d\mu < +\infty,$$

then $h < +\infty$ μ -almost everywhere.

Proof. For every $n \geq 1$, set

$$A_n := \{x \in X: h(x) > n\}.$$

Since $n\mathbf{1}_{A_n} \leq h$, comparison gives

$$n\mu(A_n) \leq \int_X h \, d\mu.$$

In particular, $\mu(A_1) < \infty$. Let

$$N := \{x \in X: h(x) = +\infty\}.$$

Then $N = \bigcap_{n \geq 1} A_n$, and $(A_n)_{n \geq 1}$ is decreasing. By continuity from above,

$$\mu(N) = \lim_{n \rightarrow \infty} \mu(A_n) \leq \lim_{n \rightarrow \infty} \frac{1}{n} \int_X h \, d\mu = 0.$$

Hence $h < +\infty$ outside the null set N . □

Let us now observe that two functions which are equal μ -almost everywhere have the same integral.

Lemma 8.9 (Equality almost everywhere and integrals). *Let $f, g: X \rightarrow [0, \infty]$ be measurable functions, or let f, g be in $\mathcal{L}^1(X, \mathcal{A}, \mu)$. We assume that $f = g$ μ -almost everywhere on X . Then for all E in $\mathcal{A}$,*

$$\int_E f \, d\mu = \int_E g \, d\mu.$$

Proof. By definition of μ -almost everywhere, there exists M in $\mathcal{A}$ such that

$$\mu(M) = 0 \quad \text{and} \quad \{x \in X: f(x) \neq g(x)\} \subset M.$$

Assume first that f and g are positive measurable. Since $f = g$ on $E \setminus M$, the comparison theorem for positive functions gives

$$\int_{E \setminus M} f \, d\mu = \int_{E \setminus M} g \, d\mu.$$

Moreover, $\mu(E \cap M) = 0$, so item 2 of Theorem 6.4 yields

$$\int_{E \cap M} f \, d\mu = 0, \quad \int_{E \cap M} g \, d\mu = 0.$$

Since

$$E = (E \setminus M) \cup (E \cap M)$$

is a disjoint union, the generalized Chasles property for positive functions gives

$$\begin{aligned} \int_E f \, d\mu &= \int_{E \setminus M} f \, d\mu + \int_{E \cap M} f \, d\mu \\ &= \int_{E \setminus M} g \, d\mu + \int_{E \cap M} g \, d\mu \\ &= \int_E g \, d\mu. \end{aligned}$$

Assume now that f and g are in $\mathcal{L}^1(\mu)$. Then $f - g$ is in $\mathcal{L}^1(\mu)$ and vanishes on $X \setminus M$. By the previous theorem,

$$\int_E (f - g) \, d\mu = 0.$$

Using linearity, we conclude that

$$\int_E f \, d\mu = \int_E g \, d\mu.$$

□

! The converse is generally false for positive measurable functions. On the other hand, the case $g = 0$ already gives a very useful characterization.

Lemma 8.10 (Characterization of zero integral). *If $f: X \rightarrow [0, \infty]$ is a measurable function, or if f is in $\mathcal{L}^1(X, \mathcal{A}, \mu)$, then the following assertions are equivalent.*

- 1) $f(x) = 0$, μ -almost everywhere on X ,
- 2) for all E in $\mathcal{A}$,

$$\int_E f \, d\mu = 0,$$

- 3)

$$\int_X |f| \, d\mu = 0.$$

Proof. We prove the implications one after another.

1) $\Rightarrow$ 2). If f is positive measurable, the conclusion follows from item 3 of Theorem 6.4. If f is in $\mathcal{L}^1(\mu)$, it follows from the theorem on vanishing on negligible sets proved above.

2) $\Rightarrow$ 3). If f is positive measurable, then $|f| = f$, so it is enough to take $E = X$ in item 2. Assume now that f is in $\mathcal{L}^1(\mu)$. We set

$$E^+ := \{x \in X: f(x) \geq 0\}, \quad E^- := \{x \in X: f(x) < 0\}.$$

These sets belong to $\mathcal{A}$, and on them we have

$$|f| = f \text{ on } E^+, \quad |f| = -f \text{ on } E^-.$$

Therefore

$$0 = \int_{E^+} f \, d\mu = \int_{E^+} |f| \, d\mu$$

and

$$0 = \int_{E^-} f \, d\mu = - \int_{E^-} |f| \, d\mu.$$

Since $|f|$ is integrable and

$$X = E^+ \cup E^-, \quad E^+ \cap E^- = \emptyset,$$

the Chasles property gives

$$\int_X |f| \, d\mu = \int_{E^+} |f| \, d\mu + \int_{E^-} |f| \, d\mu = 0.$$

3) $\Rightarrow$ 1). We set

$$g := f$$

in the positive measurable case and

$$g := |f|$$

in the integrable case. Then g is positive and measurable, and

$$\int_X g \, d\mu = 0.$$

For every $n \geq 1$, we define

$$A_n := \{x \in X: g(x) > 1/n\}.$$

Since $(1/n)\mathbf{1}_{A_n} \leq g$, comparison gives

$$\frac{1}{n}\mu(A_n) \leq \int_X g \, d\mu = 0.$$

Hence $\mu(A_n) = 0$ for every $n \geq 1$. If

$$N := \{x \in X: g(x) \neq 0\},$$

then

$$N = \bigcup_{n \geq 1} A_n,$$

so $\mu(N) = 0$. Thus $g = 0$ almost everywhere, and therefore $f = 0$ almost everywhere as well. $\square$

8.4 Lebesgue dominated convergence theorem

We can now state one of the central results of the theory.

Theorem 8.11 (Lebesgue dominated convergence theorem, pointwise form). *Let $(X, \mathcal{A}, \mu)$ be a measure space. Let g in $\mathcal{L}^1(\mu)$ satisfy $g(x) \geq 0$ for all x in X , and let*

$$f_n: X \rightarrow \mathbb{R}, \quad n \geq 1,$$

be measurable functions satisfying the following two hypotheses:

- *domination hypothesis: for all $n \geq 1$ and for all x in X ,*

$$|f_n(x)| \leq g(x),$$

- *pointwise convergence hypothesis: for all x in X , the sequence $(f_n(x))_{n \geq 1}$ converges to a limit $f(x)$.*

Then

1) *the function $f: X \rightarrow \mathbb{R}$ is in $\mathcal{L}^1(\mu)$,*

2)

$$\int_X |f_n - f| \, d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty,$$

3)

$$\int_X f_n \, d\mu \rightarrow \int_X f \, d\mu \quad \text{as } n \rightarrow +\infty.$$

Proof. 1) The function f , as a pointwise limit of measurable functions, is measurable. Moreover, passing to the limit in the domination hypothesis, we obtain

$$|f(x)| \leq g(x) \quad \text{for all } x \text{ in } X.$$

By integration, this gives

$$\int_X |f| \, d\mu \leq \int_X g \, d\mu < \infty,$$

and therefore f is in $\mathcal{L}^1(\mu)$.

2) We consider the sequence of auxiliary functions

$$h_n := 2g - |f_n - f|.$$

By the triangle inequality and the domination hypothesis,

$$|f_n - f| \leq |f_n| + |f| \leq 2g,$$

so each h_n is a positive measurable function. Fatou's lemma therefore gives

$$\int_X \liminf_{n \rightarrow \infty} h_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int_X h_n \, d\mu. \quad (8.3)$$

Moreover, by the pointwise convergence hypothesis, we have

$$|f_n - f| \rightarrow 0 \quad \text{as } n \rightarrow +\infty$$

pointwise on X , hence

$$\liminf_{n \rightarrow \infty} h_n = 2g \quad \text{pointwise on } X, \quad (8.4)$$

and therefore

$$\int_X \liminf_{n \rightarrow \infty} h_n \, d\mu = \int_X 2g \, d\mu. \quad (8.5)$$

On the other hand, for every $n \geq 1$ we have

$$h_n + |f_n - f| = 2g.$$

Since $\int_X 2g \, d\mu < \infty$, Corollary 7.3 gives

$$\int_X h_n \, d\mu = \int_X 2g \, d\mu - \int_X |f_n - f| \, d\mu.$$

Consequently,

$$\liminf_{n \rightarrow \infty} \int_X h_n \, d\mu = \int_X 2g \, d\mu - \limsup_{n \rightarrow \infty} \int_X |f_n - f| \, d\mu. \quad (8.6)$$

Combining (8.3), (8.5), and (8.6), we obtain

$$\limsup_{n \rightarrow \infty} \int_X |f_n - f| \, d\mu \leq 0.$$

Since the integrals are nonnegative, this implies

$$\int_X |f_n - f| \, d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

3) By the linearity of the integral and by item 2 of the theorem on comparisons, we have

$$\left| \int_X f_n \, d\mu - \int_X f \, d\mu \right| = \left| \int_X (f_n - f) \, d\mu \right| \leq \int_X |f_n - f| \, d\mu,$$

which tends to 0 by item 2. This proves the convergence of the integrals. □

The previous theorem is the pointwise form. Since we already know how to work with negligible sets, we can immediately pass to the almost everywhere form.

Theorem 8.12 (Lebesgue dominated convergence theorem, almost everywhere form). Let $(X, \mathcal{A}, \mu)$ be a measure space. Let g in $\mathcal{L}^1(\mu)$ satisfy $g(x) \geq 0$ for all x in X , and let

$$f_n, f: X \rightarrow \mathbb{R}, \quad n \geq 1,$$

be measurable functions such that

1) for every $n \geq 1$, we have

$$|f_n(x)| \leq g(x)$$

for μ -almost all x in X ,

2)

$$f_n(x) \rightarrow f(x) \quad \text{as } n \rightarrow +\infty$$

for μ -almost all x in X .

Then f is in $\mathcal{L}^1(\mu)$ and, for every E in $\mathcal{A}$,

$$\int_E |f_n - f| d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty,$$

and therefore

$$\int_E f_n d\mu \rightarrow \int_E f d\mu \quad \text{as } n \rightarrow +\infty.$$

Proof. **Step 1. Removing the exceptional null set.** For each $n \geq 1$, let N_n be a μ -negligible set such that

$$|f_n(x)| \leq g(x)$$

for all x in $X \setminus N_n$. Let N_∞ be a μ -negligible set such that

$$f_n(x) \rightarrow f(x) \quad \text{as } n \rightarrow +\infty$$

for all x in $X \setminus N_\infty$. By the countable stability of negligible sets, there exists M in $\mathcal{A}$ such that

$$\mu(M) = 0 \quad \text{and} \quad N_\infty \cup \bigcup_{n \geq 1} N_n \subset M.$$

Step 2. Reduction to the pointwise dominated convergence theorem. We define

$$\tilde{f}_n := f_n \mathbf{1}_{X \setminus M}, \quad \tilde{f} := f \mathbf{1}_{X \setminus M}, \quad \tilde{g} := g \mathbf{1}_{X \setminus M}.$$

Then the functions $\tilde{f}_n$, $\tilde{f}$, and $\tilde{g}$ are measurable, $\tilde{g}$ is in $\mathcal{L}^1(\mu)$, and for all x in X we have

$$|\tilde{f}_n(x)| \leq \tilde{g}(x), \quad \tilde{f}_n(x) \rightarrow \tilde{f}(x) \quad \text{as } n \rightarrow +\infty.$$

The pointwise dominated convergence theorem therefore yields

$$\int_X |\tilde{f}_n - \tilde{f}| d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

In particular, $\tilde{f}$ is in $\mathcal{L}^1(\mu)$.

Step 3. Returning to the original representatives. Now

$$|f| = |\tilde{f}| \quad \mu\text{-almost everywhere,}$$

so Lemma 8.9 applied to the positive measurable functions $|f|$ and $|\tilde{f}|$ gives

$$\int_X |f| \, d\mu = \int_X |\tilde{f}| \, d\mu < \infty.$$

Hence f is in $\mathcal{L}^1(\mu)$.

For each $n \geq 1$, the positive measurable functions $|f_n|$ and $|\tilde{f}_n|$ are equal μ -almost everywhere. Lemma 8.9 therefore gives

$$\int_X |f_n| \, d\mu = \int_X |\tilde{f}_n| \, d\mu \leq \int_X \tilde{g} \, d\mu < \infty.$$

Thus every f_n belongs to $\mathcal{L}^1(\mu)$.

Finally, for every E in $\mathcal{A}$, Lemma 8.9 gives

$$\int_E |f_n - f| \, d\mu = \int_E |\tilde{f}_n - \tilde{f}| \, d\mu,$$

and hence

$$\int_E |f_n - f| \, d\mu \leq \int_X |\tilde{f}_n - \tilde{f}| \, d\mu \rightarrow 0.$$

Also,

$$\int_E f_n \, d\mu = \int_E \tilde{f}_n \, d\mu, \quad \int_E f \, d\mu = \int_E \tilde{f} \, d\mu.$$

Therefore

$$\begin{aligned} \left| \int_E f_n \, d\mu - \int_E f \, d\mu \right| &= \left| \int_E (\tilde{f}_n - \tilde{f}) \, d\mu \right| \\ &\leq \int_E |\tilde{f}_n - \tilde{f}| \, d\mu, \end{aligned}$$

and the right-hand side tends to 0. □

A first important consequence of the dominated convergence theorem is the following absolute-continuity property of the integral. It says, first, that the integral of an integrable function over a moving tail tends to zero, and, second, that a set of sufficiently small measure carries only a small part of the integral.

Corollary 8.13 (Absolute continuity of the integral). *Let $(X, \mathcal{A}, \mu)$ be a measure space and let f be in $\mathcal{L}^1(\mu)$. Then the following two properties hold.*

1) *If $(A_n)_{n \geq 1}$ is a sequence of measurable subsets of X such that*

$$\mathbf{1}_{A_n}(x) \rightarrow 0 \quad \text{as } n \rightarrow +\infty$$

for μ -almost all x in X , then

$$\int_{A_n} |f| \, d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

In particular, this applies when (A_n) is decreasing and $\mu(\cap_{n \geq 1} A_n) = 0$.

2) *For every $\varepsilon > 0$, there exists $\delta > 0$ such that, for every measurable set $E \subset X$ satisfying $\mu(E) \leq \delta$, one has*

$$\int_E |f| \, d\mu \leq \varepsilon.$$

Proof. We first prove item 1). The functions $|f|\mathbf{1}_{A_n}$ converge to 0 μ -almost everywhere as $n \rightarrow +\infty$, and they are dominated by the integrable function $|f|$. The dominated convergence theorem gives

$$\int_X |f|\mathbf{1}_{A_n} d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty,$$

which is the desired conclusion. If (A_n) is decreasing and $\mu(\cap_{n \geq 1} A_n) = 0$, then $\mathbf{1}_{A_n} \rightarrow \mathbf{1}_{\cap_{n \geq 1} A_n} = 0$ μ -almost everywhere, so the previous argument applies.

We now prove item 2). By item 1), applied to the sets

$$A_m := \{|f| > m\}, \quad m \geq 1,$$

there exists $M \geq 1$ such that

$$\int_{\{|f| > M\}} |f| d\mu \leq \frac{\varepsilon}{2}.$$

Set $\delta := \varepsilon/(2M)$. If E is measurable and $\mu(E) \leq \delta$, then

$$\begin{aligned} \int_E |f| d\mu &= \int_{E \cap \{|f| \leq M\}} |f| d\mu + \int_{E \cap \{|f| > M\}} |f| d\mu \\ &\leq M\mu(E) + \int_{\{|f| > M\}} |f| d\mu \\ &\leq \varepsilon. \end{aligned}$$

This proves the absolute-continuity property. □

In the special case where $X = \mathbb{N}$ is endowed with the discrete σ -algebra and the enumeration measure $\#$ introduced in Chapter 1, an integrable function is an absolutely summable sequence. If $u = (u_k)_{k \in \mathbb{N}}$ and $\sum_k |u_k| < \infty$, applying item 1) to the tails $A_N := \{k \in \mathbb{N} : k \geq N\}$ gives

$$\sum_{k \geq N} |u_k| \rightarrow 0 \quad \text{as } N \rightarrow +\infty,$$

which is the familiar fact that the remainder of an absolutely convergent series tends to zero.

The following result is a partial converse of dominated convergence.

Theorem 8.14 (Characterization of L^1 convergence). *Let f_n and f be functions in $\mathcal{L}^1(X, \mathcal{A}, \mu)$ such that*

$$\int_X |f_n - f| d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Then there exists a subsequence $(f_{n_k})_{k \geq 1}$ and a function g in $\mathcal{L}^1(X, \mathcal{A}, \mu)$ such that

$$|f_{n_k}(x)| \leq g(x)$$

and

$$f_{n_k}(x) \rightarrow f(x) \quad \text{as } k \rightarrow +\infty$$

for μ -almost all x in X .

Proof. **Step 1. Choosing a rapidly convergent subsequence.** We choose an increasing sequence of integers $(n_k)_{k \geq 1}$ such that

$$\int_X |f_{n_k} - f| d\mu \leq \frac{1}{2^k}$$

for every $k \geq 1$.

Step 2. Construction of an integrable dominator. Let

$$u := \sum_{k=1}^{\infty} |f_{n_k} - f|.$$

By Corollary 7.4,

$$\int_X u \, d\mu = \sum_{k=1}^{\infty} \int_X |f_{n_k} - f| \, d\mu \leq 1.$$

For every $m \geq 1$, we therefore have

$$m \mu(\{u \geq m\}) \leq \int_{\{u \geq m\}} u \, d\mu \leq \int_X u \, d\mu \leq 1.$$

Hence

$$\mu(\{u = +\infty\}) = 0.$$

We set

$$N := \{u = +\infty\}$$

and define

$$g(x) := \begin{cases} |f(x)| + u(x), & x \in X \setminus N, \\ 0, & x \in N. \end{cases}$$

Then g is measurable and finite everywhere. Moreover, for every x in X ,

$$g(x) \leq |f(x)| + u(x).$$

By the comparison theorem for positive functions and by Corollary 7.3, we obtain

$$\int_X g \, d\mu \leq \int_X (|f| + u) \, d\mu = \int_X |f| \, d\mu + \int_X u \, d\mu \leq \int_X |f| \, d\mu + 1 < \infty,$$

so that g is in $\mathcal{L}^1(X, \mathcal{A}, \mu)$.

Step 3. Almost everywhere convergence and domination. Now let $x \in X \setminus N$. Since $u(x) < \infty$, the numerical series

$$\sum_{k=1}^{\infty} |f_{n_k}(x) - f(x)|$$

converges. In particular,

$$|f_{n_k}(x) - f(x)| \rightarrow 0 \quad \text{as } k \rightarrow +\infty,$$

hence

$$f_{n_k}(x) \rightarrow f(x) \quad \text{as } k \rightarrow +\infty.$$

Moreover, for every $k \geq 1$,

$$|f_{n_k}(x)| \leq |f(x)| + |f_{n_k}(x) - f(x)| \leq |f(x)| + u(x) = g(x).$$

Therefore

$$|f_{n_k}(x)| \leq g(x) \quad \text{and} \quad f_{n_k}(x) \rightarrow f(x) \quad \text{as } k \rightarrow +\infty$$

for every x in $X \setminus N$, that is, for μ -almost all x in X . □

Example 8.15 (A dominated-convergence example). Let $X =]0, 1[$, let $\mathcal{A} = \mathcal{B}(]0, 1[)$, and let $\mu = \lambda$. For every $n \geq 1$, define

$$f_n(x) := n \mathbf{1}_{]0, 1/n[}(x).$$

Then $f_n(x) \rightarrow 0$ as $n \rightarrow +\infty$ for every x in $]0, 1[$. Moreover, for every $\varepsilon > 0$ and every $n > \varepsilon$, we have

$$\{|f_n| > \varepsilon\} =]0, 1/n[,$$

so

$$\lambda(\{|f_n| > \varepsilon\}) = \frac{1}{n} \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Thus (f_n) converges to 0 in measure. On the other hand,

$$\int_0^1 |f_n(x)| \, dx = \int_0^{1/n} n \, dx = 1$$

for every $n \geq 1$. Hence (f_n) does not converge to 0 in $\mathcal{L}^1(]0, 1[)$. In particular, no function g in $\mathcal{L}^1(]0, 1[)$ can dominate all the f_n , because the dominated convergence theorem would then imply

$$\int_0^1 |f_n(x)| \, dx \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

8.5 The Lebesgue space $L^1(\mu)$

We now pass from $\mathcal{L}^1(\mu)$ to the quotient by almost everywhere equality. We define on $\mathcal{L}^1(\mu)$ the equivalence relation

$$f \sim_\mu g \quad \Longleftrightarrow \quad f = g \text{ } \mu\text{-almost everywhere.}$$

By Lemma 8.9, representatives which are equal μ -almost everywhere have the same integrals. Moreover,

$$V_0(\mu) := \{f \in \mathcal{L}^1(\mu) : f \sim_\mu 0\} = \{f \in \mathcal{L}^1(\mu) : f = 0 \text{ } \mu\text{-almost everywhere}\}$$

is a sub- $\mathbb{R}$ -vector space of $\mathcal{L}^1(\mu)$.

Definition 8.16 (The space $L^1(\mu)$). Let $(X, \mathcal{A}, \mu)$ be a measure space. We denote by $L^1(\mu)$ the set of equivalence classes of $\mathcal{L}^1(\mu)$ for the relation $\sim_\mu$. Equivalently,

$$L^1(\mu) = \mathcal{L}^1(\mu) / V_0(\mu).$$

If f is in $\mathcal{L}^1(\mu)$, we denote by $\bar{f}$ its class in $L^1(\mu)$:

$$\bar{f} := f + V_0(\mu) = \{f + h : h \in V_0(\mu)\}.$$

Proposition 8.17 (The L^1 norm). The mapping

$$L^1(\mu) \rightarrow \mathbb{R}_{\geq 0}, \quad \bar{f} \mapsto \|\bar{f}\|_1 := \int_X |f| \, d\mu$$

is a norm on $L^1(\mu)$.

Proof. We first check that the formula is well defined. If $\bar{f} = \bar{g}$, then $f = g$ μ -almost everywhere, hence $|f| = |g|$ μ -almost everywhere. Lemma 8.9 applied to the positive measurable functions $|f|$ and $|g|$ gives

$$\int_X |f| \, d\mu = \int_X |g| \, d\mu.$$

Positivity is clear. If $\|\bar{f}\|_1 = 0$, then

$$\int_X |f| \, d\mu = 0,$$

so Lemma 8.10 yields $f = 0$ μ -almost everywhere. Therefore $\bar{f} = 0$ in $L^1(\mu)$.

Let c in $\mathbb{R}$. Then

$$\|c\bar{f}\|_1 = \int_X |cf| \, d\mu = |c| \int_X |f| \, d\mu = |c| \|\bar{f}\|_1.$$

Finally, if $\bar{f}, \bar{g}$ are in $L^1(\mu)$, then

$$|f + g| \leq |f| + |g|$$

pointwise. By comparison and linearity,

$$\|\bar{f} + \bar{g}\|_1 = \int_X |f + g| \, d\mu \leq \int_X (|f| + |g|) \, d\mu = \int_X |f| \, d\mu + \int_X |g| \, d\mu = \|\bar{f}\|_1 + \|\bar{g}\|_1.$$

This proves the triangle inequality. □

As usual, when no confusion can arise, we identify an equivalence class with one of its representatives and we simply write $\|f\|_1$ instead of $\|\bar{f}\|_1$.

8.6 Dominated convergence theorem for series

Proposition 8.18 (Normally convergent series in $\mathcal{L}^1$). *Let $(X, \mathcal{A}, \mu)$ be a measure space and let $(f_n)_{n \geq 1}$ be a sequence in $\mathcal{L}^1(\mu)$ such that*

$$\sum_{n \geq 1} \int_X |f_n| \, d\mu < \infty.$$

Then there exists a measurable set N and a function f in $\mathcal{L}^1(\mu)$ such that $\mu(N) = 0$ and

1) *for all x in $X \setminus N$, the series $\sum_{n \geq 1} |f_n(x)|$ converges and*

$$f(x) = \sum_{n \geq 1} f_n(x),$$

2) *for all A in $\mathcal{A}$,*

$$\int_A f \, d\mu = \sum_{n \geq 1} \int_A f_n \, d\mu,$$

3)

$$\lim_{n \rightarrow \infty} \int_X \left| f - \sum_{k=1}^n f_k \right| \, d\mu = 0.$$

Proof. **Step 1. Construction of an integrable dominator.** For x in X , we define

$$\phi(x) := \sum_{k=1}^{\infty} |f_k(x)|.$$

The function ϕ is positive and measurable as the limit of the sequence of partial sums. Since each $|f_k|$ is positive and measurable, the monotone convergence theorem gives

$$\int_X \phi \, d\mu = \lim_{n \rightarrow \infty} \int_X \sum_{k=1}^n |f_k| \, d\mu = \lim_{n \rightarrow \infty} \sum_{k=1}^n \int_X |f_k| \, d\mu = \sum_{k=1}^{\infty} \int_X |f_k| \, d\mu < \infty.$$

By Lemma 8.8, the set

$$N := \{x \in X : \phi(x) = +\infty\}$$

belongs to $\mathcal{A}$ and satisfies $\mu(N) = 0$. We set

$$\tilde{\phi} := \phi \mathbf{1}_{X \setminus N}.$$

Then $\tilde{\phi}$ is real-valued, nonnegative, measurable, equal to ϕ μ -almost everywhere, and belongs to $\mathcal{L}^1(\mu)$.

Step 2. Pointwise construction of the sum. For $n \geq 1$, set

$$s_n := \sum_{k=1}^n f_k.$$

On $X \setminus N$, the numerical series $\sum_k f_k(x)$ is absolutely convergent. We define

$$f(x) := \begin{cases} 0, & x \in N, \\ \sum_{k=1}^{\infty} f_k(x), & x \in X \setminus N. \end{cases}$$

The function f is measurable because it is the pointwise limit of the measurable functions $s_n \mathbf{1}_{X \setminus N}$. Item 1 follows from the definition of N . Moreover,

$$|f| \leq \tilde{\phi}$$

on X , so f belongs to $\mathcal{L}^1(\mu)$.

Step 3. Passage to the limit under the integral on arbitrary measurable sets. On $X \setminus N$, the sequence (s_n) converges pointwise to f , and for every $n \geq 1$ we have

$$|s_n| \leq \tilde{\phi}.$$

Theorem 8.12 therefore implies that, for every A in $\mathcal{A}$,

$$\int_A s_n \, d\mu \rightarrow \int_A f \, d\mu \quad \text{as } n \rightarrow +\infty.$$

Since

$$\int_A s_n \, d\mu = \sum_{k=1}^n \int_A f_k \, d\mu$$

by linearity, item 2 follows.

Step 4. Convergence in $\mathcal{L}^1$. Finally, on $X \setminus N$ we have

$$f - s_n \rightarrow 0 \quad \text{as } n \rightarrow +\infty$$

and

$$|f - s_n| \leq |f| + |s_n| \leq 2\tilde{\phi}.$$

A second application of Theorem 8.12, now to the sequence $f - s_n$ with dominator $2\tilde{\phi}$, yields

$$\int_X |f - s_n| \, d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

This is exactly item 3. □

8.7 Completeness of L^1

We conclude by proving the completeness of $L^1(\mu)$.

Theorem 8.19 (Riesz–Fischer theorem for L^1). *Let $(f_n)_{n \geq 1}$ be a sequence in $\mathcal{L}^1(\mu)$ such that*

$$\|f_n - f_m\|_1 = \int_X |f_n - f_m| d\mu \rightarrow 0 \quad \text{as } n, m \rightarrow +\infty.$$

Then there exists f in $\mathcal{L}^1(\mu)$ such that

1) a subsequence of $(f_n)_{n \geq 1}$ converges to f μ -almost everywhere on X ,

2)

$$\|f_n - f\|_1 \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

In particular, the normed vector space $(L^1(\mu), \|\cdot\|_1)$ is a Banach space.

Proof. Step 1. Extraction of a subsequence with summable increments. Since $(f_n)_{n \geq 1}$ is Cauchy for $\|\cdot\|_1$, there exists a strictly increasing map

$$\varphi: \mathbb{N}_{\geq 1} \rightarrow \mathbb{N}_{\geq 1}$$

such that

$$\|f_{\varphi(i+1)} - f_{\varphi(i)}\|_1 < 2^{-i}$$

for every $i \geq 1$.

Step 2. Construction of the candidate limit. For $i \geq 1$, we set

$$g_i := f_{\varphi(i+1)} - f_{\varphi(i)}.$$

Then

$$\sum_{i=1}^{\infty} \int_X |g_i| d\mu \leq \sum_{i=1}^{\infty} 2^{-i} < \infty.$$

By Proposition 8.18, there exist g in $\mathcal{L}^1(\mu)$ and a measurable set N with $\mu(N) = 0$ such that, for all x in $X \setminus N$,

$$g(x) = \sum_{i=1}^{\infty} g_i(x).$$

We define

$$f := f_{\varphi(1)} + g.$$

Then f is in $\mathcal{L}^1(\mu)$. Moreover, for every x in $X \setminus N$ and every $n \geq 2$,

$$\sum_{i=1}^{n-1} g_i(x) = f_{\varphi(n)}(x) - f_{\varphi(1)}(x),$$

so the convergence of the series implies

$$f_{\varphi(n)}(x) \rightarrow f(x) \quad \text{as } n \rightarrow +\infty.$$

This proves item 1.

Step 3. L^1 convergence of the extracted subsequence. By item 3 of Proposition 8.18,

$$\left\| g - \sum_{i=1}^{n-1} g_i \right\|_1 \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Using again the identity

$$\sum_{i=1}^{n-1} g_i = f_{\varphi(n)} - f_{\varphi(1)},$$

we deduce that

$$\|f_{\varphi(n)} - f\|_1 \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Step 4. Convergence of the whole Cauchy sequence. Let $\varepsilon > 0$. Choose n_0 such that

$$\|f_{\varphi(n_0)} - f\|_1 \leq \varepsilon/2.$$

Since (f_n) is Cauchy in $\|\cdot\|_1$, there exists N_0 such that

$$\|f_n - f_m\|_1 \leq \varepsilon/2$$

for all $n, m \geq N_0$. Taking n_0 larger if necessary, we may assume that

$$\varphi(n_0) \geq N_0.$$

Then, for every $n \geq N_0$,

$$\|f_n - f\|_1 \leq \|f_n - f_{\varphi(n_0)}\|_1 + \|f_{\varphi(n_0)} - f\|_1 \leq \varepsilon.$$

Thus $\|f_n - f\|_1 \rightarrow 0$ as $n \rightarrow +\infty$, which proves item 2.

Finally, let $(u_n)_{n \geq 1}$ be a Cauchy sequence in $L^1(\mu)$. Choose representatives f_n in $\mathcal{L}^1(\mu)$ such that

$$u_n = \bar{f}_n.$$

Then

$$\|f_n - f_m\|_1 = \|u_n - u_m\|_1 \rightarrow 0 \quad \text{as } m, n \rightarrow +\infty,$$

so the first part of the theorem provides f in $\mathcal{L}^1(\mu)$ with

$$\|f_n - f\|_1 \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Equivalently,

$$\|u_n - \bar{f}\|_1 \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Hence every Cauchy sequence in $L^1(\mu)$ converges in $L^1(\mu)$, and $(L^1(\mu), \|\cdot\|_1)$ is a Banach space. $\square$

Corollary 8.20 (Nikodym metric). *Let $(X, \mathcal{A}, \mu)$ be a finite measure space. We identify two measurable sets when their symmetric difference is μ -negligible, and we denote by $\tilde{\mathcal{A}}$ the quotient set. For $[A]$ and $[B]$ in $\tilde{\mathcal{A}}$, we define*

$$\rho([A], [B]) := \mu(A \triangle B).$$

Then ρ is a metric on $\tilde{\mathcal{A}}$, and $(\tilde{\mathcal{A}}, \rho)$ is complete.

Proof. The map

$$\Phi: \tilde{\mathcal{A}} \rightarrow L^1(X, \mu), \quad \Phi([A]) := \overline{\mathbf{1}_A},$$

is well defined and satisfies

$$\|\Phi([A]) - \Phi([B])\|_{L^1(X, \mu)} = \int_X |\mathbf{1}_A - \mathbf{1}_B| d\mu = \mu(A \triangle B) = \rho([A], [B]).$$

Hence Φ is an isometry.

It remains to prove that the image of Φ is closed in $L^1(X, \mu)$. Let $(\overline{\mathbf{1}_{A_n}})_{n \geq 1}$ be a sequence in the image of Φ and assume that

$$\overline{\mathbf{1}_{A_n}} \rightarrow \bar{f} \quad \text{as } n \rightarrow +\infty$$

in $L^1(X, \mu)$, with f in $\mathcal{L}^1(X, \mu)$. The sequence $(\mathbf{1}_{A_n})$ is Cauchy after choosing the displayed representatives in $\mathcal{L}^1(X, \mu)$. By Theorem 8.19, it has a subsequence converging almost everywhere to a function which is equal to f in $L^1(X, \mu)$. After changing f on a negligible set, we may therefore assume that this subsequence converges pointwise to f . Since each $\mathbf{1}_{A_n}$ only takes the values 0 and 1, it follows that

$$f(x) \in \{0, 1\}$$

for μ -almost every x in X . Let

$$A := \{x \in X : f(x) = 1\}.$$

Then A is measurable and $\bar{f} = \overline{\mathbf{1}_A}$ in $L^1(X, \mu)$. Thus the image of Φ is closed. Since $L^1(X, \mu)$ is complete, the closed subset $\Phi(\tilde{\mathcal{A}})$ is complete, and therefore $(\tilde{\mathcal{A}}, \rho)$ is complete. $\square$

8.8 A first link with positive Radon measures

As an application of the integral developed so far, let Ω be an open subset of $\mathbb{R}^N$ and let μ be a positive Radon measure on Ω , viewed as a Borel measure on $(\Omega, \mathcal{B}(\Omega))$. We denote by $\mathcal{C}_c(\Omega)$ the space of continuous functions with compact support in Ω . Since the support of such a function is compact and μ is finite on compact sets, every function in $\mathcal{C}_c(\Omega)$ is integrable with respect to μ .

Lemma 8.21 (A Radon functional associated with an integrable function). *If μ is a positive Radon measure on Ω , then the map $L: \mathcal{C}_c(\Omega) \rightarrow \mathbb{R}$ defined by*

$$L(f) := \int_{\Omega} f \, d\mu$$

is a positive linear functional on $\mathcal{C}_c(\Omega)$. Moreover, if $K = \text{supp}(f)$, then

$$|L(f)| \leq \mu(K) \|f\|_{\infty}.$$

Proof. Let f be in $\mathcal{C}_c(\Omega)$ and let $K = \text{supp}(f)$. Since K is compact and μ is a positive Radon measure, $\mu(K) < \infty$. Moreover,

$$|f| \leq \|f\|_{\infty} \mathbf{1}_K.$$

Therefore

$$\int_{\Omega} |f| \, d\mu \leq \mu(K) \|f\|_{\infty} < \infty,$$

so $L(f)$ is well defined and the displayed estimate follows. The linearity and positivity of L are immediate consequences of the properties of the Lebesgue integral. $\square$

The converse problem, namely to reconstruct a positive Radon measure from a positive linear functional on $\mathcal{C}_c(\Omega)$, will be solved later by the Riesz representation theorem.

The next chapter gives a first direct application of dominated convergence: continuity and differentiation under the integral sign.

Chapter 9

Integrals with parameters

This chapter studies parameter-dependent integrals. The first result gives a criterion for the continuity of

$$t \mapsto \int_E f(x, t) \, d\mu(x),$$

and the second gives a criterion for differentiating such a map under the integral sign. In both cases the main tool is dominated convergence, used through sequential characterizations of continuity and differentiability.

We consider a measurable set E in a measure space $(X, \mathcal{A}, \mu)$, an interval $I \subset \mathbb{R}$, and a function

$$f: E \times I \rightarrow \mathbb{R}.$$

We are interested in the regularity, with respect to the parameter t , of the map

$$F: I \rightarrow \mathbb{R}, \quad t \mapsto \int_E f(x, t) \, d\mu(x). \quad (9.1)$$

Whenever we need to regard a function defined on E as a function on the whole space X , we extend it by 0 outside E without changing the notation. In particular, saying that a function is integrable over E means that this zero extension belongs to $\mathcal{L}^1(X, \mathcal{A}, \mu)$.

9.1 Continuity under the integral sign

Theorem 9.1 (Continuity under the integral sign). *Let t_0 in I . We assume that*

1) there exists $\eta > 0$ such that, for every t in I satisfying $|t - t_0| < \eta$, the function

$$x \mapsto f(x, t)$$

is measurable on E and integrable over E ,

2) for μ -almost all x in E ,

$$f(x, t) \rightarrow f(x, t_0) \quad \text{as } t \rightarrow t_0,$$

3) there exists a function $g: E \rightarrow [0, \infty)$, integrable over E , such that for every t in I satisfying $|t - t_0| < \eta$,

$$|f(x, t)| \leq g(x)$$

for μ -almost all x in E .

Then the function $x \mapsto f(x, t_0)$ is integrable over E and

$$\lim_{t \rightarrow t_0} \int_E f(x, t) \, d\mu(x) = \int_E f(x, t_0) \, d\mu(x).$$

Proof. Let $(t_n)_{n \geq 1}$ be a sequence in I converging to t_0 . By discarding finitely many terms, we may assume that

$$|t_n - t_0| < \eta$$

for all $n \geq 1$.

For every $n \geq 1$, hypothesis 3 provides a negligible set $N_n \subset E$ such that

$$|f(x, t_n)| \leq g(x)$$

for all x in $E \setminus N_n$. Hypothesis 2 provides a negligible set $N_\infty \subset E$ such that

$$f(x, t_n) \rightarrow f(x, t_0) \quad \text{as } n \rightarrow +\infty$$

for all x in $E \setminus N_\infty$. By countable stability of negligible sets, there exists M in $\mathcal{A}$ such that

$$N_\infty \cup \bigcup_{n \geq 1} N_n \subset M, \quad \mu(M) = 0.$$

We define functions on X by

$$u_n(x) := f(x, t_n) \mathbf{1}_E(x), \quad u(x) := f(x, t_0) \mathbf{1}_E(x), \quad v(x) := g(x) \mathbf{1}_E(x).$$

The function v is in $\mathcal{L}^1(\mu)$ and is nonnegative. Moreover, for every x in $X \setminus M$, convergence and domination hold; outside E all zero extensions vanish. Thus, for every x in $X \setminus M$, we have

$$u_n(x) \rightarrow u(x) \quad \text{as } n \rightarrow +\infty, \quad \text{and} \quad |u_n(x)| \leq v(x)$$

for all $n \geq 1$. Theorem 8.12 therefore yields

$$\int_X u_n \, d\mu \rightarrow \int_X u \, d\mu \quad \text{as } n \rightarrow +\infty.$$

By the indicator formulation of the integral, this is exactly

$$\int_E f(x, t_n) \, d\mu(x) \rightarrow \int_E f(x, t_0) \, d\mu(x) \quad \text{as } n \rightarrow +\infty.$$

Since this is true for every sequence (t_n) converging to t_0 , the sequential criterion for continuity proves the result. $\square$

Corollary 9.2 (Continuity of derivatives under the integral sign). *Let $I \subset \mathbb{R}$ be an interval. We assume that*

1) *for every t in I , the function $x \mapsto f(x, t)$ is measurable on E and integrable over E ,*

2) *for μ -almost all x in E , the map*

$$t \mapsto f(x, t)$$

is continuous on I ,

3) *there exists a measurable set $Z \subset E$ with $\mu(E \setminus Z) = 0$ and there exists a function $g: E \rightarrow [0, \infty)$, integrable over E , such that*

$$|f(x, t)| \leq g(x)$$

for every x in Z and every t in I .

Then the map

$$t \mapsto \int_E f(x, t) \, d\mu(x)$$

is continuous on I .

Proof. We apply Theorem 9.1 at each point t_0 in I . □

9.2 Differentiation under the integral sign

Theorem 9.3 (Differentiation under the integral sign). *Let $I \subset \mathbb{R}$ be an open interval and let t_0 in I . We assume that*

1) *there exists $\eta > 0$ such that, with*

$$J :=]t_0 - \eta, t_0 + \eta[,$$

one has $J \subset I$, and for every t in J the function

$$x \mapsto f(x, t)$$

is measurable on E and integrable over E ,

2) *for every t in J , the function*

$$x \mapsto \frac{\partial f}{\partial t}(x, t)$$

is measurable on E , and there exists a measurable set $Z \subset E$ with $\mu(E \setminus Z) = 0$ such that, for every x in Z , the map

$$t \mapsto f(x, t)$$

is differentiable on J ,

3) *there exists a function $g: E \rightarrow [0, \infty)$, integrable over E , such that*

$$\left| \frac{\partial f}{\partial t}(x, t) \right| \leq g(x)$$

for every x in Z and every t in J .

Then the function

$$x \mapsto \frac{\partial f}{\partial t}(x, t_0)$$

is integrable over E , the map F defined in (9.1) is differentiable at t_0 , and

$$F'(t_0) = \int_E \frac{\partial f}{\partial t}(x, t_0) d\mu(x).$$

Proof. Step 1. Sequential reduction. Let $(t_n)_{n \geq 1}$ be a sequence in $I \setminus \{t_0\}$ converging to t_0 . By discarding finitely many terms, we may assume that

$$|t_n - t_0| < \eta$$

for every $n \geq 1$.

Let Z be the full-measure subset of E given by hypotheses 2 and 3, and choose M in $\mathcal{A}$ such that

$$E \setminus Z \subset M, \quad \mu(M) = 0.$$

Step 2. Difference quotients and domination. For x in X , we define

$$u_n(x) := \frac{f(x, t_n) - f(x, t_0)}{t_n - t_0} \mathbf{1}_E(x), \quad u(x) := \frac{\partial f}{\partial t}(x, t_0) \mathbf{1}_E(x), \quad v(x) := g(x) \mathbf{1}_E(x).$$

For every x in Z , differentiability at t_0 gives

$$u_n(x) \rightarrow u(x) \quad \text{as } n \rightarrow +\infty.$$

Moreover, for every x in Z and every $n \geq 1$, the mean value theorem provides a point

$$s_n(x) \in]\min(t_n, t_0), \max(t_n, t_0)[\subset J$$

such that

$$\frac{f(x, t_n) - f(x, t_0)}{t_n - t_0} = \frac{\partial f}{\partial t}(x, s_n(x)).$$

Therefore

$$|u_n(x)| \leq v(x)$$

for every x in Z and every $n \geq 1$.

Step 3. Application of dominated convergence. On $X \setminus M$, convergence and domination hold, and outside E all zero extensions vanish. Since v is in $\mathcal{L}^1(\mu)$ and is nonnegative, Theorem 8.12 applied to the sequence $(u_n)_{n \geq 1}$ yields

$$\int_X u_n \, d\mu \rightarrow \int_X u \, d\mu \quad \text{as } n \rightarrow +\infty.$$

Step 4. Identification of the derivative. By the indicator formulation,

$$\int_X u_n \, d\mu = \frac{1}{t_n - t_0} \left(\int_E f(x, t_n) \, d\mu(x) - \int_E f(x, t_0) \, d\mu(x) \right) = \frac{F(t_n) - F(t_0)}{t_n - t_0},$$

and

$$\int_X u \, d\mu = \int_E \frac{\partial f}{\partial t}(x, t_0) \, d\mu(x).$$

Therefore

$$\frac{F(t_n) - F(t_0)}{t_n - t_0} \rightarrow \int_E \frac{\partial f}{\partial t}(x, t_0) \, d\mu(x) \quad \text{as } n \rightarrow +\infty.$$

Since this is true for every sequence (t_n) in $I \setminus \{t_0\}$ converging to t_0 , the sequential characterization of the derivative proves the result. $\square$

Corollary 9.4 (Smoothness under the integral sign). *Let $I \subset \mathbb{R}$ be an open interval. We assume that*

1) *for every t in I , the function $x \mapsto f(x, t)$ is measurable on E and integrable over E ,*

2) *for every t in I , the function*

$$x \mapsto \frac{\partial f}{\partial t}(x, t)$$

is measurable on E , and there exists a measurable set $Z \subset E$ with $\mu(E \setminus Z) = 0$ such that, for every x in Z , the map

$$t \mapsto f(x, t)$$

is differentiable on I ,

3) *there exists a function $g: E \rightarrow [0, \infty)$, integrable over E , such that*

$$\left| \frac{\partial f}{\partial t}(x, t) \right| \leq g(x)$$

for every x in Z and every t in I .

Then the map

$$F(t) = \int_E f(x, t) \, d\mu(x)$$

is differentiable on I and

$$F'(t) = \int_E \frac{\partial f}{\partial t}(x, t) \, d\mu(x)$$

for every t in I .

Proof. We apply Theorem 9.3 at each point t_0 in I . □

Example 9.5 (Gamma function). *For $y > 0$, we define*

$$\Gamma(y) := \int_0^\infty e^{-x} x^{y-1} \, dx.$$

Then Γ is well defined and of class C^∞ on $]0, \infty[$. More precisely, for every integer $k \geq 1$ and every $y > 0$,

$$\Gamma^{(k)}(y) = \int_0^\infty e^{-x} x^{y-1} (\log x)^k \, dx.$$

Proof. **Step 1. Well-definedness.** Let us first prove that Γ is well defined. Fix $y > 0$. Splitting the integral at 1, we obtain

$$\int_0^\infty e^{-x} x^{y-1} \, dx = \int_0^1 e^{-x} x^{y-1} \, dx + \int_1^\infty e^{-x} x^{y-1} \, dx.$$

On $]0, 1]$, we have $e^{-x} \leq 1$, hence

$$0 \leq e^{-x} x^{y-1} \leq x^{y-1},$$

and the function $x \mapsto x^{y-1}$ is integrable on $]0, 1]$ because $y > 0$. On $[1, \infty[$, the factor e^{-x} gives exponential decay. More explicitly, there exists $C_y > 0$ such that $x^{y-1} \leq C_y e^{x/2}$ for all $x \geq 1$, and therefore

$$0 \leq e^{-x} x^{y-1} \leq C_y e^{-x/2}$$

for all $x \geq 1$. Thus $x \mapsto e^{-x}x^{y-1}$ is integrable on $[1, \infty[$. Therefore $\Gamma(y)$ is well defined for every $y > 0$.

Step 2. Uniform dominators on compact parameter intervals. We now prove continuity and smoothness on an arbitrary compact interval

$$J = [a, b] \subset]0, \infty[.$$

For every integer $r \geq 0$, every y in J , and every $x > 0$, set

$$F_r(x, y) := e^{-x}x^{y-1}(\log x)^r.$$

For $r = 0$, this is the original integrand. If $0 < x \leq 1$, then $a \leq y$ and therefore

$$|F_r(x, y)| \leq x^{a-1}|\log x|^r.$$

The function on the right is integrable on $]0, 1]$. Indeed, by the classical one-dimensional change of variable $x = e^{-t}$,

$$\int_0^1 x^{a-1}|\log x|^r dx = \int_0^\infty e^{-at}t^r dt < \infty.$$

If $x \geq 1$, then $y \leq b$ and

$$|F_r(x, y)| \leq e^{-x}x^{b-1}(\log x)^r.$$

The function on the right is integrable on $[1, \infty[$ because the exponential factor dominates the polynomial-logarithmic growth. Thus

$$g_r(x) := \mathbf{1}_{]0,1]}(x)x^{a-1}|\log x|^r + \mathbf{1}_{[1,\infty[}(x)e^{-x}x^{b-1}(\log x)^r$$

is an integrable dominator for $F_r(\cdot, y)$, uniformly for y in J .

Step 3. Continuity of the candidate derivatives. For each fixed $x > 0$, the map $y \mapsto F_r(x, y)$ is continuous on J . Theorem 9.1 applied to F_0 gives the continuity of Γ on J . More generally, for every $r \geq 0$, Theorem 9.1 shows that the map

$$G_r(y) := \int_0^\infty F_r(x, y) dx$$

is continuous on J .

Step 4. Induction on the order of differentiation. We now prove by induction that Γ is of class C^k on the interior of J and that $\Gamma^{(k)} = G_k$ there for every $k \geq 0$. The case $k = 0$ has just been proved. Assume that Γ is of class C^k on the interior of J and that $\Gamma^{(k)} = G_k$. Since, for every $x > 0$,

$$\frac{\partial F_k}{\partial y}(x, y) = F_{k+1}(x, y),$$

and since g_{k+1} is an integrable dominator for F_{k+1} on J , Theorem 9.3 applies at each point y in the interior of J , after choosing a smaller open interval around y contained in the interior of J . Hence

$$G'_k(y) = \int_0^\infty F_{k+1}(x, y) dx = G_{k+1}(y)$$

for every y in the interior of J . Since G_{k+1} is continuous on J , this proves that Γ is of class C^{k+1} on the interior of J and that $\Gamma^{(k+1)} = G_{k+1}$ there. Hence, for every $k \geq 1$ and every y in the interior of J ,

$$\Gamma^{(k)}(y) = \int_0^\infty e^{-x}x^{y-1}(\log x)^k dx.$$

Since every point of $]0, \infty[$ lies in the interior of some such compact interval J , the function Γ is of class C^∞ on $]0, \infty[$ and the displayed formula holds for all $y > 0$. $\square$

Chapter 10

Product measures, the Fubini and Tonelli theorems

The basic method for calculating an integral of a function of two variables is to reduce it to iterated integrals of functions of a single variable. To do so, we first have to explain how to put a measure-space structure on a Cartesian product.

10.1 Definition of σ -finite measures

Definition 10.1 (σ -finite measures). Let $(X, \mathcal{A}, \mu)$ be a measure space. We say that $(X, \mathcal{A}, \mu)$ is σ -finite if there exists a sequence $(A_n)_{n \geq 1}$ of elements of $\mathcal{A}$ such that

$$\mu(A_n) < \infty \text{ for every } n \geq 1, \quad X = \bigcup_{n \geq 1} A_n.$$

This hypothesis is satisfied, for instance, when $\mu(X) < \infty$, when $X = \mathbb{N}$ is endowed with the discrete σ -algebra and the counting measure, or when $X = \mathbb{R}^n$ is endowed with the Lebesgue measure.

10.2 σ -algebra tensor product and measure tensor product

Definition 10.2 (Product σ -algebra). Let $(X_1, \mathcal{A}_1)$ and $(X_2, \mathcal{A}_2)$ be two measurable spaces. Let

$$\mathcal{R} := \{B_1 \times B_2 : B_1 \in \mathcal{A}_1, B_2 \in \mathcal{A}_2\}.$$

The product σ -algebra is

$$\mathcal{A}_1 \otimes \mathcal{A}_2 := \sigma(\mathcal{R}).$$

Proposition 10.3 (Product measure). Let $(X_1, \mathcal{A}_1, \mu_1)$ and $(X_2, \mathcal{A}_2, \mu_2)$ be two σ -finite measure spaces. Then there exists a unique measure, denoted by $\mu_1 \otimes \mu_2$, on $\mathcal{A}_1 \otimes \mathcal{A}_2$ such that

$$(\mu_1 \otimes \mu_2)(B_1 \times B_2) = \mu_1(B_1)\mu_2(B_2)$$

for all B_1 in $\mathcal{A}_1$ and all B_2 in $\mathcal{A}_2$. Moreover, the measure space $(X_1 \times X_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \mu_1 \otimes \mu_2)$ is σ -finite.

Proof. **Step 1. The algebra generated by rectangles.** Let

$$\mathcal{R} := \{B_1 \times B_2 : B_1 \in \mathcal{A}_1, B_2 \in \mathcal{A}_2\}.$$

Since

$$(B_1 \times B_2) \cap (C_1 \times C_2) = (B_1 \cap C_1) \times (B_2 \cap C_2),$$

the intersection of two rectangles is again a rectangle. Moreover, for two rectangles $B_1 \times B_2$ and $C_1 \times C_2$, we have the disjoint decomposition

$$(B_1 \times B_2) \setminus (C_1 \times C_2) = ((B_1 \setminus C_1) \times B_2) \cup ((B_1 \cap C_1) \times (B_2 \setminus C_2)).$$

It follows by induction that the difference between a rectangle and a finite disjoint union of rectangles is again a finite disjoint union of rectangles. Consequently, every finite union of rectangles can be rewritten as a finite disjoint union of rectangles. Finally, since

$$(X_1 \times X_2) \setminus (B_1 \times B_2) = ((X_1 \setminus B_1) \times X_2) \cup (B_1 \times (X_2 \setminus B_2)),$$

and since complements of finite disjoint unions can be treated recursively by repeated use of the previous difference formula, the family $\mathcal{E}$ of finite disjoint unions of elements of $\mathcal{R}$ is an algebra on $X_1 \times X_2$. It contains $\mathcal{R}$, hence

$$\sigma(\mathcal{E}) = \sigma(\mathcal{R}) = \mathcal{A}_1 \otimes \mathcal{A}_2.$$

Step 2. Definition of the premeasure through sections. Let E be in $\mathcal{E}$. We choose a disjoint decomposition

$$E = \bigcup_{i=1}^m (B_{1,i} \times B_{2,i}), \quad B_{1,i} \in \mathcal{A}_1, \quad B_{2,i} \in \mathcal{A}_2.$$

For every x in X_1 , we define the vertical section

$$E_x := \{y \in X_2 : (x, y) \in E\}.$$

Because the rectangles are pairwise disjoint, the sets $B_{2,i}$ such that x belongs to $B_{1,i}$ are pairwise disjoint. Therefore

$$\mu_2(E_x) = \sum_{i=1}^m \mu_2(B_{2,i}) \mathbf{1}_{B_{1,i}}(x).$$

In particular, the map $x \mapsto \mu_2(E_x)$ is measurable. We then set

$$\mu_0(E) := \int_{X_1} \mu_2(E_x) \, d\mu_1(x).$$

This quantity only depends on E , since it is defined from the sections E_x . If $E = B_1 \times B_2$, then

$$E_x = \begin{cases} B_2 & \text{if } x \in B_1, \\ \emptyset & \text{if } x \notin B_1, \end{cases}$$

so

$$\mu_0(B_1 \times B_2) = \int_{X_1} \mu_2(B_2) \mathbf{1}_{B_1}(x) \, d\mu_1(x) = \mu_1(B_1) \mu_2(B_2),$$

with the usual convention $0 \cdot \infty = 0$.

Step 3. Countable additivity on the algebra. Let now $(E_n)_{n \geq 1}$ be a sequence of pairwise disjoint elements of $\mathcal{E}$, and assume that

$$E := \bigcup_{n=1}^{\infty} E_n$$

still belongs to $\mathcal{E}$. For every x in X_1 , we have

$$E_x = \bigcup_{n=1}^{\infty} (E_n)_x,$$

and this union is disjoint. Hence

$$\mu_2(E_x) = \sum_{n=1}^{\infty} \mu_2((E_n)_x).$$

By the monotone convergence theorem,

$$\begin{aligned} \mu_0(E) &= \int_{X_1} \mu_2(E_x) \, d\mu_1(x) \\ &= \int_{X_1} \sum_{n=1}^{\infty} \mu_2((E_n)_x) \, d\mu_1(x) \\ &= \sum_{n=1}^{\infty} \int_{X_1} \mu_2((E_n)_x) \, d\mu_1(x) \\ &= \sum_{n=1}^{\infty} \mu_0(E_n). \end{aligned}$$

Thus μ_0 is a premeasure on $\mathcal{E}$.

Step 4. Extension and uniqueness. Since μ_1 and μ_2 are σ -finite, there exist increasing sequences $(X_{1,n})_{n \geq 1}$ in $\mathcal{A}_1$ and $(X_{2,n})_{n \geq 1}$ in $\mathcal{A}_2$ such that

$$X_1 = \bigcup_{n=1}^{\infty} X_{1,n}, \quad X_2 = \bigcup_{n=1}^{\infty} X_{2,n},$$

and

$$\mu_1(X_{1,n}) < \infty, \quad \mu_2(X_{2,n}) < \infty$$

for every $n \geq 1$. Then the sets

$$C_n := X_{1,n} \times X_{2,n}$$

belong to $\mathcal{E}$, satisfy

$$\mu_0(C_n) = \mu_1(X_{1,n})\mu_2(X_{2,n}) < \infty,$$

and form an increasing covering of $X_1 \times X_2$. Hence the premeasure μ_0 is σ -finite on $\mathcal{E}$. By the Carathéodory extension theorem, it extends to a measure on

$$\sigma(\mathcal{E}) = \mathcal{A}_1 \otimes \mathcal{A}_2.$$

By the uniqueness result proved in Chapter 3, this extension is unique. It is denoted by $\mu_1 \otimes \mu_2$. The same sequence (C_n) shows that $(X_1 \times X_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \mu_1 \otimes \mu_2)$ is σ -finite. $\square$

In the case of the Borel σ -algebras of Euclidean spaces, we record the following consequence.

Proposition 10.4 (Euclidean product structures). *For all integers $n, m \geq 1$, we have*

$$\mathcal{B}(\mathbb{R}^{n+m}) = \mathcal{B}(\mathbb{R}^n) \otimes \mathcal{B}(\mathbb{R}^m).$$

Moreover, if λ_d denotes the Lebesgue measure on $\mathbb{R}^d$, then

$$\lambda_{n+m} = \lambda_n \otimes \lambda_m.$$

Proof. The projections from $\mathbb{R}^{n+m} = \mathbb{R}^n \times \mathbb{R}^m$ onto the two factors are continuous. Hence every set $A \times B$, with A in $\mathcal{B}(\mathbb{R}^n)$ and B in $\mathcal{B}(\mathbb{R}^m)$, is Borel in $\mathbb{R}^{n+m}$, and therefore

$$\mathcal{B}(\mathbb{R}^n) \otimes \mathcal{B}(\mathbb{R}^m) \subset \mathcal{B}(\mathbb{R}^{n+m}).$$

Conversely, every open subset of $\mathbb{R}^{n+m}$ is a countable union of products $U \times V$, where U and V are open rectangles with rational endpoints in $\mathbb{R}^n$ and $\mathbb{R}^m$. Such products belong to $\mathcal{B}(\mathbb{R}^n) \otimes \mathcal{B}(\mathbb{R}^m)$, so

$$\mathcal{B}(\mathbb{R}^{n+m}) \subset \mathcal{B}(\mathbb{R}^n) \otimes \mathcal{B}(\mathbb{R}^m).$$

This proves the equality of the Borel σ -algebras.

For the measures, both λ_{n+m} and $\lambda_n \otimes \lambda_m$ are σ -finite Borel measures on $\mathbb{R}^{n+m}$. On every half-open rectangle $R \times S$, with R a half-open rectangle of $\mathbb{R}^n$ and S a half-open rectangle of $\mathbb{R}^m$, the construction of Lebesgue measure gives

$$\lambda_{n+m}(R \times S) = \lambda_n(R)\lambda_m(S) = (\lambda_n \otimes \lambda_m)(R \times S).$$

These rectangles form a generating π -system. By uniqueness of the σ -finite extension, the two measures coincide. $\square$

10.3 The Tonelli theorem for positive measurable functions

Theorem 10.5 (Tonelli theorem). *Let $(X_1, \mathcal{A}_1, \mu_1)$ and $(X_2, \mathcal{A}_2, \mu_2)$ be two σ -finite measure spaces. Let*

$$f: X_1 \times X_2 \rightarrow [0, \infty]$$

be an $\mathcal{A}_1 \otimes \mathcal{A}_2$ -measurable function. Then

1) *for every x in X_1 , the section $y \mapsto f(x, y)$ is $(\mathcal{A}_2, \mathcal{B}([0, \infty]))$ -measurable, and the map*

$$x \mapsto \int_{X_2} f(x, y) \, d\mu_2(y)$$

is $(\mathcal{A}_1, \mathcal{B}([0, \infty]))$ -measurable,

2) *for every y in X_2 , the section $x \mapsto f(x, y)$ is $(\mathcal{A}_1, \mathcal{B}([0, \infty]))$ -measurable, and the map*

$$y \mapsto \int_{X_1} f(x, y) \, d\mu_1(x)$$

is $(\mathcal{A}_2, \mathcal{B}([0, \infty]))$ -measurable,

3) *we have*

$$\begin{aligned} \int_{X_1 \times X_2} f(x, y) \, d(\mu_1 \otimes \mu_2)(x, y) &= \int_{X_1} \left(\int_{X_2} f(x, y) \, d\mu_2(y) \right) \, d\mu_1(x) \\ &= \int_{X_2} \left(\int_{X_1} f(x, y) \, d\mu_1(x) \right) \, d\mu_2(y). \end{aligned}$$

Proof. For each fixed x in X_1 , the map $y \mapsto (x, y)$ from X_2 to $X_1 \times X_2$ is measurable: the preimage of a rectangle $A \times B$ is B if x is in A and is $\emptyset$ otherwise, and the rectangles generate the product σ -algebra. Hence the section $y \mapsto f(x, y)$ is measurable. The same argument gives the measurability of the sections $x \mapsto f(x, y)$ for fixed y in X_2 . We therefore only have to prove the measurability of the partial integrals and the identity of the integrals.

Step 1: the finite case. Assume first that $\mu_1(X_1) < \infty$ and $\mu_2(X_2) < \infty$. For every E in $\mathcal{A}_1 \otimes \mathcal{A}_2$, we write

$$E_x := \{y \in X_2 : (x, y) \in E\}, \quad E^y := \{x \in X_1 : (x, y) \in E\}.$$

Let $\mathcal{C}$ be the class of sets E in $\mathcal{A}_1 \otimes \mathcal{A}_2$ such that

1) the maps $x \mapsto \mu_2(E_x)$ and $y \mapsto \mu_1(E^y)$ are measurable,

2)

$$(\mu_1 \otimes \mu_2)(E) = \int_{X_1} \mu_2(E_x) d\mu_1(x) = \int_{X_2} \mu_1(E^y) d\mu_2(y).$$

We apply Theorem 3.8. We first check that $\mathcal{C}$ is a Dynkin system. Since both measures are finite, the whole space $X_1 \times X_2$ belongs to $\mathcal{C}$. Now let E, F in $\mathcal{C}$ with $E \subset F$. Then

$$(F \setminus E)_x = F_x \setminus E_x, \quad (F \setminus E)^y = F^y \setminus E^y,$$

hence

$$\mu_2((F \setminus E)_x) = \mu_2(F_x) - \mu_2(E_x), \quad \mu_1((F \setminus E)^y) = \mu_1(F^y) - \mu_1(E^y).$$

These maps are measurable, and

$$\begin{aligned} \int_{X_1} \mu_2((F \setminus E)_x) d\mu_1(x) &= \int_{X_1} \mu_2(F_x) d\mu_1(x) - \int_{X_1} \mu_2(E_x) d\mu_1(x) \\ &= (\mu_1 \otimes \mu_2)(F) - (\mu_1 \otimes \mu_2)(E) \\ &= (\mu_1 \otimes \mu_2)(F \setminus E). \end{aligned}$$

The same computation with the horizontal sections shows that $F \setminus E$ belongs to $\mathcal{C}$.

Let now $(E_n)_{n \geq 1}$ be an increasing sequence in $\mathcal{C}$, and set

$$E := \bigcup_{n \geq 1} E_n.$$

For every x in X_1 , we have

$$(E_n)_x \uparrow E_x,$$

hence

$$\mu_2((E_n)_x) \uparrow \mu_2(E_x).$$

Therefore $x \mapsto \mu_2(E_x)$ is measurable as the increasing limit of measurable functions, and the monotone convergence theorem gives

$$\begin{aligned} \int_{X_1} \mu_2(E_x) d\mu_1(x) &= \lim_{n \rightarrow \infty} \int_{X_1} \mu_2((E_n)_x) d\mu_1(x) \\ &= \lim_{n \rightarrow \infty} (\mu_1 \otimes \mu_2)(E_n) \\ &= (\mu_1 \otimes \mu_2)(E). \end{aligned}$$

The same argument with the horizontal sections shows that E belongs to $\mathcal{C}$. Thus $\mathcal{C}$ is a Dynkin system.

If $E = A \times B$ with A in $\mathcal{A}_1$ and B in $\mathcal{A}_2$, then

$$E_x = \begin{cases} B & \text{if } x \in A, \\ \emptyset & \text{if } x \notin A, \end{cases} \quad E^y = \begin{cases} A & \text{if } y \in B, \\ \emptyset & \text{if } y \notin B. \end{cases}$$

Therefore

$$x \mapsto \mu_2(E_x) = \mu_2(B)\mathbf{1}_A(x), \quad y \mapsto \mu_1(E^y) = \mu_1(A)\mathbf{1}_B(y),$$

and

$$\begin{aligned}\int_{X_1} \mu_2(E_x) d\mu_1(x) &= \mu_2(B)\mu_1(A) = (\mu_1 \otimes \mu_2)(A \times B) \\ &= \mu_1(A)\mu_2(B) = \int_{X_2} \mu_1(E^y) d\mu_2(y).\end{aligned}$$

Hence every rectangle belongs to $\mathcal{C}$. Since the rectangles form a π -system which generates $\mathcal{A}_1 \otimes \mathcal{A}_2$, Theorem 3.8 implies that

$$\mathcal{C} = \mathcal{A}_1 \otimes \mathcal{A}_2.$$

We have therefore proved the theorem for indicator functions of measurable sets. If $s = \sum_{i=1}^r \alpha_i \mathbf{1}_{E_i}$ is a nonnegative simple measurable function, then the partial-integral maps are finite linear combinations of the measurable maps $x \mapsto \mu_2((E_i)_x)$ and $y \mapsto \mu_1((E_i)^y)$. The integral identities follow from the finite additivity and finite linearity already proved for positive integrals. Thus the theorem holds for every nonnegative simple function on $X_1 \times X_2$. Now let $(f_n)_{n \geq 1}$ be a sequence of nonnegative simple functions such that $f_n \uparrow f$ pointwise. For every n , the map

$$g_n(x) := \int_{X_2} f_n(x, y) d\mu_2(y)$$

is measurable, and

$$\int_{X_1} g_n(x) d\mu_1(x) = \int_{X_1 \times X_2} f_n d(\mu_1 \otimes \mu_2).$$

For each fixed x in X_1 , we have $f_n(x, \cdot) \uparrow f(x, \cdot)$, so by the monotone convergence theorem

$$g_n(x) \uparrow g(x) := \int_{X_2} f(x, y) d\mu_2(y).$$

Thus g is measurable as the increasing limit of measurable functions, and

$$\begin{aligned}\int_{X_1} g(x) d\mu_1(x) &= \lim_{n \rightarrow \infty} \int_{X_1} g_n(x) d\mu_1(x) \\ &= \lim_{n \rightarrow \infty} \int_{X_1 \times X_2} f_n d(\mu_1 \otimes \mu_2) \\ &= \int_{X_1 \times X_2} f d(\mu_1 \otimes \mu_2).\end{aligned}$$

The same argument with the variables exchanged gives the second iterated formula. This proves the theorem when both measures are finite.

Step 2: the σ -finite case. We now choose increasing sequences $(A_n)_{n \geq 1}$ in $\mathcal{A}_1$ and $(B_n)_{n \geq 1}$ in $\mathcal{A}_2$ such that

$$X_1 = \bigcup_{n \geq 1} A_n, \quad X_2 = \bigcup_{n \geq 1} B_n,$$

and

$$\mu_1(A_n) < \infty, \quad \mu_2(B_n) < \infty$$

for every $n \geq 1$. For every $n \geq 1$, we set

$$f_n := f \mathbf{1}_{A_n \times B_n}.$$

Apply the finite case to the finite measure spaces obtained by restricting X_1 to A_n and X_2 to B_n , with the restricted σ -algebras and the restricted measures. The product of the restricted measures agrees, by uniqueness of product measure, with the restriction of $\mu_1 \otimes \mu_2$ to $A_n \times B_n$. This gives that

$$g_n(x) := \int_{X_2} f_n(x, y) d\mu_2(y)$$

is measurable and that

$$\int_{X_1} g_n(x) d\mu_1(x) = \int_{X_1 \times X_2} f_n d(\mu_1 \otimes \mu_2).$$

For each fixed x in X_1 , we have $f_n(x, \cdot) \uparrow f(x, \cdot)$, hence

$$g_n(x) \uparrow \int_{X_2} f(x, y) d\mu_2(y).$$

Therefore the map $x \mapsto \int_{X_2} f(x, y) d\mu_2(y)$ is measurable, and the monotone convergence theorem yields

$$\begin{aligned} \int_{X_1} \left(\int_{X_2} f(x, y) d\mu_2(y) \right) d\mu_1(x) &= \lim_{n \rightarrow \infty} \int_{X_1} g_n(x) d\mu_1(x) \\ &= \lim_{n \rightarrow \infty} \int_{X_1 \times X_2} f_n d(\mu_1 \otimes \mu_2) \\ &= \int_{X_1 \times X_2} f d(\mu_1 \otimes \mu_2). \end{aligned}$$

The same argument in the other order completes the proof. □

In the particular case of Borel functions, we obtain the following corollary.

Corollary 10.6 (Tonelli theorem on Borel subsets). *Let m, p be two integers ≥ 1 , let $U \subset \mathbb{R}^m$ and $V \subset \mathbb{R}^p$ be Borel sets, and let*

$$f: U \times V \rightarrow [0, \infty]$$

be a Borel function. Then

1) *for every x in U , the section $y \mapsto f(x, y)$ is Borel on V , and the map*

$$x \mapsto \int_V f(x, y) d\lambda_p(y)$$

is Borel on U ,

2) *for every y in V , the section $x \mapsto f(x, y)$ is Borel on U , and the map*

$$y \mapsto \int_U f(x, y) d\lambda_m(x)$$

is Borel on V ,

3) *we have*

$$\begin{aligned} \int_{U \times V} f d\lambda_{m+p} &= \int_U \left(\int_V f(x, y) d\lambda_p(y) \right) d\lambda_m(x) \\ &= \int_V \left(\int_U f(x, y) d\lambda_m(x) \right) d\lambda_p(y). \end{aligned}$$

Proof. Step 1. Absolute integrability of almost all sections. We apply Tonelli's theorem to the spaces U and V endowed with their Borel σ -algebras and with the restrictions of the Lebesgue measures. These spaces are σ -finite, their product σ -algebra is the Borel σ -algebra of $U \times V$, and the corresponding product measure is the restriction of λ_{m+p} to $U \times V$. □

Example 10.7 (Area below a graph). *Let*

$$f(x, y) := \mathbf{1}_{\{x < y\}}$$

on $[0, 1]^2$. For every fixed x in $[0, 1]$, we have

$$\int_0^1 f(x, y) \, dy = \int_0^1 \mathbf{1}_{]x, 1]}(y) \, dy = 1 - x.$$

Similarly, for every fixed y in $[0, 1]$,

$$\int_0^1 f(x, y) \, dx = \int_0^1 \mathbf{1}_{[0, y[}(x) \, dx = y.$$

Tonelli's theorem therefore gives

$$\begin{aligned} \int_{[0, 1]^2} \mathbf{1}_{\{x < y\}} \, d\lambda_2 &= \int_0^1 (1 - x) \, dx \\ &= \int_0^1 y \, dy \\ &= \frac{1}{2}. \end{aligned}$$

This is a first concrete computation where the two iterated integrals are both simple and have the same value.

Exercise 10.8 (Area of the unit disk). *Let us calculate the area of the unit disk*

$$D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

By definition, the area of D is the integral of the characteristic function of D over $\mathbb{R}^2$; we therefore have

$$\begin{aligned} \text{area}(D) &= \int_{\mathbb{R}^2} \mathbf{1}_D(x, y) \, d\lambda_2(x, y) \\ &= \int_{-1}^1 \left(\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy \right) dx \\ &= 2 \int_{-1}^1 \sqrt{1-x^2} \, dx \\ &= 2 \int_{-\pi/2}^{\pi/2} \cos^2(t) \, dt \\ &= \pi. \end{aligned}$$

The last step uses the fact that the Riemann and Lebesgue integrals coincide for continuous functions on bounded intervals, followed by the classical trigonometric substitution $x = \sin(t)$ in the Riemann sense; this one-dimensional change-of-variables formula will be recalled later in the chapter on change of variables.

10.4 The Fubini theorem for integrable functions

In the case where f is not positive, Tonelli's theorem first allows us to control the integral of $|f|$. We may then apply the following statement.

Theorem 10.9 (Fubini theorem). Let $(X_1, \mathcal{A}_1, \mu_1)$ and $(X_2, \mathcal{A}_2, \mu_2)$ be two σ -finite measure spaces. Let

$$f: X_1 \times X_2 \rightarrow \mathbb{R}$$

be a function in $\mathcal{L}^1(\mu_1 \otimes \mu_2)$. Then

- 1) there exists a function G in $\mathcal{L}^1(\mu_1)$ such that, for almost every x in X_1 , the section $y \mapsto f(x, y)$ belongs to $\mathcal{L}^1(\mu_2)$ and

$$G(x) = \int_{X_2} f(x, y) \, d\mu_2(y),$$

- 2) there exists a function H in $\mathcal{L}^1(\mu_2)$ such that, for almost every y in X_2 , the section $x \mapsto f(x, y)$ belongs to $\mathcal{L}^1(\mu_1)$ and

$$H(y) = \int_{X_1} f(x, y) \, d\mu_1(x),$$

- 3) for any such choices of G and H , we have

$$\begin{aligned} \int_{X_1 \times X_2} f(x, y) \, d(\mu_1 \otimes \mu_2)(x, y) &= \int_{X_1} G(x) \, d\mu_1(x) \\ &= \int_{X_2} H(y) \, d\mu_2(y). \end{aligned}$$

Proof. **Step 1. Integrability of almost all sections.** We write

$$f = f^+ - f^-.$$

Since f belongs to $\mathcal{L}^1(\mu_1 \otimes \mu_2)$, the function $|f| = f^+ + f^-$ is integrable. Applying Tonelli's theorem to $|f|$, we obtain that

$$g(x) := \int_{X_2} |f(x, y)| \, d\mu_2(y)$$

defines a positive measurable function on X_1 and

$$\int_{X_1} g(x) \, d\mu_1(x) = \int_{X_1 \times X_2} |f(x, y)| \, d(\mu_1 \otimes \mu_2)(x, y) < \infty.$$

By Lemma 8.8, the set

$$N := \{x \in X_1 : g(x) = +\infty\}$$

satisfies $\mu_1(N) = 0$. We define the finite representative

$$\tilde{g} := g \mathbf{1}_{X_1 \setminus N}.$$

Then $\tilde{g}$ is nonnegative and belongs to $\mathcal{L}^1(\mu_1)$.

Step 2. Finite representatives for partial integrals. We now define

$$g_+(x) := \int_{X_2} f^+(x, y) \, d\mu_2(y), \quad g_-(x) := \int_{X_2} f^-(x, y) \, d\mu_2(y).$$

By Tonelli's theorem, the functions g_+ and g_- are positive and measurable, and

$$g_+(x) + g_-(x) = g(x)$$

for every x in X_1 .

We set

$$G_+ := g_+ \mathbf{1}_{X_1 \setminus N}, \quad G_- := g_- \mathbf{1}_{X_1 \setminus N}, \quad G := G_+ - G_-.$$

Then

$$0 \leq G_+ \leq \tilde{g}, \quad 0 \leq G_- \leq \tilde{g},$$

so G_+ and G_- belong to $\mathcal{L}^1(\mu_1)$, and therefore G belongs to $\mathcal{L}^1(\mu_1)$. Moreover, for every x in $X_1 \setminus N$, the section $y \mapsto f(x, y)$ belongs to $\mathcal{L}^1(\mu_2)$ and

$$G(x) = \int_{X_2} f(x, y) \, d\mu_2(y).$$

Since $G_+ = g_+$ and $G_- = g_-$ μ_1 -almost everywhere, Lemma 8.9 and Tonelli's theorem give

$$\begin{aligned} \int_{X_1} G_+(x) \, d\mu_1(x) &= \int_{X_1} g_+(x) \, d\mu_1(x) = \int_{X_1 \times X_2} f^+ \, d(\mu_1 \otimes \mu_2), \\ \int_{X_1} G_-(x) \, d\mu_1(x) &= \int_{X_1} g_-(x) \, d\mu_1(x) = \int_{X_1 \times X_2} f^- \, d(\mu_1 \otimes \mu_2). \end{aligned}$$

Therefore

$$\begin{aligned} \int_{X_1} G(x) \, d\mu_1(x) &= \int_{X_1} G_+(x) \, d\mu_1(x) - \int_{X_1} G_-(x) \, d\mu_1(x) \\ &= \int_{X_1 \times X_2} f^+ \, d(\mu_1 \otimes \mu_2) - \int_{X_1 \times X_2} f^- \, d(\mu_1 \otimes \mu_2) \\ &= \int_{X_1 \times X_2} f \, d(\mu_1 \otimes \mu_2). \end{aligned}$$

This proves item 1 and the first equality in item 3 for the function G constructed above. If $\tilde{G}$ is another function satisfying item 1, then $\tilde{G} = G$ μ_1 -almost everywhere, hence

$$\int_{X_1} \tilde{G} \, d\mu_1 = \int_{X_1} G \, d\mu_1$$

by Lemma 8.9.

Step 3. The second variable and the equality of integrals. By symmetry, the same argument applied after exchanging the roles of X_1 and X_2 proves item 2 and the second equality in item 3. The same uniqueness-up-to-almost-everywhere-equality remark applies to H . $\square$

Remark. Once measurable representatives G and H have been fixed, we usually write item 3 in the shorter form

$$\begin{aligned} \int_{X_1 \times X_2} f(x, y) \, d(\mu_1 \otimes \mu_2)(x, y) &= \int_{X_1} \left(\int_{X_2} f(x, y) \, d\mu_2(y) \right) \, d\mu_1(x) \\ &= \int_{X_2} \left(\int_{X_1} f(x, y) \, d\mu_1(x) \right) \, d\mu_2(y). \end{aligned}$$

Only the values of the inner integrals on sets of full measure matter in these formulas.

For integrable Borel functions, we obtain the following corollary.

Corollary 10.10 (Fubini theorem on Borel subsets). Let m, p be two integers ≥ 1 , let $U \subset \mathbb{R}^m$ and $V \subset \mathbb{R}^p$ be Borel sets, and let

$$f: U \times V \rightarrow \mathbb{R}$$

be a Borel function integrable with respect to the restriction of λ_{m+p} to $U \times V$. Then

- 1) there exists a function G in $\mathcal{L}^1(U, \lambda_m)$ such that, for almost every x in U , the section $y \mapsto f(x, y)$ belongs to $\mathcal{L}^1(V, \lambda_p)$ and

$$G(x) = \int_V f(x, y) \, d\lambda_p(y),$$

- 2) there exists a function H in $\mathcal{L}^1(V, \lambda_p)$ such that, for almost every y in V , the section $x \mapsto f(x, y)$ belongs to $\mathcal{L}^1(U, \lambda_m)$ and

$$H(y) = \int_U f(x, y) \, d\lambda_m(x),$$

- 3) for any such choices of G and H , we have

$$\begin{aligned} \int_{U \times V} f \, d\lambda_{m+p} &= \int_U G(x) \, d\lambda_m(x) \\ &= \int_V H(y) \, d\lambda_p(y). \end{aligned}$$

Proof. We apply Fubini's theorem to the spaces U and V endowed with their Borel σ -algebras and with the restrictions of the Lebesgue measures. These spaces are σ -finite, their product σ -algebra is the Borel σ -algebra of $U \times V$, and the corresponding product measure is the restriction of λ_{m+p} to $U \times V$. $\square$

Remark. In the particular case of a continuous function on a bounded closed rectangle $U \times V$ of $\mathbb{R}^{m+p}$, integrability is automatic, so Fubini's theorem applies directly. More generally, the relevant hypothesis is integrability, not continuity.

In practice, the theorems of Tonelli and Fubini are used jointly: Tonelli allows us to prove the integrability of a nonnegative function, which is often the key preliminary step before Fubini can be used for an actual computation.

Example 10.11 (A sine integral by Fubini theorem). We have

$$\int_0^\infty e^{-x} \frac{\sin x}{x} \, dx = \frac{\pi}{4}.$$

Proof. We consider the function

$$G(x, y) := e^{-x} \cos(xy)$$

on $]0, \infty[\times]0, 1[$. For every (x, y) in $]0, \infty[\times]0, 1[$, we have

$$|G(x, y)| \leq e^{-x}.$$

Moreover,

$$\int_{]0, \infty[\times]0, 1[} e^{-x} \, d(\lambda_1 \otimes \lambda_1)(x, y) = \int_0^\infty e^{-x} \, dx = 1,$$

so the function G is integrable on the product space. Fubini's theorem therefore applies.

For every $x > 0$, we have

$$\int_0^1 G(x, y) \, dy = e^{-x} \int_0^1 \cos(xy) \, dy = e^{-x} \frac{\sin x}{x}.$$

Hence

$$\int_0^\infty e^{-x} \frac{\sin x}{x} \, dx = \int_0^\infty \left(\int_0^1 G(x, y) \, dy \right) \, dx.$$

Applying Fubini, we obtain

$$\int_0^\infty e^{-x} \frac{\sin x}{x} \, dx = \int_0^1 \left(\int_0^\infty e^{-x} \cos(xy) \, dx \right) \, dy.$$

It remains to compute the inner integral. For fixed y in $]0, 1[$, an antiderivative of $x \mapsto e^{-x} \cos(xy)$ is

$$x \mapsto \frac{e^{-x}(-\cos(xy) + y \sin(xy))}{1 + y^2}.$$

Therefore

$$\int_0^\infty e^{-x} \cos(xy) \, dx = \left[\frac{e^{-x}(-\cos(xy) + y \sin(xy))}{1 + y^2} \right]_{x=0}^{x=\infty} = \frac{1}{1 + y^2}.$$

We conclude that

$$\int_0^\infty e^{-x} \frac{\sin x}{x} \, dx = \int_0^1 \frac{1}{1 + y^2} \, dy = \arctan(1) - \arctan(0) = \frac{\pi}{4}.$$

□

Chapter 11

Modes of convergence

We now compare the main notions of convergence for measurable functions and the logical relations between them. The first part of the chapter concerns implications which do not use equi-integrability. Equi-integrability is then introduced exactly at the point where it becomes the additional hypothesis in the two Vitali implications. We end the chapter by collecting the model counterexamples which explain why the hypotheses cannot simply be removed.

Let $(X, \mathcal{A}, \mu)$ be a measure space.

11.1 Four modes of convergence

Definition 11.1 (Almost uniform convergence). *Let f_n and f be measurable real-valued functions on $(X, \mathcal{A}, \mu)$. We say that*

- 1) (f_n) converges to f almost everywhere if there exists a measurable set $Z \subset X$ with $\mu(Z) = 0$ such that

$$f_n(x) \rightarrow f(x) \quad \text{as } n \rightarrow +\infty$$

for each x in $X \setminus Z$;

- 2) (f_n) converges to f almost uniformly if for every $\varepsilon > 0$, there exists a measurable set $E_\varepsilon \subset X$ such that $\mu(X \setminus E_\varepsilon) < \varepsilon$ and

$$\sup_{x \in E_\varepsilon} |f_n(x) - f(x)| \rightarrow 0 \quad \text{as } n \rightarrow +\infty;$$

- 3) (f_n) converges to f in $\mathcal{L}^1(X, \mu)$ if f_n and f belong to $\mathcal{L}^1(X, \mu)$ and

$$\int_X |f_n - f| d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty;$$

- 4) (f_n) converges to f in measure if for every $\delta > 0$,

$$\mu(\{|f_n - f| > \delta\}) \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

For these four modes of convergence we will respectively use the short-cut notations a.e., a.u., $\mathcal{L}^1$ and μ .

11.2 Implications not involving equi-integrability

We first prove the implications which do not require any equi-integrability hypothesis. Some of them are unconditional, while others require either the finiteness of the measure or the extraction of a subsequence.

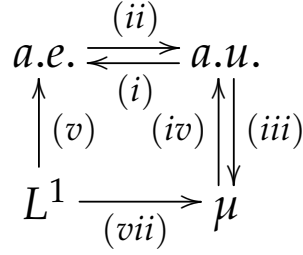


Figure 11.1: First comparison diagram. The arrows displayed here do not involve equi-integrability. Arrow (ii) requires $\mu(X) < \infty$, and arrows (iv) and (v) hold after extraction of a subsequence.

11.2.1 Almost uniform versus almost everywhere convergence

We start with implication (i), that is, almost uniform convergence implies almost everywhere convergence.

Proposition 11.2 (Almost uniform convergence implies almost everywhere convergence).
Let f_n and f be measurable functions. If (f_n) converges to f almost uniformly, then it converges to f almost everywhere.

Proof. If (f_n) converges to f almost uniformly, then, by definition, for every $\varepsilon > 0$, there exists a measurable set $E_\varepsilon \subset X$ such that

$$\mu(X \setminus E_\varepsilon) < \varepsilon \quad \text{and} \quad \sup_{x \in E_\varepsilon} |f_n(x) - f(x)| \rightarrow 0 \quad \text{as } n \rightarrow +\infty. \quad (11.1)$$

Take $\varepsilon = 1/k$ with k in $\mathbb{N}^*$, and define

$$Z := \bigcap_{k=1}^{\infty} (X \setminus E_{1/k}).$$

Then, for every k in $\mathbb{N}^*$, we have

$$\mu(Z) \leq \mu(X \setminus E_{1/k}) < 1/k.$$

Letting $k \rightarrow +\infty$, we obtain $\mu(Z) = 0$. Moreover,

$$X \setminus Z = \bigcup_{k=1}^{\infty} E_{1/k}.$$

Thus, for each x in $X \setminus Z$, there exists k in $\mathbb{N}^*$ such that x belongs to $E_{1/k}$. For this k , we have

$$|f_n(x) - f(x)| \leq \sup_{y \in E_{1/k}} |f_n(y) - f(y)|.$$

The right-hand side goes to 0 as $n \rightarrow +\infty$ by (11.1), applied with $\varepsilon = 1/k$. Hence $f_n(x) \rightarrow f(x)$ as $n \rightarrow +\infty$ for every x in $X \setminus Z$. $\square$

The converse statement is true in the case of finite measures; this is Egoroff's theorem and corresponds to arrow (ii) in Figure 11.1.

Theorem 11.3 (Egoroff theorem). Assume that $\mu(X) < \infty$. Let f_n and f be measurable functions. If (f_n) converges almost everywhere to f , then (f_n) converges almost uniformly to f .

Proof. We use a fixed summable budget. For every i, j in $\mathbb{N}^*$, define

$$E_{i,j} := \bigcup_{n=j}^{\infty} \{|f_n - f| > 2^{-i}\}.$$

For each fixed i , the sequence $(E_{i,j})_{j \geq 1}$ is decreasing. Since $\mu(X) < \infty$, Theorem 2.3, item 2), gives

$$\lim_{j \rightarrow +\infty} \mu(E_{i,j}) = \mu \left(\bigcap_{j=1}^{\infty} E_{i,j} \right).$$

If x belongs to $\bigcap_{j=1}^{\infty} E_{i,j}$, then, for every $j \geq 1$, there exists $n \geq j$ such that

$$|f_n(x) - f(x)| > 2^{-i}.$$

Thus the sequence $(f_n(x))$ cannot converge to $f(x)$. Since $f_n \rightarrow f$ almost everywhere as $n \rightarrow +\infty$, this intersection is negligible. Consequently,

$$\lim_{j \rightarrow +\infty} \mu(E_{i,j}) = 0.$$

For each $i \geq 1$, we can therefore choose an integer $N(i)$ such that

$$\mu(E_{i,N(i)}) < 2^{-i}.$$

Let $\varepsilon > 0$. Choose i_ε in $\mathbb{N}^*$ such that

$$\sum_{i=i_\varepsilon}^{\infty} 2^{-i} < \varepsilon.$$

Define

$$E_\varepsilon := X \setminus \bigcup_{i=i_\varepsilon}^{\infty} E_{i,N(i)}.$$

Then

$$\mu(X \setminus E_\varepsilon) \leq \sum_{i=i_\varepsilon}^{\infty} \mu(E_{i,N(i)}) < \sum_{i=i_\varepsilon}^{\infty} 2^{-i} < \varepsilon.$$

Moreover, for every $i \geq i_\varepsilon$, every $n \geq N(i)$, and every x in E_ε , we have $x \notin E_{i,N(i)}$, hence

$$|f_n(x) - f(x)| \leq 2^{-i}.$$

Thus

$$\sup_{x \in E_\varepsilon} |f_n(x) - f(x)| \leq 2^{-i}$$

for every $n \geq N(i)$ and every $i \geq i_\varepsilon$. Since i can be chosen arbitrarily large, $f_n \rightarrow f$ uniformly on E_ε as $n \rightarrow +\infty$. $\square$

11.2.2 Almost uniform versus convergence in measure

We now compare almost uniform convergence and convergence in measure. We start with implication (iii): almost uniform convergence is stronger than convergence in measure.

Theorem 11.4 (Almost uniform convergence implies convergence in measure). *Let f_n and f be measurable functions. If (f_n) converges to f almost uniformly, then it converges to f in measure.*

Proof. Let $\delta > 0$ and $\varepsilon > 0$. Since (f_n) converges to f almost uniformly, there exists a measurable set $E_\varepsilon \subset X$ such that

$$\mu(X \setminus E_\varepsilon) < \varepsilon \quad \text{and} \quad \sup_{x \in E_\varepsilon} |f_n(x) - f(x)| \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Thus there exists N in $\mathbb{N}^*$ such that, for every $n \geq N$,

$$\sup_{x \in E_\varepsilon} |f_n(x) - f(x)| < \delta.$$

For every such n ,

$$\{|f_n - f| > \delta\} \cap E_\varepsilon = \emptyset,$$

and therefore

$$\mu(\{|f_n - f| > \delta\}) \leq \mu(X \setminus E_\varepsilon) < \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we get

$$\mu(\{|f_n - f| > \delta\}) \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

This proves convergence in measure. □

The converse statement, which is implication (iv), only holds after extraction of a subsequence.

Theorem 11.5 (Almost uniform subsequence theorem). *Let f_n and f be measurable functions. If (f_n) converges to f in measure, then there exists a subsequence (f_{n_k}) which converges to f almost uniformly.*

Proof. We use the same summable-budget device as in the proof of Egoroff's theorem. Set $a_k := 2^{-k}$. Since (f_n) converges in measure to f , we can choose, by induction, a strictly increasing sequence of integers $(n_k)_{k \geq 1}$ such that

$$\mu\left(\left\{|f_{n_k} - f| > \frac{1}{k}\right\}\right) < a_k$$

for every k in $\mathbb{N}^*$. Define

$$E_k := \left\{|f_{n_k} - f| \leq \frac{1}{k}\right\}.$$

Let $\varepsilon > 0$, and choose k_ε in $\mathbb{N}^*$ such that

$$\sum_{k=k_\varepsilon}^{\infty} a_k < \varepsilon.$$

Set

$$E_\varepsilon := \bigcap_{k=k_\varepsilon}^{\infty} E_k.$$

Then

$$X \setminus E_\varepsilon = \bigcup_{k=k_\varepsilon}^{\infty} (X \setminus E_k),$$

so that

$$\mu(X \setminus E_\varepsilon) \leq \sum_{k=k_\varepsilon}^{\infty} \mu(X \setminus E_k) < \sum_{k=k_\varepsilon}^{\infty} a_k < \varepsilon.$$

Moreover, for every x in E_ε and every $k \geq k_\varepsilon$, we have

$$|f_{n_k}(x) - f(x)| \leq \frac{1}{k}.$$

Thus

$$\sup_{x \in E_\varepsilon} |f_{n_k}(x) - f(x)| \leq \frac{1}{k}$$

for every $k \geq k_\varepsilon$. Therefore $f_{n_k} \rightarrow f$ uniformly on E_ε as $k \rightarrow +\infty$, and the subsequence (f_{n_k}) converges to f almost uniformly. $\square$

11.2.3 One-way links from $\mathcal{L}^1$ convergence

We now record the two implications from $\mathcal{L}^1$ convergence which do not require equi-integrability. First, $\mathcal{L}^1$ convergence implies almost everywhere convergence after extraction of a subsequence.

Theorem 11.6 (Riesz–Fischer extraction theorem). *Let f_n and f be in $\mathcal{L}^1(\mu)$. If $(f_n)_{n \geq 1}$ converges to f in $\mathcal{L}^1(\mu)$ as $n \rightarrow +\infty$, then there exists a subsequence $(f_{\varphi(n)})_{n \geq 1}$ which converges to f μ -almost everywhere.*

Proof. Since $f_n \rightarrow f$ in $\mathcal{L}^1(\mu)$, the sequence (f_n) is Cauchy in $L^1(\mu)$. By the completeness theorem, Theorem 8.19, there exists a subsequence $(f_{\varphi(n)})$ which converges almost everywhere to some representative g of the L^1 limit of the sequence. The convergence in L^1 also gives

$$\|g - f\|_1 \leq \|g - f_{\varphi(n)}\|_1 + \|f_{\varphi(n)} - f\|_1 \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Thus $g = f$ μ -almost everywhere. After changing g on a negligible set, the chosen subsequence converges to f μ -almost everywhere. $\square$

Second, $\mathcal{L}^1$ convergence implies convergence in measure, by Markov's inequality.

Theorem 11.7 (L^1 convergence implies convergence in measure). *Let $(X, \mathcal{A}, \mu)$ be a measure space and let (f_n) be a sequence of integrable functions converging in $\mathcal{L}^1(X, \mu)$ to an integrable function f as $n \rightarrow +\infty$. Then (f_n) converges to f in measure.*

Proof. Let $\delta > 0$. Markov's inequality gives

$$\delta \mu(\{|f_n - f| > \delta\}) \leq \int_X |f_n - f| d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Hence $\mu(\{|f_n - f| > \delta\}) \rightarrow 0$ for every $\delta > 0$. $\square$

11.3 Equi-integrability and the Vitali implications

We now introduce the additional hypothesis needed to reverse the two links toward $\mathcal{L}^1$ convergence. The condition rules out concentration of mass on sets of very small measure.

Definition 11.8 (Equi-integrability). *A sequence of integrable functions $(f_n)_{n \in \mathbb{N}^*}$ is said to be equi-integrable if for every $\varepsilon > 0$, there exists $\delta > 0$ such that*

$$\sup_{n \in \mathbb{N}^*} \int_E |f_n| d\mu \leq \varepsilon$$

for every measurable set $E \subset X$ satisfying $\mu(E) \leq \delta$.

Proposition 11.9 (Elementary examples of equi-integrability). *The following properties hold.*

- 1) *Let f be in $\mathcal{L}^1(X, \mu)$ and let $(f_n)_{n \in \mathbb{N}^*}$ be the sequence constant equal to f . Then (f_n) is equi-integrable.*
- 2) *Let k be in $\mathbb{N}^*$ and let $f_1, \dots, f_k$ be in $\mathcal{L}^1(X, \mu)$. Then the sequence $(f_1, \dots, f_k, f_k, \dots, f_k, \dots)$ is equi-integrable.*
- 3) *Let (f_n) be a sequence of functions converging to f in $\mathcal{L}^1(X, \mu)$ as $n \rightarrow +\infty$. Then (f_n) is equi-integrable.*

Proof. We prove the three items separately.

Proof of 1). Let $\varepsilon > 0$. By Corollary 8.13, applied to the integrable function f , there exists $\delta > 0$ such that, for every measurable set $E \subset X$ satisfying $\mu(E) \leq \delta$, one has

$$\int_E |f| d\mu \leq \varepsilon.$$

For the constant sequence $f_n = f$, this gives

$$\sup_{n \in \mathbb{N}^*} \int_E |f_n| d\mu = \int_E |f| d\mu \leq \varepsilon.$$

Thus the constant sequence is equi-integrable.

Proof of 2). Let $\varepsilon > 0$. By item 1), applied to each of the finitely many functions $f_1, \dots, f_k$, there exist $\delta_1, \dots, \delta_k > 0$ such that

$$\int_E |f_i| d\mu \leq \varepsilon$$

whenever $\mu(E) \leq \delta_i$. Taking $\delta := \min_{1 \leq i \leq k} \delta_i$, we obtain, for every measurable set $E \subset X$ satisfying $\mu(E) \leq \delta$,

$$\sup_{n \in \mathbb{N}^*} \int_E |f_n| d\mu = \max_{1 \leq i \leq k} \int_E |f_i| d\mu \leq \varepsilon.$$

This proves item 2).

Proof of 3). Let $\varepsilon > 0$. Since $f_n \rightarrow f$ in $\mathcal{L}^1(X, \mu)$ as $n \rightarrow +\infty$, there exists n_0 in $\mathbb{N}^*$ such that

$$\int_X |f_n - f| d\mu \leq \frac{\varepsilon}{2}$$

for every $n \geq n_0$. By item 1), applied to the constant sequence equal to f , there exists $\delta_1 > 0$ such that

$$\int_E |f| d\mu \leq \frac{\varepsilon}{2}$$

for every measurable set $E \subset X$ satisfying $\mu(E) \leq \delta_1$. Therefore, for every $n \geq n_0$ and every such set E ,

$$\int_E |f_n| d\mu \leq \int_E |f_n - f| d\mu + \int_E |f| d\mu \leq \varepsilon.$$

If $n_0 = 1$, this already proves the claim. If $n_0 \geq 2$, item 2), applied to the finite family $f_1, \dots, f_{n_0-1}$, gives $\delta_2 > 0$ such that

$$\sup_{1 \leq n < n_0} \int_E |f_n| d\mu \leq \varepsilon$$

whenever $\mu(E) \leq \delta_2$. Taking $\delta := \min(\delta_1, \delta_2)$ in the case $n_0 \geq 2$, and $\delta := \delta_1$ in the case $n_0 = 1$, we get

$$\sup_{n \in \mathbb{N}^*} \int_E |f_n| d\mu \leq \varepsilon$$

for every measurable set $E \subset X$ satisfying $\mu(E) \leq \delta$. Thus (f_n) is equi-integrable. $\square$

A useful practical criterion is the following tail criterion. It separates equi-integrability from the mere boundedness of the $\mathcal{L}^1$ norms.

Proposition 11.10 (Tail criterion for equi-integrability). *Let $(f_n)_{n \in \mathbb{N}^*}$ be a bounded sequence in $\mathcal{L}^1(X, \mu)$. Then the sequence (f_n) is equi-integrable if and only if*

$$\lim_{R \rightarrow +\infty} \sup_{n \in \mathbb{N}^*} \int_{\{|f_n| > R\}} |f_n| d\mu = 0.$$

Proof. Let

$$M := \sup_{n \in \mathbb{N}^*} \int_X |f_n| d\mu < \infty.$$

Step 1. Assume that (f_n) is equi-integrable, and let $\varepsilon > 0$. Choose $\delta > 0$ such that

$$\sup_{n \in \mathbb{N}^*} \int_E |f_n| d\mu \leq \varepsilon$$

for every measurable set $E \subset X$ with $\mu(E) \leq \delta$. Choose $R_\varepsilon > 0$ such that

$$\frac{M}{R_\varepsilon} \leq \delta.$$

Then, for every $R \geq R_\varepsilon$ and every n in $\mathbb{N}^*$, Markov's inequality gives

$$\mu(\{|f_n| > R\}) \leq \frac{1}{R} \int_X |f_n| d\mu \leq \frac{M}{R} \leq \delta.$$

Therefore

$$\sup_{n \in \mathbb{N}^*} \int_{\{|f_n| > R\}} |f_n| d\mu \leq \varepsilon$$

for every $R \geq R_\varepsilon$.

Step 2. Conversely, suppose that

$$\lim_{R \rightarrow +\infty} \sup_{n \in \mathbb{N}^*} \int_{\{|f_n| > R\}} |f_n| d\mu = 0.$$

Fix $\varepsilon > 0$, and choose $R_\varepsilon > 0$ such that

$$\sup_{n \in \mathbb{N}^*} \int_{\{|f_n| > R_\varepsilon\}} |f_n| d\mu \leq \frac{\varepsilon}{2}.$$

Then, for every measurable set $E \subset X$ with

$$\mu(E) \leq \frac{\varepsilon}{2R_\varepsilon} =: \delta,$$

and for every n in $\mathbb{N}^*$, we have

$$\begin{aligned} \int_E |f_n| \, d\mu &= \int_{E \cap \{|f_n| > R_\varepsilon\}} |f_n| \, d\mu + \int_{E \cap \{|f_n| \leq R_\varepsilon\}} |f_n| \, d\mu \\ &\leq \frac{\varepsilon}{2} + R_\varepsilon \mu(E) \leq \varepsilon. \end{aligned}$$

This proves equi-integrability. □

11.3.1 The first Vitali theorem

The first Vitali theorem says that, on a finite-measure space, equi-integrability promotes almost everywhere convergence to $\mathcal{L}^1$ convergence.

Theorem 11.11 (First Vitali theorem). *Let μ be a finite measure. Let (f_n) be an equi-integrable sequence of integrable functions, and let f be a measurable function. Assume that $f_n \rightarrow f$ almost everywhere as $n \rightarrow +\infty$. Then f belongs to $\mathcal{L}^1(X, \mu)$ and*

$$\int_X |f_n - f| \, d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Proof. Let $\varepsilon > 0$. Since (f_n) is equi-integrable, there exists $\delta > 0$ such that

$$\sup_{n \in \mathbb{N}^*} \int_E |f_n| \, d\mu \leq \frac{\varepsilon}{4}$$

for every measurable set $E \subset X$ with $\mu(E) \leq \delta$. By Egoroff's theorem, applied with the exceptional measure δ , there exists a measurable set $E_\delta \subset X$ such that $\mu(X \setminus E_\delta) < \delta$ and (f_n) converges uniformly to f on E_δ . Choose n_0 such that

$$\sup_{x \in E_\delta} |f_{n_0}(x) - f(x)| \leq 1.$$

Then $|f| \leq |f_{n_0}| + 1$ on E_δ , and therefore f is integrable on E_δ because $\mu(E_\delta) < \infty$. By Fatou's lemma, for every measurable set $E \subset X$ with $\mu(E) \leq \delta$,

$$\int_E |f| \, d\mu \leq \liminf_{n \rightarrow +\infty} \int_E |f_n| \, d\mu \leq \frac{\varepsilon}{4}.$$

Applying this to $X \setminus E_\delta$ shows that f is also integrable on $X \setminus E_\delta$. Hence f belongs to $\mathcal{L}^1(X, \mu)$. Moreover,

$$\begin{aligned} \int_X |f_n - f| \, d\mu &= \int_{E_\delta} |f_n - f| \, d\mu + \int_{X \setminus E_\delta} |f_n - f| \, d\mu \\ &\leq \mu(X) \sup_{x \in E_\delta} |f_n(x) - f(x)| + \int_{X \setminus E_\delta} |f_n| \, d\mu + \int_{X \setminus E_\delta} |f| \, d\mu \\ &\leq \mu(X) \sup_{x \in E_\delta} |f_n(x) - f(x)| + \frac{\varepsilon}{2}. \end{aligned}$$

Letting $n \rightarrow +\infty$, we get

$$\limsup_{n \rightarrow +\infty} \int_X |f_n - f| \, d\mu \leq \frac{\varepsilon}{2}.$$

Since ε is arbitrary, $f_n \rightarrow f$ in $\mathcal{L}^1(X, \mu)$ as $n \rightarrow +\infty$. □

11.3.2 The second Vitali theorem

The second Vitali theorem promotes convergence in measure to $\mathcal{L}^1$ convergence, again under the finite-measure and equi-integrability hypotheses. We first isolate the reduction which replaces the sequence (f_n) by the sequence of differences $g_n := f_n - f$.

Lemma 11.12 (Vitali difference reduction). *Let (f_n) be an equi-integrable sequence in $\mathcal{L}^1(X, \mu)$, and let f be in $\mathcal{L}^1(X, \mu)$. Define*

$$g_n := f_n - f.$$

Then (g_n) is equi-integrable. If $f_n \rightarrow f$ in measure, then $g_n \rightarrow 0$ in measure. If, moreover, $(X_k)_{k \geq 1}$ is a finite-measure exhaustion of a σ -finite space and (f_n) has uniformly small tails along (X_k) , then (g_n) has uniformly small tails along (X_k) .

Proof. Let $\varepsilon > 0$. Since (f_n) is equi-integrable, there exists $\delta_1 > 0$ such that

$$\sup_{n \in \mathbb{N}^*} \int_E |f_n| \, d\mu \leq \frac{\varepsilon}{2}$$

whenever $\mu(E) \leq \delta_1$. Since f is integrable, Corollary 8.13 gives $\delta_2 > 0$ such that

$$\int_E |f| \, d\mu \leq \frac{\varepsilon}{2}$$

whenever $\mu(E) \leq \delta_2$. With $\delta := \min(\delta_1, \delta_2)$, we obtain

$$\int_E |g_n| \, d\mu \leq \int_E |f_n| \, d\mu + \int_E |f| \, d\mu \leq \varepsilon$$

for every measurable set $E \subset X$ satisfying $\mu(E) \leq \delta$ and every n in $\mathbb{N}^*$. Thus (g_n) is equi-integrable.

If $f_n \rightarrow f$ in measure, then, for every $\delta > 0$,

$$\mu(\{|g_n| > \delta\}) = \mu(\{|f_n - f| > \delta\}) \rightarrow 0.$$

Hence $g_n \rightarrow 0$ in measure.

Assume finally that (X_k) is a finite-measure exhaustion and that (f_n) has uniformly small tails along (X_k) . Since f is integrable,

$$\int_{X \setminus X_k} |f| \, d\mu \rightarrow 0 \quad \text{as } k \rightarrow +\infty.$$

Therefore

$$\sup_{n \geq 1} \int_{X \setminus X_k} |g_n| \, d\mu \leq \sup_{n \geq 1} \int_{X \setminus X_k} |f_n| \, d\mu + \int_{X \setminus X_k} |f| \, d\mu \rightarrow 0.$$

This proves the last assertion. □

Theorem 11.13 (Second Vitali theorem). *Let $(X, \mathcal{A}, \mu)$ be a measure space with $\mu(X) < \infty$. Let (f_n) be an equi-integrable sequence of integrable functions, and let f be a measurable function. Assume that $f_n \rightarrow f$ in measure as $n \rightarrow +\infty$. Then f belongs to $\mathcal{L}^1(X, \mu)$ and*

$$\int_X |f_n - f| \, d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Proof. By the subsequence theorem for convergence in measure, there exists a subsequence (f_{n_k}) which converges to f almost uniformly. Hence this subsequence converges to f almost everywhere. Since (f_{n_k}) is equi-integrable, the first Vitali theorem, Theorem 11.11, gives f in $\mathcal{L}^1(X, \mu)$. Define $g_n := f_n - f$. By Lemma 11.12, the sequence (g_n) is equi-integrable and converges to 0 in measure. Let $\varepsilon > 0$. Choose $\delta > 0$ such that

$$\int_E |g_n| d\mu \leq \frac{\varepsilon}{2} \quad (11.2)$$

for every measurable set $E \subset X$ with $\mu(E) \leq \delta$ and every n in $\mathbb{N}^*$. Choose $c > 0$ such that $c\mu(X) < \varepsilon/2$. Since $g_n \rightarrow 0$ in measure, there exists N such that

$$\mu(\{|g_n| \geq c\}) < \delta$$

for every $n \geq N$. For every $n \geq N$, we have

$$\begin{aligned} \int_X |g_n| d\mu &= \int_{\{|g_n| \geq c\}} |g_n| d\mu + \int_{\{|g_n| < c\}} |g_n| d\mu \\ &\leq \frac{\varepsilon}{2} + c\mu(X) < \varepsilon, \end{aligned}$$

where (11.2) has been applied to $\{|g_n| \geq c\}$. Thus $f_n \rightarrow f$ in $\mathcal{L}^1(X, \mu)$. $\square$

Combining the first implications with the two Vitali implications gives the following complete diagram.

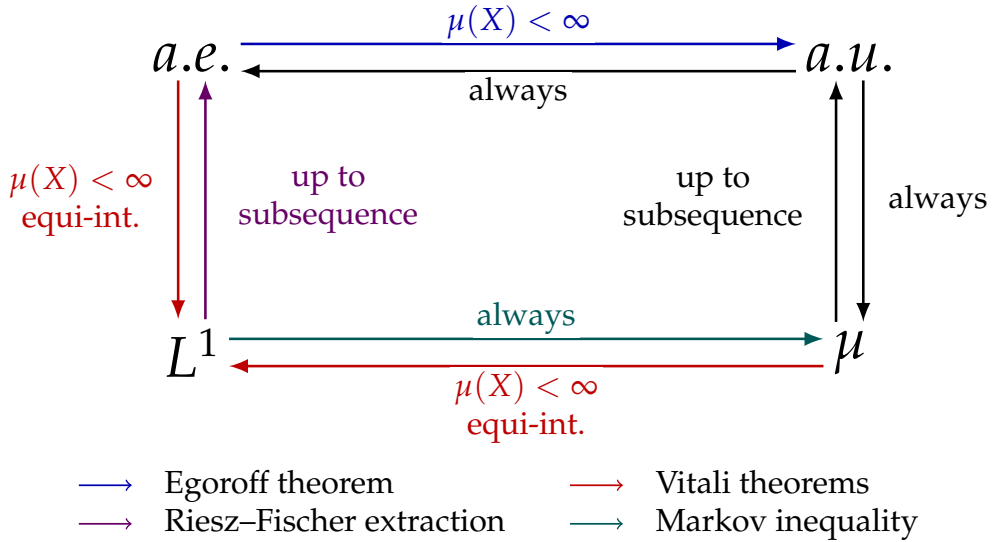


Figure 11.2: Precise comparison diagram. The labels record the main assumptions or subtleties needed for each implication.

11.3.3 Vitali theorems on σ -finite spaces

The two Vitali implications above use the finite-measure assumption. On a σ -finite space, this assumption can be replaced by a uniform tail condition along a finite-measure exhaustion.

Definition 11.14 (Uniformly small tails). Let $(X, \mathcal{A}, \mu)$ be σ -finite and let $(X_k)_{k \geq 1}$ be an increasing sequence in $\mathcal{A}$ such that $\mu(X_k) < \infty$ and $\cup_{k \geq 1} X_k = X$. A family (f_n) of integrable functions is said to have uniformly small tails along (X_k) if

$$\lim_{k \rightarrow +\infty} \sup_{n \geq 1} \int_{X \setminus X_k} |f_n| d\mu = 0.$$

Of course, every finite measure space satisfies this assumption by taking $X_k = X$ for every k . We first extend the first Vitali theorem.

Theorem 11.15 (First Vitali theorem on sigma-finite spaces). *Let $(X, \mathcal{A}, \mu)$ be σ -finite. Let (f_n) be an equi-integrable sequence of integrable functions, and let f be a measurable function. Assume that $f_n \rightarrow f$ almost everywhere as $n \rightarrow +\infty$. Assume moreover that there exists an increasing sequence $(X_k)_{k \geq 1}$ in $\mathcal{A}$ such that $\mu(X_k) < \infty$, $\cup_{k \geq 1} X_k = X$, and (f_n) has uniformly small tails along (X_k) . Then f belongs to $\mathcal{L}^1(X, \mu)$ and*

$$\int_X |f_n - f| d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Proof. Let $\varepsilon > 0$. Choose $k \geq 1$ such that

$$\sup_{n \geq 1} \int_{X \setminus X_k} |f_n| d\mu \leq \frac{\varepsilon}{3}.$$

By Fatou's lemma,

$$\int_{X \setminus X_k} |f| d\mu \leq \liminf_{n \rightarrow +\infty} \int_{X \setminus X_k} |f_n| d\mu \leq \frac{\varepsilon}{3}.$$

On X_k the measure is finite, the restriction of (f_n) is equi-integrable, and $f_n \rightarrow f$ almost everywhere. The finite-measure first Vitali theorem gives f in $\mathcal{L}^1(X_k, \mu)$ and an integer N such that, for every $n \geq N$,

$$\int_{X_k} |f_n - f| d\mu \leq \frac{\varepsilon}{3}.$$

The two previous estimates show that f belongs to $\mathcal{L}^1(X, \mu)$. Finally, for every $n \geq N$,

$$\begin{aligned} \int_X |f_n - f| d\mu &= \int_{X_k} |f_n - f| d\mu + \int_{X \setminus X_k} |f_n - f| d\mu \\ &\leq \frac{\varepsilon}{3} + \int_{X \setminus X_k} |f_n| d\mu + \int_{X \setminus X_k} |f| d\mu \leq \varepsilon. \end{aligned}$$

□

The second σ -finite extension follows by first extracting an almost everywhere convergent subsequence, then applying the preceding theorem.

Theorem 11.16 (Second Vitali theorem on sigma-finite spaces). *Let $(X, \mathcal{A}, \mu)$ be σ -finite. Let (f_n) be an equi-integrable sequence of integrable functions, and let f be a measurable function. Assume that $f_n \rightarrow f$ in measure on X as $n \rightarrow +\infty$. Assume moreover that there exists an increasing sequence $(X_k)_{k \geq 1}$ in $\mathcal{A}$ such that $\mu(X_k) < \infty$, $\cup_{k \geq 1} X_k = X$, and (f_n) has uniformly small tails along (X_k) . Then f belongs to $\mathcal{L}^1(X, \mu)$ and*

$$\int_X |f_n - f| d\mu \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Proof. The subsequence theorem for convergence in measure gives a subsequence (f_{n_j}) which converges almost uniformly, hence almost everywhere, to f . The subsequence is equi-integrable and has uniformly small tails along (X_k) . Therefore Theorem 11.15 applied to (f_{n_j}) gives f in $\mathcal{L}^1(X, \mu)$.

Define $g_n := f_n - f$. By Lemma 11.12, the sequence (g_n) is equi-integrable, converges to 0 in measure, and has uniformly small tails along (X_k) . Let $\varepsilon > 0$. Choose $k \geq 1$ such that

$$\sup_{n \geq 1} \int_{X \setminus X_k} |g_n| d\mu \leq \frac{\varepsilon}{2}.$$

On X_k the measure is finite. The restriction of (g_n) to X_k is equi-integrable and converges to 0 in measure. The finite-measure second Vitali theorem, applied with limit 0, gives an integer N such that, for every $n \geq N$,

$$\int_{X_k} |g_n| \, d\mu \leq \frac{\varepsilon}{2}.$$

Consequently, for every $n \geq N$,

$$\int_X |f_n - f| \, d\mu = \int_X |g_n| \, d\mu \leq \int_{X_k} |g_n| \, d\mu + \int_{X \setminus X_k} |g_n| \, d\mu \leq \varepsilon.$$

□

11.3.4 Three model counterexamples

The hypotheses in the comparison theorems are not cosmetic. Three elementary families explain the role of the finite-measure assumption, the equi-integrability assumption, and the uniform tail assumption. We use the following terminology.

Example 11.17 (Three model counterexamples). *The following three sequences are the basic counterexample mechanisms.*

1) On $]0, 1[$ with the Lebesgue measure, the focusing spikes are

$$u_n(x) := n \mathbf{1}_{]0, 1/n[}(x).$$

They converge to 0 almost everywhere and in measure, but

$$\int_0^1 |u_n(x)| \, dx = 1$$

for every n . They are not equi-integrable. Indeed, if $\delta > 0$, choose n such that $1/n < \delta$ and set $E_n :=]0, 1/n[$. Then $\lambda(E_n) < \delta$ and

$$\int_{E_n} |u_n| \, d\lambda = 1.$$

2) On $\mathbb{R}$ with the Lebesgue measure, the defocusing spikes are

$$v_n(x) := \frac{1}{n} \mathbf{1}_{[0, n]}(x).$$

They converge uniformly to 0, hence also almost uniformly, almost everywhere, and in measure. They are equi-integrable, because $|v_n| \leq 1$ for every n , and therefore

$$\int_E |v_n| \, d\lambda \leq \lambda(E)$$

for every measurable set E . However,

$$\|v_n\|_{L^1(\mathbb{R})} = 1$$

for every n . Moreover, if $A \subset \mathbb{R}$ is measurable and $\lambda(A) < \infty$, then

$$\int_{\mathbb{R} \setminus A} v_n \, d\lambda = 1 - \frac{\lambda(A \cap [0, n])}{n} \rightarrow 1 \quad \text{as } n \rightarrow +\infty.$$

Thus the tails are not uniformly small along any finite-measure exhaustion.

3) On $\mathbb{R}$ with the Lebesgue measure, the translated indicators are

$$w_n(x) := \mathbf{1}_{[n, n+1]}(x).$$

They converge to 0 almost everywhere, but they do not converge to 0 in measure on the whole space, since

$$\lambda(\{|w_n| > 1/2\}) = \lambda([n, n+1]) = 1$$

for every n . In particular they do not converge to 0 almost uniformly.

These examples answer the natural questions about the assumptions in the preceding theorems.

Question	Theorem concerned	Counterexample
Can we remove $\mu(X) < \infty$?	Egoroff, almost everywhere convergence implies almost uniform convergence	Translated indicators
Can we remove $\mu(X) < \infty$ without adding a tail condition?	The first Vitali theorem in finite measure	Defocusing spikes
Can we remove $\mu(X) < \infty$ without adding a tail condition?	The second Vitali theorem in finite measure	Defocusing spikes
Can we remove equi-integrability?	The first Vitali theorem in finite measure	Focusing spikes
Can we remove equi-integrability?	The second Vitali theorem in finite measure	Focusing spikes
Can we remove the uniform tail condition?	The two σ -finite Vitali theorems	Defocusing spikes

11.3.5 Complement: de la Vallée Poussin criterion

We finish with a useful characterization of equi-integrability. It is not needed in the proof of the comparison diagram, but it is often an efficient way to verify equi-integrability in applications.

Theorem 11.18 (de la Vallée Poussin criterion). *Let $(f_n)_{n \in \mathbb{N}^*}$ be a bounded sequence in $\mathcal{L}^1(X, \mu)$. Then the sequence (f_n) is equi-integrable if and only if there exists an increasing function $\theta : [0, \infty) \rightarrow [0, \infty]$ with $\theta(t)/t \rightarrow \infty$ as $t \rightarrow +\infty$ and such that*

$$\sup_{n \in \mathbb{N}^*} \int_X \theta(|f_n|) d\mu < \infty.$$

Proof. **Step 1.** Assume that the sequence (f_n) is equi-integrable. By Proposition 11.10, there is an increasing sequence of positive integers (k_i) such that

$$\sup_{n \in \mathbb{N}^*} \int_{\{|f_n| > k_i\}} |f_n| d\mu \leq \frac{1}{2^i}$$

for every $i \geq 1$. Define $b_0 = 0$ and, for each l in $\mathbb{N}^*$, let b_l be the number of indices $i \geq 1$ such that $k_i < l$. Note that $b_l \nearrow \infty$ as $l \rightarrow +\infty$. Set $\theta(t) = 0$ for t in $[0, 1)$, and define

$$\theta(t) = tb_l \quad \text{if } t \in [l, l+1) \text{ and } l \geq 1.$$

Then

$$\frac{\theta(t)}{t} \geq b_{[t]} \rightarrow \infty \quad \text{as } t \rightarrow +\infty,$$

where $[t]$ denotes the integer part of t . Moreover, for all n in $\mathbb{N}^*$,

$$\begin{aligned}
\int_X \theta(|f_n|) \, d\mu &= \sum_{l=1}^{\infty} \int_{\{l \leq |f_n| < l+1\}} \theta(|f_n|) \, d\mu \\
&= \sum_{l=1}^{\infty} b_l \int_{\{l \leq |f_n| < l+1\}} |f_n| \, d\mu \\
&= \sum_{l=1}^{\infty} \sum_{\{i \geq 1: k_i < l\}} \int_{\{l \leq |f_n| < l+1\}} |f_n| \, d\mu \\
&\leq \sum_{i=1}^{\infty} \int_{\{|f_n| > k_i\}} |f_n| \, d\mu \\
&\leq \sum_{i=1}^{\infty} \frac{1}{2^i} \leq 1.
\end{aligned}$$

Step 2. Conversely, assume that there exists an increasing function $\theta : [0, \infty) \rightarrow [0, \infty]$ with $\theta(t)/t \rightarrow \infty$ as $t \rightarrow +\infty$ and such that

$$C := \sup_{n \in \mathbb{N}^*} \int_X \theta(|f_n|) \, d\mu < \infty.$$

Since $\theta(t)/t \rightarrow \infty$ as $t \rightarrow +\infty$, for every $\varepsilon > 0$, there exists $t_\varepsilon > 0$ such that

$$\theta(t) \geq \frac{(C+1)t}{\varepsilon} \quad \text{for all } t \geq t_\varepsilon.$$

Then, for all $t \geq t_\varepsilon$,

$$\sup_{n \in \mathbb{N}^*} \int_{\{|f_n| > t\}} |f_n| \, d\mu \leq \frac{\varepsilon}{C+1} \sup_{n \in \mathbb{N}^*} \int_{\{|f_n| > t\}} \theta(|f_n|) \, d\mu \leq \varepsilon.$$

Hence, by Proposition 11.10, the sequence (f_n) is equi-integrable. □

Chapter 12

Absolute continuity, Lebesgue decomposition and the Radon–Nikodym theorem

The comparison notions needed in this chapter were introduced in Section 2.5. We use the notation $\nu \ll \mu$ for absolute continuity and $\nu \perp \mu$ for mutual singularity, as in Definitions 2.13 and 2.14. In the singular case, a set S such that $\nu(S^c) = 0$ and $\mu(S) = 0$ is called a mutually singular set associated with the ordered pair (ν, μ) .

We now decompose a positive measure into an absolutely continuous part and a singular part with respect to a fixed reference measure.

12.1 Lebesgue decomposition

Theorem 12.1 (Lebesgue decomposition theorem). *Let $(X, \mathcal{A})$ be a measurable space and let μ and ν be finite positive measures on $(X, \mathcal{A})$. Then there exist unique finite positive measures ν_a and ν_s such that*

$$\nu = \nu_a + \nu_s, \quad \nu_a \ll \mu, \quad \nu_s \perp \mu.$$

Moreover, there exists a measurable function $f: X \rightarrow [0, \infty)$ such that

$$\nu_a(E) = \int_E f \, d\mu$$

for every E in $\mathcal{A}$. The function f is unique up to equality μ -almost everywhere.

Proof. **Step 1. Uniqueness of the decomposition.** Assume that

$$\nu = \alpha + \sigma = \beta + \tau,$$

where α and β are finite positive measures absolutely continuous with respect to μ , and σ and τ are finite positive measures singular with respect to μ . Choose a mutually singular set S associated with (σ, μ) and a mutually singular set T associated with (τ, μ) , so that

$$\mu(S) = 0, \quad \sigma(S^c) = 0, \quad \mu(T) = 0, \quad \tau(T^c) = 0.$$

Set

$$C := X \setminus (S \cup T).$$

Since $\mu(S \cup T) = 0$, absolute continuity gives

$$\alpha(S \cup T) = \beta(S \cup T) = 0.$$

Moreover $\sigma(C) = \tau(C) = 0$. Therefore, for every E in $\mathcal{A}$,

$$\alpha(E) = \alpha(E \cap C) = \nu(E \cap C) = \beta(E \cap C) = \beta(E).$$

Thus $\alpha = \beta$, and then also $\sigma = \tau$.

Step 2. Uniqueness of the density. Assume that f and h are nonnegative measurable functions such that

$$\int_E f \, d\mu = \int_E h \, d\mu$$

for every E in $\mathcal{A}$. For $m \geq 1$, set

$$A_m := \left\{ f \geq h + \frac{1}{m} \right\}.$$

Then

$$0 = \int_{A_m} (f - h) \, d\mu \geq \frac{1}{m} \mu(A_m),$$

and hence $\mu(A_m) = 0$. Since

$$\{f > h\} = \bigcup_{m=1}^{\infty} A_m,$$

we get $\mu(\{f > h\}) = 0$. By symmetry, $\mu(\{h > f\}) = 0$. Thus $f = h$ μ -almost everywhere.

Step 3. Existence of the density and of the absolutely continuous part. Consider the family of candidate densities

$$\mathcal{C} := \left\{ g: X \rightarrow [0, \infty] \text{ measurable} : \int_E g \, d\mu \leq \nu(E) \text{ for every } E \in \mathcal{A} \right\}.$$

The function 0 belongs to $\mathcal{C}$. Moreover, if g and h belong to $\mathcal{C}$, then $\max(g, h)$ also belongs to $\mathcal{C}$, because for every E in $\mathcal{A}$,

$$\begin{aligned} \int_E \max(g, h) \, d\mu &= \int_{E \cap \{g \geq h\}} g \, d\mu + \int_{E \cap \{h > g\}} h \, d\mu \\ &\leq \nu(E \cap \{g \geq h\}) + \nu(E \cap \{h > g\}) = \nu(E). \end{aligned}$$

In particular, every g in $\mathcal{C}$ satisfies

$$\int_X g \, d\mu \leq \nu(X) < \infty.$$

We may therefore define

$$m := \sup \left\{ \int_X g \, d\mu : g \in \mathcal{C} \right\}.$$

Choose a sequence $(g_n)_{n \geq 1}$ in $\mathcal{C}$ such that

$$\int_X g_n \, d\mu \rightarrow m \quad \text{as } n \rightarrow +\infty.$$

For every $n \geq 1$, set

$$f_n := \max(g_1, \dots, g_n).$$

Then f_n belongs to $\mathcal{C}$, the sequence (f_n) is increasing, and

$$\int_X f_n \, d\mu \geq \int_X g_n \, d\mu.$$

Hence

$$\int_X f_n \, d\mu \rightarrow m \quad \text{as } n \rightarrow +\infty.$$

Define

$$f := \lim_{n \rightarrow +\infty} f_n.$$

By the monotone convergence theorem, for every E in $\mathcal{A}$,

$$\int_E f \, d\mu = \lim_{n \rightarrow +\infty} \int_E f_n \, d\mu \leq \nu(E).$$

Thus f belongs to $\mathcal{C}$ and

$$\int_X f \, d\mu = m.$$

Since $\int_X f \, d\mu < \infty$, the set $\{f = +\infty\}$ is μ -negligible. Changing f on this negligible set, we may assume that f takes values in $[0, \infty)$.

Define

$$\nu_a(E) := \int_E f \, d\mu, \quad \nu_s(E) := \nu(E) - \nu_a(E),$$

for every E in $\mathcal{A}$. Then ν_a and ν_s are finite positive measures, and $\nu_a \ll \mu$.

Step 4. Existence of a mutually singular set for the singular part. It remains to prove that $\nu_s \perp \mu$. For every integer $n \geq 1$, consider the finite signed measure

$$\eta_n := \nu_s - \frac{1}{n}\mu,$$

and choose a Hahn decomposition (P_n, N_n) for η_n . We claim that

$$\mu(P_n) = 0$$

for every $n \geq 1$. Indeed, if $\mu(P_n) > 0$, then P_n is positive for η_n , so for every E in $\mathcal{A}$,

$$\nu_s(E \cap P_n) = \eta_n(E \cap P_n) + \frac{1}{n}\mu(E \cap P_n) \geq \frac{1}{n}\mu(E \cap P_n).$$

Therefore

$$\begin{aligned} \int_E \left(f + \frac{1}{n}\mathbf{1}_{P_n}\right) d\mu &= \int_E f \, d\mu + \frac{1}{n}\mu(E \cap P_n) \\ &\leq \int_E f \, d\mu + \nu_s(E \cap P_n) \\ &\leq \nu(E) \end{aligned}$$

for every E in $\mathcal{A}$. Hence $f + n^{-1}\mathbf{1}_{P_n}$ belongs to $\mathcal{C}$. But

$$\int_X \left(f + \frac{1}{n}\mathbf{1}_{P_n}\right) d\mu = m + \frac{1}{n}\mu(P_n) > m,$$

which contradicts the definition of m . Thus $\mu(P_n) = 0$.

Set

$$S := \bigcup_{n=1}^{\infty} P_n.$$

Then $\mu(S) = 0$. If $E \subset S^c$ is measurable, then $E \subset N_n$ for every $n \geq 1$, hence

$$\eta_n(E) = \nu_s(E) - \frac{1}{n}\mu(E) \leq 0.$$

Thus

$$\nu_s(E) \leq \frac{1}{n}\mu(E)$$

for every $n \geq 1$. Letting $n \rightarrow +\infty$, we obtain $\nu_s(E) = 0$. In particular, $\nu_s(S^c) = 0$. Since $\mu(S) = 0$, the set S is a mutually singular set associated with the ordered pair (ν_s, μ) . Thus $\nu_s \perp \mu$. $\square$

12.2 Radon–Nikodym theorem for positive measures

The previous theorem immediately gives the positive Radon–Nikodym theorem when the singular part is absent.

Corollary 12.2 (Finite Radon–Nikodym theorem for positive measures). *Let $(X, \mathcal{A})$ be a measurable space and let μ and ν be finite positive measures on $(X, \mathcal{A})$. If $\nu \ll \mu$, then there exists a measurable function $f: X \rightarrow [0, \infty)$ such that*

$$\nu(E) = \int_E f \, d\mu$$

for every E in $\mathcal{A}$. Moreover, f is uniquely defined μ -almost everywhere. We write $f = \frac{d\nu}{d\mu}$.

Proof. We apply the previous theorem and write

$$\nu = \nu_a + \nu_s, \quad \nu_a(E) = \int_E f \, d\mu,$$

with $\nu_a \ll \mu$ and $\nu_s \perp \mu$. Since $\nu_s \leq \nu$ and $\nu \ll \mu$, we also have $\nu_s \ll \mu$. Choose S in $\mathcal{A}$ such that

$$\mu(S) = 0, \quad \nu_s(S^c) = 0.$$

Then $\nu_s(S) = 0$ by absolute continuity, so

$$\nu_s(X) = \nu_s(S) + \nu_s(S^c) = 0.$$

Hence $\nu_s = 0$ and $\nu = \nu_a$. The uniqueness of f follows from the uniqueness statement in the previous theorem. $\square$

Theorem 12.3 (Radon–Nikodym theorem for positive sigma-finite measures). *Let $(X, \mathcal{A})$ be a measurable space and let μ and ν be σ -finite positive measures on $(X, \mathcal{A})$. If $\nu \ll \mu$, then there exists a measurable function $f: X \rightarrow [0, \infty)$ such that*

$$\nu(E) = \int_E f \, d\mu$$

for every E in $\mathcal{A}$. The function f is uniquely defined μ -almost everywhere.

Proof. Since μ and ν are σ -finite, there exists an increasing sequence (X_k) in $\mathcal{A}$ such that $X = \bigcup_{k \geq 1} X_k$ and $\mu(X_k) < \infty$ and $\nu(X_k) < \infty$ for every k . We apply the finite case on each X_k to obtain a density f_k such that

$$\nu(E \cap X_k) = \int_{E \cap X_k} f_k \, d\mu.$$

By uniqueness of the Radon–Nikodym derivative on X_k , the restrictions of f_{k+1} and f_k coincide μ -almost everywhere on X_k . We define f by setting $f = f_1$ on X_1 and $f = f_k$ on $X_k \setminus X_{k-1}$ for $k \geq 2$. Then f is measurable and satisfies the desired identity on every X_k , hence on X by σ -additivity. $\square$

12.3 A Baire lemma for signed measures

For finite signed measures, absolute continuity was defined in Definition 2.29 through the total variation measure. The next lemma is a Baire-category uniform absolute-continuity statement for sequences of finite signed measures. It will be used later in the proof of Theorem 15.32.

Lemma 12.4 (Baire lemma for finite signed measures). *Let $(X, \mathcal{A}, \mu)$ be a finite measure space. Let $(\nu_n)_{n \geq 1}$ be a sequence of finite signed measures on $(X, \mathcal{A})$ such that*

$$\nu_n \ll \mu$$

for every $n \geq 1$. Assume that for every E in $\mathcal{A}$, the sequence $(\nu_n(E))_{n \geq 1}$ converges. Then for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\mu(E) < \delta \quad \implies \quad \sup_{n \geq 1} |\nu_n(E)| \leq 3\varepsilon$$

for every E in $\mathcal{A}$.

Proof. Let $\varepsilon > 0$.

Step 1. Closed sets in the Nikodym metric space. We work on the complete metric space $(\tilde{\mathcal{A}}, \rho)$ from Corollary 8.20. For every integer $N \geq 1$, define

$$F_N := \{[A] \in \tilde{\mathcal{A}} : |\nu_p(A) - \nu_q(A)| \leq \varepsilon \text{ for all } p, q \geq N\}.$$

Each F_N is closed in $\tilde{\mathcal{A}}$. Indeed, for fixed p, q , the map

$$[A] \mapsto \nu_p(A) - \nu_q(A)$$

is continuous with respect to ρ . To see this, if A and B are measurable, then

$$|\nu_p(A) - \nu_p(B)| = |\nu_p(A \setminus B) - \nu_p(B \setminus A)| \leq |\nu_p|(A \triangle B),$$

and $|\nu_p| \ll \mu$ by Definition 2.29. Lemma 2.17 applied to $|\nu_p|$ gives the desired continuity.

Step 2. Application of Baire's theorem. By assumption, the sequence $(\nu_n(A))_{n \geq 1}$ converges, hence is Cauchy, for every $[A]$ in $\tilde{\mathcal{A}}$. Therefore

$$\tilde{\mathcal{A}} = \bigcup_{N=1}^{\infty} F_N.$$

Theorem 16.17 gives an integer $N_0 \geq 1$, a class $[A_0]$ in $\tilde{\mathcal{A}}$, and a radius $r > 0$ such that

$$B_\rho([A_0], r) \subset F_{N_0}.$$

Step 3. Uniform Cauchy estimate on small sets. Let E be measurable with $\mu(E) < r$. Then

$$\rho([A_0 \cup E], [A_0]) = \mu(E \setminus A_0) \leq \mu(E) < r$$

and

$$\rho([A_0 \setminus E], [A_0]) = \mu(E \cap A_0) \leq \mu(E) < r.$$

Thus $[A_0 \cup E]$ and $[A_0 \setminus E]$ both belong to F_{N_0} . For every $p, q \geq N_0$,

$$\begin{aligned} |\nu_p(E) - \nu_q(E)| &= |\nu_p(A_0 \cup E) - \nu_p(A_0 \setminus E) - \nu_q(A_0 \cup E) + \nu_q(A_0 \setminus E)| \\ &\leq |\nu_p(A_0 \cup E) - \nu_q(A_0 \cup E)| + |\nu_p(A_0 \setminus E) - \nu_q(A_0 \setminus E)| \\ &\leq 2\varepsilon. \end{aligned}$$

Step 4. Inclusion of the finitely many first measures. The finitely many measures $|\nu_1|, \dots, |\nu_{N_0}|$ are absolutely continuous with respect to μ . Hence Lemma 2.17 gives a number δ in $]0, r[$ such that

$$\mu(E) < \delta \quad \implies \quad |\nu_n(E)| \leq \varepsilon$$

for every $n \in \{1, \dots, N_0\}$. If $n \geq N_0$ and $\mu(E) < \delta$, then

$$|\nu_n(E)| \leq |\nu_n(E) - \nu_{N_0}(E)| + |\nu_{N_0}(E)| \leq 2\varepsilon + \varepsilon = 3\varepsilon.$$

The same estimate already holds for $1 \leq n \leq N_0$. This proves the lemma. $\square$

12.4 Signed Radon–Nikodym theorem

We now deduce a signed version from the positive Radon–Nikodym theorem and the Jordan decomposition theorem. We use Definition 2.29: a finite signed measure ν is absolutely continuous with respect to μ when $|\nu| \ll \mu$.

Corollary 12.5 (Signed Radon–Nikodym theorem). *Let μ be a σ -finite positive measure and let ν be a finite signed measure such that $\nu \ll \mu$. Then there exists a function g in $L^1(X, \mu)$ such that*

$$\nu(E) = \int_E g \, d\mu$$

for every E in $\mathcal{A}$. Moreover, if $\nu = \nu^+ - \nu^-$ is the Jordan decomposition, then

$$\nu^+(E) = \int_E g^+ \, d\mu, \quad \nu^-(E) = \int_E g^- \, d\mu, \quad |\nu|(E) = \int_E |g| \, d\mu.$$

Proof. By the Jordan decomposition theorem (Theorem 2.25), we may write

$$\nu = \nu^+ - \nu^-,$$

where ν^+ and ν^- are finite positive measures. Since $\nu \ll \mu$, the measures $\nu^\pm$ are also absolutely continuous with respect to μ by Definition 2.29. Applying Theorem 12.3 to ν^+ and ν^- , we obtain nonnegative measurable functions p and q such that

$$\nu^+(E) = \int_E p \, d\mu, \quad \nu^-(E) = \int_E q \, d\mu$$

for every E in $\mathcal{A}$. Since $\nu^+(X) + \nu^-(X) < +\infty$, the functions p and q belong to $L^1(X, \mu)$. Moreover, $\nu^+ \perp \nu^-$; choosing a singularity set and using the two representations shows that $pq = 0$ μ -almost everywhere. Set

$$g := p - q.$$

Then $g^+ = p$ and $g^- = q$ μ -almost everywhere. The final identities follow from the definition $|\nu| = \nu^+ + \nu^-$ of the total variation measure. $\square$

Example 12.6 (A measure with density and atom). *Let $X = [-1, 1]$, let μ be the restriction of the Lebesgue measure to $[-1, 1]$, and define the finite positive measure ν by*

$$\nu(E) := \int_E x^2 \, d\mu(x) + \delta_0(E)$$

for every Borel set $E \subset [-1, 1]$. Then

$$\nu_a(E) = \int_E x^2 \, d\mu(x), \quad \nu_s(E) = \delta_0(E)$$

gives the Lebesgue decomposition of ν with respect to μ . Indeed, $\nu_a \ll \mu$, whereas $\nu_s \perp \mu$ because it is concentrated on the μ -negligible set $\{0\}$. The Radon–Nikodym derivative of the absolutely continuous part is

$$\frac{d\nu_a}{d\mu}(x) = x^2.$$

Chapter 13

Fundamental theorem of calculus and change of variables

This chapter studies the two main links between integration and differential calculus. We first recover antiderivatives from integrable functions through a differentiation theorem, then prove the converse statement using the Radon–Nikodym theorem, and finally explain how integrals transform under smooth changes of variables, first in dimension 1 and then in higher dimension. Unless explicitly stated otherwise, all measures in this chapter are Lebesgue measures: λ on $\mathbb{R}$ in the one-dimensional parts and λ_d on $\mathbb{R}^d$ in the multidimensional parts.

13.1 Integration and antiderivatives

The following results make the link between integration and antiderivatives.

13.1.1 The Hardy–Littlewood maximal inequality

The proof of the differentiation theorem rests on the centered Hardy–Littlewood maximal operator on $\mathbb{R}$. We isolate it first.

Definition 13.1 (Centered Hardy–Littlewood maximal operator). *Let $h: \mathbb{R} \rightarrow \mathbb{R}$ be an integrable function. For every $r > 0$ and every x in $\mathbb{R}$, we define*

$$(Mh)_r(x) := \frac{1}{2r} \int_{x-r}^{x+r} |h(t)| \, dt.$$

The centered Hardy–Littlewood maximal function of h is

$$Mh(x) := \sup_{r>0} (Mh)_r(x).$$

Lemma 13.2 (Continuity of fixed-radius averages). *Let $h: \mathbb{R} \rightarrow \mathbb{R}$ be an integrable function. For every fixed $r > 0$, the function*

$$x \mapsto (Mh)_r(x) = \frac{1}{2r} \int_{x-r}^{x+r} |h(t)| \, dt$$

is continuous on $\mathbb{R}$.

Proof. Let x_0 be in $\mathbb{R}$. For x in $\mathbb{R}$, write

$$(Mh)_r(x) = \frac{1}{2r} \int_{\mathbb{R}} \mathbf{1}_{[x-r, x+r]}(t) |h(t)| dt.$$

As $x \rightarrow x_0$, the integrands

$$t \mapsto \mathbf{1}_{[x-r, x+r]}(t) |h(t)|$$

converge pointwise for every t different from $x_0 - r$ and $x_0 + r$, hence for λ -almost every t , to

$$t \mapsto \mathbf{1}_{[x_0-r, x_0+r]}(t) |h(t)|.$$

They are dominated by the integrable function $|h|$. The continuity under the integral sign theorem, Theorem 9.1, gives the desired continuity. $\square$

Lemma 13.3 (Finite Vitali covering lemma in dimension one). *Let $I_1, \dots, I_N$ be bounded open intervals of $\mathbb{R}$. Then there exist pairwise disjoint intervals $J_1, \dots, J_m$ among $I_1, \dots, I_N$ such that*

$$\bigcup_{i=1}^N I_i \subset \bigcup_{j=1}^m 3J_j,$$

where, if $J = (c - r, c + r)$, we write $3J = (c - 3r, c + 3r)$.

Proof. Among the intervals $I_1, \dots, I_N$, we choose one of maximal length and call it J_1 . We then discard all the intervals which intersect J_1 . Among the remaining intervals, if there are any, we choose one of maximal length and call it J_2 , and we continue in the same way. The process stops after finitely many steps and yields pairwise disjoint intervals $J_1, \dots, J_m$.

Let I_i be one of the initial intervals. If I_i is itself one of the selected intervals, there is nothing to prove. Otherwise, I_i was discarded when some J_j was selected. Hence $I_i \cap J_j \neq \emptyset$ and, by construction, the length of I_i is not larger than the length of J_j . Therefore $I_i \subset 3J_j$. This proves the claim. $\square$

Theorem 13.4 (Weak Hardy–Littlewood maximal inequality). *Let $h: \mathbb{R} \rightarrow \mathbb{R}$ be an integrable function. Then, for every $\alpha > 0$,*

$$\lambda(\{Mh > \alpha\}) \leq \frac{3}{\alpha} \int_{\mathbb{R}} |h(t)| dt.$$

Proof. Let $\alpha > 0$ and set

$$E_\alpha := \{x \in \mathbb{R} : Mh(x) > \alpha\}.$$

Step 1: openness of the superlevel set. For every $r > 0$, define

$$E_{\alpha,r} := \{x \in \mathbb{R} : (Mh)_r(x) > \alpha\}.$$

By Lemma 13.2, the function $(Mh)_r$ is continuous. Since $] \alpha, +\infty[$ is open, the set $E_{\alpha,r}$ is open. Moreover,

$$E_\alpha = \bigcup_{r>0} E_{\alpha,r},$$

because the condition $Mh(x) > \alpha$ means, by definition of the supremum in Definition 13.1, that at least one radius $r > 0$ satisfies $(Mh)_r(x) > \alpha$. Thus E_α is open.

Step 2: estimate on compact subsets of E_α . Let K be a compact subset of E_α . For every x in K , by definition of $Mh(x)$ we can choose $r_x > 0$ such that, with $I_x = (x - r_x, x + r_x)$,

$$\frac{1}{|I_x|} \int_{I_x} |h(t)| dt > \alpha.$$

The family $(I_x)_{x \in K}$ is an open covering of the compact set K , so we can extract a finite subcover $I_{x_1}, \dots, I_{x_N}$. By Lemma 13.3, there exist pairwise disjoint intervals $J_1, \dots, J_m$ among them such that

$$K \subset \bigcup_{j=1}^m 3J_j.$$

Therefore

$$\begin{aligned} \lambda(K) &\leq \sum_{j=1}^m \lambda(3J_j) = 3 \sum_{j=1}^m |J_j| \\ &< \frac{3}{\alpha} \sum_{j=1}^m \int_{J_j} |h(t)| \, dt \\ &\leq \frac{3}{\alpha} \int_{\mathbb{R}} |h(t)| \, dt, \end{aligned}$$

because the intervals J_j are pairwise disjoint.

Step 3: passage from compact subsets to E_α . By Lemma 16.12, the open set E_α is the union of an increasing sequence of compact sets $(K_n)_{n \geq 1}$. The estimate of Step 2 gives, for every $n \geq 1$,

$$\lambda(K_n) \leq \frac{3}{\alpha} \int_{\mathbb{R}} |h(t)| \, dt.$$

Since $K_n \uparrow E_\alpha$, the continuity from below of measures, Theorem 2.3, gives

$$\lambda(E_\alpha) = \lim_{n \rightarrow \infty} \lambda(K_n) \leq \frac{3}{\alpha} \int_{\mathbb{R}} |h(t)| \, dt.$$

This proves the desired estimate. □

13.1.2 Lebesgue points and differentiation

We now use Theorem 13.4 to prove the Lebesgue differentiation theorem on the real line.

Definition 13.5 (Lebesgue points). *Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a measurable function that is integrable on every compact interval. A point x in $\mathbb{R}$ is called a Lebesgue point of f if*

$$\lim_{r \rightarrow 0^+} \frac{1}{2r} \int_{x-r}^{x+r} |f(t) - f(x)| \, dt = 0.$$

Theorem 13.6 (Lebesgue differentiation theorem). *Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a measurable function that is integrable on every compact interval. Then λ -almost every point x in $\mathbb{R}$ is a Lebesgue point of f .*

Proof. For every x in $\mathbb{R}$, set

$$Df(x) := \limsup_{r \rightarrow 0^+} \frac{1}{2r} \int_{x-r}^{x+r} |f(t) - f(x)| \, dt.$$

Step 1: the uniformly continuous case. Assume first that g is uniformly continuous on $\mathbb{R}$. Let x be in $\mathbb{R}$ and let $\varepsilon > 0$. There exists $\rho > 0$ such that

$$|s - x| < \rho \quad \implies \quad |g(s) - g(x)| < \varepsilon.$$

If $0 < r < \rho$, then

$$\frac{1}{2r} \int_{x-r}^{x+r} |g(t) - g(x)| \, dt \leq \varepsilon.$$

Thus $Dg(x) = 0$ for every x in $\mathbb{R}$. In particular, this applies to every function g in $\mathcal{C}_c(\mathbb{R})$. Indeed, if $\text{supp}(g) \subset [-R, R]$, then g is uniformly continuous on the compact interval $[-R-1, R+1]$ by the Heine–Cantor theorem, Theorem 16.33, and g is identically zero outside $[-R, R]$.

Step 2: the integrable case. Assume that f is integrable on $\mathbb{R}$. We fix $\alpha > 0$ and $\varepsilon > 0$. At this point we use the density of $\mathcal{C}_c(\mathbb{R})$ in $L^1(\mathbb{R}, \lambda)$. This result will be proved later in the more general form of Theorem 15.10, independently of the present differentiation theorem. Hence there exists g in $\mathcal{C}_c(\mathbb{R})$ such that

$$\int_{\mathbb{R}} |f - g| \, dt < \varepsilon.$$

For every x in $\mathbb{R}$ and every $r > 0$, we have

$$\begin{aligned} \frac{1}{2r} \int_{x-r}^{x+r} |f(t) - f(x)| \, dt &\leq \frac{1}{2r} \int_{x-r}^{x+r} |g(t) - g(x)| \, dt \\ &\quad + \frac{1}{2r} \int_{x-r}^{x+r} |f(t) - g(t)| \, dt + |f(x) - g(x)|. \end{aligned}$$

Taking the limit superior as $r \rightarrow 0^+$, and using Step 1, gives

$$Df(x) \leq M(f - g)(x) + |f(x) - g(x)|.$$

Hence

$$\{Df > 2\alpha\} \subset \{M(f - g) > \alpha\} \cup \{|f - g| > \alpha\}.$$

By the weak Hardy–Littlewood maximal inequality, Theorem 13.4, and by Markov’s inequality, Lemma 8.6, applied to $f - g$, we obtain

$$\begin{aligned} \lambda(\{Df > 2\alpha\}) &\leq \lambda(\{M(f - g) > \alpha\}) + \lambda(\{|f - g| > \alpha\}) \\ &\leq \frac{3}{\alpha} \int_{\mathbb{R}} |f - g| \, dt + \frac{1}{\alpha} \int_{\mathbb{R}} |f - g| \, dt \\ &\leq \frac{4\varepsilon}{\alpha}. \end{aligned}$$

Letting $\varepsilon \rightarrow 0$, we conclude that

$$\lambda(\{Df > 2\alpha\}) = 0$$

for every $\alpha > 0$. Since

$$\{Df > 0\} = \bigcup_{m=1}^{\infty} \left\{Df > \frac{1}{m}\right\},$$

it follows that $Df(x) = 0$ for λ -almost every x in $\mathbb{R}$. This proves the theorem when f is integrable on $\mathbb{R}$.

Step 3: the locally integrable case. We now return to the general case $f \in L^1_{\text{loc}}(\mathbb{R})$. For every integer $N \geq 1$, we define

$$f_N := f \mathbf{1}_{[-N-1, N+1]}.$$

The function f_N is integrable on $\mathbb{R}$, so almost every point of $[-N, N]$ is a Lebesgue point of f_N by Step 2. If x belongs to $[-N, N]$ and $0 < r \leq 1$, then

$$[x - r, x + r] \subset [-N - 1, N + 1],$$

so

$$\frac{1}{2r} \int_{x-r}^{x+r} |f_N(t) - f_N(x)| \, dt = \frac{1}{2r} \int_{x-r}^{x+r} |f(t) - f(x)| \, dt.$$

Hence every Lebesgue point of f_N in $[-N, N]$ is also a Lebesgue point of f . Since this holds for every $N \geq 1$, we conclude that λ -almost every point of $\mathbb{R}$ is a Lebesgue point of f . $\square$

13.1.3 Fundamental theorem of calculus

We use the notion of absolute continuity of functions introduced in Definition 3.15. This should be distinguished from absolute continuity of measures, introduced in Definition 2.13. For increasing functions on a compact interval, Proposition 3.18 explains the link: absolute continuity of the function is equivalent to absolute continuity, with respect to the Lebesgue measure, of the associated Stieltjes measure.

Theorem 13.7 (Fundamental theorem of calculus for Lebesgue integrals, direct part). *Let $f: [a, b] \rightarrow \mathbb{R}$ be a measurable function such that*

$$\int_a^b |f(t)| \, dt < \infty.$$

We define $F: [a, b] \rightarrow \mathbb{R}$ by

$$F(x) = \int_a^x f(t) \, dt.$$

Then

- 1) *F is absolutely continuous on $[a, b]$,*
- 2) *F is differentiable λ -almost everywhere on (a, b) ,*
- 3) *for λ -almost every x in (a, b) , we have $F'(x) = f(x)$.*

Proof. Step 1: absolute continuity. Let $\varepsilon > 0$. Since $|f|$ is integrable on $[a, b]$, we can choose $M \geq 1$ such that

$$\int_{\{|f|>M\}} |f(t)| \, dt \leq \varepsilon/2.$$

We set

$$\delta := \frac{\varepsilon}{2M}.$$

Let E be a measurable subset of $[a, b]$ such that $\lambda(E) \leq \delta$. Then

$$\begin{aligned} \int_E |f(t)| \, dt &= \int_{E \cap \{|f| \leq M\}} |f(t)| \, dt + \int_{E \cap \{|f| > M\}} |f(t)| \, dt \\ &\leq M \lambda(E) + \int_{\{|f| > M\}} |f(t)| \, dt \\ &\leq M\delta + \varepsilon/2 \\ &\leq \varepsilon. \end{aligned}$$

Let $\{(\alpha_k, \beta_k)\}_{1 \leq k \leq m}$ be a finite family of pairwise disjoint intervals included in $[a, b]$ such that

$$\sum_{k=1}^m (\beta_k - \alpha_k) < \delta.$$

Setting

$$E := \bigcup_{k=1}^m (\alpha_k, \beta_k),$$

we obtain

$$\begin{aligned} \sum_{k=1}^m |F(\beta_k) - F(\alpha_k)| &= \sum_{k=1}^m \left| \int_{\alpha_k}^{\beta_k} f(t) \, dt \right| \\ &\leq \sum_{k=1}^m \int_{\alpha_k}^{\beta_k} |f(t)| \, dt \\ &= \int_E |f(t)| \, dt \\ &\leq \varepsilon. \end{aligned}$$

This is the definition of absolute continuity of F .

Step 2: identification of the derivative. Let $\tilde{f}$ be the extension of f by 0 outside $[a, b]$. Then $\tilde{f}$ is locally integrable on $\mathbb{R}$. Let x be a Lebesgue point of $\tilde{f}$ in (a, b) . For $h \neq 0$ small enough so that $x + h$ belongs to $[a, b]$, we write

$$\frac{F(x+h) - F(x)}{h} - f(x) = \frac{1}{h} \int_x^{x+h} (f(t) - f(x)) \, dt.$$

Let

$$I_h := \begin{cases} [x, x+h], & h > 0, \\ [x+h, x], & h < 0. \end{cases}$$

Then

$$\begin{aligned} \left| \frac{F(x+h) - F(x)}{h} - f(x) \right| &\leq \frac{1}{|h|} \int_{I_h} |f(t) - f(x)| \, dt \\ &\leq \frac{1}{|h|} \int_{x-|h|}^{x+|h|} |f(t) - f(x)| \, dt \\ &= 2 \frac{1}{2|h|} \int_{x-|h|}^{x+|h|} |f(t) - f(x)| \, dt. \end{aligned}$$

The right-hand side tends to 0 as $h \rightarrow 0$ because x is a Lebesgue point of $\tilde{f}$ and because, for h small enough, the interval $[x - |h|, x + |h|]$ is contained in $[a, b]$. Therefore F is differentiable at x and $F'(x) = f(x)$. Since λ -almost every point is a Lebesgue point of $\tilde{f}$ by Theorem 13.6, the conclusion follows on (a, b) . $\square$

Conversely, if F is absolutely continuous, then F is an antiderivative of an L^1 function.

Theorem 13.8 (Fundamental theorem of calculus for Lebesgue integrals, converse part). *If F is absolutely continuous on $[a, b]$, then F' exists λ -almost everywhere, F' is Lebesgue integrable on $[a, b]$, and for every x in $[a, b]$ we have*

$$F(x) = F(a) + \int_a^x F'(t) \, dt.$$

The proof of the converse direction will be split into two independent lemmas. The first lemma is purely about functions of one real variable. It shows that an absolutely continuous function can be written as a difference of two increasing absolutely continuous functions. The second lemma treats the increasing case by using the Stieltjes measure constructed in Theorem 3.17, the absolute-continuity criterion of Proposition 3.18, and the Radon–Nikodym theorem.

Lemma 13.9 (Decomposition into increasing absolutely continuous functions). *Let F be absolutely continuous on $[a, b]$. Then there exist two increasing absolutely continuous functions F_1 and F_2 on $[a, b]$ such that*

$$F = F_1 - F_2.$$

Proof. **Step 1. The variation function.** For $a \leq s < t \leq b$, define the variation of F on $[s, t]$ by

$$\text{Var}_{[s,t]}(F) := \sup \left\{ \sum_{i=1}^m |F(x_i) - F(x_{i-1})| : s = x_0 < x_1 < \cdots < x_m = t \right\}.$$

We also set $V_F(a) := 0$ and

$$V_F(x) := \text{Var}_{[a,x]}(F)$$

for x in $]a, b]$.

We first prove that $V_F(b)$ is finite. Apply the absolute continuity of F with $\varepsilon = 1$ and choose $\delta > 0$ accordingly. Choose a partition

$$a = x_0 < x_1 < \cdots < x_N = b$$

such that $x_j - x_{j-1} < \delta$ for every j . If $a = t_0 < \cdots < t_m = b$ is any partition of $[a, b]$, we refine it by adding the points $x_0, \dots, x_N$. On each interval $[x_{j-1}, x_j]$, the pieces of the refined partition form a finite family of pairwise disjoint intervals whose total length is smaller than δ . The sum of the corresponding oscillations of F is therefore at most 1. Adding the contributions of the N intervals gives

$$\sum_{i=1}^m |F(t_i) - F(t_{i-1})| \leq N.$$

Taking the supremum over all partitions proves that $V_F(b) < +\infty$.

Step 2. Local variation and monotonicity. For $a \leq s < t \leq b$, concatenating partitions of $[a, s]$ and $[s, t]$ gives

$$V_F(t) - V_F(s) = \text{Var}_{[s,t]}(F). \quad (13.1)$$

In particular, V_F is increasing and

$$V_F(t) - V_F(s) \geq |F(t) - F(s)|. \quad (13.2)$$

Step 3. Absolute continuity of the variation function. It remains to prove that V_F is absolutely continuous. Let $\varepsilon > 0$. By the absolute continuity of F , there exists $\delta > 0$ such that for every finite family of pairwise disjoint intervals (α_j, β_j) in $[a, b]$ satisfying

$$\sum_{j=1}^m (\beta_j - \alpha_j) < \delta,$$

we have

$$\sum_{j=1}^m |F(\beta_j) - F(\alpha_j)| < \varepsilon.$$

Consider such a family $(\alpha_j, \beta_j)_{1 \leq j \leq m}$. Let $\eta > 0$. For each j , by (13.1), we can choose a partition

$$\alpha_j = t_{j,0} < \cdots < t_{j,\ell_j} = \beta_j$$

such that

$$V_F(\beta_j) - V_F(\alpha_j) \leq \sum_{i=1}^{\ell_j} |F(t_{j,i}) - F(t_{j,i-1})| + \frac{\eta}{m}.$$

The intervals $(t_{j,i-1}, t_{j,i})$ are pairwise disjoint when j and i vary, and their total length is smaller than δ . Thus

$$\begin{aligned} \sum_{j=1}^m (V_F(\beta_j) - V_F(\alpha_j)) &\leq \sum_{j=1}^m \sum_{i=1}^{\ell_j} |F(t_{j,i}) - F(t_{j,i-1})| + \eta \\ &< \varepsilon + \eta. \end{aligned}$$

Letting $\eta \rightarrow 0^+$ proves the absolute continuity of V_F .

Step 4. Decomposition into two increasing functions. Define

$$F_1 := \frac{V_F + F}{2}, \quad F_2 := \frac{V_F - F}{2}.$$

Since V_F and F are absolutely continuous, Proposition 3.16 shows that F_1 and F_2 are absolutely continuous. For $a \leq s < t \leq b$, inequality (13.2) gives

$$\begin{aligned} F_1(t) - F_1(s) &= \frac{V_F(t) - V_F(s) + F(t) - F(s)}{2} \geq 0, \\ F_2(t) - F_2(s) &= \frac{V_F(t) - V_F(s) - F(t) + F(s)}{2} \geq 0. \end{aligned}$$

Thus F_1 and F_2 are increasing, and $F = F_1 - F_2$. □

Lemma 13.10 (Increasing absolutely continuous functions). *Let H be increasing and absolutely continuous on $[a, b]$. Then H' exists λ -almost everywhere on (a, b) , H' belongs to $L^1([a, b])$, and*

$$H(x) = H(a) + \int_a^x H'(t) \, dt$$

for every $x \in [a, b]$.

Proof. By Proposition 3.16, absolute continuity implies continuity, hence H is right-continuous. By Theorem 3.17, the increasing function H has an associated Stieltjes measure μ_H on $[a, b]$. Since H is absolutely continuous, Proposition 3.18 gives

$$\mu_H \ll \lambda_1|_{[a,b]}.$$

By the finite Radon–Nikodym theorem, Corollary 12.2, there exists h in $L^1([a, b])$ such that

$$\mu_H(E) = \int_E h(t) \, dt$$

for every Borel set $E \subset [a, b]$. In particular, for every x in $[a, b]$,

$$H(x) - H(a) = \mu_H((a, x]) = \int_a^x h(t) \, dt.$$

By the direct part of the fundamental theorem of calculus, Theorem 13.7, the function

$$x \mapsto \int_a^x h(t) \, dt$$

is differentiable λ -almost everywhere on (a, b) , with derivative h . Therefore H' exists λ -almost everywhere, $H' = h$ almost everywhere, H' belongs to $L^1([a, b])$, and the desired identity follows. □

Proof of Theorem 13.8. Let F be absolutely continuous on $[a, b]$. By Lemma 13.9, there exist two increasing absolutely continuous functions F_1 and F_2 on $[a, b]$ such that

$$F = F_1 - F_2.$$

By Lemma 13.10, F'_1 and F'_2 exist almost everywhere, belong to $L^1([a, b])$, and satisfy

$$F_i(x) = F_i(a) + \int_a^x F'_i(t) \, dt, \quad i = 1, 2.$$

Subtracting the two identities gives

$$F(x) = F(a) + \int_a^x (F'_1(t) - F'_2(t)) dt$$

for every $x \in [a, b]$. The direct part of the fundamental theorem of calculus, Theorem 13.7, applied to the integrable function $F'_1 - F'_2$ shows that the derivative of the right-hand side is $F'_1 - F'_2$ almost everywhere. Since the right-hand side is equal to F , we obtain

$$F' = F'_1 - F'_2$$

almost everywhere. Thus F' belongs to $L^1([a, b])$ and

$$F(x) = F(a) + \int_a^x F'(t) dt$$

for every $x \in [a, b]$. □

Remark. The Cantor function is continuous and increasing, is differentiable with derivative 0 at λ -almost every point, but is not constant. Equivalently, its associated Stieltjes measure is the Cantor measure, which is singular with respect to the Lebesgue measure. This example shows that differentiability almost everywhere, even with an integrable derivative, is not enough to recover a function by integrating its derivative. The absolute continuity assumption in Theorem 13.8 is therefore essential.

13.1.4 The space $AC([a, b])$

The preceding theorem also identifies the natural structure of $AC([a, b])$. The derivative of an absolutely continuous function is understood below as an element of $L^1([a, b])$, hence up to equality almost everywhere.

Proposition 13.11 (Banach structure of absolutely continuous functions). *Let $a < b$ be real numbers. The set $AC([a, b])$ is a real vector space. Moreover,*

$$\|F\|_{AC} := |F(a)| + \int_a^b |F'(t)| dt$$

defines a norm on $AC([a, b])$, and $AC([a, b])$ is a Banach space for this norm.

Proof. By Proposition 3.16, $AC([a, b])$ is a real vector space. Consider the map

$$\Phi: AC([a, b]) \rightarrow \mathbb{R} \times L^1([a, b]), \quad \Phi(F) := (F(a), F').$$

The converse part of the fundamental theorem of calculus, Theorem 13.8, shows that Φ is well defined. The linearity of Φ follows from the linearity of differentiation almost everywhere and from the vector-space property of $AC([a, b])$. It is injective: if $F(a) = 0$ and $F' = 0$ almost everywhere, then Theorem 13.8 gives

$$F(x) = F(a) + \int_a^x F'(t) dt = 0$$

for every x in $[a, b]$.

It is also surjective. Let (c, g) be in $\mathbb{R} \times L^1([a, b])$, choose a representative of g , and define

$$F(x) := c + \int_a^x g(t) dt.$$

By the direct part of the fundamental theorem of calculus, Theorem 13.7, the function F is absolutely continuous and $F' = g$ almost everywhere. Hence $\Phi(F) = (c, g)$.

The norm $\|\cdot\|_{AC}$ is exactly the pullback by Φ of the product norm

$$\|(c, g)\| := |c| + \|g\|_{L^1([a, b])}$$

on $\mathbb{R} \times L^1([a, b])$. Therefore $\|\cdot\|_{AC}$ is a norm. Since $L^1([a, b])$ is complete by Theorem 8.19, the product space $\mathbb{R} \times L^1([a, b])$ is complete. The isometry Φ then shows that $AC([a, b])$ is complete for $\|\cdot\|_{AC}$. $\square$

Remark. The norm

$$\|F\|_{\infty} + \|F'\|_{L^1([a, b])}$$

is equivalent to $\|F\|_{AC}$. The derivative term is essential: the uniform norm alone does not capture the absolutely continuous structure.

13.2 Change of variables

We now turn to change of variables. We first recall the one-dimensional statement for the Riemann integral, then derive the corresponding Lebesgue statement in dimension 1, and finally prove the multidimensional formula. The one-dimensional proof uses a transport measure; the multidimensional proof first treats elementary diffeomorphisms and then factors a general local diffeomorphism into elementary ones.

13.2.1 One-dimensional change of variables

The first result in this subsection is the classical Riemann change-of-variables theorem. We recall it to fix the orientation convention and to connect the Lebesgue formula below with the familiar one-dimensional statement.

Theorem 13.12 (Riemann one-dimensional change-of-variables theorem). *Let $a < b$ be real numbers and let*

$$\varphi: [a, b] \rightarrow \varphi([a, b])$$

be a function of class C^1 whose derivative does not vanish on $[a, b]$. Let f be a function defined on $\varphi([a, b])$ and having only a finite number of discontinuities on this interval. Then

$$\int_{\varphi(a)}^{\varphi(b)} f(t) \, dt = \int_a^b f(\varphi(x)) \varphi'(x) \, dx.$$

Observe that if $\varphi'(x) > 0$ for all x in $[a, b]$, then $\varphi([a, b]) = [\varphi(a), \varphi(b)]$; if $\varphi'(x) < 0$ for all x in $[a, b]$, then $\varphi([a, b]) = [\varphi(b), \varphi(a)]$.

We now pass from intervals and Riemann-integrable functions to Borel sets and positive Borel functions. The next lemma packages the one-dimensional Jacobian formula as the equality between the Lebesgue measure on the target interval and a transport measure defined on that interval.

Lemma 13.13 (Transport measure for a one-dimensional diffeomorphism). *Let I, J be open intervals of $\mathbb{R}$, and let $\varphi: I \rightarrow J$ be a diffeomorphism of class C^1 . For every E in $\mathcal{B}(J)$, define*

$$\nu(E) := \int_I \mathbf{1}_E(\varphi(x)) |\varphi'(x)| dx.$$

Then ν is a measure on $\mathcal{B}(J)$ and

$$\nu(E) = \lambda(E)$$

for every E in $\mathcal{B}(J)$.

Proof. Step 1. Construction of the transport measure. For every Borel set $E \subset J$, the set $\varphi^{-1}(E)$ is Borel in I , because φ is continuous. Hence the formula defining $\nu(E)$ is meaningful. If $(E_k)_{k \geq 1}$ is a pairwise disjoint sequence in $\mathcal{B}(J)$, then

$$\mathbf{1}_{\bigcup_{k \geq 1} E_k}(\varphi(x)) |\varphi'(x)| = \sum_{k=1}^{\infty} \mathbf{1}_{E_k}(\varphi(x)) |\varphi'(x)|.$$

The integration theorem for series of positive functions, Corollary 7.4, gives

$$\nu \left(\bigcup_{k=1}^{\infty} E_k \right) = \sum_{k=1}^{\infty} \nu(E_k).$$

Thus ν is a measure.

Step 2. Computation on half-open intervals. Because φ' is continuous and never vanishes on the connected set I , it has a constant sign on I . Let $c < d$ be two points of J . If $\varphi' > 0$ on I , then

$$\varphi^{-1}((c, d]) = (\varphi^{-1}(c), \varphi^{-1}(d)],$$

and the usual fundamental theorem for C^1 functions gives

$$\nu((c, d]) = \int_{\varphi^{-1}(c)}^{\varphi^{-1}(d)} \varphi'(x) dx = d - c.$$

If $\varphi' < 0$ on I , then

$$\varphi^{-1}((c, d]) = [\varphi^{-1}(d), \varphi^{-1}(c)),$$

and

$$\nu((c, d]) = \int_{\varphi^{-1}(d)}^{\varphi^{-1}(c)} -\varphi'(x) dx = d - c.$$

Thus ν and λ coincide on all half-open intervals contained in J .

Step 3. Extension to all Borel sets. We now prove that $\nu = \lambda$ on $\mathcal{B}(J)$ by localizing on bounded half-open intervals. Choose an increasing sequence $(J_n)_{n \geq 1}$ of bounded half-open intervals contained in J such that

$$J = \bigcup_{n \geq 1} J_n.$$

For every $n \geq 1$, the restrictions of ν and λ to $\mathcal{B}(J_n)$ are finite measures, because $\nu(J_n) = \lambda(J_n) < +\infty$ by the computation above. By Lemma 3.13, the family

$$\mathcal{P}_{J_n} := \{\emptyset, J_n\} \cup \{(s, t] \subset J_n : s < t\}$$

is a π -system that generates $\mathcal{B}(J_n)$. The two finite measures coincide on this π -system, so Corollary 3.12 gives

$$\nu(F) = \lambda(F)$$

for every F in $\mathcal{B}(J_n)$. In particular,

$$\nu(E \cap J_n) = \lambda(E \cap J_n)$$

for every E in $\mathcal{B}(J)$ and every $n \geq 1$. Since $(E \cap J_n)_{n \geq 1}$ is increasing and has union E , the continuity from below of measures, Theorem 2.3, gives

$$\nu(E) = \lim_{n \rightarrow \infty} \nu(E \cap J_n) = \lim_{n \rightarrow \infty} \lambda(E \cap J_n) = \lambda(E).$$

This proves the lemma. □

The transport-measure identity now gives the Lebesgue one-dimensional change-of-variables theorem on intervals. The proof first checks the formula on nonnegative simple functions and then uses monotone convergence.

Theorem 13.14 (Lebesgue one-dimensional change-of-variables theorem). *Let I, J be open intervals of $\mathbb{R}$, and let $\varphi: I \rightarrow J$ be a diffeomorphism of class C^1 . Let $f: J \rightarrow [0, \infty]$ be a Borel function. Then the map*

$$x \mapsto f(\varphi(x))|\varphi'(x)|$$

is measurable on I , and

$$\int_J f(t) dt = \int_I f(\varphi(x))|\varphi'(x)| dx.$$

The same identity also holds for every f in $\mathcal{L}^1(J)$.

Proof. Step 1. Measurability and the simple-function case. Since f is Borel and φ is continuous, the map $x \mapsto f(\varphi(x))|\varphi'(x)|$ is measurable. Let ν be the measure constructed in Lemma 13.13. If

$$s = \sum_{k=1}^m a_k \mathbf{1}_{E_k}$$

is a nonnegative simple Borel function on J , then by the definition of ν and by Lemma 13.13,

$$\int_J s(t) dt = \int_J s d\nu = \int_I s(\varphi(x))|\varphi'(x)| dx.$$

Step 2. Passage to nonnegative Borel and integrable functions. For a general nonnegative Borel function f , choose simple functions $s_n \uparrow f$ and apply the monotone convergence theorem, Theorem 7.1, on both sides. If f is in $\mathcal{L}^1(J)$, apply the nonnegative case to f^+ and f^- . □

The next corollary removes the assumption that the open sets are intervals. The only new point is that every open subset of $\mathbb{R}$ is the disjoint union of its connected components.

Corollary 13.15 (One-dimensional change of variables for positive functions). *Let I and J be open subsets of $\mathbb{R}$, and let $\varphi: I \rightarrow J$ be a diffeomorphism of class C^1 . Let $f: J \rightarrow [0, \infty]$ be a Borel function. Then*

$$\int_J f(t) dt = \int_I f(\varphi(x))|\varphi'(x)| dx.$$

The same identity also holds for every f in $\mathcal{L}^1(J)$.

Proof. Step 1. Decomposition into connected components. The open sets I and J are countable unions of pairwise disjoint open intervals, namely their connected components. Since φ is

a homeomorphism, it induces a bijection between the connected components of I and those of J . Therefore we may write

$$I = \bigcup_{n=1}^{\infty} I_n, \quad J = \bigcup_{n=1}^{\infty} J_n,$$

where the unions are disjoint, each I_n and each J_n is an open interval, and

$$\varphi(I_n) = J_n$$

for every $n \geq 1$.

Step 2. Summation of the interval formulas. Applying Theorem 13.14 on each pair (I_n, J_n) and summing by the monotone convergence theorem, Theorem 7.1, gives the identity for nonnegative Borel functions. If $f \in \mathcal{L}^1(J)$, apply the nonnegative case to f^+ and f^- . $\square$

13.2.2 Multidimensional change of variables

The following part proves the multidimensional change-of-variables formula first in the compactly supported continuous case. The full extension to nonnegative Borel functions and then to $\mathcal{L}^1$ functions will be given later in Corollary 14.6, once the Radon-measure machinery is available.

Definition 13.16 (Elementary diffeomorphisms). *Let $d \geq 2$. A C^1 diffeomorphism $\Psi: U \rightarrow V$ between open subsets of $\mathbb{R}^d$ is called an elementary diffeomorphism if there exists k in $\{1, \dots, d\}$ and a C^1 function ψ such that*

$$\Psi(x_1, \dots, x_d) = (x_1, \dots, x_{k-1}, \psi(x_1, \dots, x_d), x_{k+1}, \dots, x_d).$$

Thus only one coordinate is changed.

The strategy is to prove the formula for elementary diffeomorphisms, then to show that a general local diffeomorphism can be decomposed, after a linear normalization, as a finite composition of such elementary diffeomorphisms.

Lemma 13.17 (Change of variables for elementary diffeomorphisms). *Let $d \geq 2$, let U and V be open subsets of $\mathbb{R}^d$, and let $\Psi: U \rightarrow V$ be an elementary diffeomorphism of class C^1 . If*

$$\Psi(x_1, \dots, x_d) = (x_1, \dots, x_{k-1}, \psi(x_1, \dots, x_d), x_{k+1}, \dots, x_d),$$

then, for every nonnegative Borel function f on V ,

$$\int_V f(y) \, dy = \int_U f(\Psi(x)) \left| \frac{\partial \psi}{\partial x_k}(x) \right| \, dx.$$

The same identity also holds for every f in $\mathcal{L}^1(V)$.

Proof. Up to permuting the coordinates, we may assume that $k = d$. We write

$$x = (x', t), \quad x' = (x_1, \dots, x_{d-1}) \in \mathbb{R}^{d-1}, \quad t \in \mathbb{R}.$$

For every x' in $\mathbb{R}^{d-1}$, define the sections

$$U_{x'} := \{t \in \mathbb{R} : (x', t) \in U\}, \quad V_{x'} := \{s \in \mathbb{R} : (x', s) \in V\}.$$

Since Ψ leaves the first $d - 1$ coordinates unchanged, the map

$$\psi_{x'}: U_{x'} \rightarrow V_{x'}, \quad \psi_{x'}(t) := \psi(x', t),$$

is a C^1 diffeomorphism between open subsets of $\mathbb{R}$. Therefore Corollary 13.15 gives, for every fixed x' and every nonnegative Borel function f ,

$$\int_{V_{x'}} f(x', s) ds = \int_{U_{x'}} f(x', \psi(x', t)) \left| \frac{\partial \psi}{\partial t}(x', t) \right| dt.$$

We integrate this identity with respect to x' and use Tonelli's theorem, Theorem 10.5. The left-hand side becomes

$$\int_{\mathbb{R}^{d-1}} \int_{V_{x'}} f(x', s) ds dx' = \int_V f(y) dy,$$

and the right-hand side becomes

$$\int_{\mathbb{R}^{d-1}} \int_{U_{x'}} f(x', \psi(x', t)) \left| \frac{\partial \psi}{\partial t}(x', t) \right| dt dx' = \int_U f(\Psi(x)) \left| \frac{\partial \psi}{\partial x_d}(x) \right| dx.$$

This proves the formula for nonnegative Borel functions. If $f \in \mathcal{L}^1(V)$, apply the nonnegative case to f^+ and f^- . $\square$

Lemma 13.18 (Linear change of variables for continuous compactly supported functions). *Let $A: \mathbb{R}^d \rightarrow \mathbb{R}^d$ be an invertible linear map. Then, for every f in $C_c(\mathbb{R}^d)$,*

$$\int_{\mathbb{R}^d} f(y) dy = \int_{\mathbb{R}^d} f(Ax) |\det A| dx.$$

Proof. If $d = 1$, this is exactly Theorem 13.14 applied to the diffeomorphism $x \mapsto Ax$. Assume now that $d \geq 2$. The formula is immediate for permutations of the coordinates. It also follows from Lemma 13.17 for diagonal maps and for shears. Since every invertible matrix is a product of elementary matrices of these three types by Gaussian elimination, and since the Jacobian determinant is multiplicative under composition, the general case follows. $\square$

Lemma 13.19 (Local factorization of diffeomorphisms). *Let $d \geq 2$, let $\varphi: U \rightarrow V$ be a C^1 diffeomorphism between open sets of $\mathbb{R}^d$, and let a be in U . Then there exists an open neighbourhood W of a , an open neighbourhood Z of $\varphi(a)$, and elementary diffeomorphisms*

$$\Psi_1, \dots, \Psi_d$$

between suitable open sets such that

$$D\varphi(a)^{-1} \circ \varphi = \Psi_d \circ \dots \circ \Psi_1$$

on W , and such that $\varphi(W) = Z$. The composition is understood between the successive open sets which occur in the construction.

Proof. **Step 1. Linear normalization.** We set

$$A := D\varphi(a)^{-1}, \quad F_0 := A \circ \varphi.$$

Then $DF_0(a) = I$. After shrinking U around a , we may assume, by the inverse function theorem, that F_0 is a diffeomorphism from an open neighbourhood W_0 of a onto an open set.

Step 2. Inductive elimination of the coordinates. We construct inductively open neighbourhoods W_k , points b_k in W_k , and diffeomorphisms F_k such that

- 1) F_k is a C^1 diffeomorphism on W_k ,
- 2) the first k coordinates of F_k are the identity on W_k ,

$$3) \quad DF_k(b_k) = I.$$

Initially, $b_0 := a$ and F_0 satisfies 1) and 3). Assume that $0 \leq k \leq d-1$ and that F_k has already been constructed. Since $DF_k(b_k) = I$, we have

$$\frac{\partial(F_k)_{k+1}}{\partial x_{k+1}}(b_k) = 1.$$

After shrinking W_k if necessary, we may therefore assume that this partial derivative never vanishes on W_k . We define

$$\Psi_{k+1}(x_1, \dots, x_d) := (x_1, \dots, x_k, (F_k)_{k+1}(x), x_{k+2}, \dots, x_d)$$

for x in W_k . By the inverse function theorem, after possibly shrinking W_k once more, the map Ψ_{k+1} is an elementary C^1 diffeomorphism from W_k onto an open set W_{k+1} . We set

$$F_{k+1} := F_k \circ \Psi_{k+1}^{-1}$$

on W_{k+1} and

$$b_{k+1} := \Psi_{k+1}(b_k).$$

For $j \leq k$, the j -th coordinate of F_k is the identity, and Ψ_{k+1} leaves the first k coordinates unchanged. Hence, for every y in W_{k+1} ,

$$(F_{k+1})_j(y) = (F_k)_j(\Psi_{k+1}^{-1}(y)) = y_j.$$

Moreover,

$$(F_{k+1})_{k+1}(y) = (F_k)_{k+1}(\Psi_{k+1}^{-1}(y)) = y_{k+1}$$

by the definition of Ψ_{k+1} . Thus the first $k+1$ coordinates of F_{k+1} are the identity on W_{k+1} . Finally, since $DF_k(b_k) = I$, the Jacobian matrix of Ψ_{k+1} at b_k is also the identity. Therefore

$$DF_{k+1}(b_{k+1}) = DF_k(b_k) D\Psi_{k+1}^{-1}(b_{k+1}) = I.$$

This completes the induction.

Step 3. Reconstruction of the factorization. After d steps, all the coordinates of F_d are the identity, so $F_d = \text{Id}$ on W_d . Tracing back the construction, we obtain

$$F_0 = \Psi_d \circ \dots \circ \Psi_1$$

on W_0 , where the composition is taken through the successive open sets $W_0, W_1, \dots, W_d$. We set $W := W_0$ and $Z := \varphi(W)$. $\square$

Corollary 13.20 (Local-to-global change of variables). *Let $d \geq 2$, let $\varphi: U \rightarrow V$ be a C^1 diffeomorphism between open sets of $\mathbb{R}^d$. For every a in U , there exists an open neighbourhood W of a and an open neighbourhood Z of $\varphi(a)$ such that $\varphi(W) = Z$ and such that, for every g in $C_c(Z)$,*

$$\int_Z g(y) \, dy = \int_W g(\varphi(x)) |\text{Jac} \varphi(x)| \, dx.$$

Proof. By Lemma 13.19, after shrinking around a if necessary, we can write on W

$$D\varphi(a)^{-1} \circ \varphi = \Psi_d \circ \dots \circ \Psi_1.$$

The formula is known for each elementary diffeomorphism Ψ_j by Lemma 13.17, and for the linear map $D\varphi(a)$ by Lemma 13.18. By composing these formulas and using the multiplicativity of the Jacobian determinant, we obtain the desired local identity. $\square$

Theorem 13.21 (Multidimensional change-of-variables theorem). *Let $d \geq 1$, let U and V be open sets of $\mathbb{R}^d$, and let $\varphi: U \rightarrow V$ be a diffeomorphism of class C^1 . Then, for every f in $\mathcal{C}_c(V)$,*

$$\int_V f(y) \, dy = \int_U f(\varphi(x)) |\text{Jac}\varphi(x)| \, dx.$$

Proof. **Step 1. Reduction to the case $d \geq 2$.** If $d = 1$, this is Corollary 13.15 applied to compactly supported continuous functions. We now assume that $d \geq 2$.

Step 2. A finite local cover of the support. Let $K := \text{supp}(f)$. For every y in K , set $a := \varphi^{-1}(y)$ and choose open neighbourhoods W_y of a and Z_y of y given by Corollary 13.20, with $\varphi(W_y) = Z_y$. Since K is compact, there exist finitely many points $y_1, \dots, y_m$ in K and open sets $\tilde{Z}_j$ such that

$$y_j \in \tilde{Z}_j, \quad \overline{\tilde{Z}_j} \subset Z_{y_j} \quad (j = 1, \dots, m),$$

and

$$K \subset \bigcup_{j=1}^m \tilde{Z}_j.$$

Step 3. A partition of the function. For every $j = 1, \dots, m$, define

$$u_j(y) := \text{dist}(y, V \setminus \tilde{Z}_j), \quad y \in V.$$

Then u_j is continuous, nonnegative, and

$$\text{supp}(u_j) \subset \overline{\tilde{Z}_j} \subset Z_{y_j}.$$

We set

$$u := \sum_{j=1}^m u_j.$$

Since the open sets $\tilde{Z}_j$ cover K , we have $u > 0$ on K . Moreover, if $u(y) = 0$, then y does not belong to $K = \text{supp}(f)$, hence $f(y) = 0$. We may therefore define, for every $j = 1, \dots, m$,

$$f_j(y) := \begin{cases} \frac{u_j(y)}{u(y)} f(y) & \text{if } u(y) > 0, \\ 0 & \text{if } u(y) = 0. \end{cases}$$

We now check that each f_j is continuous on V . Continuity is clear at points where $u > 0$. If $u(y_0) = 0$, then $y_0 \notin K = \text{supp}(f)$. Since K is closed, there is a neighbourhood of y_0 on which $f = 0$. Hence each f_j is identically zero on a neighbourhood of y_0 , and is continuous at y_0 . Moreover,

$$\text{supp}(f_j) \subset K \cap \text{supp}(u_j) \subset K \cap \overline{\tilde{Z}_j} \subset Z_{y_j},$$

so f_j belongs to $\mathcal{C}_c(Z_{y_j})$. Finally,

$$f = \sum_{j=1}^m f_j.$$

Step 4. Summation of the local formulas. Applying Corollary 13.20 to each f_j and summing, we obtain

$$\begin{aligned} \int_V f(y) \, dy &= \sum_{j=1}^m \int_V f_j(y) \, dy \\ &= \sum_{j=1}^m \int_U f_j(\varphi(x)) |\text{Jac}\varphi(x)| \, dx \\ &= \int_U f(\varphi(x)) |\text{Jac}\varphi(x)| \, dx. \end{aligned}$$

□

Chapter 14

Radon measures, Riesz representation and the Lebesgue measure

We return to measures on open subsets of $\mathbb{R}^N$ and exploit the underlying topological structure. The chapter proves the Riesz representation theorem for positive functionals on $\mathcal{C}_c(\Omega)$, uses it to recover the Lebesgue measure, and then studies bounded Radon measures and their weak convergence. The proof of the Riesz theorem uses two topological tools recalled in the appendix: Urysohn's lemma and a partition-of-unity lemma, stated there as Lemmas 16.41 and 16.42.

14.1 The converse problem and the Riesz representation theorem

In this section Ω stands for an open subset of $\mathbb{R}^N$ ($N \geq 1$). We recall from Definition 2.8 that a positive Radon measure on Ω is a Borel measure which is finite on compact sets. We already saw in Section 8.8 that every positive Radon measure on Ω defines a positive linear functional on $\mathcal{C}_c(\Omega)$ by integration. We now address the converse problem: starting from a positive linear functional on $\mathcal{C}_c(\Omega)$, reconstruct a measure. The theorem below shows that the resulting measure is not only locally finite, but also complete and regular.

Actually, the result is even stronger, for it provides a complete measure, in the following sense (not to be confused with the topological notion of completeness), representing L . The regularity properties appearing in items 3) and 4) below are precisely the notions introduced in Section 2.3. We also recall that a measure space is said to be complete if every negligible set is measurable.

Theorem 14.1 (Riesz representation theorem for positive functionals on C_c). *Let $L : C_c(\Omega) \rightarrow \mathbb{R}$ be a positive linear functional. Then there exists a σ -algebra $\mathcal{A}$ containing the Borel σ -algebra $\mathcal{B}(\Omega)$ and a measure μ on $\mathcal{A}$ such that*

$$\begin{aligned} 1) \quad & L(f) = \int_{\Omega} f d\mu, \text{ for every } f \in C_c(\Omega), \\ 2) \quad & \mu(K) < \infty \text{ for every compact set } K \subset \Omega, \\ 3) \quad & \text{for every } E \in \mathcal{A}, \\ & \mu(E) = \inf\{\mu(V) : E \subset V, V \text{ open}\}, \end{aligned} \tag{14.1}$$

$$\begin{aligned} 4) \quad & \text{for every open set } E, \text{ and every } E \in \mathcal{A} \text{ with } \mu(E) < \infty, \\ & \mu(E) = \sup\{\mu(K) : K \subset E, K \text{ compact}\}, \end{aligned} \tag{14.2}$$

5) the measure space $(X, \mathcal{A}, \mu)$ is complete,

6) for any $E \in \mathcal{B}(\Omega)$ and any $\varepsilon > 0$, there exists a closed set C and an open set V such that $C \subset E \subset V$, and

$$\mu(V \setminus C) < \varepsilon. \tag{14.3}$$

Moreover, this measure is unique in the sense that if $(\mathcal{A}_1, \mu_1)$ and $(\mathcal{A}_2, \mu_2)$ satisfy the properties 1)-2)-3)-4), then the restrictions of μ_1 and μ_2 to $\mathcal{B}(\Omega)$ are equal.

Proof. Let us start by proving uniqueness. Assume that μ_1 and μ_2 are two Radon measures satisfying the conclusion of the Riesz representation Theorem. By the regularity property (14.2), it suffices to prove that $\mu_1(K) = \mu_2(K)$ for every compact set K .

Let $\varepsilon > 0$ and K a compact set. By (14.1), there exists an open set V containing K such that $\mu_2(V) < \mu_2(K) + \varepsilon$. By Urysohn's lemma (Lemma 16.41 in the appendix) one can find a function $f \in C_c(V; [0, 1])$ such that $f = 1$ on K . In particular,

$$\mathbf{1}_K \leq f \leq \mathbf{1}_V,$$

hence

$$\mu_1(K) = \int_{\Omega} \mathbf{1}_K d\mu_1 \leq \int_{\Omega} f d\mu_1 = L(f) = \int_{\Omega} f d\mu_2 \leq \int_{\Omega} \mathbf{1}_V d\mu_2 = \mu_2(V) < \mu_2(K) + \varepsilon.$$

Thus $\mu_1(K) \leq \mu_2(K)$ and exchanging the roles of μ_1 and μ_2 we deduce that this inequality is actually an equality. This concludes the uniqueness part.

We now turn our attention to the existence. For every open set $V \subset \Omega$, we define

$$\mu(V) := \sup\{L(f) : f \in C_c(\Omega; [0, 1]), \text{ supp}(f) \subset V\}. \tag{14.4}$$

Clearly, if $V_1 \subset V_2$, then $\mu(V_1) \leq \mu(V_2)$ so that we can extend μ to any arbitrary $E \subset \Omega$ by setting

$$\mu(E) := \inf\{\mu(V) : E \subset V, V \text{ open}\}.$$

Note that both definitions are consistent for open sets, and that property (14.1) will be automatically satisfied. Moreover, μ is an increasing set function, i.e. if $E_1 \subset E_2$, then $\mu(E_1) \leq \mu(E_2)$.

Let $\mathcal{A}_F$ be the family of all sets $E \subset \Omega$ such that $\mu(E) < \infty$ and

$$\mu(E) = \sup\{\mu(K) : K \subset E, K \text{ compact}\}.$$

Finally, let $\mathcal{A}$ be the class of all $E \subset \Omega$ such that $E \cap K \in \mathcal{A}_F$ for any compact K .

Step 1. If K is compact, then $K \in \mathcal{A}_F$, and

$$\mu(K) = \inf\{L(f) : f \in \mathcal{C}_c(\Omega; [0, 1]), f = 1 \text{ on } K\}. \quad (14.5)$$

Moreover, if V is open, then V satisfies (14.2). In particular if $\mu(V) < \infty$, then $V \in \mathcal{A}_F$.

Let $f \in \mathcal{C}_c(\Omega; [0, 1])$ such that $f = 1$ on K , $\alpha \in (0, 1)$, and $V_\alpha := \{f > \alpha\}$. Then $K \subset V_\alpha$ and $\alpha g \leq f$ for every $g \in \mathcal{C}_c(V_\alpha; [0, 1])$. Hence

$$\mu(K) \leq \mu(V_\alpha) = \sup\{L(g) : g \in \mathcal{C}_c(V_\alpha; [0, 1])\} \leq \alpha^{-1}L(f).$$

Letting $\alpha \rightarrow 1$ leads to $\mu(K) \leq L(f) < \infty$. Therefore $K \in \mathcal{A}_F$ since (14.2) is immediate. Next if $\varepsilon > 0$, there exists an open set V containing K with $\mu(V) < \mu(K) + \varepsilon$. By Urysohn's lemma (Lemma 16.41 in the appendix), there exists a function $f \in \mathcal{C}_c(V; [0, 1])$ satisfying $f = 1$ on K , and thus

$$\mu(K) \leq L(f) \leq \mu(V) < \mu(K) + \varepsilon,$$

which gives (14.5).

Consider now an open set V . Then for any $\alpha < \mu(V)$, there exists $f \in \mathcal{C}_c(V; [0, 1])$ such that $\alpha < L(f)$. Hence for every open set W containing

$$K := \text{supp}(f),$$

then $f \in \mathcal{C}_c(W)$ and by definition of μ we have that

$$L(f) \leq \mu(W).$$

Taking the infimum over all such W 's leads to $L(f) \leq \mu(K)$. This proves the existence of a compact set $K \subset V$ with $\alpha < \mu(K)$ which ensures that V satisfies (14.2), and that $V \in \mathcal{A}_F$ if further $\mu(V) < \infty$.

Step 2. For every sets $E_n \subset \Omega$ for $n \in \mathbb{N}^*$, then

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} \mu(E_n).$$

We first prove that μ is finitely subadditive on open sets, i.e.,

$$\mu(V_1 \cup V_2) \leq \mu(V_1) + \mu(V_2) \quad (14.6)$$

when V_1 and V_2 are open sets. Let $g \in \mathcal{C}_c(V_1 \cup V_2; [0, 1])$. By the partition-of-unity lemma from the appendix (Lemma 16.42), there exist functions f_1 and f_2 such that $f_i \in \mathcal{C}_c(V_i; [0, 1])$ and $f_1 + f_2 = 1$ on $\text{supp}(g)$. Hence $f_i g \in \mathcal{C}_c(V_i; [0, 1])$,

$$g = f_1 g + f_2 g$$

so that by linearity of L and by definition of μ ,

$$L(g) = L(f_1 g) + L(f_2 g) \leq \mu(V_1) + \mu(V_2).$$

Taking the supremum with respect to all g as above gives

$$\mu(V_1 \cup V_2) \leq \mu(V_1) + \mu(V_2).$$

Now if $\mu(E_n) = \infty$ for some n then the result is obvious. Otherwise, if $\mu(E_n) < \infty$ for all n , then for any $\varepsilon > 0$ there exists open sets V_n such that $E_n \subset V_n$ and

$$\mu(V_n) < \mu(E_n) + 2^{-n}\varepsilon.$$

Define

$$V := \bigcup_{n=1}^{\infty} V_n,$$

and take $f \in \mathcal{C}_c(V; [0, 1])$. In particular, since $\text{supp}(f)$ is compact, one can find finitely many $V_1, \dots, V_p$ such that

$$\text{supp}(f) \subset \bigcup_{n=1}^p V_n.$$

Hence iterating (14.6),

$$L(f) \leq \mu \left(\bigcup_{n=1}^p V_n \right) \leq \sum_{n=1}^p \mu(V_n) \leq \sum_{n=1}^{\infty} \mu(E_n) + \varepsilon.$$

Since this holds for every $f \in \mathcal{C}_c(V; [0, 1])$, and since $\bigcup_{n=1}^{\infty} E_n \subset V$, it follows that

$$\mu \left(\bigcup_{n=1}^{\infty} E_n \right) \leq \mu(V) \leq \sum_{n=1}^{\infty} \mu(E_n) + \varepsilon,$$

which completes the proof of step 2 since ε is arbitrary.

Step 3. suppose that $E = \bigcup_{n=1}^{\infty} E_n$, where E_n are pairwise disjoint elements of $\mathcal{A}_F$ for all $n \in \mathbb{N}^*$. Then,

$$\mu(E) = \sum_{n=1}^{\infty} \mu(E_n). \quad (14.7)$$

If in addition, $\mu(E) < \infty$, then $E \in \mathcal{A}_F$.

We first prove that μ is finitely additive on compact sets, i.e.,

$$\mu(K_1 \cup K_2) = \mu(K_1) + \mu(K_2) \quad (14.8)$$

if K_1 and K_2 are disjoint compact sets. Let $\varepsilon > 0$. By Urysohn's lemma (Lemma 16.41 in the appendix), there exists $f \in \mathcal{C}_c(\Omega; [0, 1])$ such that $f = 1$ on K_1 and $f = 0$ on K_2 . Then by Step 1 there exists a function $g \in \mathcal{C}_c(\Omega)$ such that $g = 1$ on $K_1 \cup K_2$ and

$$L(g) \leq \mu(K_1 \cup K_2) + \varepsilon.$$

Note that $fg = 1$ on K_1 and $(1 - f)g = 1$ on K_2 . Hence by linearity of L , it follows that

$$\mu(K_1) + \mu(K_2) \leq L(fg) + L((1 - f)g) = L(g) \leq \mu(K_1 \cup K_2) + \varepsilon.$$

Since ε is arbitrary, (14.8) follows from step 2.

If $\mu(E) = \infty$, then (14.7) is an immediate consequence of step 2. Therefore, we can assume that $\mu(E) < \infty$. Let $\varepsilon > 0$, since $E_n \in \mathcal{A}_F$, there exist compact sets $H_n \subset E_n$ with

$$\mu(H_n) > \mu(E_n) - 2^{-n}\varepsilon,$$

for each $n \in \mathbb{N}^*$. Define

$$K_n := \bigcup_{k=1}^n H_k \subset E,$$

which is compact. Thus using an induction on (14.8), we deduce that

$$\mu(E) \geq \mu(K_n) = \sum_{k=1}^n \mu(H_k) > \sum_{k=1}^n \mu(E_k) - \varepsilon.$$

Since the previous relation holds for every $n \in \mathbb{N}^*$ and every $\varepsilon > 0$, using again step 2, we deduce (14.7). Moreover for n large enough (depending on ε) we have

$$\mu(E) \leq \sum_{k=1}^n \mu(E_k) + \varepsilon \leq \mu(K_n) + 2\varepsilon,$$

which proves that $E \in \mathcal{A}_F$.

Step 4. If $E \in \mathcal{A}_F$ and $\varepsilon > 0$, there is a compact set K and an open set V such that $K \subset E \subset V$ and $\mu(V \setminus K) < \varepsilon$. Moreover, if A and $B \in \mathcal{A}_F$, then $A \setminus B$, $A \cup B$ and $A \cap B \in \mathcal{A}_F$.

Our definitions prove that there exists a compact set K and an open set V with $K \subset E \subset V$ to that

$$\mu(V) - \frac{\varepsilon}{2} < \mu(E) < \mu(K) + \frac{\varepsilon}{2} < \infty,$$

where we used the fact that, from step 1, μ is finite on compact sets. Using again step 1, since $V \setminus K$ is open and $\mu(V \setminus K) < \infty$, it therefore belongs to $\mathcal{A}_F$. We apply step 3 to get that

$$\mu(K) + \mu(V \setminus K) = \mu(V) < \mu(K) + \varepsilon < \infty.$$

Now if A and $B \in \mathcal{A}_F$ and $\varepsilon > 0$, one can find compact sets K_1 and K_2 , and open sets V_1 and V_2 such that $K_1 \subset A \subset V_1$ and $K_2 \subset B \subset V_2$, and $\mu(V_i \setminus K_i) < \varepsilon$ (for $i = 1, 2$). Since

$$A \setminus B \subset V_1 \setminus K_2 \subset (V_1 \setminus K_1) \cup (K_1 \setminus V_2) \cup (V_2 \setminus K_2),$$

from the subadditivity proved in step 2, we infer that

$$\mu(A \setminus B) \leq 2\varepsilon + \mu(K_1 \setminus V_2).$$

But since $K_1 \setminus V_2$ is a compact subset of $A \setminus B$, it proves that $A \setminus B \in \mathcal{A}_F$. Then since

$$A \cup B = (A \setminus B) \cup B,$$

an application of step 3 proves that $A \cup B \in \mathcal{A}_F$. Finally, since $A \cap B = A \setminus (A \setminus B)$, we also have $A \cap B \in \mathcal{A}_F$.

Step 5. $\mathcal{A}$ is a σ -algebra which contains all Borel sets, and all sets $E \subset \Omega$ such that $\mu(E) = 0$. Moreover

$$\mathcal{A}_F = \{E \in \mathcal{A} : \mu(E) < \infty\}.$$

Finally, μ is a measure on $\mathcal{A}$ satisfying (14.1) and (14.2).

First step 1 gives that Ω is in $\mathcal{A}$.

Now let K be an arbitrary compact set in X . If $A \in \mathcal{A}$, then $A \cap K \in \mathcal{A}_F$ and thus by step 4,

$$A^c \cap K = K \setminus (A \cap K) \in \mathcal{A}_F.$$

Consequently, $A^c \in \mathcal{A}$. Now if $A_n \in \mathcal{A}$ for each $n \in \mathbb{N}^*$, define $B_1 = A_1 \cap K$ and for $n \geq 2$,

$$B_n = (A_n \cap K) \setminus \bigcup_{k=1}^{n-1} B_k.$$

Then by step 4, (B_n) is made of pairwise disjoint elements of $\mathcal{A}_F$. Thus by step 3, we have

$$K \cap \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n \in \mathcal{A}_F$$

and thus

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}.$$

This proves that $\mathcal{A}$ is a σ -algebra. If C is closed, then $K \cap C$ is compact and by step 1, it belongs to $\mathcal{A}_F$. Hence $C \in \mathcal{A}$. This proves that $\mathcal{A}$ is a σ -algebra containing all closed sets and consequently all Borel sets. If $E \subset \Omega$ is such that $\mu(E) = 0$, then clearly $E \in \mathcal{A}$ since μ is an increasing set function.

By steps 1 and 4, we clearly have that

$$\mathcal{A}_F \subset \{E \subset \mathcal{A} : \mu(E) < \infty\}.$$

Conversely, let $E \in \mathcal{A}$ such that $\mu(E) < \infty$ and let $\varepsilon > 0$. There exists an open set V containing E such that $\mu(V) < \infty$. Using again steps 1 and 4, one can find a compact set $K \subset V$ such that $\mu(V \setminus K) < \varepsilon$. Since

$$E \cap K \in \mathcal{A}_F,$$

there exists a compact set $H \subset E \cap K$ with

$$\mu(E \cap K) < \mu(H) + \varepsilon.$$

Since

$$E \subset (E \cap K) \cup (V \setminus K),$$

it follows that

$$\mu(E) \leq \mu(E \cap K) + \mu(V \setminus K) < \mu(H) + 2\varepsilon,$$

which implies that $E \in \mathcal{A}_F$. Hence properties (14.1) and (14.2) follow.

We next prove that μ is a measure on $(\Omega, \mathcal{A})$. If (E_n) is a sequence of pairwise disjoint elements of $\mathcal{A}$, then

$$E := \bigcup_{n=1}^{\infty} E_n \in \mathcal{A}.$$

If $\mu(E) = \infty$, then by Step 2, we have

$$\mu(E) = \sum_{n=1}^{\infty} \mu(E_n),$$

while if $\mu(E) < \infty$ then $\mu(E_n) < \infty$ for each $n \in \mathbb{N}^*$ and thus $E, E_n \in \mathcal{A}_F$ for all $n \in \mathbb{N}^*$. Consequently, by step 3, we have the same result which proves that μ is a measure on $\mathcal{A}$.

Step 6. *Proof of the representation property 1).*

Let $f \in \mathcal{C}_c(\Omega)$. Clearly it is enough to check the inequality $L(f) \leq \int_{\Omega} f d\mu$ because by linearity of L , it implies

$$-L(f) = L(-f) \leq \int_{\Omega} (-f) d\mu = - \int_{\Omega} f d\mu,$$

and therefore the representation property 1). Let $K := \text{supp } f$ and $[a, b]$ be an interval which contains the range of f . For $\varepsilon > 0$, let $y_0, \dots, y_n \in \mathbb{R}$ be such that $y_0 < a < y_1 < \dots < y_n = b$, and $\max_{1 \leq i \leq n} (y_i - y_{i-1}) < \varepsilon$. Define

$$E_i := \{y_{i-1} < f \leq y_i\} \cap K.$$

Since f is continuous, f is Borel measurable and the sets E_i are therefore disjoint Borel sets whose union is K . There are open sets V_i containing E_i and such that $\mu(V_i) < \mu(E_i) + \varepsilon/n$ and $f(x) <$

$y_i + \varepsilon$ for all $x \in V_i$ and all $i = 1, \dots, n$. By the partition-of-unity lemma from the appendix (Lemma 16.42), one can find functions $h_i \in \mathcal{C}_c(V_i; [0, 1])$ such that $\sum_{i=1}^n h_i = 1$ on K . Hence,

$$f = \sum_{i=1}^n h_i f$$

in Ω and by step 1, we infer that

$$\mu(K) \leq L\left(\sum_{i=1}^n h_i\right) = \sum_{i=1}^n L(h_i).$$

Note that

$$L(h_i) \leq \mu(V_i) < \mu(E_i) + \varepsilon/n,$$

$h_i f \leq (y_i + \varepsilon)h_i$ and $y_i - \varepsilon < f(x)$ for $x \in E_i$. Denoting by $M := \max_K |f| < +\infty$ we have

$$\begin{aligned} L(f) &= \sum_{i=1}^n L(h_i f) \leq \sum_{i=1}^n (y_i + \varepsilon) L(h_i) = \sum_{i=1}^n (M + y_i + \varepsilon) L(h_i) - M \sum_{i=1}^n L(h_i) \\ &\leq \sum_{i=1}^n (M + y_i + \varepsilon) \left(\mu(E_i) + \frac{\varepsilon}{n} \right) - M \mu(K) \\ &\leq \sum_{i=1}^n (y_i - \varepsilon) \mu(E_i) + 2\varepsilon \mu(K) + \frac{\varepsilon}{n} \sum_{i=1}^n (M + y_i + \varepsilon) \\ &\leq \int_{\Omega} f d\mu + \varepsilon(2\mu(K) + 2M + \varepsilon). \end{aligned}$$

Since ε is arbitrary, the proof of the representation property 1) is complete.

Step 7. For any $E \in \mathcal{B}(\Omega)$, for any $\varepsilon > 0$, there are a closed set C and an open set V such that $C \subset E \subset V$ and $\mu(V \setminus C) < \varepsilon$.

Let $E \in \mathcal{B}(\Omega)$ and let (K_n) be an increasing sequence of compact sets whose union is Ω (take e.g. $K_n := \{x \in \Omega : |x| \leq n \text{ and } \text{dist}(x, \mathbb{R}^N \setminus \Omega) \geq 1/n\}$). Since μ is a Radon measure, then $\mu(E \cap K_n) < \infty$ for all $n \geq 1$, and there exists an open set V_n such that $E \cap K_n \subset V_n$ and

$$\mu(V_n) < \mu(E \cap K_n) + 2^{-n-1}\varepsilon.$$

In particular, since $\mu(E \cap K_n) < \infty$, then

$$\mu(V_n \setminus (E \cap K_n)) < 2^{-n-1}\varepsilon.$$

Define the open set

$$V := \bigcup_{n \geq 1} V_n,$$

which satisfies $E \subset V$, and

$$V \setminus E \subset \bigcup_{n \geq 1} V_n \setminus (E \cap K_n).$$

Then we have

$$\mu(V \setminus E) < \varepsilon/2. \quad (14.9)$$

Moreover, applying this property to $F := E^c$, there exists an open set W containing F and such that

$$\mu(W \setminus F) \leq \varepsilon/2. \quad (14.10)$$

Let $C := W^c$ be closed, then $C \subset E$ and

$$\mu(E \setminus C) = \mu(E \cap W) = \mu(W \setminus (E^c)) = \mu(W \setminus F) < \varepsilon/2.$$

Therefore, combining (14.9) and (14.10), we arrive at the conclusion that for any $E \in \mathcal{B}(\Omega)$, for any $\varepsilon > 0$, there are a closed set C and an open set V such that $C \subset E \subset V$ and $\mu(V \setminus C) < \varepsilon$. The proof of the theorem is now completed. $\square$

We give a first application of the Riesz representation Theorem, which concerns the regularity of Radon measures. It will be useful to prove the density of continuous functions and the separability of Lebesgue spaces in Chapter 15.

Theorem 14.2 (Regularity theorem for Radon measures). *Any positive Radon measure λ on Ω satisfies*

$$\lambda(E) = \inf\{\lambda(V) : E \subset V, V \text{ open}\} \text{ for every Borel set } E \subset \Omega,$$

and

$$\lambda(E) = \sup\{\lambda(K) : K \subset E, K \text{ compact}\} \text{ for every Borel set } E \subset \Omega \text{ with } \lambda(E) < \infty.$$

Moreover, for any $E \in \mathcal{B}(\Omega)$ and any $\varepsilon > 0$, there exists a closed set C and an open set V such that $C \subset E \subset V$, and

$$\lambda(V \setminus C) < \varepsilon. \quad (14.11)$$

Proof. Define

$$L(f) := \int_{\Omega} f d\lambda,$$

for any $f \in \mathcal{C}_c(\Omega)$. Since $\lambda(K) < \infty$ for any compact set $K \subset \Omega$, L is a positive linear form on $\mathcal{C}_c(\Omega)$, and from the Riesz representation Theorem, there exists a unique Radon measure μ such that

$$\int_{\Omega} f d\lambda = \int_{\Omega} f d\mu.$$

Let V be an open subset of Ω . Then $V = \bigcup_{n=1}^{\infty} K_n$ for some compact sets K_n with $n \in \mathbb{N}^*$. By Urysohn's lemma (Lemma 16.41 in the appendix), we can choose $f_n \in \mathcal{C}_c(V; [0, 1])$ such that $f_n = 1$ on K_n . Define

$$g_n := \max_{1 \leq i \leq n} f_i,$$

then $g_n \in \mathcal{C}_c(V; [0, 1])$ and it increasingly converges to $\mathbf{1}_V$ pointwise in Ω . Hence by the monotone convergence Theorem,

$$\lambda(V) = \lim_{n \rightarrow \infty} \int_{\Omega} g_n d\lambda = \lim_{n \rightarrow \infty} \int_{\Omega} g_n d\mu = \mu(V).$$

Now let E be a Borel subset of Ω and let $\varepsilon > 0$. By (14.3), there exists an open set V and a closed set C such that $C \subset E \subset V$ and $\mu(V \setminus C) < \varepsilon$. But since $V \setminus C$ is open we infer that $\lambda(V \setminus C) < \varepsilon$. Consequently,

$$\lambda(E) \leq \lambda(V) = \mu(V) \leq \mu(E) + \varepsilon \leq \mu(V) + \varepsilon = \lambda(V) + \varepsilon \leq \lambda(E) + 2\varepsilon,$$

so that $\mu(E) = \lambda(E)$ by the arbitrariness of ε .

Therefore, the restriction of μ to the Borel sets matches with the given Radon measure λ , and since μ satisfies the desired regularity property by the Riesz representation theorem, we may now deduce Theorem 14.2. $\square$

14.2 Lusin's theorem

The regularity of Radon measures has an important consequence: on a set of arbitrarily large measure, a measurable function may be replaced by a continuous one.

Theorem 14.3 (Lusin theorem). *Let μ be a positive Radon measure on Ω . Let E in $\mathcal{B}(\Omega)$ be such that $\mu(E) < \infty$, and let $f: \Omega \rightarrow \mathbb{R}$ be a Borel measurable function. Then for every $\varepsilon > 0$, there exists a compact set $K \subset E$ such that*

$$\mu(E \setminus K) < \varepsilon$$

and the restriction $f|_K$ is continuous.

Proof. **Step 1: indicator functions.** Let A in $\mathcal{B}(\Omega)$ and let $\eta > 0$. By Theorem 14.2, there exists a compact set $F \subset A$ and an open set V such that

$$F \subset A \subset V, \quad \mu(V \setminus F) < \eta.$$

By Urysohn's lemma (Lemma 16.41 in the appendix), there exists $h \in \mathcal{C}_c(\Omega; [0, 1])$ such that $h = 1$ on F and $h = 0$ on $\Omega \setminus V$. Hence

$$h = \mathbf{1}_A$$

on $\Omega \setminus (V \setminus F)$.

Step 2: simple functions. Let

$$s = \sum_{j=1}^m \alpha_j \mathbf{1}_{A_j}$$

be a simple Borel function on Ω , and let $\eta > 0$. Applying Step 1 to each set A_j with η/m in place of η , we obtain functions $h_j \in \mathcal{C}_c(\Omega)$ and Borel sets $N_j \subset \Omega$ such that

$$\mu(N_j) < \frac{\eta}{m}, \quad h_j = \mathbf{1}_{A_j}$$

on $\Omega \setminus N_j$. If we define

$$g := \sum_{j=1}^m \alpha_j h_j, \quad N := \bigcup_{j=1}^m N_j,$$

then g belongs to $\mathcal{C}_c(\Omega)$, $\mu(N) < \eta$, and

$$g = s$$

on $\Omega \setminus N$.

Step 3: bounded measurable functions. Assume now that f is bounded on E . Arguing as in the proof of Theorem 6.6, we can choose a sequence $(s_n)_{n \geq 1}$ of simple Borel functions such that

$$\sup_{x \in E} |f(x) - s_n(x)| \leq 2^{-n}$$

for every $n \geq 1$. Applying Step 2 with $\eta = 2^{-n-2}\varepsilon$, we obtain $g_n \in \mathcal{C}_c(\Omega)$ and a Borel set $N_n \subset \Omega$ such that

$$\mu(N_n) < 2^{-n-2}\varepsilon, \quad g_n = s_n$$

on $\Omega \setminus N_n$. Let

$$F := E \setminus \bigcup_{n=1}^{\infty} N_n.$$

Then F is a Borel set and

$$\mu(E \setminus F) \leq \sum_{n=1}^{\infty} \mu(N_n) < \frac{\varepsilon}{2}.$$

Moreover, for every x in F and every $n \geq 1$, we have $g_n(x) = s_n(x)$, so

$$\sup_{x \in F} |f(x) - g_n(x)| \leq 2^{-n}.$$

Thus $(g_n)_{n \geq 1}$ converges uniformly to f on F . Since each g_n is continuous on Ω , the restriction $f|_F$ is continuous. By Theorem 14.2 again, there exists a compact set $K \subset F$ such that

$$\mu(F \setminus K) < \frac{\varepsilon}{2}.$$

Hence

$$\mu(E \setminus K) \leq \mu(E \setminus F) + \mu(F \setminus K) < \varepsilon,$$

and $f|_K$ is continuous.

Step 4: the general case. For every $M > 0$, we define

$$E_M := E \cap \{|f| \leq M\}.$$

Since f is finite-valued, the sequence $(E_M)_{M \geq 1}$ is increasing and its union is E . Because $\mu(E) < \infty$, continuity from below gives

$$\mu(E \setminus E_M) \rightarrow 0 \quad \text{as } M \rightarrow +\infty.$$

We fix M so large that

$$\mu(E \setminus E_M) < \frac{\varepsilon}{2}.$$

The function f is bounded on E_M , so Step 3 yields a compact set $K \subset E_M$ such that

$$\mu(E_M \setminus K) < \frac{\varepsilon}{2}$$

and $f|_K$ is continuous. Therefore

$$\mu(E \setminus K) \leq \mu(E \setminus E_M) + \mu(E_M \setminus K) < \varepsilon.$$

This concludes the proof. □

14.3 Construction of the Lebesgue measure

The Riesz representation Theorem enables one to prove the existence of the Lebesgue measure, and to prove that the Lebesgue integral with respect to the Lebesgue measure is a natural extension of the Riemann integral. Let us recall that, by definition, a F_σ is a countable union of closed sets and a G_δ is a countable intersection of open sets.

Theorem 14.4 (Existence and regularity theorem for Lebesgue measure). *There exists a σ -algebra $\mathcal{L}(\mathbb{R}^N)$ (containing the Borel σ -algebra $\mathcal{B}(\mathbb{R}^N)$) and a unique measure λ_N on $\mathcal{L}(\mathbb{R}^N)$ such that*

- 1) $\lambda_N([0, 1]^N) = 1$;
- 2) For every $E \in \mathcal{L}(\mathbb{R}^N)$ and every $x \in \mathbb{R}^N$, $\lambda_N(x + E) = \lambda_N(E)$;
- 3) $E \in \mathcal{L}(\mathbb{R}^N)$ if and only if there exists a F_σ set A and a G_δ set B such that $A \subset E \subset B$ and $\lambda_N(B \setminus A) = 0$;
- 4) For every $f \in \mathcal{C}_c(\mathbb{R}^N)$,

$$\int_{\mathbb{R}^N} f d\lambda_N = \int_{\mathbb{R}^N} f(x) dx,$$

where the integral in left hand side is the Lebesgue integral of f with respect to the measure λ_N , and the integral in the right hand side is the Riemann integral of f .

The restriction of λ_N to $\mathcal{B}(\mathbb{R}^N)$ is the *Borel Lebesgue measure*. Its completion on $\mathcal{L}(\mathbb{R}^N)$ is called the *Lebesgue measure*, and $\mathcal{L}(\mathbb{R}^N)$ is called the *Lebesgue σ -algebra*; its elements are the *Lebesgue measurable sets*.

Proof. **Step 1.** Define $L : \mathcal{C}_c(\mathbb{R}^N) \rightarrow \mathbb{R}$ by

$$L(f) = \int_{\mathbb{R}^N} f(x) dx,$$

where the integral is the Riemann integral of f . Note that since f has compact support, then the previous integral is not improper since

$$\int_{\mathbb{R}^N} f(x) dx = \int_{\text{supp}(f)} f(x) dx.$$

Clearly L is a positive linear functional on $\mathcal{C}_c(\mathbb{R}^N)$, and according to the Riesz representation Theorem, there exist a σ -algebra $\mathcal{L}(\mathbb{R}^N)$ (containing the Borel σ -algebra $\mathcal{B}(\mathbb{R}^N)$) and a measure λ_N on $\mathcal{L}(\mathbb{R}^N)$ such that $\lambda_N(K) < \infty$ for every compact set $K \subset \mathbb{R}^N$ and

$$L(f) = \int_{\mathbb{R}^N} f(x) dx = \int_{\mathbb{R}^N} f d\lambda_N$$

for every $f \in \mathcal{C}_c(\mathbb{R}^N)$.

Step 2. The conclusion of Theorem 14.1 also proves that for any $E \in \mathcal{L}(\mathbb{R}^N)$

$$\lambda_N(E) = \inf\{\lambda_N(V) : E \subset V, V \text{ open}\} \quad (14.12)$$

and any $E \in \mathcal{L}(\mathbb{R}^N)$ with $\lambda_N(E) < \infty$

$$\lambda_N(E) = \sup\{\lambda_N(K) : K \subset E, K \text{ compact}\}.$$

Moreover, from (14.3), for every $E \in \mathcal{L}(\mathbb{R}^N)$ and any $\varepsilon > 0$, there exists a closed set C and an open set V such that $C \subset E \subset V$ and $\lambda_N(V \setminus C) < \varepsilon$. Hence, for each $n \geq 1$, one can find a closed set C_n and an open set V_n such that

$$C_n \subset E \subset V_n, \text{ and } \lambda_N(V_n \setminus C_n) < 1/n.$$

Define

$$A := \bigcup_{n \geq 1} C_n \text{ and } B := \bigcap_{n \geq 1} V_n.$$

Then A is a F_σ set, B is a G_δ set, $A \subset E \subset B$ and

$$\lambda_N(B \setminus A) \leq \lambda_N(V_n \setminus C_n) < 1/n \rightarrow 0 \quad \text{as } n \rightarrow +\infty.$$

Conversely, assume that $E \subset \mathbb{R}^N$ is such that $A \subset E \subset B$ and $\lambda_N(B \setminus A) = 0$ for some F_σ set A and some G_δ set B . Then $E = A \cup (E \setminus A)$ where $A \in \mathcal{B}(\mathbb{R}^N)$ and $E \setminus A \subset B \setminus A$ with $B \setminus A \in \mathcal{B}(\mathbb{R}^N)$ and $\lambda_N(B \setminus A) = 0$. Since the measure λ_N is complete, we deduce that $E \setminus A \in \mathcal{L}(\mathbb{R}^N)$ and finally that $E \in \mathcal{L}(\mathbb{R}^N)$.

Step 3. Let us prove that for every open cartesian product

$$Q := \prod_{i=1}^N (a_i, b_i)$$

(for $a_i < b_i$ and $i \in \{1, \dots, N\}$), then

$$\lambda_N(Q) = \prod_{i=1}^N (b_i - a_i). \quad (14.13)$$

Define for each $n \geq 1$ and each $x \in \mathbb{R}^N$,

$$f_n(x) := \prod_{i=1}^N \varphi_n^{a_i, b_i}(x_i),$$

where

$$\varphi_n^{a_i, b_i}(x_i) := \begin{cases} 0 & \text{if } x_i \notin [a_i, b_i], \\ 1 & \text{if } x_i \in [a_i + \frac{1}{n}, b_i - \frac{1}{n}], \\ n(x_i - a_i) & \text{if } x_i \in [a_i, a_i + \frac{1}{n}], \\ -n(x_i - b_i) & \text{if } x_i \in [b_i - \frac{1}{n}, b_i]. \end{cases}$$

Then $f_n \in \mathcal{C}_c(\mathbb{R}^N)$ for all $n \geq 1$ and $\mathbf{1}_{\overline{Q_n}} \leq f_n \leq \mathbf{1}_Q$ where

$$\overline{Q_n} := \prod_{i=1}^N [a_i + 1/n, b_i - 1/n]$$

is a closed cube. Hence integrating the previous chain of inequalities with respect to the Lebesgue measure λ_N leads to

$$\lambda_N(\overline{Q_n}) \leq \int_{\mathbb{R}^N} f_n d\lambda_N = \int_{\mathbb{R}^N} f_n(x) dx \leq \lambda_N(Q). \quad (14.14)$$

Since $\overline{Q_n}$ is an increasing sequence of (Lebesgue) measurable sets whose union is $Q = \prod_{i=1}^N (a_i, b_i)$, we deduce from the properties of positive measures that $\lambda_N(\overline{Q_n}) \rightarrow \lambda_N(Q)$ as $n \rightarrow +\infty$. Next by construction of f_n , its Riemann integral can be exactly computed as

$$\int_{\mathbb{R}^N} f_n(x) dx = \int_Q f_n(x) dx = \prod_{i=1}^N \int_{a_i}^{b_i} \varphi_n^{a_i, b_i}(x_i) dx_i = \prod_{i=1}^N \left(b_i - a_i - \frac{1}{n} \right).$$

Taking the limit in 14.14 as $n \rightarrow \infty$ yields formula 14.13.

Step 4. Let us prove that for any $i \in \{1, \dots, N\}$ and any $a \in \mathbb{R}$,

$$\lambda_N(\{x_i = a\}) = 0. \quad (14.15)$$

Let us assume without loss of generality that $i = 1$ and $a = 0$. Then for each $k \geq 1$, we have

$$\{x_1 = 0\} \cap Q(0, n) \subset (-1/k, 1/k) \times \prod_{i=2}^N (-n, n)$$

where $Q(0, n) = \prod_{i=1}^N (-n, n)$ is the open cube of center 0 and side length $2n$. Using (14.13), we infer that

$$\lambda_N(\{x_1 = 0\} \cap Q(0, n)) \leq \frac{2}{k} (2n)^{N-1}.$$

For fixed $n \geq 1$, letting $k \rightarrow +\infty$ implies that

$$\lambda_N(\{x_1 = 0\} \cap Q(0, n)) = 0.$$

Then taking the limit as $n \rightarrow \infty$ and using the properties of positive measures yields $\lambda_N(\{x_1 = 0\}) = 0$.

Step 5. Since

$$\prod_{i=1}^N [a_i, b_i] \setminus \prod_{i=1}^N (a_i, b_i)$$

is made of finitely many subsets of hyperplane of the form $\{x_i = a\}$, we deduce from Step 4 that

$$\lambda_N \left(\prod_{i=1}^N [a_i, b_i] \setminus \prod_{i=1}^N (a_i, b_i) \right) = 0$$

so that

$$\lambda_N \left(\prod_{i=1}^N [a_i, b_i] \right) = \lambda_N \left(\prod_{i=1}^N (a_i, b_i) \right) = \prod_{i=1}^N (b_i - a_i).$$

In particular, taking $a_i = 0$ and $b_i = 1$ for all $i \in \{1, \dots, N\}$ leads to $\lambda_N([0, 1]^N) = 1$.

Step 6. Let us prove that λ_N is translation invariant. Let $x \in \mathbb{R}^N$ and $V \subset \mathbb{R}^N$ an open set. Since the translation $\tau_x : y \in \mathbb{R}^N \mapsto x + y$ is a homeomorphism (with inverse $(\tau_x)^{-1} = \tau_{-x}$), then $x + V$ is open, and by definition of λ_N on open sets (see (14.4)) we have

$$\lambda_N(x + V) = \sup \left\{ \int_{\mathbb{R}^N} f(y) dy : f \in \mathcal{C}_c(\mathbb{R}^N; [0, 1]), \text{supp}(f) \subset x + V \right\}.$$

By changing variables in the Riemann integral we obtain that

$$\begin{aligned} \lambda_N(x + V) &= \sup \left\{ \int_{\mathbb{R}^N} g(x + y) dy : g \in \mathcal{C}_c(\mathbb{R}^N; [0, 1]), \text{supp}(g) \subset V \right\} \\ &= \sup \left\{ \int_{\mathbb{R}^N} g(z) dz : g \in \mathcal{C}_c(\mathbb{R}^N; [0, 1]), \text{supp}(g) \subset V \right\} \\ &= \lambda_N(V). \end{aligned}$$

Next from the above mentioned properties of the translation, if $E \in \mathcal{B}(\mathbb{R}^N)$, then $x + E \in \mathcal{B}(\mathbb{R}^N)$. Applying the outer regularity (14.12) of the Lebesgue measure, we get that

$$\begin{aligned} \lambda_N(x + E) &= \inf \{ \lambda_N(V) : x + E \subset V, V \text{ open} \} \\ &= \inf \{ \lambda_N(-x + V) : E \subset -x + V, V \text{ open} \} \\ &= \inf \{ \lambda_N(U) : E \subset U, U \text{ open} \} \\ &= \lambda_N(E). \end{aligned}$$

Finally, from step 2, we know that if $E \in \mathcal{L}(\mathbb{R}^N)$ then there exists a F_σ set A , and a G_δ set B such that $A \subset E \subset B$ and $\lambda_N(B \setminus A) = 0$. Then

$$x + A \subset x + E \subset x + B,$$

where $x + A$ is a F_σ and $x + B$ is a G_δ . Moreover, since

$$(x + B) \setminus (x + A) = x + (B \setminus A)$$

and $B \setminus A \in \mathcal{B}(\mathbb{R}^N)$, then

$$\lambda_N((x + B) \setminus (x + A)) = \lambda_N(x + (B \setminus A)) = \lambda_N(B \setminus A) = 0.$$

Consequently, $x + E \in \mathcal{L}(\mathbb{R}^N)$ and

$$\lambda_N(x + E) = \lambda_N(x + A) = \lambda_N(A) = \lambda_N(E).$$

Step 7. It remains to prove uniqueness. Let ν be a translation invariant Radon measure on $(\mathbb{R}^N, \mathcal{B}(\mathbb{R}^N))$ such that $\nu([0, 1]^N) = 1$. We claim that $\nu = \lambda_N$ on $\mathcal{B}(\mathbb{R}^N)$.

We first prove that for every $i \in \{1, \dots, N\}$ and every $a \in \mathbb{R}$,

$$\nu(\{x_i = a\}) = 0.$$

By translation invariance it is enough to consider the case $i = 1$ and $a = 0$. Let $e_1 := (1, 0, \dots, 0)$ and, for $m \geq 1$, set

$$H_m := \{0\} \times [-m, m]^{N-1}.$$

If $F \subset \mathbb{Q} \cap [0, 1]$ is finite, then the translates $(qe_1 + H_m)_{q \in F}$ are pairwise disjoint and all contained in the compact set $[0, 1] \times [-m, m]^{N-1}$. Hence

$$|F| \nu(H_m) = \sum_{q \in F} \nu(qe_1 + H_m) = \nu\left(\bigcup_{q \in F} (qe_1 + H_m)\right) \leq \nu([0, 1] \times [-m, m]^{N-1}) < \infty.$$

Since F can have arbitrarily large cardinality, we obtain $\nu(H_m) = 0$. Because

$$\{x_1 = 0\} = \bigcup_{m=1}^{\infty} H_m,$$

continuity from below yields $\nu(\{x_1 = 0\}) = 0$, and the general statement follows by translation invariance.

As a consequence, the boundary of every axis-parallel rectangle has ν -measure 0. In particular,

$$\nu([0, 1]^N) = \nu([0, 1]^N) = 1.$$

Fix $n \geq 1$. The half-open cube $[0, 1]^N$ is the disjoint union of the n^N translates

$$\frac{k}{n} + [0, 1/n)^N, \quad k \in \{0, \dots, n-1\}^N.$$

By translation invariance,

$$1 = \nu([0, 1]^N) = n^N \nu([0, 1/n)^N),$$

so that

$$\nu([0, 1/n)^N) = n^{-N}.$$

Therefore, if

$$R = \prod_{i=1}^N [a_i, b_i)$$

has rational endpoints, then after choosing n large enough so that $a_i, b_i \in n^{-1}\mathbb{Z}$ for every i , the rectangle R is a finite disjoint union of translates of $[0, 1/n)^N$. Hence

$$\nu(R) = \prod_{i=1}^N (b_i - a_i).$$

By Steps 4 and 5, the same formula holds for λ_N . Thus ν and λ_N coincide on every half-open rectangle with rational endpoints.

For every integer $m \geq 1$, we set

$$Q_m := [-m, m)^N$$

and denote by $\mathcal{P}_m$ the family of all half-open rectangles with rational endpoints contained in Q_m . This is a π -system on Q_m , it generates $\mathcal{B}(Q_m)$, and the restrictions of ν and λ_N to $\mathcal{B}(Q_m)$ are finite measures which coincide on $\mathcal{P}_m$. Corollary 3.12 therefore implies that

$$\nu(E) = \lambda_N(E)$$

for every Borel set $E \subset Q_m$. Finally, if $E \in \mathcal{B}(\mathbb{R}^N)$, then $E \cap Q_m \uparrow E$ as $m \rightarrow \infty$, so continuity from below gives

$$\nu(E) = \lim_{m \rightarrow \infty} \nu(E \cap Q_m) = \lim_{m \rightarrow \infty} \lambda_N(E \cap Q_m) = \lambda_N(E).$$

This proves the uniqueness of the Borel Lebesgue measure. Since Step 2 shows that $\mathcal{L}(\mathbb{R}^N)$ is exactly the completion of $\mathcal{B}(\mathbb{R}^N)$ with respect to λ_N , the complete extension is unique as well. $\square$

14.4 Complement: measure-theoretic extension of the multidimensional change-of-variables formula

We now complement Theorem 13.21 by extending the formula from $\mathcal{C}_c(V)$ to nonnegative Borel functions and then to all integrable functions. The measure introduced below is the multidimensional analogue of the transport measure used in Lemma 13.13 in the one-dimensional proof.

Lemma 14.5 (Transport Radon measure associated with a diffeomorphism). *Let $d \geq 1$, let U and V be open sets of $\mathbb{R}^d$, and let $\varphi: U \rightarrow V$ be a diffeomorphism of class C^1 . For every E in $\mathcal{B}(V)$, define*

$$\nu(E) := \int_{\varphi^{-1}(E)} |\text{Jac}\varphi(x)| \, dx.$$

Then ν is a positive Radon measure on V . Moreover, for every nonnegative Borel function g on V ,

$$\int_V g \, d\nu = \int_U g(\varphi(x)) |\text{Jac}\varphi(x)| \, dx.$$

Proof. For every Borel set $E \subset V$, the set $\varphi^{-1}(E)$ is Borel in U , because φ is continuous. Hence $\nu(E)$ is well defined in $[0, +\infty]$. If $(E_n)_{n \geq 1}$ is a pairwise disjoint sequence in $\mathcal{B}(V)$, then the sets $\varphi^{-1}(E_n)$ are pairwise disjoint and

$$\mathbf{1}_{\varphi^{-1}(\cup_{n \geq 1} E_n)} |\text{Jac}\varphi| = \sum_{n=1}^{\infty} \mathbf{1}_{\varphi^{-1}(E_n)} |\text{Jac}\varphi|.$$

Corollary 7.4 therefore gives

$$\nu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \nu(E_n),$$

so ν is a positive Borel measure on V .

Let K be a compact subset of V . Since φ is a homeomorphism, $\varphi^{-1}(K)$ is a compact subset of U . The function $|\text{Jac}\varphi|$ is continuous, hence bounded on $\varphi^{-1}(K)$. Therefore

$$\nu(K) = \int_{\varphi^{-1}(K)} |\text{Jac}\varphi(x)| \, dx < +\infty,$$

because compact subsets of $\mathbb{R}^d$ have finite Lebesgue measure. Thus ν is a positive Radon measure in the sense of Definition 2.8.

The integral identity holds first for indicator functions by the definition of ν . By finite linearity it holds for nonnegative simple Borel functions. For a general nonnegative Borel function g , choose simple functions $g_n \uparrow g$ and apply the monotone convergence theorem, Theorem 7.1, to both sides. $\square$

Corollary 14.6 (Change of variables for Radon measures). *Let $d \geq 1$, let U and V be open sets of $\mathbb{R}^d$, and let $\varphi: U \rightarrow V$ be a diffeomorphism of class C^1 . Then, for every nonnegative Borel function f on V ,*

$$\int_V f(y) \, dy = \int_U f(\varphi(x)) |\text{Jac}\varphi(x)| \, dx.$$

The same identity also holds for every f in $\mathcal{L}^1(V)$.

Proof. Let ν be the measure from Lemma 14.5. For every g in $\mathcal{C}_c(V)$, that lemma gives

$$\int_V g \, d\nu = \int_U g(\varphi(x)) |\text{Jac}\varphi(x)| \, dx.$$

By Theorem 13.21, the right-hand side is equal to

$$\int_V g(y) \, dy.$$

Thus ν and λ_d define the same positive linear functional on $\mathcal{C}_c(V)$. Since both measures are positive Radon measures on V , Theorem 14.2 gives the outer and inner regularity assumptions required in the uniqueness statement of the Riesz representation theorem, Theorem 14.1. Hence $\nu = \lambda_d$ on $\mathcal{B}(V)$. For every nonnegative Borel function f on V , Lemma 14.5 then gives

$$\int_V f(y) \, dy = \int_V f \, d\nu = \int_U f(\varphi(x)) |\text{Jac}\varphi(x)| \, dx.$$

If f belongs to $\mathcal{L}^1(V)$, apply this identity to f^+ and f^- . □

Corollary 14.7 (Linear change of variables). *Let $A: \mathbb{R}^d \rightarrow \mathbb{R}^d$ be an invertible linear map. Then, for every nonnegative Borel function f on $\mathbb{R}^d$,*

$$\int_{\mathbb{R}^d} f(y) \, dy = \int_{\mathbb{R}^d} f(Ax) |\det A| \, dx.$$

The same identity also holds for every f in $\mathcal{L}^1(\mathbb{R}^d)$.

Proof. Apply Corollary 14.6 to the C^1 diffeomorphism $\varphi = A$. □

Example 14.8 (Volume of an ellipsoid). Let A be an invertible linear map on $\mathbb{R}^d$, let

$$B := \{x \in \mathbb{R}^d : |x| < 1\}, \quad E := A(B).$$

Then E is an ellipsoid. Applying Corollary 14.7 to the integrable function $\mathbf{1}_E$, we obtain

$$\begin{aligned} \lambda_d(E) &= \int_{\mathbb{R}^d} \mathbf{1}_E(y) \, d\lambda_d(y) \\ &= \int_{\mathbb{R}^d} \mathbf{1}_E(Ax) |\det A| \, d\lambda_d(x) \\ &= |\det A| \int_B 1 \, d\lambda_d(x) \\ &= |\det A| \lambda_d(B). \end{aligned}$$

In particular, if Q is a symmetric positive definite matrix and

$$E_Q := \{x \in \mathbb{R}^d : x \cdot Qx < 1\},$$

then there exists an invertible matrix S such that $Q = S^T S$. The set E_Q is the preceding ellipsoid with

$$A = S^{-1}, \quad E_Q = S^{-1}(B),$$

and therefore

$$\lambda_d(E_Q) = \frac{\lambda_d(B)}{|\det S|} = \frac{\lambda_d(B)}{\sqrt{\det Q}}.$$

For example, in dimension 2, the ellipse

$$\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} < 1$$

has area πab . In dimension 3, the ellipsoid

$$\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} + \frac{x_3^2}{c^2} < 1$$

has volume $4\pi abc/3$.

14.5 Bounded Radon measures

So far, most measures in the notes have been positive. We now use the signed-measure formalism from Section 2.6, in particular the total variation measure, in the topological setting of Radon measures. We first recall that $\mathcal{C}_0(\Omega)$ is defined, for any open subset $\Omega \subset \mathbb{R}^N$, as the closure of the space $\mathcal{C}_c(\Omega)$ for the uniform topology. Its characterization by functions vanishing at infinity will be given later in Proposition 16.43 of the appendix.

Definition 14.9 (Bounded signed Radon measures). A finite signed Borel measure λ on Ω is called a bounded signed Radon measure if its total variation $|\lambda|$ is a positive Radon measure and

$$|\lambda|(\Omega) < +\infty.$$

We denote by $\mathcal{M}(\Omega)$ the vector space of bounded signed Radon measures on Ω , endowed with the norm

$$\|\lambda\|_{\mathcal{M}(\Omega)} := |\lambda|(\Omega).$$

The next result identifies $\mathcal{M}(\Omega)$ with the dual space of $\mathcal{C}_0(\Omega)$. The equality between the operator norm and the total variation is the reason why the total variation is the natural norm on signed Radon measures.

Theorem 14.10 (Riesz–Markov representation theorem for bounded Radon measures). For every continuous linear functional $L \in \mathcal{C}_0(\Omega)'$, there exists a unique bounded signed Radon measure $\lambda \in \mathcal{M}(\Omega)$ such that

$$L(u) = \int_{\Omega} u \, d\lambda$$

for every $u \in \mathcal{C}_0(\Omega)$. Moreover,

$$\|L\|_{\mathcal{C}_0(\Omega)'} = |\lambda|(\Omega) = \|\lambda\|_{\mathcal{M}(\Omega)}.$$

Proof. **Step 1. Construction of the positive and negative parts of the functional.** We construct two positive continuous linear forms L^+ and L^- on $\mathcal{C}_0(\Omega)$ such that $L = L^+ - L^-$. Let

$$\mathcal{C}^+ := \{u \in \mathcal{C}_0(\Omega) : u \geq 0\}.$$

For $u \in \mathcal{C}^+$, define

$$L^+(u) := \sup\{L(v) : v \in \mathcal{C}^+, 0 \leq v \leq u\}.$$

This number is finite, and

$$0 \leq L^+(u) \leq \|L\| \|u\|_{\infty}.$$

We first prove that L^+ is positively homogeneous on $\mathcal{C}^+$. If $a \geq 0$ and $u \in \mathcal{C}^+$, then, for $a > 0$, the competitors $0 \leq v \leq au$ are exactly the functions $v = aw$ with $0 \leq w \leq u$, and the case $a = 0$ is immediate. Hence

$$L^+(au) = aL^+(u).$$

We next prove additivity on $\mathcal{C}^+$. Let u_1, u_2 be in $\mathcal{C}^+$. If $0 \leq v_i \leq u_i$ for $i = 1, 2$, then $0 \leq v_1 + v_2 \leq u_1 + u_2$, and therefore

$$L^+(u_1 + u_2) \geq L(v_1 + v_2) = L(v_1) + L(v_2).$$

Taking the supremum in v_1 and v_2 gives

$$L^+(u_1 + u_2) \geq L^+(u_1) + L^+(u_2).$$

Conversely, let v be in $\mathcal{C}^+$ and satisfy $0 \leq v \leq u_1 + u_2$. Set

$$v_1 := \min(u_1, v), \quad v_2 := v - v_1 = \max(v - u_1, 0).$$

Then v_1, v_2 belong to $\mathcal{C}^+$, with $v_1 \leq u_1$ and $v_2 \leq u_2$, and $v = v_1 + v_2$. Hence

$$L(v) = L(v_1) + L(v_2) \leq L^+(u_1) + L^+(u_2).$$

Taking the supremum in v gives the reverse inequality. Thus L^+ is additive on the positive cone.

We extend L^+ to $\mathcal{C}_0(\Omega)$ as follows. If $u = p - q$ with p, q in $\mathcal{C}^+$, set

$$L^+(u) := L^+(p) - L^+(q).$$

This is well defined. Indeed, if $p - q = p' - q'$ with p, q, p', q' in $\mathcal{C}^+$, then $p + q' = p' + q$, and additivity on $\mathcal{C}^+$ gives

$$L^+(p) + L^+(q') = L^+(p') + L^+(q).$$

The extension is linear by construction, and it is positive because it coincides with the original L^+ on $\mathcal{C}^+$. Moreover, if $u \in \mathcal{C}_0(\Omega)$, then

$$|L^+(u)| \leq L^+(u^+) + L^+(u^-) \leq 2\|L\|\|u\|_\infty,$$

so L^+ is continuous. Finally define

$$L^- := L^+ - L.$$

For $u \in \mathcal{C}^+$, the function u itself is an admissible competitor in the definition of $L^+(u)$, so $L^+(u) \geq L(u)$. Hence L^- is positive. It is continuous as a difference of continuous linear forms, and $L = L^+ - L^-$.

Step 2. Representation by finite positive Radon measures. By applying the positive Riesz representation theorem, Theorem 14.1, to the restrictions of L^+ and L^- to $\mathcal{C}_c(\Omega)$, we obtain positive Radon measures λ^+ and λ^- such that

$$L^\pm(u) = \int_\Omega u \, d\lambda^\pm$$

for every $u \in \mathcal{C}_c(\Omega)$. We first check that these measures are finite. For every compact set $K \subset \Omega$, Urysohn's lemma, Lemma 16.41, gives $u_K \in \mathcal{C}_c(\Omega)$ with $0 \leq u_K \leq 1$ and $u_K = 1$ on K , hence

$$\lambda^\pm(K) \leq L^\pm(u_K) \leq \|L^\pm\|.$$

By inner regularity on the open set Ω , $\lambda^\pm(\Omega) \leq \|L^\pm\| < +\infty$. Since $\lambda^\pm$ are finite and $\mathcal{C}_c(\Omega)$ is dense in $\mathcal{C}_0(\Omega)$ for the uniform norm, the identities above extend to every $u \in \mathcal{C}_0(\Omega)$. Set

$$\lambda := \lambda^+ - \lambda^-.$$

Then λ is a finite signed Borel measure. Moreover, by the partition formula for total variation, Proposition 2.27, for every Borel set $E \subset \Omega$ we have

$$|\lambda|(E) \leq \lambda^+(E) + \lambda^-(E).$$

In particular,

$$|\lambda|(\Omega) \leq \lambda^+(\Omega) + \lambda^-(\Omega) < +\infty.$$

Thus $|\lambda|$ is finite on compact subsets of Ω , hence it is a positive Radon measure in the sense of Definition 2.8. Therefore λ is a bounded signed Radon measure. Finally,

$$L(u) = \int_\Omega u \, d\lambda$$

for every $u \in \mathcal{C}_0(\Omega)$.

Step 3. Verification of the norm identity. The argument below applies to any bounded signed Radon measure which represents L . For every $u \in \mathcal{C}_0(\Omega)$,

$$|L(u)| \leq \int_\Omega |u| \, d|\lambda| \leq \|u\|_\infty |\lambda|(\Omega),$$

so

$$\|L\|_{\mathcal{C}_0(\Omega)'} \leq |\lambda|(\Omega).$$

For the reverse inequality, let (P, N) be a Hahn decomposition for λ . Since $|\lambda|$ is a finite Radon measure, the regularity theorem for Radon measures, Theorem 14.2, gives, for every $\varepsilon > 0$, compact sets

$$K_+ \subset P, \quad K_- \subset N$$

such that

$$|\lambda|(\Omega \setminus (K_+ \cup K_-)) < \varepsilon.$$

By Urysohn's lemma, Lemma 16.41, there exists $u \in \mathcal{C}_c(\Omega)$ with $\|u\|_\infty \leq 1$, $u = 1$ on K_+ , and $u = -1$ on K_- . Then

$$\begin{aligned} L(u) = \int_{\Omega} u \, d\lambda &\geq |\lambda|(K_+ \cup K_-) - |\lambda|(\Omega \setminus (K_+ \cup K_-)) \\ &\geq |\lambda|(\Omega) - 2\varepsilon. \end{aligned}$$

Taking the supremum over all u with $\|u\|_\infty \leq 1$ and then letting $\varepsilon \rightarrow 0$ gives

$$\|L\|_{\mathcal{C}_0(\Omega)'} \geq |\lambda|(\Omega).$$

Thus the norms are equal.

Step 4. Uniqueness. Finally, if two bounded signed Radon measures represent the same functional, their difference represents the zero functional. Applying the norm identity to this difference gives total variation 0, hence the two measures are equal. $\square$

Example 14.11 (Finite atomic signed Radon measures). Let $x_1, \dots, x_m$ be points of Ω , and let $a_1, \dots, a_m$ be real numbers. We define, for every u in $\mathcal{C}_0(\Omega)$,

$$L(u) := \sum_{j=1}^m a_j u(x_j).$$

Then L is a continuous linear form on $\mathcal{C}_0(\Omega)$, and Theorem 14.10 represents it by an element of $\mathcal{M}(\Omega)$. The corresponding bounded Radon measure is

$$\lambda = \sum_{j=1}^m a_j \delta_{x_j},$$

because

$$\int_{\Omega} u \, d\lambda = \sum_{j=1}^m a_j u(x_j)$$

for every u in $\mathcal{C}_0(\Omega)$. In particular, the Dirac mass δ_{x_0} at a point x_0 in Ω is the bounded Radon measure characterized by

$$\int_{\Omega} u \, d\delta_{x_0} = u(x_0) \quad \text{for every } u \in \mathcal{C}_0(\Omega).$$

14.6 Weak convergences

Because of the Riesz representation theorem for bounded Radon measures, Theorem 14.10, we identify $\mathcal{M}(\Omega)$ with $\mathcal{C}_0(\Omega)'$ and write the action of a bounded signed Radon measure ν on a test function u as

$$\nu(u) = \int_{\Omega} u \, d\nu.$$

Definition 14.12 (Weak-* convergence of bounded Radon measures). *We say that a sequence $(\lambda_n) \subset \mathcal{M}(\Omega)$ converges weak* to λ in $\mathcal{M}(\Omega)$ if for any v in $\mathcal{C}_0(\Omega)$,*

$$\int_{\Omega} v d\lambda_n \rightarrow \int_{\Omega} v d\lambda \quad \text{as } n \rightarrow \infty.$$

A proof of the separability of the space $\mathcal{C}_0(\Omega)$ for the sup norm is given later in Proposition 16.44 of the appendix. Once this is known, the sequential Banach–Alaoglu theorem from the appendix (Theorem 16.58) implies that any bounded sequence of bounded Radon measures has a subsequence which converges weak* to a bounded Radon measure.

Theorem 14.13 (Weak-star sequential compactness theorem for bounded Radon measures). *Let Ω be an open subset of $\mathbb{R}^N$ and let $(\lambda_n)_n$ be a bounded sequence in $\mathcal{M}(\Omega)$. Then there is a subsequence $(\lambda_{n_k})_{k \geq 1}$ converging weak* to a bounded Radon measure λ .*

Proof. By Proposition 16.44 of the appendix, the Banach space $\mathcal{C}_0(\Omega)$ is separable for the uniform norm. The sequential Banach–Alaoglu theorem from the appendix (Theorem 16.58) therefore applies to the dual space $(\mathcal{C}_0(\Omega))'$ and yields a weak* convergent subsequence. By the Riesz representation theorem for bounded Radon measures (Theorem 14.10), $\mathcal{M}(\Omega)$ is isometrically identified with $(\mathcal{C}_0(\Omega))'$. Transporting the weak* convergence through this identification gives the conclusion. $\square$

If μ is a positive Radon measure, we observe that the space $L^1(\Omega, \mu)$ can be injected into $\mathcal{M}(\Omega)$ through the map

$$u \in L^1(\Omega, \mu) \mapsto Tu \in \mathcal{M}(\Omega),$$

with

$$Tu : v \in \mathcal{C}_0(\Omega) \mapsto \int_{\Omega} uv d\mu.$$

As a consequence, if (u_n) is a uniformly bounded sequence in $L^1(\Omega, \mu)$, then one can extract a subsequence which converges weak* in $\mathcal{M}(\Omega)$ to a bounded Radon measure, i.e., there exist $(u_{n_k}) \subset (u_n)$ and $\lambda \in \mathcal{M}(\Omega)$ such that for every $v \in \mathcal{C}_0(\Omega)$,

$$\int_{\Omega} u_{n_k} v d\mu \rightarrow \int_{\Omega} v d\lambda \quad \text{as } k \rightarrow +\infty.$$

However, one may wonder if the limit $\lambda \in \mathcal{M}(\Omega)$ is also of the form Tu , with u in $L^1(\Omega, \mu)$. This leads to the following notion, which is induced by the embedding of $L^1(\Omega, \mu)$ into $\mathcal{M}(\Omega)$.

Definition 14.14 (Weak-* convergence in $\mathcal{C}_0(\Omega)'$). *Let μ be a positive Radon measure. We say that a sequence $(u_n)_n$ in $L^1(\Omega, \mu)$ converges to u against $\mathcal{C}_0(\Omega)$ test functions if for every v in $\mathcal{C}_0(\Omega)$,*

$$\int_{\Omega} u_n v d\mu \rightarrow \int_{\Omega} u v d\mu \quad \text{as } n \rightarrow \infty.$$

It is straightforward that strong convergence in $L^1(\Omega, \mu)$ implies convergence against $\mathcal{C}_0(\Omega)$ test functions. The genuine weak topology of the Banach space $L^1(\Omega, \mu)$ will be introduced later, in Section 15.7, once the dual of $L^1(\Omega, \mu)$ has been identified with $L^\infty(\Omega, \mu)$. On finite-measure spaces, the corresponding compactness criterion is the compactness part of the Dunford–Pettis theorem, which will also be stated there.

Chapter 15

Lebesgue spaces $L^p(\mu)$

We complete the measure-theoretic core with the scale of Lebesgue spaces L^p . After the basic definitions, the chapter develops completeness, density, and separability, and then moves to several standard applications such as convolution, duality, and weak convergence. We now consider the more general case of the Lebesgue spaces L^p of all measurable functions whose absolute value raised to the p -th power has finite integral. Below μ is a measure over a measurable space $(\Omega, \mathcal{A})$. Unless explicitly stated otherwise, the first definitions and abstract properties are given in this general framework. Later, for density, duality, and weak convergence, we will specialize to positive Radon measures on open subsets of $\mathbb{R}^N$, and in the section on convolution and mollifiers we will further specialize to the Lebesgue measure.

15.1 First definitions and properties

Definition 15.1 (Lebesgue spaces L^p). *Let $1 \leq p < \infty$. We define*

$$\mathcal{L}^p(\Omega, \mu) := \left\{ u : \Omega \rightarrow \mathbb{R} \text{ measurable} : \|u\|_{\mathcal{L}^p(\Omega, \mu)} := \left(\int_{\Omega} |u|^p d\mu \right)^{1/p} < +\infty \right\},$$

and

$$\mathcal{L}^\infty(\Omega, \mu) := \left\{ u : \Omega \rightarrow \mathbb{R} \text{ measurable} : \|u\|_{\mathcal{L}^\infty(\Omega, \mu)} := \operatorname{ess\,sup}_{x \in \Omega} |u(x)| < +\infty \right\},$$

where

$$\operatorname{ess\,sup}_{x \in \Omega} |u(x)| := \inf \{ C \in \mathbb{R} : |u(x)| \leq C \text{ } \mu\text{-a.e. in } \Omega \}.$$

There are two points to keep in mind after this definition.

First, the sequence spaces ℓ^p are examples of $\mathcal{L}^p$ spaces. Indeed, if $\Omega = \mathbb{N}$ and if μ is the counting measure on $\mathbb{N}$, then integration is summation and the condition above becomes

$$\sum_{k \in \mathbb{N}} |u(k)|^p < +\infty.$$

Second, one has to distinguish functions from equivalence classes of functions. On $\mathcal{L}^p(\Omega, \mu)$ itself, the map

$$u \mapsto \|u\|_{\mathcal{L}^p(\Omega, \mu)}$$

is only a seminorm: it can vanish for a function which is not identically zero, but only zero almost everywhere. Therefore we identify functions which are equal almost everywhere, exactly as in the construction of L^1 .

More precisely, for measurable functions $u, v : \Omega \rightarrow [-\infty, +\infty]$, we write $u \sim v$ when

$$u(x) = v(x) \quad \text{for } \mu\text{-a.e. } x \in \Omega.$$

This is an equivalence relation. The space $L^p(\Omega, \mu)$ is defined as the quotient of $\mathcal{L}^p(\Omega, \mu)$ by this relation. If $u \in \mathcal{L}^p(\Omega, \mu)$, we denote by $[u]$ its equivalence class.

From now on, we use the standard convention of writing u instead of $[u]$. This convention is harmless: by Lemma 8.10, if $\|u\|_{\mathcal{L}^p(\Omega, \mu)} = 0$, then $u = 0$ μ -a.e. Thus the following definition depends only on the equivalence class and not on the chosen representative.

Definition 15.2 (Norms in L^p). Let $u \in L^p(\Omega, \mu)$, we denote

$$\begin{aligned} \|u\|_p &:= \left(\int_{\Omega} |u|^p d\mu \right)^{1/p} \quad \text{if } 1 \leq p < \infty, \\ \|u\|_{\infty} &:= \operatorname{ess\,sup}_{x \in \Omega} |u(x)| \quad \text{if } p = \infty. \end{aligned}$$

Example 15.3 (Power singularities in L^p). We consider the Lebesgue measure on $B(0, 1) \subset \mathbb{R}^N$ and, for $\alpha > 0$, the function

$$u_{\alpha}(x) := |x|^{-\alpha}.$$

Let $1 \leq p < \infty$. By the polar-coordinate formula,

$$\begin{aligned} \int_{B(0,1)} |u_{\alpha}(x)|^p dx &= \int_{B(0,1)} |x|^{-\alpha p} dx \\ &= |\mathbb{S}^{N-1}| \int_0^1 r^{N-1-\alpha p} dr. \end{aligned}$$

The last one-dimensional integral is finite if and only if

$$N - 1 - \alpha p > -1,$$

that is, if and only if $\alpha p < N$. Thus

$$u_{\alpha} \in L^p(B(0, 1)) \quad \text{if and only if} \quad \alpha p < N.$$

For $p = \infty$, the same function is not in $L^{\infty}(B(0, 1))$, since it is unbounded near the origin.

Due to this identification, the Lebesgue spaces $L^p(\Omega, \mu)$ are normed spaces. Before proving this property, we state two important inequalities which will be instrumental in the sequel.

Proposition 15.4 (Hölder's inequality). Let $1 \leq p \leq \infty$ and $p' \geq 1$ its conjugate exponent defined by $1/p + 1/p' = 1$ (by convention $p' = 1$ if $p = \infty$, and $p' = \infty$ if $p = 1$). If $u \in L^p(\Omega, \mu)$ and $v \in L^{p'}(\Omega, \mu)$ then $uv \in L^1(\Omega, \mu)$ and

$$\|uv\|_1 \leq \|u\|_p \|v\|_{p'}.$$

Proof. Since the logarithm is concave on $\mathbb{R}_+^*$, for $a, b > 0$ and $1 \leq p, p' < \infty$ with $1/p + 1/p' = 1$, we have

$$\log(ab) = \log(a) + \log(b) = \frac{1}{p} \log(a^p) + \frac{1}{p'} \log(b^{p'}) \leq \log\left(\frac{1}{p} a^p + \frac{1}{p'} b^{p'}\right).$$

Taking the exponential we obtain Young's inequality:

$$ab \leq \frac{1}{p}a^p + \frac{1}{p'}b^{p'}.$$

Using this with $a = \lambda u$ and $b = \lambda^{-1}v$ and integrating over Ω yields

$$\int_{\Omega} |uv| d\mu \leq \frac{1}{p} \lambda^p \|u\|_p^p + \frac{1}{p'} \lambda^{-p'} \|v\|_{p'}^{p'}.$$

The last term is minimized by choosing $\lambda = \|v\|_{p'}^{1/p} \|u\|_p^{(1-p)/p}$.

If $p = \infty$ or $p' = \infty$, the inequality is immediate. □

Remark 15.5 (Finite-measure embeddings). From Hölder's inequality, it follows that if $\mu(\Omega) < \infty$ and $1 \leq q \leq p \leq \infty$ then

$$L^\infty(\Omega, \mu) \subset L^p(\Omega, \mu) \subset L^q(\Omega, \mu) \subset L^1(\Omega, \mu).$$

However, if $\mu(\Omega) = \infty$, nothing can be said in general.

Proposition 15.6 (Minkowski inequality). *For $1 \leq p \leq \infty$, and for any $u, v \in L^p(\Omega, \mu)$, we have*

$$\|u + v\|_p \leq \|u\|_p + \|v\|_p.$$

Proof. We start with the case where $p = \infty$. By definition of the essential supremum, $|u(x)| \leq \|u\|_\infty$ and $|v(x)| \leq \|v\|_\infty$ for μ -a.e. $x \in \Omega$, hence

$$|u(x) + v(x)| \leq |u(x)| + |v(x)| \leq \|u\|_\infty + \|v\|_\infty,$$

for μ -a.e. in Ω . Using again the definition of the essential supremum of $u + v$, we get

$$\|u + v\|_\infty \leq \|u\|_\infty + \|v\|_\infty.$$

Similarly, for $p = 1$, we have

$$\|u + v\|_1 = \int_{\Omega} |u + v| d\mu \leq \int_{\Omega} (|u| + |v|) d\mu \leq \|u\|_1 + \|v\|_1.$$

For $1 < p < \infty$,

$$\begin{aligned} \|u + v\|_p^p &= \int_{\Omega} |u + v|^p d\mu = \int_{\Omega} |u + v| |u + v|^{p-1} d\mu \\ &\leq \int_{\Omega} |u| |u + v|^{p-1} d\mu + \int_{\Omega} |v| |u + v|^{p-1} d\mu \\ &\leq (\|u\|_p + \|v\|_p) \left(\int_{\Omega} (|u + v|^{p-1})^{p/(p-1)} d\mu \right)^{(p-1)/p} \end{aligned}$$

by Hölder's inequality. Then it is sufficient to observe that the last quantity is equal to

$$(\|u\|_p + \|v\|_p) \|u + v\|_p^{p-1},$$

and to simplify the inequality by $\|u + v\|_p^{p-1}$. □

We are now in a position to state that the Lebesgue spaces are normed spaces:

Proposition 15.7 (Normed structure of L^p). *For $1 \leq p \leq \infty$, the map $u \in L^p(\Omega, \mu) \mapsto \|u\|_p$ defines a norm over $L^p(\Omega, \mu)$.*

Proof. We already have mentioned that if $\|u\|_p = 0$, then $u = 0$ μ -a.e. in Ω , and thus $u = 0$ in $L^p(\Omega, \mu)$. Clearly, $\|\lambda u\|_p = |\lambda| \|u\|_p$ for every $\lambda \in \mathbb{R}$ and $u \in L^p(\Omega, \mu)$. Finally the triangle inequality is a consequence of the Minkowski inequality. $\square$

Observe that it follows from Theorem 11.18 that if (f_n) is a sequence bounded both in $\mathcal{L}^1(X, \mu)$ and in $\mathcal{L}^p(\Omega, \mu)$ for some $1 < p < \infty$ then the sequence (f_n) is equi-integrable.

15.2 Completeness

We saw in Proposition 15.7 that the Lebesgue spaces $L^p(\Omega, \mu)$ are normed spaces. We will prove now that they are actually Banach spaces as a consequence of the following completeness result.

Theorem 15.8 (Riesz–Fischer theorem for L^p). *For $1 \leq p \leq \infty$, the space $L^p(\Omega, \mu)$ is complete.*

Proof. It is enough to prove that any normally converging series in $L^p(\Omega, \mu)$ is converging in $L^p(\Omega, \mu)$ (see Proposition 16.24). Let $(u_n)_n \subset L^p(\Omega, \mu)$ be such that $\sum_{n=1}^{\infty} \|u_n\|_p < \infty$. Then the sequence of partial sums

$$v_n := \sum_{k=1}^n |u_k|$$

is increasing, and from the Minkowski inequality, for any $n \in \mathbb{N}$,

$$\int_{\Omega} v_n^p d\mu = \|v_n\|_p^p \leq \left(\sum_{k=1}^{\infty} \|u_k\|_p \right)^p < \infty, \quad (15.1)$$

if $p < \infty$, while

$$v_n(x) \leq \sum_{k=1}^{\infty} \|u_k\|_{\infty} \quad (15.2)$$

for μ -a.e. $x \in \Omega$ if $p = \infty$. We deduce from the monotone convergence Theorem that (v_n) converges μ -a.e. in Ω to the function $\sum_{k=1}^{\infty} |u_k|$ which, by (15.1) and (15.2), belongs to $L^p(\Omega, \mu)$. In particular, we deduce that the series $\sum_{k=1}^{\infty} |u_k(x)|$ converges for μ -a.e. $x \in \Omega$, and from the completeness of $\mathbb{R}$, using the converse part of Proposition 16.24, that the series $\sum_{k=1}^{\infty} u_k(x)$ converges as well. Moreover,

$$\sum_{k=1}^{\infty} u_k \in L^p(\Omega, \mu),$$

and for $n \geq m$, we have

$$\left\| \sum_{k=1}^n u_k - \sum_{k=1}^m u_k \right\|_p = \left\| \sum_{k=m+1}^n u_k \right\|_p \leq \sum_{k=m+1}^n \|u_k\|_p$$

where we used again the Minkowski inequality. Letting $n \rightarrow +\infty$, we infer that

$$\left\| \sum_{k=1}^{\infty} u_k - \sum_{k=1}^m u_k \right\|_p \leq \sum_{k=m+1}^{\infty} \|u_k\|_p$$

and since $\sum_{n=1}^{\infty} \|u_n\|_p < \infty$, then $\sum_{k=m+1}^{\infty} \|u_k\|_p \rightarrow 0$ as $m \rightarrow \infty$, and the conclusion follows. $\square$

15.3 Density

Theorem 15.9 (Density of simple functions). *For any $1 \leq p \leq \infty$ the space of all measurable simple functions is dense in $L^p(\Omega, \mu)$.*

Proof. Let u in $L^p(\Omega, \mu)$. Decomposing $u = u_+ - u_-$, we can assume without loss of generality that $u \geq 0$. Let (s_n) be the sequence of simple functions constructed as in the proof of Theorem 6.6. We distinguish two cases.

- If $p = \infty$, then for any $n \geq \|u\|_\infty$, we have $|s_n(x) - u(x)| \leq 2^{-n}$ for μ -a.e. $x \in \Omega$, and thus $\|s_n - u\|_\infty \leq 2^{-n} \rightarrow 0$ as $n \rightarrow +\infty$.
- If $1 \leq p < \infty$, since $s_n(x) \rightarrow u(x)$ as $n \rightarrow +\infty$ for μ -a.e. $x \in \Omega$, we get that $|u(x) - s_n(x)|^p \rightarrow 0$ as $n \rightarrow +\infty$ for μ -a.e. $x \in \Omega$ and

$$|u(x) - s_n(x)|^p \leq 2^p u(x)^p$$

for μ -a.e. $x \in \Omega$. Hence, by the dominated convergence Theorem $\|u - s_n\|_p \rightarrow 0$ as $n \rightarrow +\infty$. □

From now on, we assume that Ω is an open subset of $\mathbb{R}^N$, with $N \geq 1$, and that μ is a positive Radon measure.

Theorem 15.10 (Density of compactly supported continuous functions). *For any $1 \leq p < \infty$ the space $\mathcal{C}_c(\Omega)$ is dense in $L^p(\Omega, \mu)$.*

Proof. Let u in $L^p(\Omega, \mu)$ and let

$$K_n := \{x \in \Omega : \text{dist}(x, \Omega^c) \geq 1/n \text{ and } \|x\|_{\mathbb{R}^N} \leq n\}.$$

Then (K_n) is an increasing sequence of compact sets such that $\cup_n K_n = \Omega$. Let $\varepsilon > 0$. By the dominated convergence Theorem, for n large enough, we have

$$\|u - \mathbf{1}_{K_n} u\|_{L^p(\Omega, \mu)} < \frac{\varepsilon}{2}.$$

Fix such an n and set

$$u_n := \mathbf{1}_{K_n} u.$$

The restricted measure $\mu_n := \mu \llcorner K_n$ is finite because μ is Radon and K_n is compact. By Theorem 15.9, applied to the finite measure space $(K_n, \mathcal{B}(K_n), \mu_n)$, it is enough to approximate u_n in $L^p(K_n, \mu_n)$ by continuous functions with compact support in Ω . By linearity, it therefore suffices to approximate characteristic functions of Borel subsets of K_n .

Let $E \subset K_n$ be a Borel set. Choose $\delta > 0$. By Theorem 14.2, there exist a compact set K and an open set V such that

$$K \subset E \subset V \quad \text{and} \quad \mu(V \setminus K) < \delta.$$

By Urysohn's lemma (Lemma 16.41 in the appendix), there exists a function $v \in \mathcal{C}_c(V; [0, 1])$ such that $v = 1$ on K . Then

$$|v - \mathbf{1}_E|^p \leq \mathbf{1}_{V \setminus K}$$

on Ω , and therefore

$$\|v - \mathbf{1}_E\|_{L^p(\Omega, \mu)}^p \leq \mu(V \setminus K) < \delta.$$

Choosing δ small enough, we approximate u_n arbitrarily well in $L^p(\Omega, \mu)$ by elements of $\mathcal{C}_c(\Omega)$. Since

$$\|u - u_n\|_{L^p(\Omega, \mu)} < \frac{\varepsilon}{2},$$

this proves that $\mathcal{C}_c(\Omega)$ is dense in $L^p(\Omega, \mu)$. $\square$

15.4 Separability

Recalling the definition of separability from Definition 16.37, we have the following two results regarding the separability of the spaces L^p , which distinguish the case where p is finite from the case $p = \infty$.

Corollary 15.11 (Separability of finite-exponent L^p spaces). *For any $1 \leq p < \infty$ the space $L^p(\Omega, \mu)$ is separable.*

Proof. With the notation of the proof of Theorem 15.10, for every n in $\mathbb{N}^*$, define

$$X_n := \{u \in \mathcal{C}_c(\Omega) : \text{supp}(u) \subset K_n\}.$$

Since every compact subset of Ω is contained in K_n for n large enough, we have

$$\mathcal{C}_c(\Omega) = \bigcup_{n=1}^{\infty} X_n.$$

Fix now $n \in \mathbb{N}^*$. The restriction map $u \mapsto u|_{K_n}$ identifies X_n with a subspace of $\mathcal{C}(K_n)$. By Proposition 16.39, the space $\mathcal{C}(K_n)$ is separable for the uniform norm, hence X_n is also separable for the uniform norm. Moreover, if $u, v \in X_n$, then

$$\|u - v\|_{L^p(\Omega, \mu)}^p = \int_{K_n} |u - v|^p d\mu \leq \mu(K_n) \sup_{x \in K_n} |u(x) - v(x)|^p.$$

Therefore uniform convergence on K_n implies convergence in $L^p(\Omega, \mu)$ on X_n , so X_n is separable for the topology induced by $L^p(\Omega, \mu)$. Since $\mathcal{C}_c(\Omega)$ is a countable union of separable subsets of the metric space $L^p(\Omega, \mu)$, it is itself separable for the L^p topology. The conclusion follows from Theorem 15.10. $\square$

The space $L^\infty(\Omega, \mu)$ is more specific to the Radon measure μ . Indeed, if μ is a Dirac mass, then the space $L^\infty(\Omega, \mu)$ can be identified to $\mathbb{R}$ which is therefore separable. However, this property fails in general.

Theorem 15.12 (Nonseparability of L^∞ for Lebesgue measure). *If $\mu = \lambda_N$ is the Lebesgue measure, then $L^\infty(\Omega, \lambda_N)$ is not separable.*

Proof. If $x \in \Omega$ and $R > 0$ are such that $B(x, R) \subset \Omega$, then the family $\{\mathbf{1}_{B(x, r)} : 0 < r < R\}$ is uncountable and if $0 < r \neq r' < R$, then

$$\|\mathbf{1}_{B(x, r)} - \mathbf{1}_{B(x, r')}\|_\infty = 1.$$

The conclusion follows from the non-separability criterion (Proposition 16.40). $\square$

15.5 Convolution

In this section and in the next one, the ambient space is $\mathbb{R}^N$ endowed with the Lebesgue measure λ_N . For simplicity, we write $L^p(\mathbb{R}^N)$ simply as L^p when no confusion is possible.

Definition 15.13 (Convolution). *Let u and $v \in \mathcal{C}_c(\mathbb{R}^N)$, the convolution product $u * v$ is defined by*

$$(u * v)(x) := \int_{\mathbb{R}^N} u(x - y)v(y) dy$$

for all $x \in \mathbb{R}^N$.

The next lemma asserts that $u * v$ belongs to $\mathcal{C}_c(\mathbb{R}^N)$. If one factor is smoother, one can expect more regularity on the convolution product. We recall that for any open set $U \subset \mathbb{R}^N$,

$$\mathcal{D}(U) := \mathcal{C}_c^\infty(U) := \{u \in \mathcal{C}^\infty(U) : \text{supp}(u) \text{ is a compact subset of } U\}.$$

Lemma 15.14 (Support of a convolution). *If $u, v \in \mathcal{C}_c(\mathbb{R}^N)$ then $u * v \in \mathcal{C}_c(\mathbb{R}^N)$ and*

$$\text{supp}(u * v) \subset \overline{\text{supp}(u) + \text{supp}(v)}.$$

Proof. For every $x \in \mathbb{R}^N$, the function $y \mapsto u(x - y)v(y)$ is integrable. Since u is uniformly continuous and v has compact support, the dominated convergence Theorem shows that the function $x \mapsto (u * v)(x)$ is continuous. Moreover, if $x \notin \overline{\text{supp}(u) + \text{supp}(v)}$, then $x - y \notin \text{supp}(u)$ for every y in $\text{supp}(v)$, hence

$$u(x - y)v(y) = 0$$

for every $y \in \mathbb{R}^N$. Therefore $(u * v)(x) = 0$, which proves the announced support inclusion. Since the right-hand side is compact, we conclude that $u * v \in \mathcal{C}_c(\mathbb{R}^N)$. $\square$

Lemma 15.15 (Convolution is well-defined almost everywhere). *If $u \in L^1(\mathbb{R}^N)$ and $v \in \mathcal{C}_c^\infty(\mathbb{R}^N)$, then $u * v \in \mathcal{C}^\infty(\mathbb{R}^N)$ and for any multi-index $\alpha \in \mathbb{N}^N$,*

$$\partial^\alpha(u * v) = u * (\partial^\alpha v).$$

If moreover $u = 0$ a.e. in $\mathbb{R}^N \setminus K$ for some compact set $K \subset \mathbb{R}^N$, then

$$\text{supp}(u * v) \subset K + \text{supp}(v),$$

*so that $u * v \in \mathcal{C}_c^\infty(\mathbb{R}^N)$.*

Proof. It suffices to consider the case $|\alpha| = 1$, the general case then follows by induction. Writing

$$(u * v)(x) = \int_{\mathbb{R}^N} u(y)v(x - y) dy,$$

we infer that

$$\frac{(u * v)(x + h\alpha) - (u * v)(x)}{h} = \int_{\mathbb{R}^N} u(y) \frac{v(x + h\alpha - y) - v(x - y)}{h} dy.$$

By the mean value theorem,

$$\left| \frac{v(x + h\alpha - y) - v(x - y)}{h} \right| \leq \|\partial^\alpha v\|_\infty$$

for all $y \in \mathbb{R}^N$ and all $h \neq 0$. Since $u \in L^1(\mathbb{R}^N)$, the dominated convergence Theorem yields

$$\partial^\alpha(u * v)(x) = \int_{\mathbb{R}^N} u(y) \partial^\alpha v(x - y) dy = (u * (\partial^\alpha v))(x).$$

This proves that $u * v$ is of class $\mathcal{C}^\infty$.

Assume now that $u = 0$ a.e. in $\mathbb{R}^N \setminus K$ with K compact, and let $x \notin K + \text{supp}(v)$. If $y \in K$, then $x - y \notin \text{supp}(v)$, hence $v(x - y) = 0$. If $y \notin K$, then $u(y) = 0$ for a.e. such y . Therefore $u(y)v(x - y) = 0$ for a.e. $y \in \mathbb{R}^N$, which gives $(u * v)(x) = 0$. Thus

$$\text{supp}(u * v) \subset K + \text{supp}(v).$$

Since the right-hand side is compact, $u * v$ belongs to $\mathcal{C}_c^\infty(\mathbb{R}^N)$. □

Lemma 15.16 (Young convolution inequality). *For every $1 \leq p \leq \infty$,*

$$\|u * v\|_p \leq \|u\|_p \|v\|_1. \quad (15.3)$$

Proof. Thanks to Fubini's Theorem

$$\begin{aligned} \int_{\mathbb{R}^N} |(u * v)(x)| dx &= \int_{\mathbb{R}^N} \left| \int_{\mathbb{R}^N} u(x - y) v(y) dy \right| dx \\ &\leq \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} |u(x - y)| |v(y)| dy dx \\ &= \int_{\mathbb{R}^N} \left(\int_{\mathbb{R}^N} |u(x - y)| dx \right) |v(y)| dy \\ &= \|u\|_1 \|v\|_1, \end{aligned}$$

which completes the case $p = 1$.

If now $1 < p < \infty$, thanks to Hölder's inequality and Fubini's Theorem, we get

$$\begin{aligned} \int_{\mathbb{R}^N} |(u * v)(x)|^p dx &= \int_{\mathbb{R}^N} \left| \int_{\mathbb{R}^N} u(x - y) v(y) dy \right|^p dx \\ &\leq \int_{\mathbb{R}^N} \left| \left(\int_{\mathbb{R}^N} |u(x - y)|^p |v(y)| dy \right)^{1/p} \left(\int_{\mathbb{R}^N} |v(y)| dy \right)^{1-1/p} \right|^p dx \\ &= \|v\|_1^{p-1} \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} |u(x - y)|^p |v(y)| dy dx \\ &= \|u\|_p^p \|v\|_1^p. \end{aligned}$$

Finally, if $p = \infty$, we simply write

$$|(u * v)(x)| \leq \int_{\mathbb{R}^N} |u(x - y)| |v(y)| dy \leq \|u\|_\infty \|v\|_1$$

for all $x \in \mathbb{R}^N$. □

Corollary 15.17 (Extension of convolution to $L^p \times L^1$). For $1 \leq p < \infty$, the convolution product $(u, v) \mapsto u * v$ from $\mathcal{C}_c(\mathbb{R}^N) \times \mathcal{C}_c(\mathbb{R}^N)$ to $\mathcal{C}_c(\mathbb{R}^N)$ extends uniquely into a continuous bilinear map from $L^p(\mathbb{R}^N) \times L^1(\mathbb{R}^N)$ to $L^p(\mathbb{R}^N)$. For $p = \infty$, the formula

$$(u * v)(x) := \int_{\mathbb{R}^N} u(x - y)v(y) dy$$

defines a continuous bilinear map from $L^\infty(\mathbb{R}^N) \times L^1(\mathbb{R}^N)$ to $L^\infty(\mathbb{R}^N)$. In all cases one has

$$\|u * v\|_p \leq \|u\|_p \|v\|_1.$$

Proof. For $1 \leq p < \infty$, the existence and uniqueness of the continuous extension follow from the density of $\mathcal{C}_c(\mathbb{R}^N)$ in $L^p(\mathbb{R}^N)$ (see Theorem 15.10) together with the estimate (15.3) proved in Lemma 15.16.

For $p = \infty$, let $u \in L^\infty(\mathbb{R}^N)$ and $v \in L^1(\mathbb{R}^N)$. For every $x \in \mathbb{R}^N$, the function $y \mapsto u(x - y)v(y)$ belongs to $L^1(\mathbb{R}^N)$. The value of the integral does not depend on the chosen representative of u , because translations preserve Lebesgue-negligible sets. Moreover

$$|(u * v)(x)| \leq \|u\|_\infty \|v\|_1.$$

Hence $u * v$ is well defined and belongs to $L^\infty(\mathbb{R}^N)$, with

$$\|u * v\|_\infty \leq \|u\|_\infty \|v\|_1.$$

The bilinearity is immediate from the definition. □

Corollary 15.18 (Extension of convolution with compactly supported smooth functions). Moreover, if $u \in L^p(\mathbb{R}^N)$ is such that $u = 0$ a.e. in $\mathbb{R}^N \setminus K$, for some compact set $K \subset \mathbb{R}^N$, and if $v \in \mathcal{C}_c^\infty(\mathbb{R}^N)$, then $u * v \in \mathcal{C}_c^\infty(\mathbb{R}^N)$ and $\text{supp}(u * v) \subset K + \text{supp}(v)$.

Proof. Since K is compact, it has finite Lebesgue measure. Therefore $u \in L^1(\mathbb{R}^N)$. The conclusion now follows from Lemma 15.15. □

Definition 15.19 (Mollifiers). Let $\rho \in \mathcal{C}_c^\infty(\mathbb{R}^N; [0, +\infty))$ be such that $\text{supp}(\rho) \subset B(0, 1)$ and $\int_{\mathbb{R}^N} \rho(x) dx = 1$. For all $n \in \mathbb{N}$, we set $\rho_n(x) := n^N \rho(nx)$. We say that the sequence $(\rho_n)_{n \in \mathbb{N}}$ is a sequence of mollifiers.

We observe that any sequence of mollifiers $(\rho_n)_{n \in \mathbb{N}}$ satisfies $\int_{\mathbb{R}^N} \rho_n(x) dx = 1$ and $\text{supp}(\rho_n) \subset B(0, \frac{1}{n})$, for all n .

Lemma 15.20 (Uniform approximation by mollifiers). If $u \in \mathcal{C}_c(\mathbb{R}^N)$, then $u * \rho_n$ converges uniformly to u as $n \rightarrow +\infty$.

Proof. Assume that $u \in \mathcal{C}_c(\mathbb{R}^N)$. Then

$$\begin{aligned} |(u * \rho_n)(x) - u(x)| &= \left| \int_{\mathbb{R}^N} (u(x - y) - u(x)) \rho_n(y) dy \right| \\ &\leq \int_{B(0, \frac{1}{n})} |u(x - y) - u(x)| \rho_n(y) dy \\ &\leq \sup\{|u(z_1) - u(z_2)| : z_1, z_2 \in \mathbb{R}^N, |z_1 - z_2| \leq 1/n\} \rightarrow 0 \quad \text{as } n \rightarrow +\infty \end{aligned}$$

since u is uniformly continuous. Hence $u * \rho_n$ converges uniformly to u . □

Lemma 15.21 (L_p approximation by mollifiers). *If $u \in L^p(\mathbb{R}^N)$ for $1 \leq p < \infty$, then $u * \rho_n \rightarrow u$ in $L^p(\mathbb{R}^N)$ as $n \rightarrow +\infty$.*

Proof. We first deal with the case where $u \in \mathcal{C}_c(\mathbb{R}^N)$. Then, we use Hölder's inequality with

$$f_x(y) := (u(x-y) - u(x))(\rho_n(y))^{\frac{1}{p}}$$

and $g(y) := (\rho_n(y))^{\frac{1}{p'}}$, we get

$$|(u * \rho_n)(x) - u(x)| = \left| \int_{\mathbb{R}^N} f_x(y) g(y) dy \right| \leq \|f_x\|_p \|g\|_{p'}.$$

But

$$\|f_x\|_p = \left(\int_{\mathbb{R}^N} |u(x-y) - u(x)|^p \rho_n(y) dy \right)^{\frac{1}{p}} \text{ and } \|g\|_{p'} = \left(\int_{\mathbb{R}^N} \rho_n(y) dy \right)^{\frac{1}{p'}} = 1,$$

so that by Fubini's Theorem, we get

$$\begin{aligned} \int_{\mathbb{R}^N} |(u * \rho_n)(x) - u(x)|^p dx &\leq \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} |u(x-y) - u(x)|^p \rho_n(y) dy dx \\ &= \int_{B(0, \frac{1}{n})} \left(\int_{\mathbb{R}^N} |u(x-y) - u(x)|^p dx \right) \rho_n(y) dy \\ &\leq \sup_{y \in B(0, \frac{1}{n})} \|\tau_y u - u\|_p^p, \end{aligned}$$

where $\tau_y u(x) := u(x-y)$ is the translation of u by y . On $B(0, 1) + \text{supp}(u)$ the uniform convergence implies the convergence in $L^p(\mathbb{R}^N)$, hence $\sup_{y \in B(0, \frac{1}{n})} \|\tau_y u - u\|_p^p$ and thus $\|u * \rho_n - u\|_p$ converge to 0 as $n \rightarrow +\infty$.

The general case $u \in L^p(\mathbb{R}^N)$ follows from the density of $\mathcal{C}_c(\mathbb{R}^N)$ in $L^p(\mathbb{R}^N)$ (Theorem 15.10). More precisely, let $\varepsilon > 0$. Choose $\tilde{u} \in \mathcal{C}_c(\mathbb{R}^N)$ such that

$$\|u - \tilde{u}\|_p < \varepsilon.$$

By the compact-support case proved above, there exists n_0 such that

$$\|\tilde{u} * \rho_n - \tilde{u}\|_p < \varepsilon$$

for every $n \geq n_0$. Then, for $n \geq n_0$,

$$\begin{aligned} \|u * \rho_n - u\|_p &\leq \|(u - \tilde{u}) * \rho_n\|_p + \|\tilde{u} * \rho_n - \tilde{u}\|_p + \|\tilde{u} - u\|_p \\ &\leq \|u - \tilde{u}\|_p \|\rho_n\|_1 + \|\tilde{u} * \rho_n - \tilde{u}\|_p + \|\tilde{u} - u\|_p \\ &< 3\varepsilon, \end{aligned}$$

where we used (15.3) and the fact that $\|\rho_n\|_1 = 1$. This proves that $\|u * \rho_n - u\|_p \rightarrow 0$ as $n \rightarrow +\infty$. $\square$

Remark 15.22 (Continuity of translations in L^p). *Let us observe that proceeding as in the proof above we can prove that for any $1 \leq p < \infty$, for any $u \in L^p(\mathbb{R}^N)$, $\sup_{y \in B(0, \delta)} \|\tau_y u - u\|_p$ goes to 0 as $\delta \rightarrow 0$.*

Corollary 15.23 (Density of smooth compactly supported functions). *For every open set $\Omega \subset \mathbb{R}^N$ and every $1 \leq p < \infty$, the space $\mathcal{C}_c^\infty(\Omega)$ is dense in $L^p(\Omega)$.*

Proof. From Theorem 15.10, for every $u \in L^p(\Omega)$ and every $\varepsilon > 0$, there exist $v \in \mathcal{C}_c(\Omega)$ such that

$$\left(\int_{\Omega} |u - v|^p dx \right)^{1/p} < \frac{\varepsilon}{2}.$$

Extend v by zero outside Ω and denote by $\tilde{v}$ this extension. Then $\tilde{v} \in \mathcal{C}_c(\mathbb{R}^N)$ and $\text{supp}(\tilde{v}) \subset \Omega$. By Lemma 15.20, we can choose $n_0 \in \mathbb{N}$ large enough so that for all $n \geq n_0$,

$$\left(\int_{\mathbb{R}^N} |\tilde{v} - (\tilde{v} * \rho_n)|^p dx \right)^{1/p} < \frac{\varepsilon}{2}.$$

By Corollary 15.18, we have

$$\text{supp}(\tilde{v} * \rho_n) \subset \overline{\text{supp}(\tilde{v}) + \text{supp}(\rho_n)} \subset \overline{\text{supp}(\tilde{v}) + B(0, 1/n)}.$$

Hence, for

$$n > \max \left\{ n_0, \frac{1}{\text{dist}(\text{supp}(\tilde{v}), \mathbb{R}^N \setminus \Omega)} \right\},$$

we have $\text{supp}(\tilde{v} * \rho_n) \subset \Omega$, so that $\tilde{v} * \rho_n$, viewed as a function on Ω , belongs to $\mathcal{C}_c^\infty(\Omega)$ and

$$\left(\int_{\Omega} |u - (\tilde{v} * \rho_n)|^p dx \right)^{1/p} < \varepsilon.$$

□

15.6 Duality in the Lebesgue spaces

In the present chapter, μ is a positive Radon measure on an open subset Ω of $\mathbb{R}^N$.

15.6.1 Uniform convexity and smoothness of the L^p spaces

Proposition 15.24 (Uniform convexity of L^p). *For every $1 < p < \infty$, the space $L^p(\Omega, \mu)$ is uniformly convex.*

The proof is essentially based on the strict convexity of the map $s \mapsto s^p$. More precisely we will rely on the following scalar lemma.

Lemma 15.25 (Key lemma for L^p duality). *For any $\varepsilon > 0$, there exists $\delta > 0$ such that for any s, t in $\mathbb{R}$ with $|s| \leq 1$, $|t| \leq 1$ and $|s - t| \geq 2\varepsilon$, then*

$$\left| \frac{s + t}{2} \right|^p \leq (1 - \delta) \frac{|s|^p + |t|^p}{2}.$$

Proof. Let $\varepsilon > 0$. We first reduce the proof to the normalized case. Let s, t in $\mathbb{R}$ be such that $|s| \leq 1$, $|t| \leq 1$, and $|s - t| \geq 2\varepsilon$. Set

$$A := \left(\frac{|s|^p + |t|^p}{2} \right)^{1/p}.$$

Since $|s - t| > 0$, at least one of the numbers s and t is nonzero, hence $A > 0$. Moreover $A \leq 1$. Define

$$\sigma := \frac{s}{A} \quad \text{and} \quad \tau := \frac{t}{A}.$$

Then

$$\frac{|\sigma|^p + |\tau|^p}{2} = 1 \quad \text{and} \quad |\sigma - \tau| = \frac{|s - t|}{A} \geq |s - t| \geq 2\varepsilon.$$

Therefore it is enough to prove the lemma under the additional normalization

$$\frac{|s|^p + |t|^p}{2} = 1.$$

Consider the set

$$E_\varepsilon := \left\{ (a, b) \in \mathbb{R}^2 : \frac{|a|^p + |b|^p}{2} = 1 \text{ and } |a - b| \geq 2\varepsilon \right\}.$$

It is a closed and bounded subset of $\mathbb{R}^2$, hence it is compact. The function

$$F(a, b) := 1 - \left| \frac{a + b}{2} \right|^p$$

is continuous on E_ε . It is thus enough to show that F is everywhere positive on E_ε .

Assume by contradiction that $F(a, b) = 0$ for some (a, b) in E_ε . Then

$$1 = \left| \frac{a + b}{2} \right| \leq \frac{|a| + |b|}{2} \leq \left(\frac{|a|^p + |b|^p}{2} \right)^{1/p} = 1.$$

Therefore both inequalities are equalities. Since $p > 1$, the map $r \mapsto r^p$ is strictly convex on $[0, \infty)$, so equality in the second inequality implies $|a| = |b|$. Together with the normalization, this gives $|a| = |b| = 1$. Equality in the triangle inequality now implies that a and b have the same argument, hence $a = b$. This contradicts the condition $|a - b| \geq 2\varepsilon$. We have thus proved that $F > 0$ on E_ε .

By compactness, the minimum

$$\delta := \min_{(a, b) \in E_\varepsilon} F(a, b)$$

is well defined and satisfies $\delta > 0$. Returning to the original pair (s, t) , we obtain

$$\left| \frac{s + t}{2} \right|^p = A^p \left| \frac{\sigma + \tau}{2} \right|^p \leq (1 - \delta) A^p = (1 - \delta) \frac{|s|^p + |t|^p}{2},$$

which is the desired estimate. □

Nevertheless, one needs to pass from pointwise inequalities to uniform integral inequalities.

Proof. Let $\varepsilon > 0$. It suffices to prove that there exists $\delta > 0$ such that for any u, v in $L^p(\Omega, \mu)$ with $\|u\|_{L^p(\Omega, \mu)} \leq 1$, $\|v\|_{L^p(\Omega, \mu)} \leq 1$ and $\|u - v\|_{L^p(\Omega, \mu)} \geq 2\varepsilon$, then

$$\left\| \frac{u + v}{2} \right\|_{L^p(\Omega, \mu)} \leq 1 - \delta.$$

Let us define

$$A := \left\{ x \in \Omega : |u(x) - v(x)|^p > \frac{\varepsilon^p}{2} (|u(x)|^p + |v(x)|^p) \right\}.$$

Let us define $r(x) := \max(|u(x)|, |v(x)|)$. By definition of A , $r(x) > 0$ for any $x \in A$, and

$$\left| \frac{u(x)}{r(x)} - \frac{v(x)}{r(x)} \right| \geq \frac{\varepsilon}{2^{1/p}}.$$

Thanks to Lemma 15.25 we deduce the existence of $\delta > 0$ such that

$$\left| \frac{u(x) + v(x)}{2} \right|^p \leq (1 - \delta) \left(\frac{|u(x)|^p + |v(x)|^p}{2} \right)$$

for any $x \in A$. On the other hand, for any $x \in \Omega \setminus A$, by convexity of the map $s \mapsto |s|^p$,

$$\left| \frac{u(x) + v(x)}{2} \right|^p \leq \frac{|u(x)|^p + |v(x)|^p}{2}.$$

Integrating over the corresponding domains and setting

$$I := \int_{\Omega} \frac{|u(x)|^p + |v(x)|^p}{2} d\mu - \int_{\Omega} \left| \frac{u(x) + v(x)}{2} \right|^p d\mu,$$

we obtain

$$\begin{aligned} I &\geq \int_A \frac{|u(x)|^p + |v(x)|^p}{2} d\mu - \int_A \left| \frac{u(x) + v(x)}{2} \right|^p d\mu \\ &\geq \int_A \frac{|u(x)|^p + |v(x)|^p}{2} d\mu - (1 - \delta) \int_A \frac{|u(x)|^p + |v(x)|^p}{2} d\mu \\ &\geq \frac{\delta}{2} \max(\|u\|_{L^p(A, \mu)}^p, \|v\|_{L^p(A, \mu)}^p). \end{aligned}$$

Since $\|u\|_{L^p(\Omega, \mu)} \leq 1$ and $\|v\|_{L^p(\Omega, \mu)} \leq 1$, the quantity I is less than

$$1 - \left\| \frac{u + v}{2} \right\|_{L^p(\Omega, \mu)}^p.$$

On the other hand, by definition of A ,

$$\begin{aligned} \left(\int_{\Omega \setminus A} |u - v|^p d\mu \right)^{1/p} &\leq \frac{\varepsilon}{2^{1/p}} \left(\int_{\Omega \setminus A} |u|^p d\mu + \int_{\Omega \setminus A} |v|^p d\mu \right)^{1/p} \\ &\leq \varepsilon. \end{aligned}$$

As a consequence,

$$\|u - v\|_{L^p(A, \mu)} \geq \|u - v\|_{L^p(\Omega, \mu)} - \|u - v\|_{L^p(\Omega \setminus A, \mu)} \geq \varepsilon,$$

so that, using the triangle inequality, we get

$$\max(\|u\|_{L^p(A, \mu)}, \|v\|_{L^p(A, \mu)}) \geq \frac{\varepsilon}{2}.$$

As a consequence,

$$1 - \left\| \frac{u + v}{2} \right\|_{L^p(\Omega, \mu)}^p \geq \frac{\delta}{2} \left(\frac{\varepsilon}{2} \right)^p$$

so that

$$\left\| \frac{u + v}{2} \right\|_{L^p(\Omega, \mu)} \leq \left(1 - \frac{\delta}{2} \left(\frac{\varepsilon}{2} \right)^p \right)^{1/p} \leq 1 - \delta',$$

for $\delta' = \frac{\delta}{2^p} \left(\frac{\varepsilon}{2} \right)^p$ and the conclusion follows. □

Proposition 15.26 (Duality map for L^p). *For every $1 < p < \infty$, the space $L^p(\Omega, \mu)$ is smooth and for any u in $L^p(\Omega, \mu) \setminus 0$ and any v in $L^p(\Omega, \mu)$,*

$$D\|\cdot\|_p(u)(v) = \|u\|_p^{1-p} \int_{\Omega} |u|^{p-2} uv d\mu.$$

Proof. Note that $u \mapsto \|u\|_{L^p(\Omega, \mu)}^p$ is the composition of $u \mapsto \|u\|_{L^p(\Omega, \mu)}^p$ and the map $g : s \mapsto s^{1/p}$. Hence, it is enough to prove that for any u in $L^p(\Omega, \mu) \setminus 0$ and for any v in $L^p(\Omega, \mu)$, the function

$$\varepsilon \mapsto \int_{\Omega} |u + \varepsilon v|^p d\mu$$

admits a derivative at 0. For μ -a.e. x in Ω ,

$$\lim_{\varepsilon \rightarrow 0} \frac{|u(x) + \varepsilon v(x)|^p - |u(x)|^p}{\varepsilon} = p|u(x)|^{p-2}u(x)v(x).$$

Moreover, for every x in Ω and every ε with $0 < |\varepsilon| < 1$, the mean value theorem gives some $\theta_\varepsilon(x)$ between 0 and ε such that

$$\frac{|u(x) + \varepsilon v(x)|^p - |u(x)|^p}{\varepsilon} = p|u(x) + \theta_\varepsilon(x)v(x)|^{p-2}(u(x) + \theta_\varepsilon(x)v(x))v(x).$$

Therefore

$$\begin{aligned} \left| \frac{|u(x) + \varepsilon v(x)|^p - |u(x)|^p}{\varepsilon} \right| &\leq p|u(x) + \theta_\varepsilon(x)v(x)|^{p-1}|v(x)| \\ &\leq C_p(|u(x)|^{p-1}|v(x)| + |v(x)|^p), \end{aligned}$$

where $C_p > 0$ only depends on p , because $|a + b|^{p-1} \leq C_p(|a|^{p-1} + |b|^{p-1})$ for all a, b in $\mathbb{R}$. The right-hand side belongs to $L^1(\Omega, \mu)$ by Hölder's inequality. Hence the dominated convergence theorem yields

$$\lim_{\varepsilon \rightarrow 0} \frac{\|u + \varepsilon v\|_{L^p(\Omega, \mu)}^p - \|u\|_{L^p(\Omega, \mu)}^p}{\varepsilon} = p \int_{\Omega} |u|^{p-2}uv d\mu.$$

By the chain rule formula, we obtain that

$$\begin{aligned} D(g \circ \|\cdot\|_{L^p}^p)(u).v &= Dg(\|u\|_{L^p}^p) \cdot (D\|\cdot\|_{L^p}^p)(u).v \\ &= g'(\|u\|_{L^p}^p) (D\|\cdot\|_{L^p}^p)(u).v \\ &= \|u\|_{L^p(\Omega, \mu)}^{1-p} \int_{\Omega} |u|^{p-2}uv d\mu. \end{aligned}$$

□

Remark 15.27 (Duality map and equality cases). For $\mu = \lambda_N$, the previous results fail for $p = 1$ or ∞ , i.e., neither $L^1(\Omega, \lambda_N)$ nor $L^\infty(\Omega, \lambda_N)$ are uniformly convex nor smooth.

15.6.2 Duality in the Lebesgue spaces

We now use the smoothness result above together with the factorization theorem stated in the appendix.

Theorem 15.28 (Lp duality theorem). For $1 < p < \infty$, the dual of the space $L^p(\Omega, \mu)$ is isometrically isomorphic to $L^{p'}(\Omega, \mu)$, where

$$p' := \frac{p}{p-1}.$$

More precisely, for any f in $(L^p(\Omega, \mu))'$, there exists a unique v in $L^{p'}(\Omega, \mu)$ such that

$$f(u) = \int_{\Omega} uv \, d\mu$$

for all u in $L^p(\Omega, \mu)$, and

$$\|f\|_{(L^p(\Omega, \mu))'} = \|v\|_{L^{p'}(\Omega, \mu)}.$$

Proof. **Step 1. Local signed measures on compact exhaustions.** If $f = 0$, the conclusion is immediate with $v = 0$. Assume now that $f \in (L^p(\Omega, \mu))' \setminus \{0\}$. Thanks to Theorem 16.52, there exists a unique $w \in L^p(\Omega, \mu)$ with $\|w\|_{L^p(\Omega, \mu)} = 1$ such that for all $u \in L^p(\Omega, \mu)$,

$$f(u) = \|f\|_{(L^p(\Omega, \mu))'} D \| \cdot \| (w)(u).$$

Using Proposition 15.26, we obtain

$$f(u) = \|f\|_{(L^p(\Omega, \mu))'} \|w\|_p^{1-p} \int_{\Omega} u |w|^{p-2} w \, d\mu = \int_{\Omega} uv \, d\mu,$$

with

$$v := \|f\|_{(L^p(\Omega, \mu))'} \|w\|_p^{1-p} |w|^{p-2} w.$$

Since $w \in L^p(\Omega, \mu)$ with $\|w\|_{L^p(\Omega, \mu)} = 1$, we have that $v \in L^{p'}(\Omega, \mu)$ and

$$\|v\|_{L^{p'}(\Omega, \mu)} = \|f\|_{(L^p(\Omega, \mu))'}.$$

It remains to prove uniqueness. Let v in $L^{p'}(\Omega, \mu)$ be such that

$$\int_{\Omega} uv \, d\mu = 0$$

for all $u \in L^p(\Omega, \mu)$. We choose

$$w := |v|^{p'-1} \operatorname{sgn}(v).$$

where $\operatorname{sgn}(t) = 1$ if $t > 0$, $\operatorname{sgn}(t) = -1$ if $t < 0$, and $\operatorname{sgn}(0) = 0$. Since

$$|w|^p = |v|^{(p'-1)p} = |v|^{p'},$$

the function w belongs to $L^p(\Omega, \mu)$. Therefore

$$0 = \int_{\Omega} wv \, d\mu = \int_{\Omega} |v|^{p'} \, d\mu,$$

which implies $v = 0$ almost everywhere. Thus the representing function is unique. Since the function constructed above already satisfies

$$\|v\|_{L^{p'}(\Omega, \mu)} = \|f\|_{(L^p(\Omega, \mu))'},$$

the proof is complete. □

15.7 Weak convergence in the Lebesgue spaces L^p

We now return to weak convergence, this time in the genuine Banach-space sense. For $1 < p < \infty$, Theorem 15.28 identifies the dual of $L^p(\Omega, \mu)$ with $L^{p'}(\Omega, \mu)$. This should be distinguished from the earlier convergence against $\mathcal{C}_0(\Omega)$ test functions introduced in Definition 14.14, which came from the embedding of $L^1(\Omega, \mu)$ into $\mathcal{M}(\Omega)$. We first consider the case $1 < p < \infty$.

Definition 15.29 (Weak convergence in L^p). *Let $1 < p < \infty$, we say that a sequence $(u_n) \subset L^p(\Omega, \mu)$ converges weakly to u in $L^p(\Omega, \mu)$ if for any v in $L^{p'}(\Omega, \mu)$,*

$$\int_{\Omega} u_n v \, d\mu \rightarrow \int_{\Omega} u v \, d\mu \quad \text{as } n \rightarrow +\infty.$$

Since $1 < p' < \infty$, the separability result proved above shows that $L^{p'}(\Omega, \mu)$ is separable. Applying the sequential Banach–Alaoglu theorem from the appendix (Theorem 16.58) to $X = L^{p'}(\Omega, \mu)$ and using Theorem 15.28 with the exponent p' to identify $L^p(\Omega, \mu)$ with $(L^{p'}(\Omega, \mu))'$, we deduce that every bounded sequence in $L^p(\Omega, \mu)$ admits a subsequence which converges weakly in the sense of Definition 15.29.

Theorem 15.30 (L1 duality theorem). *The dual of the space $L^1(\Omega, \mu)$ is isometrically isomorphic to $L^\infty(\Omega, \mu)$. More precisely, for any $f \in (L^1(\Omega, \mu))'$, there exists a unique $v \in L^\infty(\Omega, \mu)$ such that*

$$f(u) = \int_{\Omega} u v \, d\mu, \tag{15.4}$$

for all $u \in L^1(\Omega, \mu)$, and

$$\|f\|_{(L^1(\Omega, \mu))'} = \|v\|_{L^\infty(\Omega, \mu)}.$$

Proof. **Step 1. Local Radon–Nikodym representations.** If $f = 0$, the conclusion is immediate with $v = 0$. Let $f \in (L^1(\Omega, \mu))'$ and, for $n \geq 1$, set

$$K_n := \{x \in \Omega : \text{dist}(x, \Omega^c) \geq 1/n \text{ and } \|x\|_{\mathbb{R}^N} \leq n\}.$$

Then each K_n is compact, $(K_n)_{n \geq 1}$ is increasing, $\cup_{n \geq 1} K_n = \Omega$, and $\mu(K_n) < \infty$ because μ is a positive Radon measure.

For every $n \geq 1$, let

$$\mathcal{A}_n := \{E \in \mathcal{A} : E \subset K_n\}$$

and define

$$\nu_n(E) := f(\mathbf{1}_E)$$

for every $E \in \mathcal{A}_n$. Since $\mu(E) \leq \mu(K_n) < \infty$, the function $\mathbf{1}_E$ belongs to $L^1(\Omega, \mu)$, so ν_n is well defined. It is immediate that ν_n is a finite signed measure on $(K_n, \mathcal{A}_n)$. Moreover, if $\mu(E) = 0$, then $\mathbf{1}_E = 0$ in $L^1(\Omega, \mu)$, hence $\nu_n(E) = 0$. Thus ν_n is absolutely continuous with respect to the restriction of μ to $\mathcal{A}_n$.

Step 2. Local Radon–Nikodym densities and uniform bounds. By Corollary 12.5, there exists $g_n \in L^1(K_n, \mu)$ such that

$$\nu_n(E) = \int_E g_n \, d\mu$$

for every $E \in \mathcal{A}_n$. **Step 2. Uniform L^∞ bound on the local densities.** We claim that

$$\|g_n\|_{L^\infty(K_n, \mu)} \leq \|f\|_{(L^1(\Omega, \mu))'}.$$

Assume on the contrary that there exist $\varepsilon > 0$ and a measurable set $A \subset K_n$ with $\mu(A) > 0$ such that

$$|g_n| > \|f\|_{(L^1(\Omega, \mu))'} + \varepsilon$$

almost everywhere on A . Set

$$u := \mathbf{1}_A \operatorname{sgn}(g_n).$$

Then $u \in L^1(\Omega, \mu)$ and

$$|f(u)| = \left| \int_A g_n \operatorname{sgn}(g_n) d\mu \right| = \int_A |g_n| d\mu > (\|f\|_{(L^1(\Omega, \mu))'} + \varepsilon) \mu(A),$$

whereas

$$|f(u)| \leq \|f\|_{(L^1(\Omega, \mu))'} \|u\|_{L^1(\Omega, \mu)} = \|f\|_{(L^1(\Omega, \mu))'} \mu(A).$$

This contradiction proves the claim.

Step 3. Compatibility and construction of the global density.

Step 3. Gluing the local densities. If $E \in \mathcal{A}_n$, then

$$\int_E g_{n+1} d\mu = f(\mathbf{1}_E) = \int_E g_n d\mu.$$

By the uniqueness statement in the Radon–Nikodym theorem, $g_{n+1} = g_n$ almost everywhere on K_n . We now define a measurable function v on Ω by

$$v = g_1 \text{ on } K_1, \quad v = g_n \text{ on } K_n \setminus K_{n-1} \text{ for } n \geq 2.$$

From the uniform L^∞ bound above, we infer that

$$\|v\|_{L^\infty(\Omega, \mu)} \leq \|f\|_{(L^1(\Omega, \mu))'}.$$

Step 4. Representation of the functional.

Step 4. Representation first for compactly supported functions, then for all L^1 functions. Let now u in $L^1(\Omega, \mu)$ be supported in some K_n . If u is a simple function, say

$$u = \sum_{j=1}^m a_j \mathbf{1}_{E_j}$$

with $E_j \in \mathcal{A}_n$, then

$$f(u) = \sum_{j=1}^m a_j f(\mathbf{1}_{E_j}) = \sum_{j=1}^m a_j \int_{E_j} g_n d\mu = \int_\Omega uv d\mu.$$

Since simple functions are dense in $L^1(K_n, \mu)$, the same identity holds for every $u \in L^1(\Omega, \mu)$ supported in K_n .

For a general $u \in L^1(\Omega, \mu)$, we set

$$u_n := u \mathbf{1}_{K_n}.$$

Then $u_n \rightarrow u$ in $L^1(\Omega, \mu)$ as $n \rightarrow +\infty$ by the dominated convergence theorem. Since each u_n is supported in K_n , we have

$$f(u_n) = \int_\Omega u_n v d\mu.$$

Passing to the limit and using $v \in L^\infty(\Omega, \mu)$, we obtain

$$f(u) = \int_\Omega uv d\mu$$

for every $u \in L^1(\Omega, \mu)$. This proves the existence of a representing function v .

Step 5. Uniqueness and norm identity.

Step 5. Uniqueness of the representing function. To prove uniqueness, let $v_1, v_2 \in L^\infty(\Omega, \mu)$ be such that

$$\int_{\Omega} uv_1 d\mu = \int_{\Omega} uv_2 d\mu$$

for every $u \in L^1(\Omega, \mu)$. If

$$A := \{x \in \Omega : v_1(x) \neq v_2(x)\}$$

had positive measure, there would exist $n \geq 1$ such that $\mu(A \cap K_n) > 0$. Taking

$$u := \mathbf{1}_{A \cap K_n} \operatorname{sgn}(v_1 - v_2),$$

we would get

$$0 = \int_{\Omega} u(v_1 - v_2) d\mu = \int_{A \cap K_n} |v_1 - v_2| d\mu > 0,$$

a contradiction. Hence $v_1 = v_2$ almost everywhere.

Step 6. Norm identity. Finally, we already know that

$$\|v\|_{L^\infty(\Omega, \mu)} \leq \|f\|_{(L^1(\Omega, \mu))'}.$$

Conversely, let $\varepsilon > 0$ and set

$$A_\varepsilon := \{x \in \Omega : |v(x)| > \|v\|_{L^\infty(\Omega, \mu)} - \varepsilon\}.$$

By the definition of the essential supremum, $\mu(A_\varepsilon) > 0$. Since $\Omega = \cup_{n \geq 1} K_n$, there exists $n \geq 1$ such that $\mu(A_\varepsilon \cap K_n) > 0$. Taking

$$u := \mathbf{1}_{A_\varepsilon \cap K_n} \operatorname{sgn}(v),$$

we get

$$\begin{aligned} \|f\|_{(L^1(\Omega, \mu))'} \mu(A_\varepsilon \cap K_n) &\geq |f(u)| = \int_{A_\varepsilon \cap K_n} |v| d\mu \\ &\geq (\|v\|_{L^\infty(\Omega, \mu)} - \varepsilon) \mu(A_\varepsilon \cap K_n). \end{aligned}$$

Hence

$$\|f\|_{(L^1(\Omega, \mu))'} \geq \|v\|_{L^\infty(\Omega, \mu)} - \varepsilon.$$

Letting $\varepsilon \rightarrow 0$, we conclude that

$$\|f\|_{(L^1(\Omega, \mu))'} \geq \|v\|_{L^\infty(\Omega, \mu)}.$$

Therefore

$$\|f\|_{(L^1(\Omega, \mu))'} = \|v\|_{L^\infty(\Omega, \mu)}.$$

□

Thanks to Theorem 15.30, we can now define weak convergence in $L^1(\Omega, \mu)$ in the usual Banach-space sense.

Definition 15.31 (Weak convergence in L^1). We say that a sequence $(u_n)_n$ in $L^1(\Omega, \mu)$ converges weakly to u in $L^1(\Omega, \mu)$ if for every v in $L^\infty(\Omega, \mu)$,

$$\int_{\Omega} u_n v d\mu \rightarrow \int_{\Omega} u v d\mu \quad \text{as } n \rightarrow \infty.$$

Since $\mathcal{C}_0(\Omega) \subset L^\infty(\Omega, \mu)$, weak convergence in $L^1(\Omega, \mu)$ implies convergence against $\mathcal{C}_0(\Omega)$ test functions in the sense of Definition 14.14.

Theorem 15.32 (Dunford–Pettis theorem: weak convergence implies equi-integrability). Assume that Ω is an open subset of $\mathbb{R}^N$ such that $\mu(\Omega) < \infty$. Let (u_n) be a sequence in $L^1(\Omega, \mu)$, and let u be in $L^1(\Omega, \mu)$. If

$$u_n \rightharpoonup u$$

weakly in $L^1(\Omega, \mu)$, then the sequence (u_n) is equi-integrable.

Proof. **Step 1. Pointwise convergence of the associated signed measures.** For every $n \geq 1$, define a finite signed measure ν_n on $\mathcal{B}(\Omega)$ by

$$\nu_n(E) := \int_E u_n \, d\mu.$$

For every Borel set $E \subset \Omega$, the characteristic function $\mathbf{1}_E$ belongs to $L^\infty(\Omega, \mu)$. The weak convergence of u_n to u in $L^1(\Omega, \mu)$ gives

$$\nu_n(E) \rightarrow \int_E u \, d\mu \quad \text{as } n \rightarrow +\infty$$

for every E in $\mathcal{B}(\Omega)$. Moreover,

$$\nu_n \ll \mu$$

for every $n \geq 1$.

Step 2. Uniform absolute continuity of the signed measures. Let $\varepsilon > 0$. Lemma 12.4, applied with $\varepsilon/6$ in place of ε , gives $\delta > 0$ such that

$$\mu(B) < \delta \quad \implies \quad \sup_{n \geq 1} \left| \int_B u_n \, d\mu \right| \leq \frac{\varepsilon}{2}$$

for every Borel set $B \subset \Omega$.

Step 3. Passage from signed estimates to estimates of absolute values. Let E be a Borel set such that $\mu(E) < \delta$. For every $n \geq 1$, set

$$E_n^+ := E \cap \{u_n \geq 0\}, \quad E_n^- := E \cap \{u_n < 0\}.$$

Then $\mu(E_n^+) < \delta$ and $\mu(E_n^-) < \delta$. Hence

$$\begin{aligned} \int_E |u_n| \, d\mu &= \int_{E_n^+} u_n \, d\mu - \int_{E_n^-} u_n \, d\mu \\ &\leq \left| \int_{E_n^+} u_n \, d\mu \right| + \left| \int_{E_n^-} u_n \, d\mu \right| \\ &\leq \varepsilon. \end{aligned}$$

Taking the supremum in n proves that (u_n) is equi-integrable. □

Theorem 15.33 (Dunford–Pettis theorem: weak compactness in finite measure). Assume that Ω is an open subset of $\mathbb{R}^N$ such that $\mu(\Omega) < \infty$. Let (u_n) be a bounded and equi-integrable sequence in $L^1(\Omega, \mu)$. Then there exist a subsequence (u_{n_k}) and a function u in $L^1(\Omega, \mu)$ such that

$$u_{n_k} \rightharpoonup u$$

weakly in $L^1(\Omega, \mu)$.

Proof. **Step 1. Diagonal extraction on a countable generating algebra.** Let $\mathcal{E}_0$ be the algebra of all finite unions of sets of the form

$$\Omega \cap]a_1, b_1] \times \cdots \times]a_N, b_N],$$

where $a_1, \dots, a_N, b_1, \dots, b_N$ are rational numbers with $a_i < b_i$ for every i . The algebra $\mathcal{E}_0$ is countable and generates $\mathcal{B}(\Omega)$. Set

$$M := \sup_{n \geq 1} \|u_n\|_{L^1(\Omega, \mu)} < +\infty.$$

For every A in $\mathcal{E}_0$ and every $n \geq 1$, we have

$$\left| \int_A u_n \, d\mu \right| \leq \|u_n\|_{L^1(\Omega, \mu)} \leq M.$$

Thus, for each fixed A in $\mathcal{E}_0$, the sequence of real numbers $(\int_A u_n \, d\mu)_{n \geq 1}$ is bounded. By a diagonal extraction on the countable family $\mathcal{E}_0$, we can choose a subsequence $(u_{n_k})_{k \geq 1}$ of (u_n) such that for every A in $\mathcal{E}_0$, the numerical sequence

$$\left(\int_A u_{n_k} \, d\mu \right)_{k \geq 1}$$

converges.

Step 2. Convergence of the integrals on every Borel set. We fix a Borel set $E \subset \Omega$ and prove that

$$\left(\int_E u_{n_k} \, d\mu \right)_{k \geq 1}$$

converges. Let $\varepsilon > 0$. By equi-integrability, there exists $\delta > 0$ such that

$$\mu(B) < \delta \quad \implies \quad \sup_{k \geq 1} \int_B |u_{n_k}| \, d\mu \leq \varepsilon$$

for every Borel set B . By Corollary 3.11, there exists A in $\mathcal{E}_0$ such that

$$\mu(E \triangle A) < \delta.$$

Then, for all $k, \ell \geq 1$,

$$\begin{aligned} \left| \int_E u_{n_k} \, d\mu - \int_E u_{n_\ell} \, d\mu \right| &\leq \left| \int_A u_{n_k} \, d\mu - \int_A u_{n_\ell} \, d\mu \right| + \int_{E \triangle A} |u_{n_k}| \, d\mu + \int_{E \triangle A} |u_{n_\ell}| \, d\mu \\ &\leq \left| \int_A u_{n_k} \, d\mu - \int_A u_{n_\ell} \, d\mu \right| + 2\varepsilon. \end{aligned}$$

The first term tends to 0 as $k, \ell \rightarrow +\infty$, by the choice of the subsequence. Thus the sequence of integrals over E is Cauchy, and therefore converges.

Step 3. Construction of the limit signed measure. For every Borel set $E \subset \Omega$, define

$$\lambda(E) := \lim_{k \rightarrow \infty} \int_E u_{n_k} \, d\mu.$$

The measure is finite-valued. Indeed, with the constant M introduced in Step 1, we have

$$|\lambda(E)| \leq \limsup_{k \rightarrow \infty} \int_E |u_{n_k}| \, d\mu \leq M$$

for every Borel set $E \subset \Omega$. The set function λ is finitely additive because each map

$$E \mapsto \int_E u_{n_k} \, d\mu$$

is finitely additive. We prove that λ is countably additive. Let $(E_j)_{j \geq 1}$ be a sequence of pairwise disjoint Borel sets, and set

$$E := \bigcup_{j=1}^{\infty} E_j, \quad R_m := E \setminus \bigcup_{j=1}^m E_j.$$

Then (R_m) is decreasing and $\bigcap_{m \geq 1} R_m = \emptyset$. Since $R_1 \subset \Omega$ and $\mu(\Omega) < \infty$, the continuity from above statement, namely item 2 of Theorem 2.3, gives

$$\mu(R_m) \rightarrow 0 \quad \text{as } m \rightarrow +\infty.$$

Given $\varepsilon > 0$, equi-integrability gives $\delta > 0$ such that every Borel set B with $\mu(B) < \delta$ satisfies

$$\sup_{k \geq 1} \int_B |u_{n_k}| \, d\mu \leq \varepsilon.$$

Since $\mu(R_m) \rightarrow 0$, there exists an integer m_0 such that $\mu(R_m) < \delta$ for every $m \geq m_0$. Therefore

$$\sup_{k \geq 1} \int_{R_m} |u_{n_k}| \, d\mu \leq \varepsilon$$

for every $m \geq m_0$. For such an m , finite additivity gives

$$\begin{aligned} \left| \lambda(E) - \sum_{j=1}^m \lambda(E_j) \right| &= |\lambda(R_m)| \\ &= \lim_{k \rightarrow \infty} \left| \int_{R_m} u_{n_k} \, d\mu \right| \\ &\leq \sup_{k \geq 1} \int_{R_m} |u_{n_k}| \, d\mu \\ &\leq \varepsilon. \end{aligned}$$

Hence

$$\lambda(E) = \sum_{j=1}^{\infty} \lambda(E_j).$$

Thus λ is a finite signed Borel measure on Ω .

If $\mu(E) = 0$, then equi-integrability gives

$$\sup_{k \geq 1} \int_E |u_{n_k}| \, d\mu = 0,$$

hence $\lambda(E) = 0$. Therefore

$$\lambda \ll \mu.$$

Step 4. Identification by Radon–Nikodym. By Corollary 12.5, there exists u in $L^1(\Omega, \mu)$ such that

$$\lambda(E) = \int_E u \, d\mu$$

for every Borel set $E \subset \Omega$.

Step 5. Weak convergence of the extracted subsequence. Let v be in $L^\infty(\Omega, \mu)$. We choose a measurable representative, still denoted by v , such that

$$|v(x)| \leq \|v\|_{L^\infty(\Omega, \mu)}$$

for μ -almost every x in Ω . By Theorem 15.9, there exists a sequence $(s_m)_{m \geq 1}$ of bounded simple measurable functions such that

$$|s_m(x) - v(x)| \leq 2^{-m}$$

for μ -almost every x in Ω . For every $m \geq 1$ and every $k \geq 1$, we have

$$\begin{aligned} \left| \int_{\Omega} (u_{n_k} - u)v \, d\mu \right| &\leq \left| \int_{\Omega} (u_{n_k} - u)s_m \, d\mu \right| + \int_{\Omega} |u_{n_k} - u| |v - s_m| \, d\mu \\ &\leq \left| \int_{\Omega} (u_{n_k} - u)s_m \, d\mu \right| + 2^{-m} \left(\|u_{n_k}\|_{L^1(\Omega, \mu)} + \|u\|_{L^1(\Omega, \mu)} \right). \end{aligned}$$

Since (u_n) is bounded in $L^1(\Omega, \mu)$, the second term is bounded above by

$$2^{-m} \left(\sup_{n \geq 1} \|u_n\|_{L^1(\Omega, \mu)} + \|u\|_{L^1(\Omega, \mu)} \right).$$

For fixed m , the function s_m is a finite linear combination of characteristic functions of Borel sets. By the definition of λ , we therefore have

$$\int_{\Omega} u_{n_k} s_m \, d\mu \rightarrow \int_{\Omega} u s_m \, d\mu \quad \text{as } k \rightarrow +\infty.$$

We first let $k \rightarrow +\infty$, and then let $m \rightarrow +\infty$. This proves that

$$\int_{\Omega} u_{n_k} v \, d\mu \rightarrow \int_{\Omega} u v \, d\mu \quad \text{as } k \rightarrow +\infty$$

for every v in $L^\infty(\Omega, \mu)$. Hence $u_{n_k} \rightharpoonup u$ weakly in $L^1(\Omega, \mu)$. $\square$

Remark. On infinite-measure spaces, equi-integrability alone is not enough for the compactness conclusion of Theorem 15.33. We also need the same spatial tail condition as in Definition 11.14: mass must not escape to infinity. In the present Euclidean Radon setting, this condition is naturally written as compact tightness. For every $\varepsilon > 0$, there should exist a compact set $K_\varepsilon \subset \Omega$ such that

$$\sup_{n \geq 1} \int_{\Omega \setminus K_\varepsilon} |u_n| \, d\mu < \varepsilon.$$

Equivalently, for any standard compact exhaustion $(K_j)_{j \geq 1}$ of Ω which eventually contains every compact subset of Ω , the condition reads

$$\lim_{j \rightarrow +\infty} \sup_{n \geq 1} \int_{\Omega \setminus K_j} |u_n| \, d\mu = 0.$$

For example, if $\Omega \neq \mathbb{R}^N$, one may take

$$K_j := \{x \in \Omega : |x| \leq j \text{ and } d(x, \mathbb{R}^N \setminus \Omega) \geq 1/j\},$$

and if $\Omega = \mathbb{R}^N$, one may take $K_j = \overline{B}(0, j)$. This is exactly the compact-set form of the uniform tail condition used in the σ -finite Vitali theorems.

Let us stress that a bounded sequence of $L^1(\Omega, \mu)$ functions may converge against $\mathcal{C}_0(\Omega)$ test functions to a measure which is not given by an element of $L^1(\Omega, \mu)$, as in the case of approximations to the identity.

Chapter 16

Appendix. Topology and functional analysis reminders

In this appendix we recall a few definitions from general topology and functional analysis, including compactness, separability, completeness, duality, and weak* compactness. An important particular case is that of metric spaces, for which most of the definitions can be rephrased in terms of sequences. We also recall the Baire category theorem, the extension of uniformly continuous functions, the uniform boundedness principle, and several standard results from Banach space theory.

16.1 Basic definitions

16.1.1 General topology

We start by recalling a few basic definitions of general topology.

Definition 16.1 (Topological spaces). *Given a set X , we say that a subset τ of $\mathcal{P}(X)$ is a topology on X if*

- 1) $\emptyset$ and X are in τ .
- 2) τ is stable by finite intersection.
- 3) τ is stable under arbitrary unions.

Then we say that (X, τ) is a topological space. The elements of τ are called open sets, and their complementary sets are the closed sets.

Definition 16.2 (Interior and closure). *Given a topological space (X, τ) and a set $A \subset X$, we define*

- 1) *the interior of A by $A^\circ := \{x \in A : \text{there exists } U \in \tau \text{ such that } x \in U \subset A\}$;*
- 2) *the closure of A by $\overline{A} := \{x \in X : \text{for any } U \in \tau \text{ with } x \in U, \text{ then } U \cap A \neq \emptyset\}$.*

We say that $x \in \overline{A}$ is an adherent point and $x \in A^\circ$ an interior point.

Observe that $A^\circ \subset A \subset \overline{A}$.

Definition 16.3 (Dense subsets). Given a topological space (X, τ) and a set $A \subset X$, we say that A is dense in X for the topology τ if $\overline{A} = X$.

Definition 16.4 (Sequential convergence). Given a topological space (X, τ) , we say that a sequence $(x_n)_{n \in \mathbb{N}} \subset X$ converges to x in X if for any open set $U \in \tau$ with $x \in U$, there exists $n_0 \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq n_0$.

Definition 16.5 (Continuity). Given two topological spaces (X_1, τ_1) and (X_2, τ_2) , we say that a map $f : X_1 \rightarrow X_2$ is continuous at $x_1 \in X_1$ if for every open set $U \in \tau_2$ such that $f(x_1) \in U$, there exists an open set $V \in \tau_1$ such that $x_1 \in V$ and $f(V) \subset U$. We say that f is continuous if it is continuous at every point of X_1 .

For extended real-valued functions the following notion of lower semicontinuity is weaker than continuity.

Definition 16.6 (Lower semicontinuity). Given a topological space (X, τ) and x_0 in X , we say that a function $f : X \rightarrow \mathbb{R} \cup \{-\infty, +\infty\}$ is lower semicontinuous at x_0 if for every $\alpha < f(x_0)$, there exists a neighborhood $U \in \tau$ of x_0 such that $f(x) > \alpha$ for all x in U . We say that f is lower semicontinuous if it is lower semicontinuous at every point of X .

It is not difficult to check that a function is lower semicontinuous if and only if $\{x \in X : f(x) > \alpha\}$ is an open set for every $\alpha \in \mathbb{R}$.

16.1.2 Metric spaces

An important case of topological spaces is given by metric spaces that we now introduce.

Definition 16.7 (Metric spaces). Given a set X , we say that a function $d : X \times X \rightarrow \mathbb{R}^+$ is a distance on X if

- 1) $d(x, y) = 0$ if and only if $x = y$;
- 2) $d(x, y) = d(y, x)$;
- 3) For any $x, y, z \in X$, $d(x, y) \leq d(x, z) + d(y, z)$.

Then we say that (X, d) is a metric space.

For any x in X and any $r > 0$, we denote

$$B(x, r) := \{y \in X : d(x, y) < r\} \quad (\text{resp. } \overline{B}(x, r) := \{y \in X : d(x, y) \leq r\})$$

the open (resp. closed) ball of center x and radius r . A subset is said to be bounded if it is contained in a ball of finite radius.

Proposition 16.8 (Topology induced by a metric). Given a metric space (X, d) , the family τ of all subsets $U \subset X$ such that for each $x \in U$, there exists $r > 0$ satisfying $B(x, r) \subset U$ defines a topology on X .

The following propositions, whose proofs are left to the reader, highlight the role of the sequences in metric spaces.

Proposition 16.9 (Sequential characterization of metric topology). *Given a metric space (X, d) and the topology τ given by Proposition 16.8. The following statements hold:*

- 1) *A sequence $(x_n)_{n \in \mathbb{N}} \subset X$ converges to x in X if and only if for every $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, then $d(x, x_n) < \varepsilon$;*
- 2) *A subset F of X is closed if and only if for every sequence $(x_n)_{n \in \mathbb{N}} \subset F$ converging to x in X , then $x \in F$.*

Proposition 16.10 (Continuity in metric spaces). *Let (X_1, d_1) and (X_2, d_2) be two metric spaces, $f : X_1 \rightarrow X_2$ and $x_1 \in X_1$. Then the following statements are equivalent:*

- 1) *f is continuous at x_1 ;*
- 2) *For any $\varepsilon > 0$, there exists $\delta > 0$ such that if $x \in X_1$ is such that $d_1(x_1, x) < \delta$, then $d_2(f(x_1), f(x)) < \varepsilon$;*
- 3) *For any sequence $(x_n)_{n \in \mathbb{N}} \subset X_1$ converging to x_1 , the sequence $(f(x_n))_{n \in \mathbb{N}}$ converges to $f(x_1)$ in X_2 .*

Proposition 16.11 (Sequential lower semicontinuity). *Given a metric space (X, d) and x_0 in X . A function $f : X \rightarrow \mathbb{R} \cup \{-\infty, +\infty\}$ is lower semicontinuous at x_0 if and only if for any sequence $(x_n)_{n \in \mathbb{N}} \subset X$ converging to x_0 in X , then*

$$f(x_0) \leq \liminf_{n \rightarrow \infty} f(x_n).$$

Lemma 16.12 (Compact exhaustion of open Euclidean sets). *Let $d \geq 1$ and let Ω be an open subset of $\mathbb{R}^d$. Then there exists an increasing sequence $(K_n)_{n \geq 1}$ of compact subsets of Ω such that*

$$\Omega = \bigcup_{n=1}^{\infty} K_n.$$

More precisely, with the convention $\text{dist}(x, \emptyset) = +\infty$, one may take

$$K_n := \left\{ x \in \Omega : |x| \leq n \text{ and } \text{dist}(x, \mathbb{R}^d \setminus \Omega) \geq \frac{1}{n} \right\}.$$

Proof. The function $x \mapsto \text{dist}(x, \mathbb{R}^d \setminus \Omega)$ is continuous when $\mathbb{R}^d \setminus \Omega$ is nonempty, and the convention for the empty complement makes the case $\Omega = \mathbb{R}^d$ immediate. Thus each K_n is closed and bounded in $\mathbb{R}^d$, hence compact. The inclusion $K_n \subset \Omega$ is clear, and $(K_n)_{n \geq 1}$ is increasing. Let x be in Ω . Since Ω is open, there exists $\rho > 0$ such that $B(x, \rho) \subset \Omega$, hence $\text{dist}(x, \mathbb{R}^d \setminus \Omega) \geq \rho$. Choosing n large enough so that $n \geq |x|$ and $1/n \leq \rho$ gives $x \in K_n$. Therefore the union of the K_n is Ω . $\square$

Definition 16.13 (Uniform continuity). *Let (X_1, d_1) and (X_2, d_2) be two metric spaces. We say that a map $f : X_1 \rightarrow X_2$ is uniformly continuous if for any $\varepsilon > 0$, there exists $\delta > 0$ such that if x and $y \in X_1$ satisfy $d_1(x, y) < \delta$, then $d_2(f(x), f(y)) < \varepsilon$.*

It is clear from the definitions above that uniform continuity implies continuity. We will see in Theorem 16.33 that the converse statement holds true when the space (X_1, d_1) is compact.

16.2 Completeness

16.2.1 Definition

Completeness is an important notion in general topology and in functional analysis because it enables one to characterize converging sequences without the knowledge of their limit. We first define the Cauchy property.

Definition 16.14 (Cauchy sequences). *Given a metric space (X, d) , we say that a sequence $(u_n)_{n \in \mathbb{N}} \subset X$ is a Cauchy sequence if for any $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that for all $n, n' \geq n_0$ then $d(u_n, u_{n'}) < \varepsilon$.*

Definition 16.15 (Complete metric spaces). *A metric space (X, d) is complete if any Cauchy sequence converges in X .*

Let us give as a first example the set $\mathbb{R}$ endowed with the usual metric $d(x, y) := |x - y|$. It is also useful to notice that a closed subset of a complete metric space is complete.

16.2.2 Baire category theorem

The following theorem is used in the proof of Lemma 12.4 and in the proof of the Uniform Boundedness Principle below.

Definition 16.16 (Nowhere dense sets). *Let (X, d) be a metric space. A subset A of X is called nowhere dense if $(\bar{A})^\circ = \emptyset$.*

Theorem 16.17 (Baire category theorem). *Let (X, d) be a nonempty complete metric space. If $(F_n)_{n \geq 1}$ is a sequence of closed subsets of X such that*

$$X = \bigcup_{n \geq 1} F_n,$$

then there exists $n_0 \geq 1$ such that F_{n_0} has nonempty interior. Equivalently, if $(U_n)_{n \geq 1}$ is a sequence of dense open subsets of X , then $\bigcap_{n \geq 1} U_n$ is dense in X .

Proof. We prove first the dense-open formulation. Let U be a nonempty open subset of X . Since U_1 is dense and open, the set $U \cap U_1$ is nonempty and open. Hence there exist x_1 in X and $r_1 > 0$ such that

$$\bar{B}(x_1, r_1) \subset U \cap U_1 \quad \text{and} \quad r_1 \leq \frac{1}{2}.$$

Assume that x_n and r_n have been constructed. Since U_{n+1} is dense and open, the set $B(x_n, r_n/2) \cap U_{n+1}$ is nonempty and open. We can therefore choose x_{n+1} in X and $r_{n+1} > 0$ such that

$$\bar{B}(x_{n+1}, r_{n+1}) \subset B(x_n, r_n/2) \cap U_{n+1} \quad \text{and} \quad r_{n+1} \leq 2^{-(n+1)}.$$

The sequence $(x_n)_{n \geq 1}$ is Cauchy. Indeed, for every $p, q \geq n$, the points x_p and x_q belong to $B(x_n, r_n)$, hence $d(x_p, x_q) \leq 2r_n$, and $r_n \rightarrow 0$. Since X is complete, x_n converges to some x in X . For each fixed n , all the points x_m with $m \geq n$ belong to the closed ball $\bar{B}(x_n, r_n)$. Passing to the limit gives x in $\bar{B}(x_n, r_n)$. Thus x belongs to U and to every U_n . Therefore every nonempty open subset U of X meets $\bigcap_{n \geq 1} U_n$, so this intersection is dense.

Now let $(F_n)_{n \geq 1}$ be a sequence of closed subsets whose union is X . If every F_n had empty interior, then $U_n := X \setminus F_n$ would be dense and open for every n . The dense-open formulation would give that $\bigcap_{n \geq 1} U_n$ is dense, hence nonempty, which contradicts $X = \bigcup_{n \geq 1} F_n$. Therefore at least one F_n has nonempty interior. $\square$

16.2.3 Banach fixed point theorem for contraction mapping

An important application of the notion of completeness is given by the following theorem.

Theorem 16.18 (Banach fixed point theorem). *Given a complete metric space (X, d) and $f : X \rightarrow X$. Assume that f is a contraction, i.e. that there exists a constant $\theta \in (0, 1)$ such that for all x and $y \in X$, then $d(f(x), f(y)) \leq \theta d(x, y)$. Then there exists a unique fixed point $x^* \in X$ such that $f(x^*) = x^*$.*

Proof. Let $x_0 \in X$ and let $(x_n)_{n \in \mathbb{N}}$ the associated sequence defined by the relation

$$x_{n+1} = f(x_n). \quad (16.1)$$

By iteration we have

$$d(x_{n+1}, x_n) \leq \theta^n d(x_1, x_0).$$

For any $n' > n$,

$$d(x_{n'}, x_n) \leq \sum_{k=1}^{n'-n} d(x_{n+k}, x_{n+k-1}) \leq d(x_1, x_0) \sum_{k=1}^{n'-n} \theta^{n+k-1} \leq \frac{\theta^n}{1-\theta} d(x_1, x_0).$$

Therefore $(x_n)_{n \in \mathbb{N}}$ is a Cauchy sequence, and by completeness, it converges to an element $x^* \in X$. Since f is continuous, we have $f(x^*) = x^*$ by passing to the limit in (16.1). The uniqueness follows from the contraction assumption. $\square$

The previous theorem is useful in the proof of the Cauchy-Lipschitz theorem in the theory of ordinary differential equations, and also in proof the local inversion theorem.

16.2.4 Extension of uniformly continuous functions

Theorem 16.19 (Extension theorem for uniformly continuous maps). *Given two metric spaces (X_1, d_1) and (X_2, d_2) , the latter being complete, a dense subset Y of X_1 , and a map $f : Y \rightarrow X_2$ which is uniformly continuous. Then there exists a unique uniformly continuous extension $g : X_1 \rightarrow X_2$ of f .*

Proof. The uniqueness is straightforward: indeed for any $x \in X_1$, since Y is dense in X_1 , there exists a sequence $(x_n)_{n \in \mathbb{N}} \subset Y$ which converges to x . If $g : X_1 \rightarrow X_2$ is a uniformly continuous extension of f , then $g(x)$ must be the limit of the sequence $(f(x_n))_{n \in \mathbb{N}}$.

Now to prove the existence of such an extension, observe that the sequence $(f(x_n))_{n \in \mathbb{N}}$ has the Cauchy property, since $(x_n)_{n \in \mathbb{N}}$ has the Cauchy property (because it converges) and f is uniformly continuous. Since (X_2, d_2) is complete, it yields that the sequence $(f(x_n))_{n \in \mathbb{N}}$ converges to an element z in X_2 . This z does not depend on the choice of the sequence $(x_n)_{n \in \mathbb{N}}$ in X_1 . Indeed, if $(x'_n)_{n \in \mathbb{N}}$ is another sequence converging to x , then $d_1(x_n, x'_n) \rightarrow 0$ as $n \rightarrow +\infty$, so that, since f is uniformly continuous, $d_2(f(x_n), f(x'_n)) \rightarrow 0$ as $n \rightarrow +\infty$. Hence the sequence $(f(x'_n))_{n \in \mathbb{N}}$ converges to z as well. Therefore, it makes sense to define $g(x) := z$.

Let us now prove that g is uniformly continuous. Let $\varepsilon > 0$. Since f is uniformly continuous there exists $\delta > 0$ such that for any $x, x' \in Y$ with $d_1(x, x') < \delta$, there holds $d_2(f(x), f(x')) < \varepsilon/3$. Let y and $y' \in X_1$ with $d_1(y, y') < \delta/3$. There exists $x, x' \in Y$ such that

$$d_1(x, y) < \delta/3, \quad d_1(x', y') < \delta/3, \quad d_2(f(x), g(y)) < \varepsilon/3 \text{ and } d_2(f(x'), g(y')) < \varepsilon/3.$$

Therefore, thanks to the triangle inequality, we have $d_1(x, x') < \delta$ and therefore $d_2(f(x), f(x')) < \varepsilon/3$. Using again the triangle inequality, we get $d_2(g(y), g(y')) < \varepsilon$. Hence g is uniformly continuous. $\square$

16.2.5 Banach spaces

Let us recall a few definitions:

Definition 16.20 (Normed vector spaces). A normed vector space over $\mathbb{R}$ is a pair $(V, \|\cdot\|)$ where V is a real vector space and $\|\cdot\|$ is a norm on V that is a function from V to $\mathbb{R}_+$ satisfying

- 1) $\|u\| = 0$ if and only if $u = 0$ (positive definiteness),
- 2) for any u in V , for any $\lambda \in \mathbb{R}$, $\|\lambda u\| = |\lambda| \|u\|$ (positive homogeneity),
- 3) for any u, v in V , $\|u + v\| \leq \|u\| + \|v\|$ (triangle inequality or subadditivity).

If the first item above is not satisfied then $\|\cdot\|$ is called a seminorm.

Remark 16.21 (Complex normed spaces). We define in a similar way a normed vector space over $\mathbb{C}$ by considering a complex vector space and by extending the second property above to any $\lambda \in \mathbb{C}$.

Remark 16.22 (Distance induced by a norm). We can easily associate a distance to the norm of a normed vector space, through the formula $d(u, v) := \|u - v\|$.

Definition 16.23 (Banach spaces). If the topology defined by this distance is complete then we say that $(V, \|\cdot\|)$ is a Banach space.

Next proposition proves that in a normed vector space, completeness can be characterized thanks to the series.

Proposition 16.24 (Normally convergent series in Banach spaces). Let $(V, \|\cdot\|)$ be a normed vector space. Then $(V, \|\cdot\|)$ is a Banach space if and only if the series normally converging actually converge in $(V, \|\cdot\|)$.

Let us recall that for a sequence $(u_n)_n$ in V , we say that the series $\sum_n u_n$ is normally converging in V if $\sum_n \|u_n\|$ converges in $\mathbb{R}$.

Proof. suppose that V is complete, and let $(u_n)_n$ be a sequence in V such that $\sum_n \|u_n\|$ converges in $\mathbb{R}$. Let $\varepsilon > 0$ be given and put $v_n := \sum_{k=0}^n u_k$. Then for $n > m$,

$$\|v_n - v_m\| = \left\| \sum_{k=m+1}^n u_k \right\| \leq \sum_{k=m+1}^n \|u_k\| < \varepsilon$$

for all sufficiently large m and n , since $\sum_{k=0}^n \|u_k\| < \infty$. Hence $(v_n)_n$ is a Cauchy sequence and so converges since V is complete, by hypothesis.

Conversely, assume that $\sum_n u_n$ converges in V whenever $\sum_n \|u_n\| < \infty$. Let $(v_n)_n$ be a Cauchy sequence in V . We will prove that $(v_n)_n$ converges in V . By iteration there exists a subsequence $(v_{n_k})_k$ of $(v_n)_n$ such that for any k , $\|v_{n_{k+1}} - v_{n_k}\| \leq 2^{-k}$. Therefore, setting $u_k := v_{n_{k+1}} - v_{n_k}$, we have:

$$\sum_{k=0}^N \|u_k\| \leq \sum_{k=0}^N \|v_{n_{k+1}} - v_{n_k}\| \leq \sum_{k=0}^N 2^{-k} < \infty.$$

We therefore get that $\sum_{k=0}^N u_k$ converges to some u in V as $N \rightarrow +\infty$. But

$$\sum_{k=0}^N u_k = \sum_{k=0}^N v_{n_{k+1}} - v_{n_k} = v_{n_{N+1}} - v_{n_0},$$

so that $v_{n_N} \rightarrow u + v_{n_0}$ as $N \rightarrow +\infty$. Thus the Cauchy sequence $(v_n)_n$ has a convergent subsequence and so must itself converge. $\square$

Let us now give a result to quotient spaces defined as follows.

Definition 16.25 (Quotient normed spaces). *Let X be a vector space, and let M be a vector subspace of X . We define an equivalence relation $\sim$ on X by setting $x \sim y$ if and only if $x - y \in M$. It is straightforward to check that this is an equivalence relation on X . For x in X we denote by $[x]$ the equivalence class containing the element x and we denote the set of equivalence classes by X/M , that is, we call X/M the quotient space of X by M .*

We define on X/M a sum law and a scalar multiplication law by $[x] + [y] := [x + y]$ and $\alpha[x] := [\alpha x]$.

Let us stress that these definitions are meaningful since M is a linear subspace of X . For example, if $x \sim x'$ and $y \sim y'$ then $x + y \sim x' + y'$, so that the definition is independent of the particular representatives taken from the various equivalence classes. Then it is straightforward to get the following.

Proposition 16.26 (Linear structure of quotients). *The quotient space X/M is a linear space.*

Let us now define

$$\|[x]\| := \inf\{\|y\| / y \in [x]\}. \quad (16.2)$$

Proposition 16.27 (Quotient seminorm). *The equality 16.2 defines a seminorm on X/M .*

Proof. Let $\alpha \in \mathbb{R}^*$ and $[x]$ be in X/M . Then

$$\begin{aligned} \|\alpha[x]\| &:= \|[\alpha x]\| \\ &= \inf\{\|y\| / y \in [\alpha x]\} \\ &= \inf\{\|\alpha x + m\| / m \in M\} \\ &= \inf\{\|\alpha x + \alpha m\| / m \in M\}, \end{aligned}$$

since the mapping $m \mapsto \alpha m$ is a bijection. Thus

$$\|\alpha[x]\| = |\alpha| \inf\{\|x + m\| / m \in M\} = |\alpha| \|[x]\|.$$

On the other hand, since $[0] = M$ we have that $\|[0]\| = 0$. Next, we consider the triangle inequality:

$$\begin{aligned}\|[x] + [y]\| &:= \|[x + y]\| \\ &= \inf\{\|x + y + m\| / m \in M\} \\ &= \inf\{\|x + y + m' + m''\| / m', m'' \in M\} \\ &\leq \inf\{\|x + m'\| + \|y + m''\| / m', m'' \in M\} \\ &\leq \|[x]\| + \|[y]\|.\end{aligned}$$

□

To see whether or not it is a norm, all that remains is to see if $\|[x]\| = 0$ implies $[x] = 0$. This may be wrong in general but next proposition proves that it is true if M is closed.

Proposition 16.28 (Quotient norm). *Assume that M is a closed linear subspace of the normed space X . Then 16.2 defines a norm on X/M , called the quotient norm.*

Proof. Assume that $\|[x]\| = 0$. Then for any $n \in \mathbb{N}^*$ there exists $m_n \in M$ such that $\|x + m_n\| \leq 1/n$. Thus the sequence $(m_n)_n$ is converging to $-x$. Since M is closed this yields that $-x$ is in M , which is a linear space, so x is in M too. □

In the case where M is closed we also have the two following properties.

Proposition 16.29 (Continuity of the quotient projection). *Let M is a closed linear subspace of a normed space X . Then the canonical projection $\pi : x \in X \mapsto [x] \in X/M$ is continuous.*

Proof. Assume that $(x_n)_n$ is a sequence in X converging to x in X . Then

$$\|\pi(x_n) - \pi(x)\| = \inf\{\|x_n - x + m\| / m \in M\} \leq \|x_n - x\|,$$

since $0 \in M$. Therefore the sequence $(\pi(x_n))_n$ is converging to $\pi(x)$ in X/M . □

Proposition 16.30 (Completeness of quotients). *Let M is a closed linear subspace of a Banach space X . Then X/M is a Banach space.*

In order to prove Proposition 16.30 it only remains to prove that X/M is complete. We are going to use the criterion of Proposition 16.24. Actually the method we will use is quite general and we therefore give first a general statement. Then we will go back to the proof of Proposition 16.30.

Proposition 16.31 (Quotient by the kernel). *Let X be a Banach space and Y be a normed vector space. Let T be a linear continuous surjective mapping from X to Y . Assume there exists some constant $C > 0$ such that for any y in Y there exists x in X such that*

$$T(x) = y \text{ and } \|x\| \leq C\|y\|. \quad (16.3)$$

Then Y is a Banach space.

Proof. Assume that $(y_n)_n$ is a sequence in Y such that $\sum_{k \geq 0} \|y_k\| < \infty$. Define the sequence $(w_n)_n$ by setting $w_n := \sum_{k=0}^n y_k$. We want to prove that the sequence $(w_n)_n$ is converging in Y so that thanks to Proposition 16.24 we will get that Y is complete. Thanks to the assumption there exists a sequence $(x_n)_n$ in X such that $T(x_n) = y_n$ and $\|x_n\| \leq C\|y_n\|$. As a consequence we get

$$\sum_{k \geq 0} \|x_k\| \leq C \sum_{k \geq 0} \|y_k\| < \infty.$$

Using Proposition 16.24 for X which is assumed to be a Banach space we get that the sequence $(z_n)_n$ defined by $z_n := \sum_{k=0}^n x_k$ is convergent to some x in X . Since T is linear we have $T(z_n) = w_n$, and T is also continuous so that $T(z_n)$ is converging to $T(x)$. This proves that the sequence $(w_n)_n$ is converging in Y and therefore the proof of the proposition is done. $\square$

Let us now go back to the proof of Proposition 16.30.

Proof. We are going to apply Proposition 16.31 with $Y = X/M$ and $T = \pi$. We know that π is linear, continuous, and surjective. Let us prove that the last assumption of Proposition 16.31 is satisfied with $C = 2$. Let $y = [x]$ be in X/M . If $y = [x] = 0$ then (16.3) holds true with $x = 0$. If $y \neq 0$ then $\|y\| > 0$ and since by definition, $\|y\| = \inf\{\|x\| / x \in y\}$ we have that there exists $x \in X$ such that $\|x\| \leq 2\|y\|$. $\square$

16.3 Compactness

Several notions of compactness are available. The following one can be formulated in a general setting.

Definition 16.32 (Compact metric spaces). *We say that a topological space (X, τ) is compact if any open cover has a finite subcover, i.e. for every arbitrary collection $\{U_i\}_{i \in I}$ of open subsets of X such that $X \subset \cup_{i \in I} U_i$, there is a finite subset $J \subset I$ such that $X \subset \cup_{i \in J} U_i$.*

It is a good exercise to prove the following theorem in order to understand the power of the previous definition.

Theorem 16.33 (Compactness theorem for continuous maps and Heine–Cantor theorem). *Every continuous image of a compact set is compact. Moreover a continuous function on a compact set is uniformly continuous.*

In a metric space, compactness can be formulated in terms of sequences. Let us first recall a few facts about the notion of limit points. Let S be a subset of a topological space X . We say that a point $x \in X$ is a limit point of S if every open set containing x also contains a point of S other than x itself. In a metric space, it is equivalent to requiring that every neighbourhood of x contains infinitely many points of S .

Let us also define what we mean by a totally bounded space.

Definition 16.34 (Totally bounded metric spaces). *We say that a metric space (X, d) is totally bounded if for every $\varepsilon > 0$, there exists a finite cover of X by open balls of radius less than ε .*

Since for every $\varepsilon > 0$, $\cup_{x \in X} B(x, \varepsilon)$ is an open cover of X , it follows from Definitions 16.32 and 16.34 that a compact metric space is totally bounded.

Proposition 16.35 (Compactness and limit points). *A metric space is compact if and only if every sequence admits a convergent subsequence.*

Proof. We start by proving the necessary condition. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in a compact metric space X . We argue by contradiction and assume that it does not admit any convergent subsequence. Then no point of X can occur infinitely many times in the sequence, otherwise a constant subsequence would converge. Moreover, the set $\{x_n : n \in \mathbb{N}\}$ has no limit point in the sense defined above: indeed, if y were a limit point of that set, we could choose inductively n_k such that $n_{k+1} > n_k$ and $x_{n_k} \in B(y, 1/k)$, which would give a subsequence converging to y . Therefore, for every y in X , there exists $r(y) > 0$ such that the ball $B(y, r(y))$ contains only finitely many elements of the sequence. The collection of these balls is an open cover of X , from which we extract a finite subcover. This would imply that the sequence is contained in the union of finitely many balls, each of them containing only finitely many elements of the sequence, which is impossible.

Let us now prove the sufficient condition. We therefore consider a metric space X such that every sequence admits a convergent subsequence. We first prove that X is totally bounded. Proceeding by contradiction, this would yield the existence of some $\varepsilon > 0$ and some sequence $(x_n)_{n \in \mathbb{N}}$ such that for any $m, n \in \mathbb{N}$ with $m \neq n$, $d(x_m, x_n) > \varepsilon$. Such a sequence cannot have a convergent subsequence, which is a contradiction. Hence X is totally bounded.

Let us now prove that X is compact. We consider an open cover $\{U_i\}_{i \in I}$ of X . We define the mapping

$$R : x \in X \mapsto R(x) := \sup\{r > 0 : \exists i \in I \text{ with } B(x, r) \subset U_i\} > 0,$$

which is lower semicontinuous. Indeed, if not, there would exist a sequence $(x_n)_{n \in \mathbb{N}} \subset X$ converging to x in X such that

$$R(x) > \liminf_{n \rightarrow \infty} R(x_n),$$

and we could choose ρ and ρ' such that $R(x) > \rho' > \rho > \liminf_n R(x_n)$. Let $i \in I$ be such that $B(x, \rho') \subset U_i$. Since the sequence $(x_n)_{n \in \mathbb{N}}$ converges to x , for n large enough one has $B(x_n, \rho) \subset B(x, \rho') \subset U_i$, which contradicts the fact that $\rho > \liminf_n R(x_n)$.

Let us now consider $\varepsilon := \inf_{x \in X} R(x)$ and $(x_n)_{n \in \mathbb{N}}$ a minimizing sequence for R , that is such that

$$\lim_{n \rightarrow \infty} R(x_n) = \varepsilon.$$

By assumption, the sequence $(x_n)_{n \in \mathbb{N}}$ admits a subsequence $(x_{n_k})_{k \in \mathbb{N}}$ converging to some x in X . Since R is lower semicontinuous, we have

$$0 < R(x) \leq \liminf_{k \rightarrow \infty} R(x_{n_k}) = \varepsilon.$$

Hence $\varepsilon > 0$. We already know that X is totally bounded. Thus there exist $x_1, \dots, x_n$ such that

$$X \subset \bigcup_{i=1}^n B(x_i, \varepsilon/2).$$

By definition of ε , for every $i = 1, \dots, n$ one has $\varepsilon \leq R(x_i)$. Therefore, for each i , we can choose r_i and $j_i \in I$ such that

$$\frac{\varepsilon}{2} < r_i < R(x_i) \quad \text{and} \quad B(x_i, r_i) \subset U_{j_i}.$$

In particular,

$$B(x_i, \varepsilon/2) \subset U_{j_i}$$

for every $i = 1, \dots, n$, and finally

$$X \subset \bigcup_{i=1}^n U_{j_i},$$

which gives a finite subcover of X . □

Next proposition gives another criterion of compactness for metric spaces.

Proposition 16.36 (Compactness, completeness and total boundedness). *A metric space is compact if and only if it is complete and totally bounded.*

Proof. According to Proposition 16.35, in a compact metric space every sequence admits a convergent subsequence. In particular, every Cauchy sequence has a convergent subsequence, hence converges to the same limit. We deduce that a compact metric space is complete. Moreover, we have already seen that a compact metric space is totally bounded. Therefore, it only remains to prove the converse statement.

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in a complete and totally bounded metric space X . We are going to prove that $(x_n)_{n \in \mathbb{N}}$ admits a convergent subsequence by Cantor's diagonal argument. Since X is totally bounded, there exists a ball B_1 of radius 1 which contains infinitely many elements of the sequence. Extract from them a subsequence that we denote by $(x_n^1)_n$. By iteration we obtain that for any $k \geq 2$, there exists a ball B_k of radius $1/k$ which contains infinitely many elements of the subsequence $(x_n^{k-1})_n$; extracting again, we obtain a subsequence $(x_n^k)_n$ of $(x_n^{k-1})_n$ contained in B_k . Then the diagonal sequence $(x_n^n)_n$ has the Cauchy property, because for any $k \geq 1$ and any $m, n \geq k$, the points x_m^m and x_n^n both belong to B_k , hence their distance is at most $2/k$. Since X is complete, the sequence $(x_n^n)_n$ converges. Thus every sequence in X admits a convergent subsequence, and Proposition 16.35 shows that X is compact. $\square$

16.4 Separability

Definition 16.37 (Separable metric spaces). *We say that a topological space (X, τ) is separable if it contains a countable dense subset, i.e., there exists a sequence $(x_n)_{n \in \mathbb{N}}$ of elements of X such that every nonempty open subset of the space contains at least one element of the sequence.*

Any topological space which is itself finite or countable is separable. An important example of uncountable separable space is the real line (with its usual topology), in which the rational numbers form a countable dense subset.

Proposition 16.38 (Compact metric spaces are separable). *Every compact metric space is separable.*

Proof. Let (X, d) be a compact metric space. For any $k \in \mathbb{N}^*$, the family $\{B(x, 1/k) : x \in X\}$ is an open cover of X . By compactness, there exist $x_1^k, \dots, x_{n_k}^k$ in X such that

$$X = \bigcup_{j=1}^{n_k} B(x_j^k, 1/k).$$

Then the collection

$$\bigcup_{k \in \mathbb{N}^*} \bigcup_{j=1}^{n_k} \{x_j^k\}$$

is a countable dense subset of X . $\square$

Proposition 16.39 (Separability of $C(K)$). *Let (K, d) be a compact metric space. Then the space $C(K)$, endowed with the uniform norm, is separable.*

Proof. By Proposition 16.38, there exists a countable dense subset

$$D = \{a_j : j \in \mathbb{N}^*\} \subset K.$$

For every m, r in $\mathbb{N}^*$, every choice of indices $j_1, \dots, j_r$ in $\mathbb{N}^*$, and every choice of rational numbers $q_1, \dots, q_r$ in $\mathbb{Q}$, define the function

$$g(x) := \max_{1 \leq i \leq r} (q_i - md(x, a_{j_i})), \quad x \in K.$$

Each such function is continuous, as a finite maximum of continuous functions. Let $\mathcal{F}$ be the collection of all functions obtained in this way. Since each function in $\mathcal{F}$ is determined by finitely many integers and rational numbers, the set $\mathcal{F}$ is countable.

We claim that $\mathcal{F}$ is dense in $\mathcal{C}(K)$ for the uniform norm. Let $f \in \mathcal{C}(K)$ and let $\varepsilon > 0$. Since K is compact, the function f is bounded and uniformly continuous. Set

$$M := \|f\|_\infty.$$

Choose $\eta > 0$ such that

$$d(x, y) < \eta \implies |f(x) - f(y)| < \frac{\varepsilon}{3}.$$

Next choose m in $\mathbb{N}^*$ so large that

$$m\eta > 2M + \frac{\varepsilon}{3}.$$

Since D is dense in K and K is compact, we can find finitely many points $a_{j_1}, \dots, a_{j_r}$ in D such that

$$K \subset \bigcup_{i=1}^r B\left(a_{j_i}, \frac{\varepsilon}{3m}\right).$$

For each $i = 1, \dots, r$, choose a rational number q_i such that

$$|q_i - f(a_{j_i})| < \frac{\varepsilon}{3}.$$

Define

$$g(x) := \max_{1 \leq i \leq r} (q_i - md(x, a_{j_i})), \quad x \in K.$$

Then g belongs to $\mathcal{F}$.

Fix $x \in K$. Choose i_0 in $\{1, \dots, r\}$ such that $d(x, a_{j_{i_0}}) < \varepsilon/(3m)$. Then

$$\begin{aligned} g(x) &\geq q_{i_0} - md(x, a_{j_{i_0}}) \\ &> f(a_{j_{i_0}}) - \frac{\varepsilon}{3} - \frac{\varepsilon}{3} \\ &> f(x) - \varepsilon. \end{aligned}$$

On the other hand, for every $i = 1, \dots, r$, if $d(x, a_{j_i}) < \eta$, then

$$q_i - md(x, a_{j_i}) \leq q_i \leq f(a_{j_i}) + \frac{\varepsilon}{3} < f(x) + \frac{2\varepsilon}{3},$$

whereas if $d(x, a_{j_i}) \geq \eta$, then

$$q_i - md(x, a_{j_i}) \leq M + \frac{\varepsilon}{3} - m\eta < -M \leq f(x) < f(x) + \frac{2\varepsilon}{3},$$

because $f(x) \geq -M$. Hence

$$g(x) \leq f(x) + \frac{2\varepsilon}{3}.$$

Since this holds for every x in K , we conclude that

$$\|f - g\|_\infty \leq \varepsilon.$$

Thus $\mathcal{F}$ is dense in $\mathcal{C}(K)$, which proves that $\mathcal{C}(K)$ is separable. □

Let us finish with the following useful criterion for a metric space to be not separable.

Proposition 16.40 (Nonseparability of $\ell^\infty(\mathbb{N})$). *If a metric space (X, d) contains a uncountable subset Y such that*

$$\delta := \inf\{d(y, y') : y, y' \in Y, y \neq y'\} > 0,$$

then X is not separable.

Proof. We argue by contradiction. Let us assume that (X, d) is separable and therefore contains a countable dense subset $(x_n)_{n \in \mathbb{N}}$. We can then define a map by associating to any $y \in Y$ the smallest $n \in \mathbb{N}$ such that $d(y, x_n) < \delta/3$. This map turns out to be injective because if $d(y, x_n) < \delta/3$ and $d(y', x_n) < \delta/3$, then $d(y, y') < 2\delta/3$ which is possible (when y and $y' \in Y$) only if $y = y'$. We deduce that Y is countable which is absurd. $\square$

For the rest of this section, Ω denotes an open subset of $\mathbb{R}^N$. The following lemma is the elementary Urysohn-type separation result that will be used below.

Lemma 16.41 (Urysohn lemma, compact-support form). *Let K be a compact subset of Ω and let V be an open subset of Ω such that $K \subset V$. Then there exists a function $f \in \mathcal{C}_c(V; [0, 1])$ such that $f = 1$ on K .*

Proof. If $K = \emptyset$, the function $f = 0$ is suitable. We therefore assume that K is nonempty. For every x in K , since V is open and contains x , there exists $r_x > 0$ such that

$$\overline{B(x, 2r_x)} \subset V.$$

The family $(B(x, r_x))_{x \in K}$ is an open covering of the compact set K , so we can extract a finite subcover

$$K \subset \bigcup_{j=1}^m B(x_j, r_{x_j}).$$

We set

$$\delta := \min_{1 \leq j \leq m} r_{x_j} > 0.$$

Then

$$\{y \in \Omega : \text{dist}(y, K) \leq \delta\} \subset V.$$

We define

$$W := \{x \in \Omega : \text{dist}(x, K) < \delta/3\}.$$

Since K is compact, the set W is open and bounded, and

$$K \subset W \subset \overline{W} \subset \{y \in \Omega : \text{dist}(y, K) \leq \delta/3\} \subset V.$$

Now set

$$f(x) := \frac{\text{dist}(x, \Omega \setminus W)}{\text{dist}(x, \Omega \setminus W) + \text{dist}(x, K)}.$$

The denominator is always positive, so f is continuous on Ω . Moreover $0 \leq f \leq 1$, the equality $f = 1$ holds on K , and $f = 0$ on $\Omega \setminus W$. Therefore

$$\text{supp}(f) \subset \overline{W} \subset V,$$

and $\overline{W}$ is compact. Hence f belongs to $\mathcal{C}_c(V; [0, 1])$. $\square$

A consequence of the previous lemma is the existence of a partition of unity in a fairly general context.

Lemma 16.42 (Finite partition-of-unity lemma). *Let $V_1, \dots, V_n$ be open subsets of Ω and let K be a compact set such that $K \subset \bigcup_{i=1}^n V_i$. Then, for each $i = 1, \dots, n$, there exist functions $f_i \in \mathcal{C}_c(V_i; [0, 1])$ such that $\sum_{i=1}^n f_i = 1$ on K .*

Proof. For each $x \in K$, there exists an open ball B_x centered at x and such that $\overline{B_x} \subset V_i$ for some i (depending on x). Hence $K \subset \bigcup_{x \in K} B_x$, and since K is compact, one can extract a finite covering $K \subset \bigcup_{j=1}^p B_{x_j}$. Define K_i as the union of those closed balls $\overline{B_{x_j}}$ which are contained in V_i . Then K_i is a compact subset of V_i and by Lemma 16.41, there exists a function $g_i \in \mathcal{C}_c(V_i; [0, 1])$ such that $g_i = 1$ on K_i . Set

$$G := \sum_{j=1}^n g_j \quad \text{and} \quad W := \{x \in \Omega : G(x) > 0\}.$$

Then W is an open subset of Ω and $K \subset W$. By Lemma 16.41, there exists a function $h \in \mathcal{C}_c(W; [0, 1])$ such that $h = 1$ on K . For each $i = 1, \dots, n$, define

$$f_i(x) := \begin{cases} \frac{h(x)g_i(x)}{G(x)} & \text{if } x \in W, \\ 0 & \text{if } x \in \Omega \setminus W. \end{cases}$$

Since $\text{supp}(h) \subset W$, each f_i is continuous on Ω . Moreover, $0 \leq f_i \leq 1$ and

$$\text{supp}(f_i) \subset \text{supp}(g_i) \subset V_i,$$

so $f_i \in \mathcal{C}_c(V_i; [0, 1])$. Finally, on K we have $h = 1$ and $G > 0$, hence

$$\sum_{i=1}^n f_i = \frac{h \sum_{i=1}^n g_i}{G} = 1.$$

Therefore the functions f_i fulfill the conclusion of the lemma. $\square$

Another useful consequence of Lemma 16.41 is the following characterization of the functional space $\mathcal{C}_0(\Omega)$, defined for any open subset $\Omega \subset \mathbb{R}^N$ as the closure of the space $\mathcal{C}_c(\Omega)$ for the uniform topology.

Proposition 16.43 (The space $\mathcal{C}_0(X)$). *Let $f \in \mathcal{C}(\Omega)$. Then f belongs to $\mathcal{C}_0(\Omega)$ if and only if for any $\varepsilon > 0$, there exists a compact set $K_\varepsilon \subset \Omega$ such that $|f| < \varepsilon$ on $\Omega \setminus K_\varepsilon$.*

Proof. First, consider $f \in \mathcal{C}_0(\Omega)$. Then there exists a sequence $(f_n) \subset \mathcal{C}_c(\Omega)$ such that $f_n \rightarrow f$ uniformly in Ω as $n \rightarrow +\infty$. Let $\varepsilon > 0$ and choose $n_\varepsilon \in \mathbb{N}$ such that $\|f_{n_\varepsilon} - f\|_\infty < \varepsilon/2$. Define

$$K := \{x \in \Omega : |f_{n_\varepsilon}(x)| \geq \varepsilon/2\}.$$

Then K is a compact subset of Ω and for all $x \in \Omega \setminus K$,

$$|f(x)| \leq |f(x) - f_{n_\varepsilon}(x)| + |f_{n_\varepsilon}(x)| < \varepsilon.$$

Conversely, let $\varepsilon > 0$ and let $K \subset \Omega$ be a compact set such that $|f| < \varepsilon$ on $\Omega \setminus K$. By Lemma 16.41 one can find $g \in \mathcal{C}_c(\Omega; [0, 1])$ such that $g = 1$ on K . Set $h := fg$. Since f is continuous and g has compact support, we have $h \in \mathcal{C}_c(\Omega)$. Moreover, $h = f$ on K and therefore

$$\|f - h\|_\infty = \sup_{x \in \Omega \setminus K} |f(x)(1 - g(x))| \leq \sup_{x \in \Omega \setminus K} |f(x)| < \varepsilon.$$

Thus $f \in \mathcal{C}_0(\Omega)$. $\square$

Proposition 16.44 (Separability of $C_0(X)$). *The space $C_0(\Omega)$, endowed with the uniform norm, is separable.*

Proof. For each $n \in \mathbb{N}^*$, let

$$K_n := \left\{ x \in \Omega : |x| \leq n \text{ and } \text{dist}(x, \mathbb{R}^N \setminus \Omega) \geq \frac{1}{n} \right\}.$$

Then K_n is a compact subset of Ω for every n , and $\bigcup_{n \geq 1} K_n = \Omega$. We also introduce the open sets

$$V_n := \left\{ x \in \Omega : |x| < n \text{ and } \text{dist}(x, \mathbb{R}^N \setminus \Omega) > \frac{1}{n} \right\}.$$

For each n , define

$$Y_n := \{f \in C_0(\Omega) : \text{supp}(f) \subset K_n\}.$$

The restriction map $f \mapsto f|_{K_n}$ is an isometric embedding of Y_n into $C(K_n)$. Since $C(K_n)$ is separable by Proposition 16.39, the metric space Y_n is also separable.

We now prove that $\bigcup_{n \geq 1} Y_n$ is dense in $C_0(\Omega)$. Let $f \in C_0(\Omega)$ and let $\varepsilon > 0$. By Proposition 16.43, there exists a compact set $K \subset \Omega$ such that $|f| < \varepsilon/2$ on $\Omega \setminus K$. Since K is compact and contained in Ω , for n large enough we have $K \subset V_n$. By Lemma 16.41, there exists $g \in C_c(V_n; [0, 1])$ such that $g = 1$ on K . Then $\text{supp}(g) \subset K_n$, so the function $h := fg$ belongs to Y_n . Moreover, $h = f$ on K , and for $x \in \Omega \setminus K$ we have

$$|f(x) - h(x)| = |f(x)|(1 - g(x)) \leq |f(x)| < \frac{\varepsilon}{2}.$$

Hence $\|f - h\|_\infty < \varepsilon/2 < \varepsilon$, which proves the density of $\bigcup_{n \geq 1} Y_n$.

Since $C_0(\Omega)$ is the closure of a countable union of separable subsets of a metric space, it is separable. $\square$

16.5 Space of continuous linear maps

This section is devoted to $\mathcal{L}_c(X, Y)$, the space of bounded linear operators from a normed linear space X to a normed linear space Y . Let us recall that a linear operator T is bounded if one of the following assertions is satisfied:

- 1) T is bounded on every ball,
- 2) T is bounded on some ball,
- 3) T is continuous at every point,
- 4) T is continuous at some point,
- 5) T is uniformly continuous,
- 6) T is Lipschitz.

Theorem 16.45 (Banach space of continuous linear maps). *If X and Y are some normed linear spaces, then $\mathcal{L}_c(X, Y)$ is a normed linear space with the norm*

$$\begin{aligned}\|T\|_{\mathcal{L}_c(X, Y)} &:= \sup_{x \neq x' \in X} \frac{\|Tx - Tx'\|_Y}{\|x - x'\|_X} \\ &= \sup_{x \in X \setminus \{0\}} \frac{\|Tx\|_Y}{\|x\|_X}, \\ &= \sup_{\|x\|_X \leq 1} \|Tx\|_Y, \\ &= \sup_{\|x\|_X = 1} \|Tx\|_Y.\end{aligned}$$

If moreover Y is a Banach space then $\mathcal{L}_c(X, Y)$ is a Banach space.

Proof. It is easy to check that $\mathcal{L}_c(X, Y)$ is a normed linear space, so we only have to prove completeness. When no confusion is possible, we simply denote by $\|\cdot\|$ the operator norm on $\mathcal{L}_c(X, Y)$.

Let (T_n) be a Cauchy sequence in $\mathcal{L}_c(X, Y)$. For every x in X , the sequence $(T_n x)$ is Cauchy in the Banach space Y , hence converges to some element that we denote by Tx . The map $T: X \rightarrow Y$ is clearly linear. Since $(\|T_n\|)_n$ is a Cauchy sequence in $\mathbb{R}$, it is bounded: there exists $K \geq 0$ such that $\|T_n\| \leq K$ for every n . Passing to the limit in

$$\|T_n x\|_Y \leq K\|x\|_X$$

gives

$$\|Tx\|_Y \leq K\|x\|_X$$

for every x in X . Thus T belongs to $\mathcal{L}_c(X, Y)$.

It remains to prove that $T_n \rightarrow T$ in operator norm as $n \rightarrow +\infty$. For every $n \geq 1$, we have

$$\begin{aligned}\|T_n - T\| &= \sup_{\|x\|_X \leq 1} \|T_n x - Tx\|_Y \\ &\leq \limsup_{m \rightarrow \infty} \sup_{\|x\|_X \leq 1} \|T_n x - T_m x\|_Y \\ &\leq \limsup_{m \rightarrow \infty} \|T_n - T_m\|_{\mathcal{L}_c(X, Y)}.\end{aligned}$$

Since (T_n) is Cauchy in $\mathcal{L}_c(X, Y)$, the right-hand side tends to 0 as $n \rightarrow \infty$. Therefore $\|T_n - T\| \rightarrow 0$ as $n \rightarrow +\infty$. $\square$

If $T \in \mathcal{L}_c(X, Y)$ and $U \in \mathcal{L}_c(Y, Z)$, then $UT = U \circ T \in \mathcal{L}_c(X, Z)$ and

$$\|UT\|_{\mathcal{L}_c(X, Z)} \leq \|U\|_{\mathcal{L}_c(Y, Z)} \|T\|_{\mathcal{L}_c(X, Y)}.$$

In particular, $\mathcal{L}_c(X) := \mathcal{L}_c(X, X)$ is an algebra, *i.e.*, it has an additional “multiplication” operation which makes it a non-commutative algebra, and this operation is continuous.

The dual space of X is $X' := \mathcal{L}_c(X, \mathbb{R})$ (or $\mathcal{L}_c(X, \mathbb{C})$ for complex vector spaces). According to the previous proposition it is a Banach space (whether X is or not).

Theorem 16.46 (Uniform Boundedness Principle, or Banach–Steinhaus theorem). *Let X be a Banach space and let Y be a normed vector space. Let $(T_i)_{i \in I}$ be a family in $\mathcal{L}_c(X, Y)$. Assume that for every x in X ,*

$$\sup_{i \in I} \|T_i x\|_Y < +\infty.$$

Then

$$\sup_{i \in I} \|T_i\|_{\mathcal{L}_c(X, Y)} < +\infty.$$

Proof. For every $m \geq 1$, define

$$E_m := \left\{ x \in X : \sup_{i \in I} \|T_i x\|_Y \leq m \right\}.$$

Each E_m is closed. Indeed, if $(x_n)_n$ is a sequence in E_m which converges to x in X , then for every i in I ,

$$\|T_i x\|_Y = \lim_{n \rightarrow \infty} \|T_i x_n\|_Y \leq m,$$

and therefore x belongs to E_m . The pointwise boundedness assumption gives

$$X = \bigcup_{m \geq 1} E_m.$$

By the Baire category theorem, Theorem 16.17, there exist $m_0 \geq 1$, x_0 in X , and $r > 0$ such that

$$B(x_0, r) \subset E_{m_0}.$$

Let h in X with $\|h\|_X < r$. Then $x_0 + h$ and x_0 belong to E_{m_0} , and for every i in I ,

$$\|T_i h\|_Y \leq \|T_i(x_0 + h)\|_Y + \|T_i x_0\|_Y \leq 2m_0.$$

Let x in X with $\|x\|_X \leq 1$. Applying the previous estimate to $h = (r/2)x$ gives, for every i in I ,

$$\|T_i x\|_Y \leq \frac{4m_0}{r}.$$

Hence $\|T_i\|_{\mathcal{L}_c(X, Y)} \leq 4m_0/r$ for every i in I . □

Definition 16.47 (Weak-* topology). *Given some normed linear spaces X and Y , and a sequence $(u_n)_n$ in $\mathcal{L}_c(X, Y)$, we say that*

- 1) $(u_n)_n$ converges strongly to u in $\mathcal{L}_c(X, Y)$ if $\|u_n - u\|_{\mathcal{L}_c(X, Y)} \rightarrow 0$ as $n \rightarrow \infty$,
- 2) $(u_n)_n$ converges pointwise to u in $\mathcal{L}_c(X, Y)$ if for any $x \in X$, $(u_n(x))_n$ converges to $u(x)$ in Y .

16.6 Geometry of Banach spaces and identification of their dual

We will slightly restrict our attention to spaces which enjoy two properties described in the definitions below.

Definition 16.48 (Uniform convergence on compact sets). *We say that a normed vector space X is uniformly convex if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for any $x, y \in X$ with $\|x\|_X \leq 1$, $\|y\|_X \leq 1$ and $\left\|\frac{x+y}{2}\right\|_X > 1 - \delta$, then $\left\|\frac{x-y}{2}\right\|_X < \varepsilon$.*

In particular the unit sphere of a uniformly convex normed vector space contains no segments. As a first application, let us give the following proposition.

Proposition 16.49 (Metritzation of weak-* convergence on bounded sets). *Let X be a uniformly convex Banach space, and let f in $X' \setminus \{0\}$. Then there exists a unique u in X such that $\|u\|_X = 1$ and $f(u) = \|f\|_{X'}$.*

Proof. Let $(u_n)_n$ be a maximizing sequence. We assume without loss of generality that $f(u_n) \geq 0$. Let $\varepsilon > 0$ and δ be given as in Definition 16.48. Since the sequence $(u_n)_n$ maximizes $|f(v)|$ over $\{v \in X : \|v\|_X = 1\}$, there exists $N \in \mathbb{N}$ such that for any $n \geq N$, $f(u_n) > (1 - \delta)\|f\|_{X'}$. Therefore for any $n, m \geq N$, we have

$$\frac{1}{2}(f(u_n) + f(u_m)) > (1 - \delta)\|f\|_{X'}.$$

But since

$$\frac{1}{2}(f(u_n) + f(u_m)) = f\left(\frac{1}{2}(u_n + u_m)\right) \leq \|f\|_{X'} \left\|\frac{1}{2}(u_n + u_m)\right\|,$$

it follows, by combining both previous inequalities and dividing by $\|f\|_{X'}$, that

$$\left\|\frac{1}{2}(u_n + u_m)\right\| > 1 - \delta. \quad (16.4)$$

Moreover, since $\|u_n\|_X = 1$, then $\frac{f(u_n)}{\|f\|_{X'}} \leq 1$, and using the uniform convexity of X , we can infer from (16.4) that $\|u_n - u_m\|_X < 2\varepsilon$, that is the sequence $(u_n)_n$ satisfies the Cauchy property. Since X is complete, there exists $u \in X$ such that $u_n \rightarrow u$ as $n \rightarrow +\infty$. Using the continuity of the norm and of the function f we get that $f(u) = \|f\|_{X'}$.

Uniqueness is proved by proceeding in the same way. $\square$

Definition 16.50 (Smooth normed spaces). *We say that a normed vector space X is smooth if for any linearly independent elements x and y , the function $t \in \mathbb{R} \mapsto \|x + ty\|_X$ is differentiable for all $t \in \mathbb{R}$.*

Let us stress that this is equivalent to the fact that for any $u \in X \setminus \{0\}$, then the limit

$$\lim_{\varepsilon \rightarrow 0} \frac{\|u + \varepsilon v\|_X - \|u\|_X}{\varepsilon}$$

exists for any $v \in X$.

Proposition 16.51 (Gateaux differentiability of the norm). *Let X be a smooth normed vector space over $\mathbb{R}$. Then the norm is Gateaux-differentiable outside 0, in the following sense: for any u in $X \setminus \{0\}$, there exists L in X' , that we denote $D\|\cdot\|_X(u)$, such that for any v in X ,*

$$\lim_{\varepsilon \rightarrow 0} \frac{\|u + \varepsilon v\|_X - \|u\|_X}{\varepsilon} = L(v).$$

Moreover

$$\|D\|\cdot\|_X(u)\|_{X'}\|u\|_X = D\|\cdot\|_X(u)(u) = \|u\|_X. \quad (16.5)$$

Proof. Since X is a smooth normed vector space, for any $u \in X \setminus \{0\}$, for any $v \in X$,

$$D\|\cdot\|_X(u)(v) := \lim_{\varepsilon \rightarrow 0} \frac{\|u + \varepsilon v\|_X - \|u\|_X}{\varepsilon}$$

exists. We first prove that $D\|\cdot\|_X(u)$ is linear. For any $\lambda \in \mathbb{R} \setminus \{0\}$ and any $u, v \in X$,

$$\frac{\|u + \varepsilon \lambda v\|_X - \|u\|_X}{\varepsilon} = \lambda \frac{\|u + \varepsilon \lambda v\|_X - \|u\|_X}{\lambda \varepsilon}$$

so that

$$D\|\cdot\|_X(u)(\lambda v) = \lambda D\|\cdot\|_X(u)(v).$$

Now, using the triangle inequality, we have for any v_1, v_2 in X , that

$$\frac{\|u + \varepsilon(v_1 + v_2)\|_X - \|u\|_X}{\varepsilon} \leq \frac{\|u + 2\varepsilon v_1\|_X - \|u\|_X}{2\varepsilon} + \frac{\|u + 2\varepsilon v_2\|_X - \|u\|_X}{2\varepsilon}$$

so that, passing to the limit $\varepsilon \rightarrow 0$,

$$D\|\cdot\|_X(u)(v_1 + v_2) \leq D\|\cdot\|_X(u)(v_1) + D\|\cdot\|_X(u)(v_2). \quad (16.6)$$

Applying now (16.6) to $-v_1$ and $-v_2$ implies that the previous inequality is actually an equality. Hence the map $D\|\cdot\|_X(u)$ is linear from X to $\mathbb{R}$.

We next prove that it is continuous. Indeed using the triangle inequality we have that for any $v \in X$,

$$\left| \frac{\|u + \varepsilon v\|_X - \|u\|_X}{\varepsilon} \right| \leq \|v\|_X,$$

so that

$$|D\|\cdot\|_X(u)(v)| \leq \|v\|_X.$$

Therefore $D\|\cdot\|_X(u)$ is a continuous linear map from X to $\mathbb{R}$ with norm $\|D\|\cdot\|_X(u)\|_{X'} \leq 1$. Moreover since $D\|\cdot\|_X(u)(u) = \|u\|$, then $\|D\|\cdot\|_X(u)\|_{X'} = 1$ which yields (16.5). $\square$

Theorem 16.52 (Factorization theorem for smooth uniformly convex Banach spaces). *Let X be a smooth and uniformly convex Banach space, and let f in $X' \setminus \{0\}$. Then there exists a unique $u \in X$ such that $\|u\|_X = 1$ and $f = \|f\|_{X'} D\|\cdot\|_X(u)$.*

Proof. Let u be given by Proposition 16.49, and let $v \in X$. The function

$$g : \varepsilon \mapsto f(u + \varepsilon v) - \|f\|_{X'} \|u + \varepsilon v\|_X$$

is nonpositive, vanishes for $\varepsilon = 0$, and has a derivative at $\varepsilon = 0$, since f is linear continuous and X is smooth. Therefore $g'(0) = 0$, i.e.

$$f(v) - \|f\|_{X'} D\|\cdot\|_X(u)(v) = 0.$$

Let us now prove the uniqueness part of Theorem 16.52. If $u \in X$ is such that $\|u\|_X = 1$ and $f = \|f\|_{X'} D\|\cdot\|_X(u)$, then in particular $f(u) = \|f\|_{X'} D\|\cdot\|_X(u)(u) = \|f\|_{X'}$, according to (16.5). It then suffices to recall the uniqueness part of Proposition 16.49. $\square$

An important corollary is the following constructive version of Hahn-Banach theorem which allows the extension of continuous linear forms defined on a subspace of a smooth uniformly convex Banach space to the whole space without increasing the norm.

Corollary 16.53 (Constructive Hahn-Banach theorem in the smooth uniformly convex case). *Let X be a smooth uniformly convex Banach space, Y be a subspace of X and $f \in Y'$. There exists a unique extension $\tilde{f}$ of f to X such that $\|\tilde{f}\|_{X'} = \|f\|_{Y'}$.*

Proof. Let f in Y' . Only the case $f \neq 0$ deserves a proof. According to Theorem 16.19 there exists a unique extension $\bar{f}$ of f to the closure $\bar{Y}$ of Y in X and $\|\bar{f}\|_{\bar{Y}'} = \|f\|_{Y'}$. Since $\bar{Y}$ is a closed subspace of the Banach space X , the space $\bar{Y}$ is a smooth uniformly convex Banach space. Thanks to Theorem 16.52 there exists a unique $u \in \bar{Y}$ such that $\|u\|_{\bar{Y}} = 1$ and $\bar{f} = \|\bar{f}\|_{\bar{Y}'} D \cdot \|_{\bar{Y}}(u)$. Let us now consider

$$\tilde{f} := \|f\|_{Y'} D \cdot \|_X(u),$$

which is in X' . Then $\tilde{f}$ is an extension of $\bar{f}$ and $\|\tilde{f}\|_{X'} = \|f\|_{Y'}$, according to (16.5).

It remains to prove the uniqueness part. Let $\check{f}$ be an extension of f to X such that $\|\check{f}\|_{X'} = \|f\|_{Y'}$. Then evaluating $\check{f}$ at the vector u above yields

$$\check{f}(u) = \bar{f}(u) = \|f\|_{Y'} = \|\check{f}\|_{X'}.$$

According to (the proof of) Theorem 16.52, $\check{f} = \|\check{f}\|_{X'} D \cdot \|_X(u) = \tilde{f}$. □

The previous result can actually be extended to a more general setting with a non-constructive proof using Zorn's lemma.

Theorem 16.54 (Hahn–Banach theorem, norm-preserving form). *If f is a bounded linear functional on a subspace of a normed linear space, then f extends to the whole space with preservation of norm.*

Note that, unlike the previous result, Theorem 16.54 does not contain any uniqueness part. It is named for Hans Hahn and Stefan Banach who proved this theorem independently in the late 1920s, though it was proved earlier (in 1912) by Eduard Helly. A key consequence is that the dual space of a nontrivial normed linear space is itself nontrivial. (Note: the norm is important for this. There exist topological vector spaces, e.g., L^p for $0 < p < 1$, with no non-zero continuous linear functionals.) Note that there are virtually no hypotheses beyond linearity and existence of a norm. In fact for some purposes a weaker version is useful. For X a vector space, we say that $p : X \rightarrow \mathbb{R}$ is sublinear if $p(x + y) \leq p(x) + p(y)$ and $p(\alpha x) = \alpha p(x)$ for $x, y \in X, \alpha \geq 0$.

Theorem 16.55 (Analytic Hahn–Banach theorem). *Let X be a vector space, $p : X \rightarrow \mathbb{R}$ a sublinear functional, S a subspace of X , and $f : S \rightarrow \mathbb{R}$ a linear function satisfying $f(x) \leq p(x)$ for all $x \in S$, then f can be extended to X so that the same inequality holds for all $x \in X$.*

Proof. It suffices to extend f to the space spanned by S and one element $x_0 \in X \setminus S$, preserving the inequality, since if we can do that we can complete the proof with Zorn's lemma.

We need to define $f(x_0)$ such that $f(tx_0 + s) \leq p(tx_0 + s)$ for all $t \in \mathbb{R}, s \in S$. The case $t = 0$ is known and it is easy to use homogeneity to restrict to $t = \pm 1$. Thus we need to find a value $f(x_0) \in \mathbb{R}$ such that

$$f(s) - p(-x_0 + s) \leq f(x_0) \leq p(x_0 + s) - f(s) \quad \text{for all } s \in S.$$

Now it is easy to check that for any $s_1, s_2 \in S$, $f(s_1) - p(-x_0 + s_1) \leq p(x_0 + s_2) - f(s_2)$, and so such an x_0 exists. □

16.7 Pointwise convergence under a uniform bound and a sequential Banach–Alaoglu theorem

We start this section with the following result.

Proposition 16.56 (Extension of pointwise convergence from a dense set). *Let X be a normed vector space and Y be a Banach space. Consider a dense subset A of X and $(u_n)_n$ a sequence in $\mathcal{L}_c(X, Y)$ such that*

- 1) $\sup_n \|u_n\|_{\mathcal{L}_c(X, Y)} < \infty$,
- 2) for any x in A , $(u_n(x))_n$ converges in Y .

Then there exists a unique u in $\mathcal{L}_c(X, Y)$ such that

$$u_n(x) \rightarrow u(x) \quad \text{as } n \rightarrow +\infty$$

in Y for every $x \in X$. Moreover

$$\|u\|_{\mathcal{L}_c(X, Y)} \leq \liminf_{n \rightarrow \infty} \|u_n\|_{\mathcal{L}_c(X, Y)}. \quad (16.7)$$

Proof. We can assume without loss of generality that $\sup_n \|u_n\| > 0$, otherwise the proof is straightforward. Let x in X and $\varepsilon > 0$. Since A is a dense subset of X , there exists $x' \in A$ such that

$$\|x - x'\|_X \leq \delta := \frac{\varepsilon}{3 \sup_n \|u_n\|_{\mathcal{L}_c(X, Y)}}.$$

Since $(u_n(x'))_n$ converges, it has the Cauchy property and there exists $n_\varepsilon \in \mathbb{N}$ such that

$$\|u_n(x') - u_m(x')\|_Y \leq \frac{\varepsilon}{3},$$

for any $n, m \geq n_\varepsilon$. For any $n, m \geq n_\varepsilon$, we have

$$\begin{aligned} \|u_n(x) - u_m(x)\|_Y &\leq \|u_n(x) - u_n(x')\|_Y + \|u_n(x') - u_m(x')\|_Y + \|u_m(x') - u_m(x)\|_Y, \\ &\leq (\|u_n\|_{\mathcal{L}_c(X, Y)} + \|u_m\|_{\mathcal{L}_c(X, Y)})\|x - x'\|_X + \frac{\varepsilon}{3}, \\ &\leq 2\delta \sup_n \|u_n\|_{\mathcal{L}_c(X, Y)} + \frac{\varepsilon}{3} \\ &\leq \varepsilon. \end{aligned}$$

Therefore the sequence $(u_n(x))_n$ has the Cauchy property. Since Y is a Banach space, it converges. We denote by $u(x)$ its limit. It is straightforward that u is linear. Moreover for any $x \in X$,

$$\|u(x)\|_Y = \lim_{n \rightarrow \infty} \|u_n(x)\|_Y \leq \liminf_{n \rightarrow \infty} \|u_n\|_{\mathcal{L}_c(X, Y)} \|x\|_X,$$

which yields (16.7). The uniqueness follows from the pointwise convergence on X . □

In the particular case where Y is $\mathbb{R}$ or $\mathbb{C}$, we denote by $X' := \mathcal{L}_c(X, Y)$, and we call X' the *topological dual space* of X . In that case, pointwise convergence on X is exactly weak* convergence in X' .

Corollary 16.57 (Boundedness of weak-star convergent sequences). *Let X be a Banach space. Then any weak* converging sequence in X' is bounded.*

Proof. If $(u_n)_n$ converges weak* in X' , then for every x in X the sequence $(u_n(x))_n$ converges, hence is bounded. Applying the Uniform Boundedness Principle, Theorem 16.46, to the family $(u_n)_n$ gives

$$\sup_n \|u_n\|_{X'} < \infty.$$

□

One advantage of the weak* convergence is that the following partial converse is available:

Theorem 16.58 (Sequential Banach–Alaoglu theorem). *Let X be a separable Banach space. Then any bounded sequence in X' admits a weak* converging subsequence.*

Proof. Let $(u_n)_n$ be a bounded sequence in X' . Let $(x_k)_{k \in \mathbb{N}}$ be a dense sequence in X . Using Cantor's diagonal argument, there exists a subsequence $(u_{n_j})_{j \geq 1}$ of $(u_n)_n$ such that

$$(u_{n_j}(x_k))_{j \geq 1}$$

converges for every k in $\mathbb{N}$. We then apply Proposition 16.56. □

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